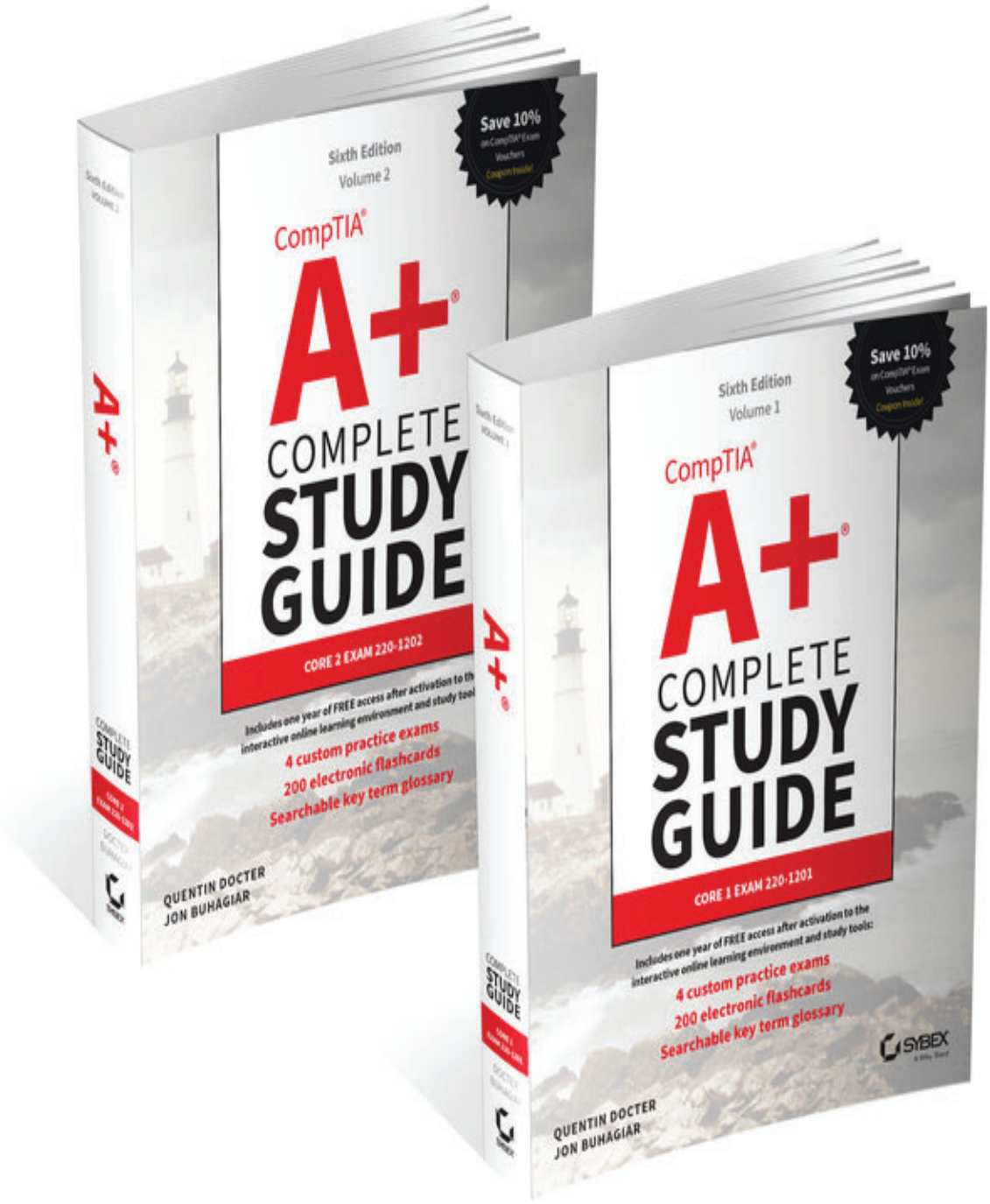

Save 10%
Sixth Edition
Volume 2
CompTIA®
A+®
COMPLETE
STUDY
GUIDE
CORE 2 EXAM 220-1202
4 custom practice exams
200 electronic flashcards
Searchable key term glossary
QUENTIN DOCTER
JON BUHAGIAR
Save 10%
Sixth Edition
Volume 1
CompTIA®
A+®
COMPLETE
STUDY
GUIDE
CORE 1 EXAM 220-1201
Includes one year of FREE access after activation to the interactive online learning environment and study tools:
4 custom practice exams
200 electronic flashcards
Searchable key term glossary
SYBEX
QUENTIN DOCTER
JON BUHAGIAR

About the pagination of this eBook

This eBook contains a multi-volume set.

To navigate this eBook by page number, you will need to use the volume number and the page number, separated by a hyphen.

For example, to go to page 5 of volume 1, type “1-5” in the Go box at the bottom of the screen and click "Go."

To go to page 5 of volume 2, type “2-5”… and so forth.

Take the Next Step in Your IT Career

Save 10% on Exam Vouchers*

(up to a $35 value)

*Some restrictions apply. See web page for details.

CompTIA®

Use coupon code WILEY10 during checkout. Redeeming the coupon code is easy:

1. **Go to www.comptiastore.com.**
2. **Browse Certification Vouchers and select the exam voucher you want.**
3. **Add the voucher to the cart (note that for A+ you will need a separate voucher for each exam).**
4. **Enter the code WILEY10 on the purchase screen, click Apply and then click Proceed to Checkout to continue and complete the payment process.**

SYBEX®

CompTIA®

A+® Complete

Study Guide

Sixth Edition

Volume 1: Core 1 Exam 220-1201

CompTIA®

A+® Complete

Study Guide

Volume 1: Core 1 Exam 220-1201

Sixth Edition

Quentin Docter

Jon Buhagiar

SYBEX®
A Wiley Brand

Copyright © 2025 by John Wiley & Sons, Inc. All rights reserved, including rights for text and data mining and training of artificial intelligence technologies or similar technologies.

Published by John Wiley & Sons, Inc., Hoboken, New Jersey.
Published simultaneously in Canada and the United Kingdom.

ISBNs for the 2-volume set: 9781394330034 (paperback), 9781394330058 (ePDF), 9781394330041 (ePub)

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning, or otherwise, except as permitted under Section 107 or 108 of the 1976 United States Copyright Act, without either the prior written permission of the Publisher, or authorization through payment of the appropriate per-copy fee to the Copyright Clearance Center, Inc., 222 Rosewood Drive, Danvers, MA 01923, (978) 750-8400, fax (978) 750-4470, or on the web at www.copyright.com. Requests to the Publisher for permission should be addressed to the Permissions Department, John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, (201) 748-6011, fax (201) 748-6008, or online at www.wiley.com/go/permission.

The manufacturer's authorized representative according to the EU General Product Safety Regulation is Wiley-VCH GmbH, Boschstr. 12, 69469 Weinheim, Germany, e-mail: Product_Safety@wiley.com.

Trademarks: Wiley, the Wiley logo, and Sybex are trademarks or registered trademarks of John Wiley & Sons, Inc. and/or its affiliates in the United States and other countries and may not be used without written permission. CompTIA, A+, and the CompTIA logo are trademarks or registered trademarks of CompTIA, Inc. All other trademarks are the property of their respective owners. John Wiley & Sons, Inc. is not associated with any product or vendor mentioned in this book.

Limit of Liability/Disclaimer of Warranty: While the publisher and authors have used their best efforts in preparing this book, they make no representations or warranties with respect to the accuracy or completeness of the contents of this book and specifically disclaim any implied warranties of merchantability or fitness for a particular purpose. No warranty may be created or extended by sales representatives or written sales materials. The advice and strategies contained herein may not be suitable for your situation. You should consult with a professional where appropriate. Further, readers should be aware that websites listed in this work may have changed or disappeared between when this work was written and when it is read. Neither the publisher nor authors shall be liable for any loss of profit or any other commercial damages, including but not limited to special, incidental, consequential, or other damages.

For general information on our other products and services, please contact our Customer Care Department within the United States at (800) 762-2974, outside the United States at (317) 572-3993 or fax (317) 572-4002. For product technical support, you can find answers to frequently asked questions or reach us via live chat at https://sybexsupport.wiley.com.

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print may not be available in electronic formats. For more information about Wiley products, visit our web site at www.wiley.com.

Library of Congress Control Number: 2025938071

Cover Design: Wiley
Cover Image: © Jeremy Woodhouse/Getty Images

For the fam.
—Quentin Docter

For my wife and son.
—Jon Buhagiar

Acknowledgments

As we were putting together this book, I was reminded of the proverb that begins "It takes a village . . ." That beginning definitely holds true for creating a book of this scope and size. From beginning to end, scores of dedicated professionals have focused on delivering the best book possible to you, the readers.

First, I need to thank my coauthor, Jon Buhagiar. I appreciate him diving in and dedicating himself to helping to produce an excellent book. I also need to give special thanks to our technical editor, Chris Crayton. He was meticulous and thorough, and challenged me to always find new and better ways to communicate complex concepts. His entire focus was on providing the best training material possible, and I doubt there's better in the business. Now, on to the rest of the team.

Kenyon Brown and Kim Wimpsett kept us on track and moving forward, which was a challenge at times. Saravanan Dakshinamurthy had the fun job of keeping us organized, which is akin to herding cats. Copyeditor Liz Welch reminded me yet again that I am no master of the English language and saved me from butchering it (too badly). Many thanks also go out to our proofreader, Nancy Carrasco, and our indexer, Tom Dinse. Without their great contributions, this book would not have made it into your hands.

On a personal note, I need to thank my family. My wife and kids are all incredibly supportive. Unfortunately, book writing as a side hustle while holding down a full-time job takes up a lot of time. I end up hiding in my office a lot, but they're always there for me, and I couldn't do it without them. Another huge thanks goes to my late grandpa, Joe, who got me into computers and taught me so many lessons I will never be able to repay. Finally, thanks to my friends who keep me relatively sane—Sean, Kurtis, Tim, John, Cory, Kevin, and others—and laugh at me when I tell them I spent my weekend writing about network cables and protocols.

—Quentin Docter

I would like to first thank my coauthor, Quentin Docter. Throughout the writing of this book, he helped me with his insight and his expertise in writing technical books. Without his words of wisdom and guidance, this book would not be the product it stands to be. I would also like to give special thanks to our technical editor, Chris Crayton. His thorough review of the material helped to identify many areas for us to elaborate on and polish.

I would also like to thank the many people who made this book possible: Kenyon Brown at Wiley Publishing, for giving me the opportunity to write this book and work with this wonderful team; Kim Wimpsett, for keeping us on track during the writing process; Pete Gaughan, who also kept us on track and organized during the publishing process; and our copyeditor, Liz Welch, for helping me use proper English. I'd also like to thank the many other people I've never met but who worked behind the scenes to make this book a success.

During the writing of this book, many others in my life supported me, and I'd like to take this opportunity to thank them as well. First and foremost, thanks to my wife and son for their support during the many evenings and weekends spent in my office at the computer. Finally, thanks to my friend and coworker, Bill, for encouraging me daily with his insight and jokes, as well as to all my other coworkers and friends. Thank you.

—Jon Buhagiar

About the Authors

Quentin Docter (A+, Network+, Tech+, Cloud Essentials +, MCSE, CCNA) is an IT author and consultant who started as a tech support agent for Packard Bell in 1994. Since then, he's worked in tech support, network administration, consulting, training, web development, and project management. During his career, he has achieved certifications from CompTIA, Microsoft, Cisco, Novell, and Sun Microsystems. He has written more than a dozen books for Sybex, including books on A+, Tech+, Cloud Essentials+, Server+, Windows, and Solaris 9 certifications, as well as PC hardware and maintenance.

Jon Buhagiar (Network+, A+, CCNA, MCSA, MCSE, BS/ITM) is an IT professional with two decades of experience in higher education. During the past 22 years he has been responsible for network operations at Pittsburgh Technical College and has led several projects, such as virtualization (server and desktop), VoIP, Microsoft 365, and many other projects supporting the quality of education at the college. He has achieved several certifications from Cisco, CompTIA, and Microsoft, and he has taught many of the certification paths. He is the author of several books, including Sybex's *CompTIA Network+ Review Guide: Exam N10-008* (2021) and *CCNA Certification Practice Tests: Exam 200-301* (2020).

About the Technical Editor

Chris Crayton is a technical consultant, trainer, author, and industry-leading technical editor. He has worked as a computer technology and networking instructor, information security director, network administrator, network engineer, and PC specialist. Chris has authored several print and online books on PC repair, CompTIA A+, CompTIA Security+, and Microsoft Windows. He has also served as technical editor and content contributor on numerous technical titles for several of the leading publishing companies. He holds numerous industry certifications, has been recognized with many professional and teaching awards, and has served as a state-level SkillsUSA final competition judge.

Contents at a Glance

Introduction		*xxvii*
Assessment Test for Exam 220-1201		*lv*
Answers to Assessment Test 220-1201		*lxiv*
Volume 1	**CompTIA A+ Core 1 Exam (220-1201)**	**1**
Chapter 1	Motherboards, Processors, and Memory	3
Chapter 2	Expansion Cards, Storage Devices, and Power Supplies	79
Chapter 3	Peripherals, Cables, and Connectors	141
Chapter 4	Laptop and Mobile Device Hardware	191
Chapter 5	Printers and Multifunction Devices	255
Chapter 6	Networking Fundamentals	323
Chapter 7	Introduction to TCP/IP	389
Chapter 8	Network Services	437
Chapter 9	Wireless and SOHO Networking	487
Chapter 10	Mobile Connectivity and Application Support	561
Chapter 11	Virtualization and Cloud Computing	647
Chapter 12	Core Hardware and Network Troubleshooting	697
Chapter 13	Mobile Device, Display, and Printer Troubleshooting	771
Appendix A	Answers to Review Questions	833
Appendix B	Answers to Performance-Based Questions	859
Index		*869*

Contents

Introduction *xxvii*

Assessment Test for Exam 220-1201 *lv*

Answers to Assessment Test 220-1201 *lxiv*

Volume 1 CompTIA A+ Core 1 Exam (220-1201) 1

Chapter 1 Motherboards, Processors, and Memory 3

Understanding Motherboards 6
- Motherboard Form Factors 6
- System Board Components 8

Understanding Processors 42
- CPU Architecture 43
- CPU Characteristics 46

Understanding Memory 50
- Important Memory Terms 50
- Types of Memory 53
- Memory Packaging 59

Understanding Cooling Systems 61
- Case Cooling and Fans 63
- CPU Cooling 66
- Cooling Other Components 70

Summary 71
Exam Essentials 71
Review Questions 73
Performance-Based Question 1 77
Performance-Based Question 2 77

Chapter 2 Expansion Cards, Storage Devices, and Power Supplies 79

Installing and Configuring Expansion Cards 81
- Video 82
- Multimedia 84
- Network Interface Card 86
- Input/Output 88
- Adapter Configuration 89

Understanding Storage Devices 94
- Conventional Hard Disk Drives 94
- Solid-State Drives 99
- RAID 109

Removable Storage and Media 111
Installing, Removing, and Configuring Storage Devices 116
Understanding Power Supplies 119
Power Supply Input 120
Power Supply Output and Wattage Ratings 121
Energy Efficiency 122
Power Connectors 123
Modular Power Supplies 126
Redundant Power Supplies 127
Replacing Power Supplies 130
AC Adapters as Power Supplies 131
Summary 132
Exam Essentials 132
Review Questions 134
Performance-Based Question 139

Chapter 3 Peripherals, Cables, and Connectors 141

Understanding Peripheral Devices 143
Video Devices 144
Audio Devices 157
Input and Output Devices 159
Storage Devices 162
Understanding Cables and Connectors 162
Peripheral Cables and Connectors 163
Serial Ports 172
Video Cables and Connectors 173
Hard Drive Cables and Connectors 181
Summary 184
Exam Essentials 185
Review Questions 186
Performance-Based Question 190

Chapter 4 Laptop and Mobile Device Hardware 191

Working with Laptop and Mobile Device Hardware 194
Understanding the Differences Between Device Types 194
Disassembling and Reassembling Laptops 199
Installing and Configuring Laptop Hardware 202
Setting Up and Configuring Accessories and Ports 244
Summary 249
Exam Essentials 249
Review Questions 250
Performance-Based Question 254

Chapter 5 Printers and Multifunction Devices 255

Understanding Print Technologies and Imaging Processes 257
Impact Printers 258
Inkjet Printers 260
Laser Printers 269
Thermal Printers 281
Installing and Maintaining Printers 282
Installing and Sharing Local Printers 282
Installing and Sharing Networked Printers 298
Performing Appropriate Printer Maintenance 310
Summary 315
Exam Essentials 316
Review Questions 318
Performance-Based Question 322

Chapter 6 Networking Fundamentals 323

Understanding Networking Principles 325
Network Types 326
Primary Network Components 333
Network Resource Access 337
Network Topologies 340
Rules of Communication 344
Identifying Common Network Hardware 348
Network Interface Cards 348
Cables and Connectors 353
Networking Components 371
Summary 381
Exam Essentials 382
Review Questions 384
Performance-Based Question 388

Chapter 7 Introduction to TCP/IP 389

Understanding TCP/IP 391
TCP/IP Structure 392
Understanding IP Addressing 403
IPv6 420
Understanding Virtual Networks 423
Virtual Local Area Networks 423
Virtual Private Networks 425
Summary 429
Exam Essentials 430
Review Questions 432
Performance-Based Question 436

Chapter 8 Network Services 437

Understanding Network Services 439
Server Roles 439
Internet Appliances 465
Legacy/Embedded Systems 473
Internet of Things Devices 475
Summary 480
Exam Essentials 480
Review Questions 482
Performance-Based Question 486

Chapter 9 Wireless and SOHO Networking 487

Understanding Wireless Networking Technologies 489
Understanding 802.11 Networking 490
Bluetooth Networking 505
Radio Frequency Networking Standards 508
Installing and Configuring SOHO Networks 511
Keys to Planning a Network 511
Choosing an Internet Connection 514
Choosing Internal Network Connections 526
Installing the Network Infrastructure 530
Configuring Wireless Routers and Access Points 539
Summary 554
Exam Essentials 554
Review Questions 556
Performance-Based Question 560

Chapter 10 Mobile Connectivity and Application Support 561

Understanding Mobile Connectivity 563
Understanding Cellular Networking Standards 565
Using Cellular Data Connections 567
Establishing Wi-Fi Connectivity 577
Establishing Bluetooth Connectivity 593
Understanding Mobile App Support 606
Understanding Location Services 606
Understanding Mobile Device and App Management 613
Configuring Mobile Device Synchronization 626
Summary 639
Exam Essentials 639
Review Questions 641
Performance-Based Question 645

Chapter 11 Virtualization and Cloud Computing 647

Understanding Virtualization 649
The Purpose of Virtual Machines 650

The Hypervisor 652
Containers 655
Client-Side Virtualization Requirements 656
Understanding Cloud Computing 668
Types of Clouds 668
Cloud Services 669
Important Cloud Characteristics 671
Using Cloud-Based Services 674
Summary 688
Exam Essentials 689
Review Questions 691
Performance-Based Question 695

Chapter 12 Core Hardware and Network Troubleshooting 697

Understanding Troubleshooting Best Practices 701
Troubleshooting Tricks and Tips 701
Structured Troubleshooting Methodology 702
Troubleshooting Motherboards, RAM, CPUs, and Power Problems 704
Identifying General Hardware Symptoms and Causes 705
Identifying BIOS/UEFI and POST Routine Problems 710
Identifying Motherboard and CPU Problems 714
Identifying Memory Issues 716
Identifying Power Supply Problems 719
Troubleshooting Drives and RAID Arrays 722
Lights and Sounds 723
Devices Not Found 724
Performance Issues 725
S.M.A.R.T. Diagnostics 730
RAID Issues 733
Optical Drive Issues 734
Troubleshooting Networking Problems 735
Using Network Troubleshooting Tools 735
Resolving Connectivity Issues 756
Summary 763
Exam Essentials 763
Review Questions 766
Performance-Based Question 770

Chapter 13 Mobile Device, Display, and Printer Troubleshooting 771

Troubleshooting Video, Projector, and Display Issues 774
Video Input Issues 775
Video Image Problems 776

Other Display Issues 781
Troubleshooting Common Mobile Device Issues 783
Power and Heat Issues 784
Input Problems 786
Connectivity Issues 788
Physical Damage and Malware 792
Other Mobile Device Issues 796
Troubleshooting Printer Issues 799
Impact Printer Issues 800
Inkjet Printer Problems 803
Laser Printer Problems 807
Managing Print Jobs 817
Summary 823
Exam Essentials 824
Review Questions 827
Performance-Based Question 831

Appendix A Answers to Review Questions 833

Chapter 1: Motherboards, Processors, and Memory 834
Chapter 2: Expansion Cards, Storage Devices, and Power Supplies 836
Chapter 3: Peripherals, Cables, and Connectors 838
Chapter 4: Laptop and Mobile Device Hardware 839
Chapter 5: Printers and Multifunction Devices 841
Chapter 6: Networking Fundamentals 843
Chapter 7: Introduction toTCP/IP 845
Chapter 8: Network Services 847
Chapter 9: Wireless and SOHO Networking 849
Chapter 10: Mobile Connectivity and Application Support 851
Chapter 11: Virtualization and Cloud Computing 853
Chapter 12: Core Hardware and Network Troubleshooting 854
Chapter 13: Mobile Device, Display, and Printer Troubleshooting 856

Appendix B Answers to Performance-Based Questions 859

Chapter 1: Motherboards, Processors, and Memory 860
Chapter 2: Expansion Cards, Storage Devices, and Power Supplies 861
Chapter 3: Peripherals, Cables, and Connectors 861
Chapter 4: Laptop and Mobile Device Hardware 862
Chapter 5: Printers and Multifunction Devices 863
Chapter 6: Networking Fundamentals 863
Chapter 7: Introduction to TCP/IP 864

Chapter 8: Network Services 865
Chapter 9: Wireless and SOHO Networking 866
Chapter 10: Mobile Connectivity and Application Support 866
Chapter 11: Virtualization and Cloud Computing 867
Chapter 12: Core Hardware and Network Troubleshooting 867
Chapter 13: Mobile Device, Display, and Printer Troubleshooting 868

Index *869*

Table of Exercises

Exercise 1.1 Entering the UEFI from Windows 11 34
Exercise 2.1 Updating a Device Driver in Windows 11. 91
Exercise 2.2 Removing an Internal Storage Device 117
Exercise 2.3 Installing an Internal Storage Device. 118
Exercise 2.4 Removing a Power Supply 131
Exercise 3.1 Changing the Refresh Rate in Windows 11 147
Exercise 3.2 Changing the Settings for Multiple Monitors 151
Exercise 4.1 Removing Speakers from a Laptop 204
Exercise 4.2 Removing the Display Assembly 210
Exercise 4.3 Removing the Display Panel. 211
Exercise 4.4 Removing the Motherboard from a Laptop. 214
Exercise 4.5 Replacing Laptop Memory 218
Exercise 4.6 Removing an M.2 SSD from a Laptop 222
Exercise 4.7 Removing a Laptop Keyboard 225
Exercise 4.8 Disabling a Touchpad in Windows 11 227
Exercise 4.9 Disabling a Track Point in Windows 11 229
Exercise 4.10 Removing an Internal Laptop Battery 234
Exercise 4.11 Removing the System Fan 236
Exercise 4.12 Removing the CPU Heat Sink 237
Exercise 4.13 Removing the Wireless NIC 238
Exercise 4.14 Removing the CMOS Battery 240
Exercise 4.15 Flashing the System BIOS 241
Exercise 5.1 Identifying the Parts of an Inkjet Printer 267
Exercise 5.2 Installing a USB Printer in Windows 11 297
Exercise 5.3 Installing a TCP/IP Printer in Windows 11. 299
Exercise 5.4 Determining if Bonjour Is Installed in Windows 301
Exercise 5.5 Scanning a Document to Google Drive 308
Exercise 5.6 Using an Inkjet Cleaning Solution 314
Exercise 6.1 Pricing Network Cables. 371
Exercise 7.1 Changing IP Address Assignment in Windows 11. 403
Exercise 7.2 Setting Up Windows 11 VPN 427
Exercise 8.1 Configuring Windows 11 to Use a Proxy Server 470
Exercise 9.1 The Cost of Networking 529

Exercise 9.2 Installing an Internal NIC in Windows 11 530
Exercise 10.1 Connecting an iPhone to a Wi-Fi Network 578
Exercise 10.2 Connecting an Android Phone to a Wi-Fi Network 580
Exercise 10.3 Disabling Cellular Use for Data Networking on an iPhone 582
Exercise 10.4 Disabling Cellular Use for Data Networking in Android OS 585
Exercise 10.5 Setting Up a VPN in Android 590
Exercise 10.6 Pairing an Android Phone with a Windows Laptop 594
Exercise 10.7 Configuring Location Services in iOS 610
Exercise 10.8 Configuring Location Services in Android 611
Exercise 10.9 Email Account Configuration on an iPhone 616
Exercise 10.10 Email Account Configuration in Android 619
Exercise 11.1 Enabling Hyper-V in Windows 11 653
Exercise 11.2 Installing VirtualBox and Lubuntu on Windows 11 661
Exercise 11.3 Using Google's Cloud Services 676
Exercise 11.4 Creating a Free AWS Account 678
Exercise 11.5 Creating a Virtual Server in AWS 681
Exercise 12.1 Troubleshooting Practice 721
Exercise 12.2 Viewing and Managing Storage Space 726
Exercise 12.3 Renewing an IP Address in Windows 11 743
Exercise 12.4 Renewing an IP Address from the Command Line 745
Exercise 12.5 Using the `net share` Command in Windows 751
Exercise 13.1 Stopping and Restarting the Print Spooler in Windows 11 820

Introduction

Welcome to the *CompTIA® A+® Complete Study Guide Volume 1: Core 1 Exam 220-1201 Sixth Edition*. This is the sixth edition of our best-selling study guide for the A+ certification sponsored by CompTIA (Computing Technology Industry Association). Thank you for choosing us to help you on your journey toward certification!

This book is written at an intermediate technical level; we assume that you already know how to *use* a personal computer and its basic peripherals, such as USB devices and printers, but we also recognize that you may be learning how to *service* some of that computer equipment for the first time. The exams cover basic computer service topics as well as more advanced issues, and they cover topics that anyone already working as a technician should be familiar with. The exams are designed to test you on these topics in order to certify that you have enough knowledge to fix and upgrade some of the most widely used types of personal computers and operating systems.

In addition to the information in the chapters, we've included a lot of extra material to help your study prep. At the end of each chapter is a list of exam essentials to know as well as 20 review questions to give you a taste of what it's like to take the exams. In addition, there are eight bonus exams of at least 75 questions each. Finally, there are flashcards designed to help your recall. Before you dive into those, though, we recommend you take the assessment test at the end of this introduction to gauge your current knowledge level.

Don't just study the questions and answers—the questions on the actual exams will be different from the practice ones included with this book. The exams are designed to test your knowledge of a concept or objective, so use this book to learn the objective behind the question. That said, we're confident that if you can do well on our quizzes, you will be well equipped to take the real exam.

This book covers more than just the exams, however. We believe in providing our students with a foundation of IT knowledge that will prepare them for real jobs, not just to pass a test. After all, life is not a multiple-choice test with the answers clearly laid out in front of you!

For experienced IT professionals, you can use the book to fill in the gaps in your current computer service knowledge. You may find, as many PC technicians have, that being well versed in all the technical aspects of hardware and operating systems is not enough to provide a satisfactory level of support—you must also have customer-relations skills, understand safety concepts, and be familiar with change management and environmental impacts and controls. We include helpful hints in all of these areas.

What Is CompTIA A+ Certification?

The A+ certification program was developed by CompTIA to provide an industrywide means of certifying the competency of computer service technicians. The A+ certification is granted to those who have attained the level of knowledge and troubleshooting skills that are needed

to provide capable support in the field of personal computers. It is similar to other certifications in the computer industry, such as the Cisco Certified Support Technician (CCST) program and the Microsoft certification programs. The theory behind these certifications is that if you need to have service performed on any of their products, you would sooner call a technician who has been certified in one of the appropriate certification programs than just call the first "expert" in the phone book.

The A+ certification program was created to offer a wide-ranging certification, in the sense that it is intended to certify competence with personal computers and mobile devices from many different makers/vendors. You must pass two tests to become A+ certified:

- The Core 1 220-1201 exam, which covers basic computer concepts, PC hardware, basic networking, mobile devices, and hardware troubleshooting
- The Core 2 220-1202 exam, which covers operating systems, security, software troubleshooting, and operational procedures

You don't have to take the 220-1201 and the 220-1202 exams at the same time. However, the A+ certification is not awarded until you've passed both tests.

Why Become A+ Certified?

There are several good reasons to get your A+ certification. The CompTIA Candidate's Information packet lists five major benefits:

- It demonstrates proof of professional achievement.
- It increases your marketability.
- It provides greater opportunity for advancement in your field.
- It is increasingly a requirement for some types of advanced training.
- It raises customer confidence in you and your company's services.

Provides Proof of Professional Achievement The A+ certification is a status symbol in the computer service industry. Organizations that include members of the computer service industry recognize the benefits of A+ certification and push for their members to become certified. And more people every day are putting the "A+ Certified Technician" emblem on their business cards.

Increases Your Marketability A+ certification makes individuals more marketable to potential employers. A+ certified employees also may receive a higher base salary because employers won't have to spend as much money on vendor-specific training.

Provides Opportunity for Advancement Most raises and advancements are based on performance. A+ certified employees work faster and more efficiently and are thus more productive. The more productive employees are, the more money they make for their company. And, of course, the more money they make for the company, the more valuable they are to the company. So, if an employee is A+ certified, their chances of being promoted are greater.

Fulfills Training Requirements Most major computer hardware vendors recognize A+ certification. Some of these vendors apply A+ certification toward prerequisites in their own respective certification programs, which has the side benefit of reducing training costs for employers.

Raises Customer Confidence As the A+ Certified Technician moniker becomes better known among computer owners, more of them will realize that the A+ technician is more qualified to work on their computer equipment than a noncertified technician.

How to Become A+ Certified

A+ certification is available to anyone who passes two exams. You don't have to work for any particular company. It's not a secret society. It is, however, an elite group. To become A+ certified, you must do two things:

- Pass the Core 1 220-1201 exam.
- Pass the Core 2 220-1202 exam.

The exams can be taken at any Pearson VUE testing center. If you pass both exams, you will get a certificate in the mail from CompTIA saying that you have passed, and you will also receive a lapel pin and business card.

To register for the tests, go to `www.pearsonvue.com/comptia`. You'll be asked for your name, Social Security number (an optional number may be assigned if you don't wish to provide your Social Security number), mailing address, phone number, employer, when and where you want to take the test, and your credit card number. (Payment arrangements must be made at the time of registration.)

Tips for Taking the A+ Exam

Here are some general tips for taking your exam successfully:

- Bring two forms of ID with you. One must be a photo ID, such as a driver's license. The other can be a major credit card or a passport. Both forms must include a signature.
- Arrive early at the exam center so that you can relax and review your study materials, particularly tables and lists of exam-related information. When you enter the testing room, you will need to leave everything outside; you won't be able to bring any materials into the testing area.
- Read the questions carefully. Don't be tempted to jump to an early conclusion. Make sure that you know exactly what each question is asking.
- Don't leave any unanswered questions. Unanswered questions are scored against you. Some questions will have multiple correct responses. When a question has more than one correct answer, a message at the bottom of the screen will prompt you to, for example, choose two. Be sure to read the messages displayed to know how many correct answers you must choose.

- When answering multiple-choice questions that you're not sure about, use a process of elimination to get rid of the obviously incorrect answers first. Doing so will improve your odds if you need to make an educated guess.
- For the hands-on questions where you need to complete a task, double-check that you have finished all parts of it, including checking all tabs available to you if they give you an interface to click on. Also, understand that these questions will take longer than multiple-choice ones. Spend the time needed on them; you will be able to make up time on the shorter multiple-choice questions.
- On form-based tests (nonadaptive), because the hard questions will take the most time, save them for last. You can move forward and backward through the exam.
- For the latest pricing on the exams and updates to the registration procedures, visit CompTIA's website at `www.comptia.org`.

Who Should Read This Book?

If you are one of the many people who want to pass the A+ exams, and pass them confidently, then you should buy this book and use it to study for the exams.

This book was written to prepare you for the challenges of the real IT world, not just to pass the A+ exams. This study guide will do that by describing in detail the concepts on which you'll be tested.

What Does This Book Cover?

CompTIA A+ Complete Study Guide: Exam 220-1201 and Exam 220-1202 covers everything you need to know to pass the CompTIA A+ exams.

Volume 1 has 13 chapters, and it covers all the topics on which you will be tested for Exam 220-1201:

Chapter 1: Motherboards, Processors, and Memory Chapter 1 details the characteristics of motherboards and their built-in components. The CPU, RAM, and BIOS, which are attached to the motherboard, are also presented in Chapter 1.

Chapter 2: Expansion Cards, Storage Devices, and Power Supplies Chapter 2 presents internal expansion options and popular forms of storage devices in use today, including traditional hard drives, solid-state drives, flash drives, and memory cards. Capacities, form factors, and the makeup of these components are also discussed. Finally, this chapter looks at providing power to computer components.

Chapter 3: Peripherals, Cables, and Connectors Chapter 3 covers installation and characteristics of external peripheral devices, the ports they use, cables required, and the connectors used to attach them.

Chapter 4: Laptop and Mobile Device Hardware Chapter 4 covers topics such as mobile device–specific hardware, components within a laptop display, and mobile device features.

Chapter 5: Printers and Multifunction Devices Chapter 5 starts by discussing different printing technologies, such as impact, inkjet, laser, and thermal printers. It then moves on to cover installing printers and performing printer maintenance.

Chapter 6: Networking Fundamentals Chapter 6 covers characteristics of networking cable types and connectors, network devices, networking concepts, and network components.

Chapter 7: Introduction to TCP/IP Chapter 7 details the most common network protocol in use today. It covers TCP/IP structure, addressing (including IPv6), common protocols in the suite, and virtual networks.

Chapter 8: Network Services Chapter 8 focuses on the various services used in networks. It covers the roles of DNS, DHCP, Syslog, file servers, and print servers. In addition, it covers Internet appliances, legacy/embedded systems, and Internet of Things (IoT) devices.

Chapter 9: Wireless and SOHO Networking Chapter 9 contains two main sections. The first is on wireless networking standards and security, and the second discusses setting up a small office, home office (SOHO) network and choosing an Internet connection type.

Chapter 10: Mobile Connectivity and Application Support Chapter 10 details the specific features of mobile devices that are unique to their formats. This chapter provides extensive hands-on steps for configuring a variety of connectivity options and application support on these devices.

Chapter 11: Virtualization and Cloud Computing Chapter 11 starts off by covering virtualization concepts, as well as the components related to virtualization such as hypervisors and containers. The chapter then covers the concepts and components of cloud computing.

Chapter 12: Core Hardware and Network Troubleshooting Chapter 12 introduces some tips and tricks for resolving computer problems, which every technician should know. It then covers troubleshooting core hardware issues related to the motherboard, CPU, RAM, and power. Then, it explores troubleshooting storage drives and RAID arrays, as well as networking tools and troubleshooting.

Chapter 13: Mobile Device, Display, and Printer Troubleshooting Chapter 13 finishes the troubleshooting section, including video issues, mobile device issues, and printers.

Volume 2 includes 10 chapters, and it covers all the topics on which you will be tested for Exam 220-1202.

What's Included in the Book

We've included several learning tools throughout the book:

Assessment Tests At the end of this introduction is an assessment test that you can use to check your readiness for the exam. Take this before you start reading the book; it will help you determine the areas on which you might need to brush up. The answers to the assessment test questions appear on a separate page after the last question of the test. Each answer includes an explanation and the chapter the material appears in.

Objective Map and Opening List of Objectives In this introduction, we have included a detailed exam objective map that shows you where each exam objective is covered. In addition, each chapter opens with a list of the exam objectives it covers. Use these resources to see exactly where each exam topic is covered.

Exam Essentials Each chapter, just after the summary, includes a number of "exam essentials." These are the key topics that you should focus on when preparing for the exam.

Chapter Review Questions To test your knowledge as you progress through the book, each chapter ends with review questions. As you finish each chapter, answer the review questions and then check your answers—the correct answers and explanations are in Appendix A. You can go back to reread the section that deals with each question you got wrong to ensure that you answer correctly the next time that you're tested on the material.

Performance-Based Questions The A+ exams may ask you to perform tasks to correctly answer a question. This may mean dragging and dropping items into the correct order or clicking the correct area of an image or screenshot. While it's hard to simulate those in a paper book, the end of each chapter features a performance-based question. These questions will have you think through a topic presented in the chapter, and then perform a task. The correct answers are in Appendix B.

Interactive Online Learning Environment and Test Bank

The interactive online learning environment that accompanies *CompTIA A+ Complete Study Guide: Exam 220-1201 and Exam 220-1202* provides a test bank with study tools to help you prepare for the certification exams and increase your chances of passing them the first time! The test bank includes the following elements:

Sample Tests All the questions and answers in this book are provided, including the assessment tests, which you'll find at the end of this introduction, and the review questions at the end of each chapter. In addition, there are eight practice exams. Use these questions to test your knowledge of the study guide material. The online test bank runs on multiple devices.

Flashcards Four sets of questions are provided in digital flashcard format (a question followed by a single correct answer). You can use the flashcards to reinforce your learning and provide last-minute test prep before the exam.

Glossary The key terms from this book and their definitions are available as a fully searchable PDF.

eBooks Read eBook versions of this Study Guide in multiple formats on your favorite device.

Audio Instruction Over 2 hours of recorded instruction from the authors to help you study for the certification exam with confidence.

Go to `www.wiley.com/go/sybextestprep` to register and gain access to this interactive online learning environment and test bank with study tools.

Like all exams, the CompTIA A+ certification is updated periodically and may eventually be retired or replaced. At some point after CompTIA is no longer offering this exam, the old editions of our books and online tools will be retired. If you have purchased this book after the exam was retired, or are attempting to register in the Sybex online learning environment after the exam was retired, please know that we make no guarantees that this exam's online Sybex tools will be available once the exam is no longer available.

How to Use This Book

If you want a solid foundation for preparing for the A+ exams, this is the book for you. We've spent countless hours putting together this book with the intention of helping you prepare for the exams.

This book is loaded with valuable information, and you will get the most out of your study time if you understand how we put the book together. Here's a list that describes how to approach studying:

1. Take the assessment tests immediately following this introduction. It's okay if you don't know any of the answers—that's what this book is for. Carefully read over the explanations for any questions you get wrong, and make note of the chapters where that material is covered.
2. Study each chapter carefully, making sure you fully understand the information and the exam objectives listed at the beginning of the chapter. Again, pay extra-close attention to any chapter that includes material covered in questions you missed on the assessment test.

3. Read over the Summary and Exam Essentials. These will highlight the sections from the chapter you need to be familiar with before sitting for the exam.
4. Answer all the review questions at the end of each chapter. Specifically note any questions that confuse you, and study the corresponding sections of the book again. Don't just skim these questions. Make sure that you understand each answer completely.
5. Go over the electronic flashcards. These help you to prepare for the latest A+ exams, and they're really great study tools.
6. Take the practice exams.

Performance-Based Questions

CompTIA includes performance-based questions on the A+ exams. These are not the traditional multiple-choice questions with which you're probably familiar. These questions require the candidate to know how to perform a specific task or series of tasks. The candidate will be presented with a scenario and will be asked to complete a task. They will be taken to a simulated environment where they will have to perform a series of steps and will be graded on how well they complete the task.

The Sybex test engine does not include performance-based questions. However, at the end of each chapter, we have included a section called "Performance-Based Question," which is designed to measure how well you understood the chapter's topics. Some simply ask you to complete a task for which there is only one correct response. Others are more subjective, with multiple ways to complete the task. We provide the most logical or practical solution in Appendix B. Note that these questions may cover topic areas not covered in the actual A+ performance-based questions. However, we feel that being able to think logically is a great way to learn.

The CompTIA A+ Exam Objectives for Core 1 220-1201

The A+ exams consist of the Core 1 220-1201 exam and the Core 2 220-1202 exam. The following are the detailed exam objectives for the Core 1 220-1201 exam.

Exam objectives are subject to change at any time without prior notice and at CompTIA's sole discretion. Please visit the A+ Certification page of CompTIA's website (`https://comptia.org/certifications/a`) for the most current listing of exam objectives.

A+ Certification Exam Objectives: 220-1201

The following table lists the domains measured by this examination and the extent to which they are represented on the exam:

Domain	Percentage of exam
1.0 Mobile Devices	13%
2.0 Networking	23%
3.0 Hardware	25%
4.0 Virtualization and Cloud Computing	11%
5.0 Hardware and Network Troubleshooting	28%
Total	100%

Objective Map

The following table lists where you can find the objectives covered in this book.

Objective	Chapter(s)
1.0 Mobile Devices	
1.1 Given a scenario, monitor mobile device hardware and use appropriate replacement techniques.	4

- Battery
- Keyboard/keys
- Random-access memory (RAM)
- Hard disk drive (HDD)/solid-state drive (SSD)
- Wireless cards
- Physical privacy and security components
  - Biometrics
  - Near-field scanner feature
- Wi-Fi antenna connector/placement
- Camera/webcam
- Microphone

Objective	Chapter(s)
1.2 Compare and contrast accessories and connectivity options for mobile devices.	4

▪ Connection methods
  ▪ Universal Serial Bus (USB)/USB-C/microUSB/miniUSB
  ▪ Lightning
  ▪ Near-field communication (NFC)
  ▪ Bluetooth
  ▪ Tethering/hotspot
▪ Accessories
  ▪ Stylus
  ▪ Headsets
  ▪ Speakers
  ▪ Webcam
▪ Docking station
▪ Port replicator
▪ Trackpad/drawing pad/track points	

1.3 Given a scenario, configure basic mobile device network connectivity and provide application support.	10

▪ Wireless/cellular data network (enable/disable)
  ▪ 3G/4G/5G
  ▪ Hotspot
  ▪ Wi-Fi
  ▪ Subscriber Identity Module (SIM)/eSIM	

Objective	Chapter(s)

▪ Bluetooth
  ▪ Enable Bluetooth
  ▪ Enable pairing
  ▪ Find a device for pairing
  ▪ Enter the appropriate personal identification number (PIN) code
  ▪ Test connectivity
▪ Location services
  ▪ Global positioning system (GPS) services
  ▪ Cellular location services
▪ Mobile device management (MDM)
  ▪ Device configurations
  ▪ Corporate
  ▪ Bring your own device (BYOD)
  ▪ Policy enforcement
  ▪ Corporate applications
▪ Mobile device synchronization
  ▪ Recognizing data caps
  ▪ Calendar
  ▪ Contacts
  ▪ Business applications
  ▪ Mail
  ▪ Cloud storage	

Objective	Chapter(s)
2.0 Networking	
2.1 Compare and contrast Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) ports, protocols, and their purposes.	7

▪ Ports and protocols
  ▪ 20-21 – File Transfer Protocol (FTP)
  ▪ 22 – Secure Shell (SSH)
  ▪ 23 – Telnet
  ▪ 25 – Simple Mail Transfer Protocol (SMTP)
  ▪ 53 – Domain Name System (DNS)
  ▪ 67/68 – Dynamic Host Configuration Protocol (DHCP)
  ▪ 80 – Hypertext Transfer Protocol (HTTP)
  ▪ 110 – Post Office Protocol 3 (POP3)
  ▪ 143 – Internet Mail Access Protocol (IMAP)
  ▪ 137-139 Network Basic Input/Output System (NetBIOS)/NetBIOS over TCP/IP (NetBT)
  ▪ 389 – Lightweight Directory Access Protocol (LDAP)
  ▪ 443 – Hypertext Transfer Protocol Secure (HTTPS)
  ▪ 445 – Server Message Block (SMB)/Common Internet File System (CIFS)
  ▪ 3389 – Remote Desktop Protocol (RDP)
▪ TCP vs. UDP	

2.2 Explain wireless networking technologies.	9

▪ Frequencies
  ▪ 2.4GHz
  ▪ 5GHz
  ▪ 6GHz
▪ Channels
  ▪ Regulations
  ▪ Channel selection	

Objective	Chapter(s)
▪ Widths ▪ Frequencies ▪ Bands ▪ Bluetooth ▪ 802.11 standards ▪ NFC ▪ Radio-frequency identification (RFID)	
2.3 Summarize services provided by networked hosts. ▪ Server roles ▪ DNS ▪ DHCP ▪ Fileshare ▪ Print servers ▪ Mail servers ▪ Syslog ▪ Web servers ▪ Authentication, Authorization, and Accounting (AAA) ▪ Database servers ▪ Network Time Protocol (NTP) ▪ Internet appliances ▪ Spam gateways ▪ Unified threat management (UTM) ▪ Load balancers ▪ Proxy servers ▪ Legacy/embedded systems ▪ Supervisory control and data acquisition (SCADA) ▪ Internet of Things (IoT) devices	8

Objective	Chapter(s)
2.4 Explain common network configuration concepts.	7, 8

▪ DNS
  ▪ A
  ▪ AAAA
  ▪ Canonical Name (CNAME)
  ▪ Mail exchanger (MX)
  ▪ Text (TXT)
  ▪ Spam management
  ▪ DomainKeys Identified Mail (DKIM)
  ▪ Sender Policy Framework (SPF)
  ▪ Domain-based Message Authentication, Reporting, and Conformance (DMARC)
▪ DHCP
  ▪ Leases
  ▪ Reservations
  ▪ Scope
  ▪ Exclusions
▪ Virtual LAN [local area network] (VLAN)
▪ Virtual private network (VPN)	

2.5 Compare and contrast common networking hardware devices.	6

▪ Routers
▪ Switches
  ▪ Managed
  ▪ Unmanaged	

Objective	Chapter(s)
▪ Access points	
▪ Patch panel	
▪ Firewall	
▪ Power over Ethernet (PoE)	
▪ Injectors	
▪ Switch	
▪ PoE standards	
▪ Cable modem	
▪ Digital subscriber line (DSL)	
▪ Optical network terminal (ONT)	
▪ Network interface card (NIC)	
▪ Physical media access control (MAC) address	
2.6 Given a scenario, configure basic wired/wireless small office/home office (SOHO) networks.	7
▪ Internet Protocol (IP) addressing	
▪ IPv4	
▪ Private addresses	
▪ Public addresses	
▪ IPv6	
▪ Automatic Private IP Addressing (APIPA)	
▪ Static	
▪ Dynamic	
▪ Subnet mask	
▪ Gateway	

Objective	Chapter(s)
2.7 Compare and contrast internet connection types, network types, and their characteristics.	6, 9

▪ Internet connection types
  ▪ Satellite
  ▪ Fiber
  ▪ Cable
  ▪ DSL
  ▪ Cellular
  ▪ Wireless internet service provider (WISP)
▪ Network types
  ▪ LAN
  ▪ Wide area network (WAN)
  ▪ Personal area network (PAN)
  ▪ Metropolitan area network (MAN)
  ▪ Storage area network (SAN)
  ▪ Wireless local area network (WLAN)	

2.8 Explain networking tools and their purposes.	12
▪ Crimper ▪ Cable stripper ▪ Wi-Fi analyzer ▪ Toner probe ▪ Punchdown tool ▪ Cable tester ▪ Loopback plug ▪ Network tap	

Objective	Chapter(s)
3.0 Hardware	
3.1 Compare and contrast display components and attributes.	3

- Types
  - Liquid crystal display (LCD)
    - In-plane switching (IPS)
    - Twisted nematic (TN)
    - Vertical alignment (VA)
  - Organic light-emitting diode (OLED)
  - Mini light-emitting diode (Mini-LED)
- Touch screen/digitizer
- Inverter
- Attributes
  - Pixel density
  - Refresh rates
  - Screen resolution
  - Color gamut

3.2 Summarize basic cable types and their connectors, features, and purposes. 3, 6

- Network cables
  - Copper
    - Categories
      - T568A/T568B standards
    - Coaxial
    - Shielded twisted pair
      - Direct burial
    - Unshielded twisted pair
  - Plenum-rated
  - Optical
    - Single-mode
    - Multimode

Objective	Chapter(s)

- Peripheral cables
  - USB 2.0
  - USB 3.0
  - Serial
  - Thunderbolt
- Video cables
  - High-definition Multimedia Interface (HDMI)
  - DisplayPort
  - Digital Visual Interface (DVI)
  - Video Graphics Array (VGA)
  - USB-C
- Hard drive cables
  - Serial Advanced Technology Attachment (SATA)
  - External SATA (eSATA)
- Adapters
- Connector types
  - RJ11
  - RJ45
  - F-type
  - Straight tip (ST)
  - Subscriber connector (SC)
  - Lucent connector (LC)
  - Punchdown block
  - MicroUSB
  - MiniUSB
  - USB-C
  - Molex
  - Lightning
  - DB9

Objective	Chapter(s)
3.3 Compare and contrast RAM characteristics.	1

▪ Form factors
  ▪ Small Outline Dual In-line Memory Module (SODIMM)
  ▪ Dual In-line Memory Module (DIMM)
▪ Double Data Rate (DDR) iterations
▪ Error-correcting code (ECC) vs. non-ECC RAM
▪ Channel configurations	

3.4 Compare and contrast storage devices.	2

▪ Hard drives
  ▪ Spindle speeds
  ▪ Form factors
  ▪ 2.5-inch
  ▪ 3.5-inch
▪ Solid-state drives
  ▪ Communications interfaces
  ▪ Non-volatile Memory Express (NVMe)
  ▪ SATA
  ▪ Peripheral Component Interconnect Express (PCIe)
  ▪ Serial Attached SCSI [Small Computer System Interface] (SAS)
  ▪ Form factors
  ▪ M.2
  ▪ Mini-serial Advanced Technology Attachment (mSATA)
▪ Drive configurations
  ▪ Redundant Array of Independent Disks (RAID) 0, 1, 5, 6, 10
▪ Removable storage
  ▪ Flash drives
  ▪ Memory cards
▪ Optical drives	

Objective	Chapter(s)
3.5 Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards.	1, 2

- Motherboard form factors
  - Advanced Technology Extended (ATX)
  - microATX
  - Information Technology eXtended (ITX)
- Motherboard connector types
  - Peripheral Component Interconnect (PCI)
  - PCIe
  - Power connectors
  - SATA
  - eSATA
  - Headers
  - M.2
- Motherboard compatibility
  - CPU socket types
    - Advanced Micro Devices, Inc. (AMD)
    - Intel
  - Multisocket
- BIOS/Unified Extensible Firmware Interface (UEFI) settings
  - Boot options
  - USB permissions
  - Trusted Platform Module (TPM) security features
  - Fan considerations

Objective	Chapter(s)

- Secure Boot
- Boot password
- BIOS password
- Temperature monitoring

- Virtualization support
- Encryption
  - TPM
  - Hardware security module (HSM)
- CPU architecture
  - x86/x64
  - Advanced RISC [Reduced Instruction Set Computer] Machine (ARM)
  - Core configurations
- Expansion cards
  - Sound card
  - Video card
  - Capture card
  - Network interface card
- Cooling
  - Fans
  - Heat sink
  - Thermal paste/pads
  - Liquid

Objective	Chapter(s)
3.6 Given a scenario, install the appropriate power supply.	2
▪ Input 110–120 VAC vs. 220–240 VAC ▪ Output 3.3V vs. 5V vs. 12V ▪ 20+4 pin motherboard connector ▪ Redundant power supply ▪ Modular power supply ▪ Wattage rating ▪ Energy efficiency	
3.7 Given a scenario, deploy and configure multifunction devices/ printers and settings.	5

▪ Properly unbox device and consider set-up location
▪ Use appropriate drivers for a given operating system
  ▪ Printer Control Language (PCL) vs. postscript
▪ Firmware
▪ Device connectivity
  ▪ USB
  ▪ Ethernet
  ▪ Wireless
▪ Public/shared devices
  ▪ Printer share
  ▪ Print server
▪ Configuration settings
  ▪ Duplex
  ▪ Orientation
  ▪ Tray settings
  ▪ Quality
▪ Security
  ▪ User authentication	

Objective	Chapter(s)
▪ Badging ▪ Audit logs ▪ Secured prints ▪ Network scan services ▪ Email ▪ SMB ▪ Cloud services ▪ Automatic document feeder (ADF)/flatbed scanner	
3.8 Given a scenario, perform appropriate printer maintenance.	5
▪ Laser ▪ Maintenance: Replace toner, apply maintenance kit, calibrate, and clean ▪ Inkjet ▪ Ink cartridge, printhead, roller, and feeder ▪ Maintenance: Clean printheads, replace cartridges, calibrate, and clear jams ▪ Thermal ▪ Feed assembly ▪ Special thermal paper ▪ Maintenance: Replace paper, clean heating element, and remove debris ▪ Impact ▪ Multipart paper ▪ Maintenance: Replace ribbon, printhead, and paper	

Objective	Chapter(s)
4.0 Virtualization and Cloud Computing	
4.1 Explain virtualization concepts.	11

▪ Purpose of virtual machines
  ▪ Sandbox
  ▪ Test development
  ▪ Application virtualization
  ▪ Legacy software/OS
  ▪ Cross-platform virtualization
▪ Requirements
  ▪ Security
  ▪ Network
  ▪ Storage
▪ Desktop virtualization
  ▪ Virtual Desktop Infrastructure (VDI)
▪ Containers
▪ Hypervisors
  ▪ Type 1
  ▪ Type 2	

4.2 Summarize cloud computing concepts.	11

▪ Common cloud models
  ▪ Private cloud
  ▪ Public cloud
  ▪ Hybrid cloud
  ▪ Community cloud
  ▪ Infrastructure as a service (IaaS)	

Objective	Chapter(s)
▪ Software as a service (SaaS) ▪ Platform as a service (PaaS) ▪ Cloud characteristics ▪ Shared resources vs. dedicated resources ▪ Metered utilization ▪ Ingress/egress ▪ Elasticity ▪ Availability ▪ File synchronization ▪ Multitenancy	
5.0 Hardware and Network Troubleshooting	
5.1 Given a scenario, troubleshoot motherboards, RAM, CPUs, and power. ▪ Common symptoms ▪ Power-on self-test (POST) beeps ▪ Proprietary crash screens ▪ Blank screen ▪ No power ▪ Sluggish performance ▪ Overheating ▪ Burning smell ▪ Random shutdown ▪ Application crashes ▪ Unusual noise ▪ Capacitor swelling ▪ Inaccurate system date/time	12

Objective	Chapter(s)
5.2 Given a scenario, troubleshoot drive and RAID issues.	12

▪ Common symptoms
  ▪ Light-emitting diode (LED) status indicators
  ▪ Grinding noises
  ▪ Clicking sounds
  ▪ Bootable device not found
  ▪ Data loss/corruption
  ▪ RAID failure
  ▪ Self-monitoring and Reporting Technology (S.M.A.R.T.) failure
  ▪ Extended read/write times
  ▪ Low performance input/output operations per second (IOPS)
  ▪ Missing drives in OS
  ▪ Array missing
  ▪ Audible alarms	

5.3 Given a scenario, troubleshoot video, projector, and display issues.	13

▪ Common symptoms
  ▪ Incorrect input source
  ▪ Physical cabling issues
  ▪ Burnt-out bulb
  ▪ Fuzzy image
  ▪ Display burn-in
  ▪ Dead pixels
  ▪ Flashing screen
  ▪ Incorrect color display
  ▪ Audio issues
  ▪ Dim image
  ▪ Intermittent projector shutdown
  ▪ Sizing issues
  ▪ Distorted image	

Objective	Chapter(s)
5.4 Given a scenario, troubleshoot common mobile device issues.	13

- Common symptoms
  - Poor battery health
  - Swollen battery
  - Broken screen
  - Improper charging
  - Poor/no connectivity
  - Liquid damage
  - Overheating
  - Digitizer issues
  - Physically damaged ports
  - Malware
  - Cursor drift/touch calibration
  - Unable to install new applications
  - Stylus does not work
  - Degraded performance

5.5 Given a scenario, troubleshoot network issues.	12

- Common symptoms
  - Intermittent wireless connectivity
  - Slow network speeds
  - Limited connectivity
  - Jitter
  - Poor Voice over Internet Protocol (VoIP) quality
  - Port flapping
  - High latency
  - External interference
  - Authentication failures
  - Intermittent internet connectivity

Objective	Chapter(s)
5.6 Given a scenario, troubleshoot printer issues.	13

- Lines down the printed pages
- Garbled print
- Paper jams
- Faded prints
- Paper not feeding
- Multipage misfeed
- Multiple prints pending in queue
- Speckling on printed pages
- Double/echo images on the print
- Grinding noise
- Finishing issues
  - Staple jams
  - Hole punch
- Incorrect page orientation
- Tray not recognized
- Connectivity issues
- Frozen print queue

Exam objectives are subject to change at any time without prior notice at CompTIA's sole discretion. Please visit CompTIA's website (`www.comptia.org`) for the most current listing of exam objectives.

How to Contact the Publisher

If you believe you have found a mistake in this book, please bring it to our attention. At John Wiley & Sons, we understand how important it is to provide our customers with accurate content, but even with our best efforts an error may occur.

In order to submit your possible errata, please email it to our Customer Service Team at `wileysupport@wiley.com` with the subject line "Possible Book Errata Submission."

Assessment Test for Exam 220-1201

1. Which of the following is not considered a system component that can be found inside a computer?
   - **A.** CPU
   - **B.** RAM
   - **C.** PCIe graphics adapter
   - **D.** Motherboard
2. Which of the following is a physical memory format installed directly in today's desktop computer systems?
   - **A.** DIMM
   - **B.** HDD
   - **C.** SSD
   - **D.** eMMC
3. Which of the following are components that can commonly be found on a motherboard? (Choose all that apply.)
   - **A.** Headers
   - **B.** Fan connectors
   - **C.** Gyroscope
   - **D.** Scanner
   - **E.** HDD
4. Which of the following statements is true regarding ECC RAM?
   - **A.** ECC RAM has full functionality in a non-ECC motherboard.
   - **B.** ECC RAM will not fit in a non-ECC motherboard.
   - **C.** ECC RAM will fit but not work in a non-ECC motherboard.
   - **D.** ECC RAM will fit and function without error-correction in a non-ECC motherboard.
5. Which of the following hard drives will fit into an M.2 slot with a B-key and an M-key?
   - **A.** B-key
   - **B.** M-key
   - **C.** B+M-key
   - **D.** All of the above

6. You are replacing a power supply with a new one. What is the primary characteristic you should look for to ensure all internal components can be supported?
   - **A.** Amperage rating
   - **B.** 20-pin to 24-pin motherboard adapter
   - **C.** Wattage rating
   - **D.** Voltage rating

7. A user has asked you to install a RAID array. Except in the case of RAID 0, which two things do all types of RAID offer? (Choose two.)
   - **A.** Faster read speeds
   - **B.** Faster write speeds
   - **C.** Redundancy
   - **D.** Fault tolerance
   - **E.** Automatic restores from backup after a drive failure

8. You are installing a new SATA hard drive. Which of the following is true, without using an adapter?
   - **A.** It will connect to a SATA controller only.
   - **B.** It will connect to a SATA controller or SAS controller.
   - **C.** It will connect to a SATA controller or PATA controller.
   - **D.** It will connect to a SATA controller or a PCIe controller.

9. Which of the following types of connectors would you expect to see on the end of a serial cable?
   - **A.** DB-9
   - **B.** RS-232
   - **C.** Molex
   - **D.** VGA

10. Which of the following video connectors has an analog variety and a digital variety?
   - **A.** VGA
   - **B.** HDMI
   - **C.** DisplayPort
   - **D.** DVI

11. Which of the following connectors will you not find on the end of a USB cable? (Choose two.)
   - **A.** Lightning
   - **B.** USB-C
   - **C.** Molex
   - **D.** DB-9
   - **E.** eSATA

12. In an LCD panel, what is the name of the component that provides power to the backlight?

A. Diffuser

B. Inverter

C. Converter

D. Digitizer

13. You are installing a new display for a laptop. Which LCD component is responsible for providing brightness?

A. Backlight

B. Inverter

C. Screen

D. Digitizer

14. Your laptop has 8 GB of installed memory and uses shared video memory. If the video card is using 512 MB, how much is left for the rest of the system?

A. 8 GB

B. 7.5 GB

C. 512 MB

D. Cannot determine

15. On a typical laptop, which of the following components runs through the hinge from the base of the laptop up into the display unit?

A. Webcam connector

B. Speaker wires

C. Microphone connector

D. Wi-Fi antenna

16. Which of the following laptop accessories is most likely to have a full-sized internal hard drive bay?

A. Docking station

B. Port replicator

C. SAN

D. NAS

17. A technician is preparing to perform inkjet printer maintenance. Which of the following should they consider?

A. Replace toner, apply maintenance kit, calibrate, clean.

B. Replace paper, clean heating element, remove debris.

C. Replace ribbon, replace printhead, replace paper.

D. Clean printheads, replace cartridges, calibrate, clear jams.

18. What is the function of the laser in a laser printer?

- **A.** It heats up the toner so that it adheres to the page.
- **B.** It charges the paper so that it will attract toner.
- **C.** It creates an image of the page on the drum.
- **D.** It cleans the drum before a page is printed.

19. What is the component called that stores the material that ends up printed to the page in a laser printer?

- **A.** Toner cartridge
- **B.** Ink cartridge
- **C.** Laser module
- **D.** Laser cartridge

20. When used with network scanning services, what does SMB allow for?

- **A.** Scan to the cloud
- **B.** Scan to email
- **C.** Scan to a network folder
- **D.** Scan to secured print

21. Which network connectivity device stops broadcasts from being sent to computers on a different network segment?

- **A.** Hub
- **B.** Switch
- **C.** Router
- **D.** Firewall

22. Which of the following cable types is made with a plastic coating that does not produce toxic fumes when it burns?

- **A.** Plenum
- **B.** Fiber
- **C.** Coaxial
- **D.** UTP

23. Which of the following are standards used by coaxial cable? (Choose two.)

- **A.** RG-6
- **B.** RG-59
- **C.** RJ11
- **D.** RJ45

24. You have a PoE switch that says it supports PoE Type 3. How many watts can it provide to attached devices?

A. 30 watts

B. 60 watts

C. 100 watts

D. 120 watts

25. On which port does FTP run by default?

A. 21

B. 25

C. 53

D. 80

26. Which of the following protocols can be used by a client to access email on a server?

A. DNS

B. FTP

C. SMTP

D. IMAP

27. Which of the following protocols provide secure network transmissions? (Choose two.)

A. SMTP

B. SNMP

C. SSH

D. HTTPS

28. Which of the following IP addresses is not routable on the Internet?

A. 192.16.33.220

B. 172.18.1.101

C. 33.33.33.33

D. 192.186.33.220

29. One of your network users was recently caught at work browsing pornographic websites. Which of the following servers could be installed to prohibit this activity?

A. Web

B. Security

C. Proxy

D. DNS

30. Which of the following Internet appliances will include services provided by an IDS or IPS?

A. UTM

B. Spam gateway

C. Load balancer

D. Proxy server

31. For security purposes, your network needs to ensure that all devices have synchronized time stamps for login events. Which network service will help provide this?

A. DNS

B. DHCP

C. AAA

D. NTP

32. Which DNS record type is used for IPv4 addresses?

A. AAAA

B. A

C. CNAME

D. MX

33. Which of the following is a company that provides direct access to the Internet for home and business computer users?

A. ASP

B. ISP

C. DNS

D. DNP

34. What is the maximum range for a Bluetooth Class 2 device?

A. 100 meters

B. 10 meters

C. 1 meter

D. 0.5 meters

35. Which of the following 802.11 technologies can communicate in the 2.4 GHz and 5 GHz frequencies? (Choose two.)

A. 802.11ax

B. 802.11ac

C. 802.11n

D. 802.11g

36. Which wireless networking technology has the longest range?

A. Bluetooth

B. 802.11

C. NFC

D. RFID

37. Which of the following is used to remotely manage corporate mobile devices on a network?

A. Location services

B. Synchronization

C. BYOD

D. MDM

38. What term refers to copying data between a mobile device and a computer system to mirror such things as contacts, programs, pictures, and music?

A. Calibration

B. Remote wipe

C. Pairing

D. Synchronization

39. You want to send emails from your smartphone using a secure connection. Which port will enable that?

A. 25

B. 110

C. 143

D. 587

40. A network administrator wants ten users in three different countries to be able to access all of the same data for a project. Which solution is best?

A. MDM

B. Data cap

C. Synchronization

D. Cloud storage

41. When using cloud services, which of the following may incur an additional charge as part of metered utilization?

A. Elasticity

B. Availability

C. Egress

D. Ingress

42. Google Docs is an example of what type of cloud service?

A. SaaS

B. IaaS

C. PaaS

D. GaaS

43. Which type of software is required to run client-side virtualization on your home network?

A. Terminal emulation

B. Process replication

C. Hyperthreading

D. Hypervisor

44. What type of cloud has elements from a public cloud run by a CSP and others owned by the client company on its internal network?

A. Private

B. Public

C. Hybrid

D. Community

45. A desktop computer has recently started exhibiting slow performance, particularly when opening and saving files to the hard drive. What should you try first?

A. Replace the hard drive.

B. Remove old files and applications to free up disk space.

C. Delete and reinstall the OS.

D. Defragment the hard drive.

46. A network cable has not been working properly. Another technician suggests testing the cable. Which tool should be used to do this? (Choose two.)

A. Loopback plug

B. Network tap

C. Cable tester

D. Toner probe

47. You are troubleshooting a network issue. What feature do switches most likely have to help troubleshoot connectivity issues?

A. Port toggles

B. Beep codes

C. Indicator lights

D. Log entries

48. When you turn a computer on, you immediately hear a loud screeching sound coming from it. What components could be making this noise? (Choose two.)

A. SSD

B. HDD

C. CPU

D. PSU

49. Your laser printer has recently starting printing vertical white lines on documents it prints. What is the most likely cause of the problem?

A. The print driver is faulty.

B. The fuser is not heating properly.

C. There is toner on the transfer corona wire.

D. There is a scratch on the EP drum.

50. A laptop's display has several spots where it does not light up. If the screen is set to a solid white background, there are at least a dozen dots that will still be black. What's the solution to this problem?

A. Change the screen resolution to a lower one.

B. Replace the laptop display.

C. Degauss the laptop display.

D. Replace the laptop display inverter.

51. Which of the following should a user do to avoid malware on an Android phone?

A. Use a screen protector.

B. Block the digitizer from receiving external input.

C. Install apps from only the Google Play store.

D. None of the above. Android phones are not susceptible to malware.

Answers to Assessment Test 220-1201

1. C. System components are essential for the basic functionality of a computer system. Many of the landmarks found on the motherboard can be considered system components, even expansion slots to a degree. What you plug into those slots, however, must be considered peripheral to the basic operation of the system. See Volume 1 Chapter 1 for more information.

2. A. Except for DIMMs, all options represent some form of secondary storage, all of which are covered in Volume 1 Chapter 2. See Volume 1 Chapter 1 for more information.

3. A, B. Motherboards commonly have headers for lights and buttons and for powering cooling fans. Gyroscopes are most commonly found in mobile devices. Scanners are external devices. Although there might be one or more types of HDD interfaces built into the motherboard, the HDD itself is not. See Volume 1 Chapter 1 for more information.

4. D. Error-correcting code (ECC) RAM will fit and work in a non-ECC motherboard, but it will not have error-correcting capabilities. See Volume 1 Chapter 1 for more information.

5. A. An M.2 slot with B- and M-keys has posts in the B- and M- positions. Therefore, the drive must have notches in both the B- and M- positions or it will not fit. See Volume 1 Chapter 2 for more information.

6. C. Wattage is the unit of output for power supplies. Make sure that the power supply has enough wattage to support all components. See Volume 1 Chapter 2 for more information.

7. C, D. Except for RAID 0, all implementations of RAID offer a way to recover from the failure of at least one drive, which is an example of fault tolerance, through the implementation of some mechanism that stores redundant information for that purpose. Some RAID types offer faster read and/or write performance. RAID 1, for instance does not guarantee either. RAID also does not automatically restore from backup. See Volume 1 Chapter 2 for more information.

8. B. SATA drives will of course connect to SATA controllers. They will connect to SAS controllers as well, although they will be much slower than SAS hard drives. See Volume 1 Chapter 2 for more information.

9. A. Serial cables use a DB-9 connector. Some people will call them an RS-232 connector, but RS-232 is the transmission standard used by serial cables. Molex are power connectors for PATA hard drives and optical drives. VGA is a video connector. See Volume 1 Chapter 3 for more information.

10. D. DVI comes in both analog (DVI-A) and digital (DVI-D) varieties. VGA is always analog; HDMI and DisplayPort are always digital. See Volume 1 Chapter 3 for more information.

11. C, D. USB cables are multipurpose cables that connect many types of peripherals. Connectors include USB-C, Lightning, and eSATA. Molex connectors provide power to PATA hard drives and optical drives. A DB-9 connector is found on the end of a serial cable. See Volume 1 Chapter 3 for more information.

12. B. A backlight is given power by an inverter, which converts DC power from the power supply to AC power for the backlight. There are no diffusers or converters in an LCD panel. A digitizer takes input from a stylus or fingertip and turns it into digital signals. See Volume 1 Chapter 3 for more information.

13. A. The backlight provides light to the LCD screen. The inverter provides power to the backlight, and the screen displays the picture. Digitizers take physical input (such as touch) and turn it into electronic data. See Volume 1 Chapter 4 for more information.

14. B. If the laptop is using shared video memory, then the system memory is shared with the video card. If the video card is using 512 MB (half a gigabyte), then there is 7.5 GB left for the system. See Volume 1 Chapter 4 for more information.

15. D. In most laptops, the Wi-Fi antenna runs from the base of the unit through the hinge and into the display. The webcam and microphone are generally part of the display unit only. Speakers are most often in the base of the laptop, not the display. See Volume 1 Chapter 4 for more information.

16. A. A docking station might have an internal full-sized hard drive bay. Port replicators only have additional ports, such as HDMI, DisplayPort, and USB. Storage area networks (SAN) and network attached storage (NAS) are not laptop accessories. See Volume 1 Chapter 4 for more information.

17. D. Inkjet printer maintenance includes clean printheads, replace cartridges, calibrate, clear jams. Laser printer maintenance includes replace toner, apply maintenance kit, calibrate, clean. Thermal printer maintenance includes replace paper, clean heating element, remove debris. Impact printer maintenance includes replace ribbon, replace printhead, replace paper. See Volume 1 Chapter 5 for more information.

18. C. The laser creates an image on the photosensitive drum that is then transferred to the paper by the transfer corona. The fuser heats up the toner so that it adheres to the page. The transfer corona charges the page, and the eraser lamp cleans the drum before a page is printed. A rubber blade is also used to remove toner physically from the drum. See Volume 1 Chapter 5 for more information.

19. A. Laser printers use toner, which they melt to the page in the image of the text and graphics being printed. A toner cartridge holds the fine toner dust until it is used in the printing process. See Volume 1 Chapter 5 for more information.

20. C. The Server Message Block (SMB) protocol allows for the saving of scanned documents onto a network folder. See Volume 1 Chapter 5 for more information.

21. C. A router does not pass along broadcasts to computers on other segments. Hubs and switches send broadcasts along because they do not segment traffic at the logical network address level. See Volume 1 Chapter 6 for more information.

22. A. The plenum is an air space within buildings used to circulate breathable air for occupants of a building. Fiber, coaxial, and UTP cables are normally enclosed with a PVC plastic coating, which produces toxic fumes if burned. For air ventilation spaces, plenum grade cable is required. All cable types have plenum-rated versions. See Volume 1 Chapter 6 for more information.

23. A, B. RG-6 and RG-59 are coaxial cable standards. RJ11 and RJ45 are connectors used with twisted pair cabling. See Volume 1 Chapter 6 for more information.

24. B. Power over Ethernet Type 3 can provide 60 W of power. PoE is 15.4 W, PoE+ is 30 W, and PoE++ Type 4 is 100 W. See Volume 1 Chapter 6 for more information.

25. A. FTP operates on ports 20 and 21. Port 25 is used by SMTP, port 53 is DNS, and port 80 is HTTP. See Volume 1 Chapter 7 for more information.

26. D. The IMAP and POP3 protocols can be used to retrieve email from mail servers. DNS is used to resolve host names to IP addresses. FTP is for file transfers, and SMTP is used to send email. See Volume 1 Chapter 7 for more information.

27. C, D. Secure Shell (SSH) and Hypertext Transfer Protocol Secure (HTTPS) are both secure transport protocols. Simple Network Management Protocol (SNMP) and Simple Mail Transfer Protocol (SMTP) are not inherently secure. See Volume 1 Chapter 7 for more information.

28. B. Private IP addresses are not routable on the Internet. The Private IP address ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. See Volume 1 Chapter 7 for more information.

29. C. A proxy server can be configured to block access to websites containing potentially objectionable material. Web servers host websites. There is no specific security server, although there are authentication servers which provide security. A DNS server resolves hostnames to IP addresses. See Volume 1 Chapter 8 for more information.

30. A. Unified threat management (UTM) is a security-based Internet appliance that will likely contain the services of an intrusion detection system (IDS) or intrusion protection system (IPS). See Volume 1 Chapter 8 for more information.

31. D. A Network Time Protocol (NTP) server can ensure that all devices on the network are operating on the same time. DNS, DHCP, and AAA do not provide timekeeping services. See Volume 1 Chapter 8 for more information.

32. B. IPv4 addresses are identified with an A DNS record type. AAAA is for IPv6, CNAME is an alias, and MX is a mail server. See Volume 1 Chapter 8 for more information.

33. B. An Internet service provider (ISP) provides direct access to the Internet. ASP is a programming language. DNS resolves hostnames to IP addresses. DNP is not a computer networking acronym. See Volume 1 Chapter 9 for more information.

34. B. A Bluetooth class 2 device (which includes headsets, keyboards, and mice) is 10 meters. Class 1 devices can transmit up to 100 meters, class 3 one meter, and class 4 half a meter. See Volume 1 Chapter 9 for more information.

35. A, C. 802.11ax (Wi-Fi 6) and 802.11n were designed to communicate at 2.4 GHz and 5 GHz. 802.11ac runs at 5 GHz, and 802.11g runs at 2.4 GHz. See Volume 1 Chapter 9 for more information.

36. B. 802.11 has the longest range. Bluetooth is next longest, and then the two shortest are the radio frequency standards NFC and RFID. See Volume 1 Chapter 9 for more information.

37. D. Mobile device management (MDM) controls which devices are on the network and can wipe entire devices. Location services does not manage software. Synchronization can back up apps and data but does not help with updates or deleting apps. BYOD is when a user owns the device that they use on a corporate network. For more information, see Volume 1 Chapter 10.

38. D. Synchronizing a mobile device with a computer system allows you to mirror personal data between the devices, regardless of which one contains the most current data. Calibration refers to matching the device's and user's perceptions of where the user is touching the screen. Remote wipes allow you to remove personal data from a lost or stolen device. Pairing is what must be done in Bluetooth for two Bluetooth devices to connect and communicate. For more information, see Volume 1 Chapter 10.

39. D. Port 587 is used by SMTP (an email sending protocol) secured with TLS. Ports 25 (SMTP), 110 (POP), and 143 (IMAP) are not secure. See Volume 1 Chapter 10 for more information.

40. D. Cloud storage is the best solution to allow geographically dispersed users to access the same data. MDM is mobile device management, which configures the devices allowed to join a network. Synchronization is the act of mirroring all unique changes and additions from one device to the other. See Volume 1 Chapter 10 for more information.

41. C. Metered utilization means client usage is tracked and they are billed for their usage. Many service providers charge extra for downloading data, which is data egress (uploading, or ingress, is generally free). Elasticity and availability are separate cloud characteristics from metered utilization. See Volume 1 Chapter 11 for more information.

42. A. Google Docs is software, so it is an example of software as a service (SaaS). IaaS is infrastructure as a service. PaaS is platform as a service. GaaS is not a test objective and is not currently a cloud service type. See Volume 1 Chapter 11 for more information.

43. D. The hypervisor is the key piece of software needed for virtualization. Terminal emulation, process replication, and Hyper-Threading are not related to client-side virtualization. See Volume 1 Chapter 11 for more information.

44. C. A hybrid cloud is a combination of a public cloud (owned and operated by a CSP) and a private cloud (owned and operated by an individual company). A community cloud is one created by multiple organizations with similar interests. See Volume 1 Chapter 11 for more information.

45. B. Hard drive performance can slow down if the drive gets short on space, particularly under 10 percent free space. Try removing old files and applications to free up disk space first. If that doesn't resolve it, then defragment. If neither of those work, it could indicate a failing hard drive. See Volume 1 Chapter 12 for more information.

46. C. Cable testers are for verifying that a cable works properly. Network taps create copies of network traffic for analysis. A loopback plug is for testing the ability of a network adapter to send and receive. A toner probe is used to trace a cable from one location to another. See Volume 1 Chapter 12 for more information.

47. C. Switches and other connectivity devices have indicator lights to help indicate when things are working normally and when something isn't working right. There are no port toggles and switches don't use beep codes. A switch might make a log entry into an event logger, but this is less likely than it having indicator lights. See Volume 1 Chapter 12 for more information.

48. B, D. A screeching sound can be made only by components that have moving parts. This includes conventional hard drives (HDDs) and power supplies (PSUs) with fans. Solid-state drives (SSDs) and CPUs have no moving parts. See Volume 1 Chapter 12 for more information.

49. C. White streaks on printouts are most likely caused by toner on the transfer corona wire. Vertical black lines are caused by a scratch or a groove in the EP drum. If the fuser was not heating properly, toner would not bond to the paper and you would have smearing. Faulty print drivers will cause garbage to print or there will be no printing at all. See Volume 1 Chapter 13 for more information.

50. B. The symptom being described is called dead pixels. If they will not light up, the only solution is to replace the display. Changing the resolution won't help. Degaussing was done on old CRT monitors. The inverter provides power to the laptop display backlight, but if it were failing, the screen would be dim or flickering. See Volume 1 Chapter 13 for more information.

51. C. Android OS can be subjected to malware. To avoid malware, install apps only from the Google Play store and avoid tapping on suspicious links. See Volume 1 Chapter 13 for more information.

CompTIA A+ Core 1 Exam (220-1201)

VOLUME 1

Chapter 1

Motherboards, Processors, and Memory

THE FOLLOWING COMPTIA A+ 220-1201 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **3.0 Hardware**

✓ **3.3 Compare and contrast RAM characteristics.**

- Form factors
  - Small Outline Dual In-line Memory Module (SODIMM)
  - Dual In-line Memory Module (DIMM)
- Double Data Rate (DDR) iterations
- Error-correcting code (ECC) vs. non-ECC RAM
- Channel configurations

✓ **3.5 Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards.**

- Motherboard form factors
  - Advanced Technology Extended (ATX)
  - microATX
  - Information Technology eXtended (ITX)
- Motherboard connector types
  - Peripheral Component Interconnect (PCI)
  - PCIe
  - Power connectors
  - SATA
  - eSATA

- Headers
- M.2

- Motherboard compatibility
  - CPU socket types
    - Advanced Micro Devices, Inc. (AMD)
    - Intel
  - Multisocket
- BIOS/Unified Extensible Firmware Interface (UEFI) settings
  - Boot options
  - USB permissions
  - Trusted Platform Module (TPM) security features
  - Fan considerations
  - Secure Boot
  - Boot password
  - BIOS password
  - Temperature monitoring
- Virtualization support
- Encryption
  - TPM
  - Hardware security module (HSM)
- CPU architecture
  - x86/x64
  - Advanced RISC [Reduced Instruction Set Computer] Machine (ARM)
  - Core configurations
- Cooling
  - Fans
  - Heat sink
  - Thermal paste/pads
  - Liquid

The computers we use daily, from the largest servers to the smallest smart watches and everything in between, are collections of different electronic components and software working together in a system. Digital computers have been around since the late 1930s, so they aren't new news. As you would imagine, though, their looks and functionality have evolved considerably since then.

As technology improved over the years, computers got smaller and faster, and inventors added features that required new hardware devices. Inspired by typewriters, keyboards were added for ease of input. Visual displays required a monitor, a video card, and a standard interface between them. Sound was provided by a different expansion card and speakers. Because of the newly added features, PCs were modular by necessity. That is, if a new feature or functionality were needed, a new component could be added. Or if a part failed, it could be replaced by a new one. Along the way in the late 1960s, the term *personal computer (PC)* was coined to differentiate between computers designed to be used by one person versus other options, such as a mainframe or one where multiple users share a processor. This book, and the CompTIA A+ exams, focus on PC hardware and software.

Unless specifically mentioned otherwise, the terms *PC* and *computer* are used interchangeably throughout this book.

Much of the computing industry today is focused on smaller devices, such as laptops, tablets, smartphones, and Internet of Things (IoT) devices. Laptops have outsold desktop computers since 2005, and it seems that everyone has a smartphone glued to their hand today. IoT devices (such as smart home automation devices and virtual assistants like Amazon Echo and Google Nest, which we will cover more in depth in Volume 1 Chapter 8, "Network Services,") have exploded in popularity from about 10 billion in 2019 to an expected 40 billion by 2030. Smaller devices require the same components as do their bigger desktop-sized cousins. Of course, the components are smaller, and by necessity most of the time they are integrated into the same circuit board. The functionality of the individual parts is still critical, though, so what you learn here will serve you well regardless of the type of device you're working on.

Even though all parts inside a computer case are important, some are more important than others. You'll sometimes hear people refer to the "big three" computer parts, which are the motherboard, processor, and memory. Without these, a computer won't work, whereas if a sound card fails, it's probably not going to make the entire system inoperable. In this chapter, you will learn how to identify, install, and configure the big three, which are critical

to computer functionality. You'll also learn about cooling systems, because too much heat will cause components to melt, which could make for a very bad day for the computer user.

With small computing devices being far more prevalent, you might wonder why we are starting the book talking about components that are primarily in desktop and larger systems. We intentionally cover this material first because it's important to understand what each component does individually. It helps you isolate the functionality of each piece of the system, which is critical for troubleshooting and fixing problems. Besides, if you are working (or want to be working) as a computer technician, you will likely see plenty of desktop-sized PCs in your travels!

Understanding Motherboards

The spine of the computer is the *motherboard*, otherwise known as the *system board* or *mainboard*. This is the *printed circuit board (PCB)*, which is a conductive series of pathways laminated to a nonconductive substrate that lines the bottom of the computer and is often of a uniform color, such as green, brown, blue, black, or red. It is the most important component in the computer because it connects all the other components together. Figure 1.1 shows a typical PC system board, as seen from above. All other components are attached to this circuit board. On the system board, you will find the central processing unit (CPU) slot or integrated CPU, underlying circuitry, expansion slots, video components, random access memory (RAM) slots, and a variety of other chips. We will be discussing each of these components throughout this book.

Motherboard Form Factors

You can find hundreds, if not thousands, of different motherboards on the market today. It can be overwhelming trying to figure out which one is needed. To help, think of motherboards in terms of which type they are, based on a standard classification. System boards are classified by their *form factor* (the design of the component), such as ATX and ITX. Exercise care and vigilance when acquiring a motherboard and components separately. Motherboards will have different expansion slots, support certain processors and memory, and fit into some cases but not others. Be sure that the other parts are physically compatible with the motherboard you choose.

The CompTIA A+ exam 220-1201 objective 3.5, "Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards," specifically calls out ATX, microATX, and ITX motherboard form factors. Be sure to understand the differences between the three standards.

FIGURE 1.1 A typical motherboard

Advanced Technology Extended

Intel developed the *Advanced Technology Extended (ATX)* motherboard in the mid-1990s to improve on the classic AT-style motherboard architecture that had ruled the PC world for many years. The ATX motherboard has the processor and memory slots at right angles to the expansion cards, like the one in Figure 1.1. This arrangement puts the processor and memory in line with the fan output of the power supply, allowing the processor to run cooler. And because those components are not in line with the expansion cards, you can

install full-length expansion cards—adapters that extend the full length of the inside of a standard computer case—in an ATX motherboard machine. ATX (and its derivatives, such as microATX) is the primary PC motherboard form factor in use today. Standard ATX motherboards measure 12″ × 9.6″ (305mm × 244mm). A slightly smaller version of an ATX motherboard is *microATX*. These boards are 9.6″ (244mm) square, and because of their smaller size typically have fewer connectors for plugging in expansion cards and peripherals. ATX and microATX motherboards use the same case mounting points, so a microATX board will fit into a standard ATX case. The opposite might not be true, however, depending on case dimensions.

We will discuss expansion cards in more depth (and cover that part of Exam Objective 3.5) in Volume 1 Chapter 2, "Expansion Cards, Storage Devices, and Power Supplies."

Information Technology Extended

The *Information Technology Extended (ITX)* line of motherboard form factors was developed by VIA Technologies in the early 2000s as a low-power, small form factor (SFF) board for specialty uses, including home-theater systems, compact desktop systems, gaming systems, and embedded components. ITX itself is not an actual form factor but a family of form factors. The family consists of the following form factors:

- **Mini-ITX:** 6.7″ × 6.7″ (170mm × 170mm)
- **Nano-ITX:** 4.7″ × 4.7″ (120mm × 120mm)
- **Pico-ITX:** 3.9″ × 2.8″ (100mm × 72mm)
- **Mobile-ITX:** 2.4″ × 2.4″ (60mm × 60mm)

The mini-ITX motherboard has four mounting holes that line up with three or four of the holes in the ATX and micro-ATX form factors. In mini-ITX boards, the rear interfaces are placed in the same location as those on the ATX motherboards. These features make mini-ITX boards compatible with ATX cases. This is where the mounting compatibility ends, because despite the PC compatibility of the other ITX form factors, they are used in embedded systems, such as set-top boxes, home entertainment systems, and smartphones, and lack the requisite mounting and interface specifications. Figure 1.2 shows the three larger forms of ITX motherboards, next to two ATX motherboards for comparison.

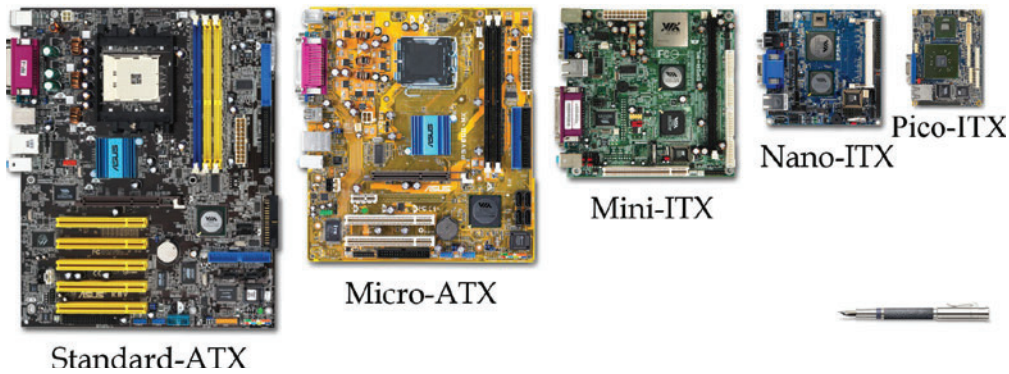
System Board Components

Now that you understand the basic types of motherboards and their form factors, it's time to look at the key characteristics and components of the motherboard and, where applicable, their locations relative to each other. The following list summarizes key concepts you need to know about motherboards:

- Bus architecture
- Chipsets

- Connector types (expansion slots)
- Memory slots and cache
- Central processing units and processor sockets
- Power connectors
- Onboard nonvolatile storage connectors (such as for hard drives)
- Motherboard headers
- BIOS/UEFI/firmware
- CMOS and CMOS battery
- Front-panel connectors and headers

FIGURE 1.2 ITX motherboards



Source: VIA Mini-ITX Form Factor Comparison / VIA Gallery / Wikimedia Commons / CC BY 2.0

In the following sections, you will learn about some of the most common components of a motherboard, what they do, and where they are located on the motherboard. We'll show what each component looks like so that you can identify it on most any motherboard that you run across. In the case of some components, this chapter provides only a brief introduction, with more detail to come in later chapters.

Bus Architecture

In a PC, data is sent from one component to another via a *bus*, which is a common collection of signal pathways. In the very early days, PCs used serial buses, which sent one bit at a time and were painfully slow. Brilliant engineers realized that they could redesign the bus and send 8 bits at a time (over synchronized separate lines), which resulted in a big speed increase. This was known as a *parallel bus*.

The downside of parallel communications is the loss of circuit length (how long the circuit could be) and throughput (how much data could move at one time). The signal could travel only a short distance, and the amount of data was limited due to the careful synchronization needed between separate lines, the speed of which must be controlled to limit skewing the arrival of the individual signals at the receiving end.

What was once old is new again, as engineers have discovered methods to make serial transmissions work at data rates that are many times faster than parallel signals. Therefore, nearly everything you see today uses a *serial bus*. The only limitation of serial circuits is in the capability of the transceivers, which tends to grow over time due to technical advancements. Examples of specifications that have heralded the dominance of serial communications are Serial Advanced Technology Attachment (Serial ATA, or SATA), Universal Serial Bus (USB), and Peripheral Component Interconnect Express (PCIe).

The term *bus* is also used in any parallel or bit-serial wiring implementation where multiple devices can be attached at the same time in parallel or in series (daisy-chained). Examples include Small Computer System Interface (SCSI), USB, and Ethernet.

On a motherboard, several different buses are used. Motherboard connector types of various architectures, such as PCIe, are included to allow for the insertion of external devices or adapters. Other types of buses exist within the system to allow communication between the CPU, RAM, and other components with which data must be exchanged. Except for CPU sockets and memory slots, there are no insertion points for devices in many closed bus systems because no expansion cards exist for those features.

The various buses throughout a given computer system can be rated by their bus speeds. The higher the bus speed, the higher its performance. In some cases, various buses must be synchronized for proper performance, such as the system bus and any expansion buses that run at the front-side bus speed. Other times, one bus will reference another for its own speed. The internal bus speed of a CPU is derived from the front-side bus clock, for instance. The buses presented throughout this chapter are accompanied by their speeds, where appropriate.

Chipsets

A *chipset* is a collection of chips or circuits that perform interface and peripheral functions for the processor. This collection of chips is usually the circuitry that provides interfaces for memory, expansion cards, and onboard peripherals, and it generally dictates how a motherboard will communicate with the installed peripherals.

Chipsets are usually given a name and model number by the original manufacturer. For example, B550 and X570 are chipsets that support Advanced Micro Devices, Inc. (AMD) processors, and Z490 and H410 are Intel motherboard chipsets. Typically, the manufacturer and model tell you that your particular chipset has a certain set of features, such as:

- Type of CPU and RAM supported
- Onboard video capabilities

- Motherboard expansion slot types, versions, and speeds
- Number of input/output ports and versions
- Support for BIOS/UEFI security

Don't worry if you don't understand what those features are just yet—we'll cover them throughout this chapter. Also don't worry about memorizing any chipset names—you can look them up online to understand their features. For now, know that the motherboard chipset and CPU must be designed to work together.

Motherboard Connector Types (Expansion Slots)

The most visible parts of any motherboard are the connectors for internal expansion cards, also called *expansion slots*. These are small plastic slots, usually from 1 to 6 inches long and approximately 1/2 inch wide. As their name suggests, these slots are used to install various devices in the computer to expand its capabilities. Some expansion devices that might be installed in these slots include video, network, sound, and disk interface cards.

If you look at the motherboard in your computer, you will more than likely see one of the main types of expansion slots used in computers today, which are PCI and PCIe. The exam objectives mention several other motherboard connector types as well such as SATA and M.2. We will cover all of them throughout this chapter, but for now, we will focus on how to visually identify the different expansion slots on the motherboard.

Exam Objective 3.5 lists several motherboard connector types you should be familiar with. They include PCI and PCIe.

PCI Expansion Slots

It's now considered an old technology, but many motherboards in use today still contain 32-bit *Peripheral Component Interconnect (PCI)* slots (32-bit means that data is handled 32 bits at a time). They are easily recognizable because they are only around 3″ long and classically white, although modern boards take liberties with the color. PCI slots became extremely popular with the advent of Pentium-class processors in the mid-1990s. Although popularity has shifted from PCI to PCIe, the PCI slot's service to the industry cannot be ignored; it has been an incredibly prolific architecture for many years.

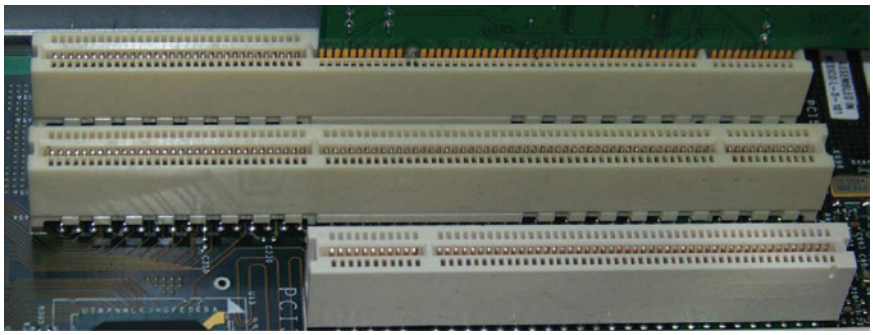
PCI expansion buses operate at 33 MHz or 66 MHz (version 2.1) over a 32-bit (4-byte) channel, resulting in data rates of 133 MBps and 266 MBps, respectively, with 133 MBps being the most common, server architectures excluded. PCI is a shared-bus topology, however, so mixing 33 MHz and 66 MHz adapters in a 66 MHz system will slow all adapters to 33 MHz. Older servers might have featured 64-bit PCI slots as well, since version 1.0, which double the 32-bit data rates. See the sidebar "Arriving at the Exact Answer" for help with understanding the math involved in frequencies and bit rates.

Arriving at the Exact Answer

To get the math exactly right when dealing with frequencies and data rates ending in 33 and 66, you have to realize that every 33 has an associated one-third (1/3), and every 66 has an associated two-thirds (2/3). The extra quantities are left off of the final result but must be added back on to get the math exactly right. The good news is that omitting these small values from the equation still gets you close, and a bit of experience with the numbers leads to being able to make the connection on the fly.

PCI slots and adapters are manufactured in 3.3V and 5V versions. Universal adapters are keyed to fit in slots based on either of the two voltages. The notch in the card edge of the common 5V slots and adapters is oriented toward the front of the motherboard and the notch in the 3.3V adapters toward the rear. Figure 1.3 shows several PCI expansion slots. Note the shorter 5V 32-bit slot in the foreground and the longer 3.3V 64-bit slots. Also notice that a universal 32-bit card, which has notches in both positions, is inserted into and operates fine in the 64-bit 3.3V slot in the background.

FIGURE 1.3 PCI expansion slots



PCIe Expansion Slots

The most common expansion slot architecture used on motherboards is *PCI Express (PCIe)*. It was designed to be a replacement for PCI, as well as an older video card standard called accelerated graphics port (AGP). PCIe has the advantage of being faster than AGP while maintaining the flexibility of PCI. PCIe has no plug compatibility with either AGP or PCI. Some modern PCIe motherboards can be found with regular PCI slots for backward compatibility, but AGP slots have not been included for many years.

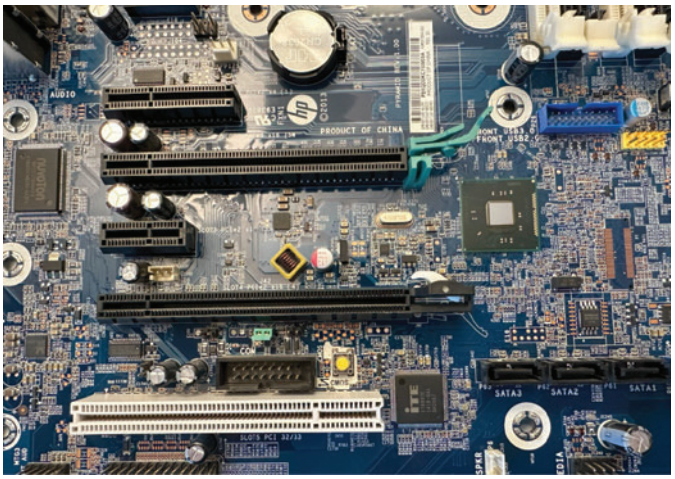
AGP was a video card–only standard introduced in 1997 and became obsolete in the late 2000s. If you see an old motherboard with a brown expansion slot parallel to white PCI slots and set back a bit from the rear of the case, it's probably AGP.

PCIe is casually referred to as a bus architecture to simplify its comparison with other bus technologies. True expansion buses share total bandwidth among all slots, each of which taps into different points (aka multipoint) along the common bus lines. In contrast, PCIe uses a switching component with point-to-point connections to slots, which gives each component full use of the corresponding bandwidth and making connections faster. Furthermore, unlike other expansion buses, which have parallel architectures, PCIe is a serial technology, striping data packets across multiple serial paths to achieve higher data rates.

PCIe uses the concept of *lanes*, which are the switched point-to-point signal paths between any two PCIe components. Each lane that the switch interconnects between any two intercommunicating devices comprises a separate pair of wires for both directions of traffic. Each PCIe pairing between cards requires a negotiation for the highest mutually supported number of lanes. The single lane, or combined collection of lanes that the switch interconnects between devices, is referred to as a *link*.

There are seven different link widths supported by PCIe, designated x1 (pronounced "by 1"), x2, x4, x8, x12, x16, and x32, with x1, x4, and x16 being the most common. A slot that supports a particular link width is of a physical size related to that width because the width is based on the number of lanes supported, requiring a related number of wires. As a result, a x8 slot is longer than a x1 slot but shorter than a x16 slot. Every PCIe slot has a 22-pin portion in common toward the rear of the motherboard, which you can see in Figure 1.4, in which the rear of the motherboard is to the left. (Figure 1.4 shows, from top to bottom, PCIe x4, PCIe x16, PCIe x1, PCIe x16, and PCI slots.) These 22 pins comprise mostly voltage and ground leads.

FIGURE 1.4 PCIe expansion slots



In addition to PCIe link widths, you should be aware of PCIe standards. The oldest version, 1.X, was released in 2003, running at 250 MBps per direction, or 500 Mbps bidirectionally. Use the doubled rate when comparing PCIe to other expansion buses because

those other rates are for bidirectional communication. This means that the original, slowest 500 MBps bidirectional link of a x1 slot in the first version of PCIe was comparable to PCI's best, a 64-bit slot running at 66 MHz and producing a throughput of 533 MBps.

Bidirectional means that data flows in both directions, often simultaneously. *Unidirectional* means data flows in only one direction.

Combining lanes simply results in a linear multiplication of these rates. For example, a PCIe 1.1 x16 slot is capable of 4 GBps of throughput in each direction, which is 16 times the 250 MBps x1 rate. Bidirectionally, this fairly common slot produces a throughput of 8 GBps. Each subsequent PCIe specification doubles this throughput. Following this math through the versions, PCIe 6.0 has a single-lane data rate of 8 GBps, for a total x16 bidirectional bandwidth of 256 GBps. It's really fast.

Real World Scenario

The Speed of Innovation

Newer standards always offer faster data transfer rates and greater throughput. Usually when the standard is released, people shake their heads at how mind-bogglingly fast they are, and then the cycle repeats itself a few years later. Hardware devices that support the standard are often one or two years behind the release of the standard.

For example, PCIe 4.0 was formally ratified by the PCI Special Interest Group (PCI-SIG) in October 2017, seven years after PCIe 3.0. It doubled the data rates of the previous standard and provided good performance for virtual and augmented reality applications.

PCIe 5.0 was released in May 2019 and of course doubled up on PCIe 4.0. This helped support machine learning and artificial intelligence, as well as high-speed data transfers needed in data centers.

Continuing the journey is PCIe 6.0, released in January 2022. Devices supporting this standard started to hit the market in 2024. And PCIe 7.0 has a 2025 release date and devices are expected in 2027.

The key thing to remember about PCIe versions is that each generation doubles the speed before it. So if you remember that PCIe 1.0 x1 was 500 Mbps, you can use a calculator to figure out any combination of standard and link width. If you want to read further on PCIe details, visit `https://pcisig.com/specifications`.

The good news is that you don't have to worry about compatibility between cards and slots of different standards. A PCIe 5.0 x1 card will fit into a PCIe 4.0 x1 slot and vice versa. The device will simply operate at the speed of the slower standard.

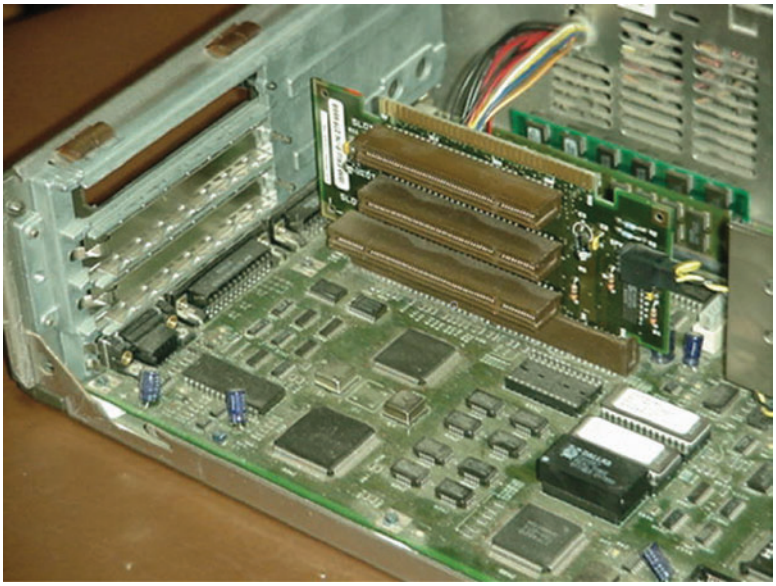
Refer back to Figure 1.4, which shows two PCIe x16 slots. It might be a bit hard to see in the picture, but on the right side of the slots there is a latch and tab that secures the x16 adapter in place by its hooks. Any movement of these high-performance adapters can result in temporary failure or poor performance, so the latch is critical. It can also make removing the card a little tricky if the latch isn't released first.

Using Riser Cards

Most PC expansion cards plug directly into the motherboard. A special type of expansion card, called a *riser card*, provides additional slots that other expansion cards plug into. The other expansion cards are then parallel to the motherboard, as opposed to perpendicular. Figure 1.5 shows an example of a riser card. It's an older example, but it illustrates the concept well.

Riser cards aren't often found in desktop PCs today but do still find some use in rack-mounted servers with low-profile cases. The motherboard must be designed to accept a riser card, and the case needs to be built for it as well, for the external portion of the expansion card to be accessible.

FIGURE 1.5 Riser card in a motherboard



Source: Dale Mahalko / Wikimedia Commons / CC BY SA 3.0

Memory Slots and Cache

Memory, or *random access memory (RAM)*, slots are the next most notable slots on a motherboard. These slots are designed for the modules that hold memory chips that make up primary memory, which is used to store currently used data and instructions for the CPU. Many types of memory are available for PCs today. In this chapter, you will become familiar with the appearance and specifications of the slots on the motherboard so that you can identify them and appropriately install or replace RAM.

For the most part, PCs today use memory chips arranged on a small circuit board. A *dual in-line memory module (DIMM)* is one type of circuit board. Today's DIMMs differ in the number of conductors, or pins, that each particular physical form factor uses. Some common examples include 168-, 184-, 240-, and 288-pin configurations. In addition, laptop memory comes in smaller form factors known as small outline DIMMs (SODIMMs) and Micro-DIMMs. More detail on memory packaging and the technologies that use them can be found later in this chapter in the section "Understanding Memory."

Memory slots are easy to identify on a motherboard. Classic DIMM slots were usually black and, like all memory slots, were placed very close together. DIMM slots with color-coding are more common these days, however. The color-coding of the slots acts as a guide to the installer of the memory. See the section "Channel Configurations," later in this chapter, for more on the purpose of this color-coding. Consult the motherboard's documentation to determine the specific modules allowed as well as their required orientation. The number of memory slots varies from motherboard to motherboard, but the structure of the different slots is similar. Metal pins in the bottom make contact with the metallic pins on each memory module. Small metal or plastic tabs on each side of the slot keep the memory module securely in its slot. Figure 1.6 shows four memory slots, with the CPU socket included for reference.

Another type of memory common in PCs is *cache memory*, which is small and fast and logically sits between the CPU and RAM. Cache is a very fast form of memory forged from static RAM, which is discussed in detail in the section "Understanding Memory," later in this chapter. Cache improves system performance by predicting what the CPU will ask for next and prefetching this information before being asked. This paradigm allows the cache to be smaller in size than the RAM itself. Only the most recently used data and code or that which is expected to be used next is stored in cache.

You'll see three different cache designations:

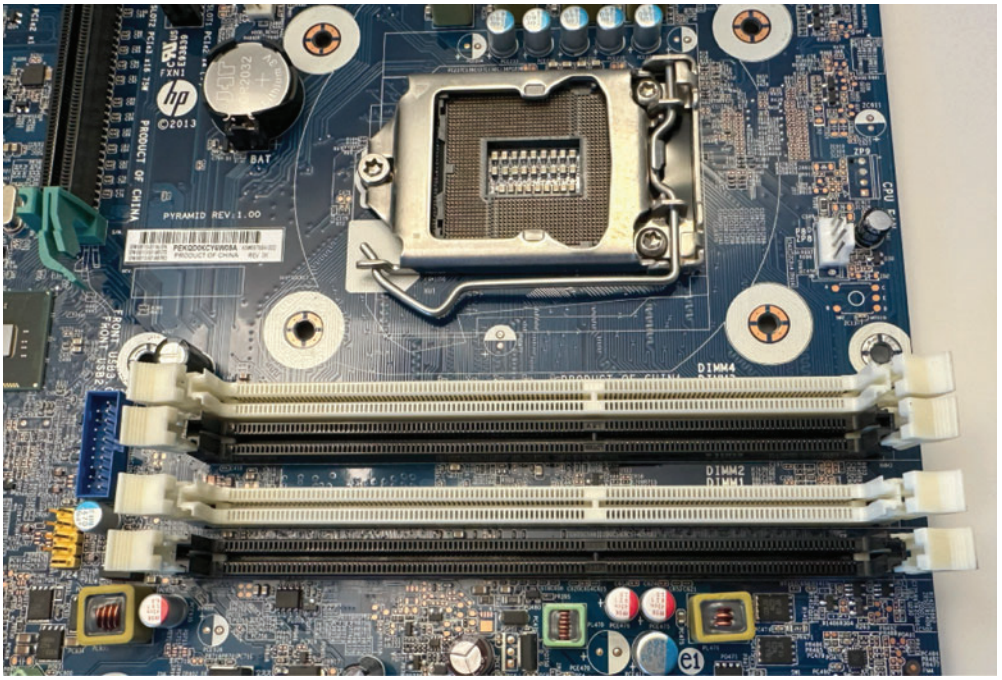
Level 1 Cache *L1 cache* is the smallest and fastest, and it's on the processor die itself. In other words, it's an integrated part of the manufacturing pattern that's used to stamp the processor pathways into the silicon chip. You can't get any closer to the processor than that.

Though the definition of L1 cache has not changed much over the years, the same is not true for other cache levels. L2 and L3 cache used to be on the motherboard but now have moved on-die in most processors as well. The biggest differences are the speed and whether they are shared.

Level 2 Cache *L2 cache* is larger but a little slower than L1 cache. For processors with multiple cores, each core will generally have its own dedicated L1 and L2 caches. A few processors share a common L2 cache among the cores.

Level 3 Cache *L3 cache* is larger and slower than L1 or L2, and is usually shared among all processor cores.

FIGURE 1.6 Double data rate (DDR) memory slots



The typical increasing order of capacity and distance from the processor die is L1 cache, L2 cache, L3 cache, RAM, and HDD/SSD (hard disk drive and solid-state drive—more on these in Volume 1 Chapter 2). This is also the typical decreasing order of speed. The following list includes representative capacities of these memory types. The cache capacities are for each core of the 10th generation Intel Core i7 processor. The other capacities are simply modern examples.

- **L1 cache:** 80 KB (32 KB for instructions and 48 KB for data)
- **L2 cache:** 512 KB
- **L3 cache:** 8–16 MB
- **RAM:** 16–256 GB
- **HDD/SSD:** 100s of GB to several TB

One way to find out how much cache your system has is to use a utility such as CPU-Z, as shown in Figure 1.7. CPU-Z is freeware that can show you the amount of cache, processor name and number, motherboard and chipset, and memory specifications. It can be found at `www.cpuid.com`.

FIGURE 1.7 Cache in a system

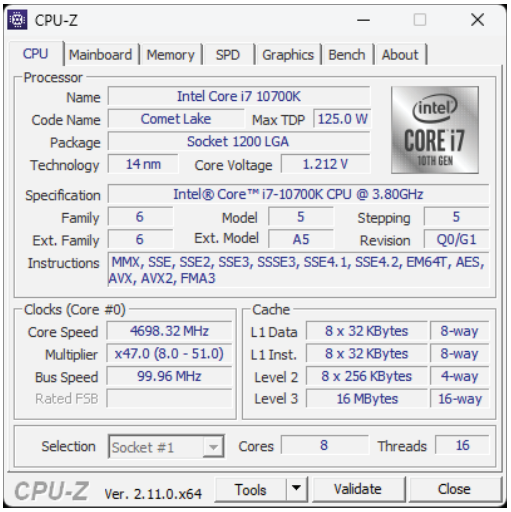


Figure 1.7 shows L1 Data and L1 Inst. caches. Many CPUs will split L1 cache into cache for data (L1D) or instructions (L1I). It's highly unlikely you'll be tested on this, but it's interesting to know.

Central Processing Unit and Processor Socket

The "brain" of any computer is the *central processing unit (CPU)*. There's no computer without a CPU. There are many different types of processors for computers—so many, in fact, that you will learn about them later in this chapter in the section "Understanding Processors." Typically, in today's computers, the processor is the easiest component to identify on the motherboard. It is usually the component that has either a fan or a heat sink (usually both) attached to it (as shown in Figure 1.8). These devices are used to draw away and disperse the heat that a processor generates. This is done because heat is the enemy of microelectronics. Today's processors generate enough heat that, without heat dispersal, they would permanently damage themselves and the motherboard in a matter of minutes, if not seconds.

CPU sockets are almost as varied as the processors that they hold. Sockets are basically flat and have several columns and rows of holes or pins arranged in a square, as shown in Figure 1.9. The left socket is known as Socket AM4, made for AMD processors such as

the Ryzen, and has holes to receive the pins on the CPU. This is known as a *pin grid array (PGA)* arrangement for a CPU socket. The holes and pins are in a row/column orientation, an array of pins. The right socket is known as LGA 1200, and there are spring-loaded pins in the socket and a grid of lands on the CPU. The *land grid array (LGA)* is a newer technology that places the delicate pins (yet more sturdy than those on chips) on the cheaper motherboard instead of on the more expensive CPU, opposite to the way that the aging PGA does. The device with the pins has to be replaced if the pins become too damaged to function. The PGA and LGA are mentioned again later in this chapter in the section "Understanding Processors."

FIGURE 1.8 Two heat sinks, one with a fan

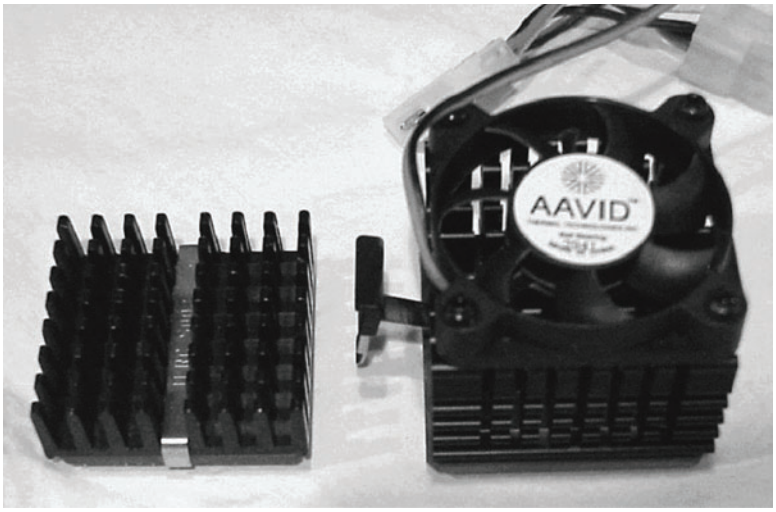
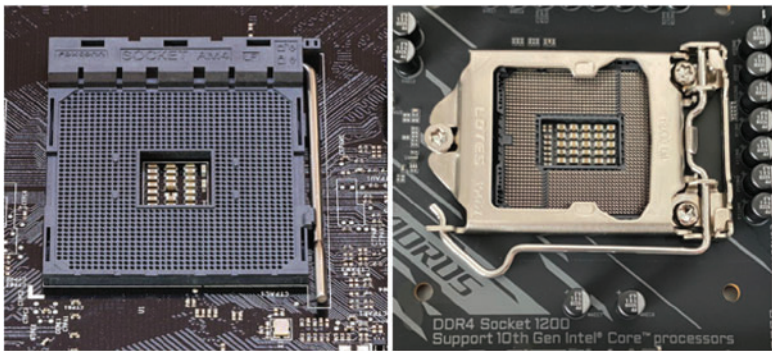


FIGURE 1.9 CPU socket examples



Modern CPU sockets have a mechanism in place that reduces the need to apply considerable force to the CPU to install a processor, which was necessary in the early days of personal computing. Given the extra surface area on today's processors, excessive pressure applied in the wrong manner could damage the CPU packaging, its pins, or the motherboard itself. For CPUs based on the PGA concept, *zero insertion force (ZIF)* sockets are exceedingly popular. ZIF sockets use a plastic or metal lever on one of the two lateral edges to lock or release the mechanism that secures the CPU's pins in the socket. The CPU rides on the mobile top portion of the socket, and the socket's contacts that mate with the CPU's pins are in the fixed bottom portion of the socket. The image of Socket AM4 shown on the left in Figure 1.9 illustrates the ZIF locking mechanism at the right edge of the socket.

For processors based on the LGA concept, a socket with a different locking mechanism is used. Because there are no receptacles in either the motherboard or the CPU, there is no opportunity for a locking mechanism that holds the component with the pins in place. LGA-compatible sockets, as they're called despite the misnomer, have a lid that closes over the CPU and is locked in place by an L-shaped arm that borders two of the socket's edges. The nonlocking leg of the arm has a bend in the middle that latches the lid closed when the other leg of the arm is secured. The right image in Figure 1.9 shows an LGA socket with no CPU installed and the locking arm secured over the lid's tab, along the bottom edge.

Listing out all the desktop PC socket types you might encounter would take a long time. Instead, we'll give you a sampling of some that you might see. The first thing you might notice is that sockets are made for Intel or AMD processors, but not both. Keep that compatibility in mind when replacing a motherboard or a processor. Make sure that the processor and motherboard were designed for each other (even within the Intel or AMD families); otherwise, they won't fit each other and won't work. Table 1.1 lists some common desktop socket/CPU relationships. Servers and laptops/tablets generally have different sockets altogether, although some CPU sockets will support processors designed for desktops or servers.

TABLE 1.1 Desktop PC socket types and the processors they support

Socket	Released	Type	Pin count	Processors
Socket sTR5	2023	LGA	4844	AMZ Ryzen Threadripper
LGA 1700	2021	LGA	1700	Intel Alder Lake and Raptor Lake
LGA 1200	2020	LGA	1200	Intel Comet Lake and Rocket Lake
Socket AM4	2017	PGA	1331	AMD Ryzen 3, Ryzen 5, Ryzen 7, Ryzen 9, Athlon 200GE
LGA 2066 (Socket R4)	2017	LGA	2066	Intel Skylake-X and Kaby Lake-X
LGA 1151 (Socket H4)	2015	LGA	1151	Intel Skylake, Kaby Lake, and Coffee Lake

Socket	Released	Type	Pin count	Processors
Socket FM2+	2014	PGA	906	AMD Kaveri and Godavari
Socket AM1	2014	PGA	721	AMD Athlon and Sempron

Real World Scenario

Playing the Name Game

Back in simpler times, Intel marketed only one or two processors at a time. For example, when the 80486DX was on the market, they didn't have much else. Then the Pentium came along, followed by the Celeron (which everyone knew was a stripped-down Pentium), and the Pentium M, which was the mobile version of the Pentium. The active roster of processors started to expand.

Now, there are a lot of active chips, including the Xeon and the Core series (i9, i7, i5, and i3), and even the Pentium and Celeron are still kicking around. And with new innovation coming nearly every year, it's difficult to keep track of what's current.

In about 2013, Intel began to publicize its project code names a bit more to make it easier to keep track of the new technology bundles. No longer were techs able to remember that an i7 required a specific type of socket, but they needed to know *which* i7 they were dealing with. For example, a Comet Lake (2020) i7 requires an LGA 1200 socket, as does the Rocket Lake, whereas the Alder Lake (2021) i7s use an LGA 1700.

You aren't going to need to memorize all possible Intel chip and motherboard socket combinations for the exam or the real world. In the real world, the Internet's always there to remind you of which socket you need for a specific processor. To give you a flavor, though, Table 1.2 covers a few of Intel's recent CPU architecture codenames and the associated processor model numbers.

TABLE 1.2 Select Intel desktop processors

Name (Year)	Gen	Socket	Core i9	Core i7	Core i5	Core i3
Raptor Lake-S Refresh (2023)	14th	LGA1700	149xx	147xx	146xx 145xx 144xx	141xx

TABLE 1.2 Select Intel desktop processors *(continued)*

Name (Year)	Gen	Socket	Core i9	Core i7	Core i5	Core i3
Raptor Lake (2022)	13th	LGA 1700	139xx	137xx	136xx 135xx 134xx	131xx
Alder Lake (2021)	12th	LGA 1700	129xx	127xx	126xx	n/a
Rocket Lake (2020)	11th	LGA 1200	119xx	117xx	116xx 115xx 114xx	n/a

To make things even more interesting, you might hear references to the *n*th generation of the i5, i7, or i9 processor. For example, Raptor Lake is considered 13th generation. Intel codenames over the past several years have had some variation of a "Lake" name, such as Coffee Lake (2017), Comet Lake (2019), and Rocket Lake (2020). Perhaps the names keep it interesting for the engineers.

When you go to the parts store, though, you're probably not going to ask for a Rocket Lake i7. You'll ask for an i7, see what they have in stock—usually they will have the current generation and possibly the previous one, and then figure out what motherboard you need. (Or alternatively, based on your motherboard, you'll figure out which i7 you can use.) For example, the motherboard may advertise that it supports "12th and 13th generation" Intel processors. If your processor number starts with a 12xxx or a 13xxx, you should be good to go.

What does all of this mean you need to do? First, get comfortable knowing that a new architecture may require a different motherboard. Second, feel free to memorize some of the more common combinations of motherboards and processors if you find it helpful, although you won't be tested on it. It may be handy in the real world to know that if you have an 11th gen Intel chip, you need a board with an LGA 1200 socket. (Or you can always Google it!)

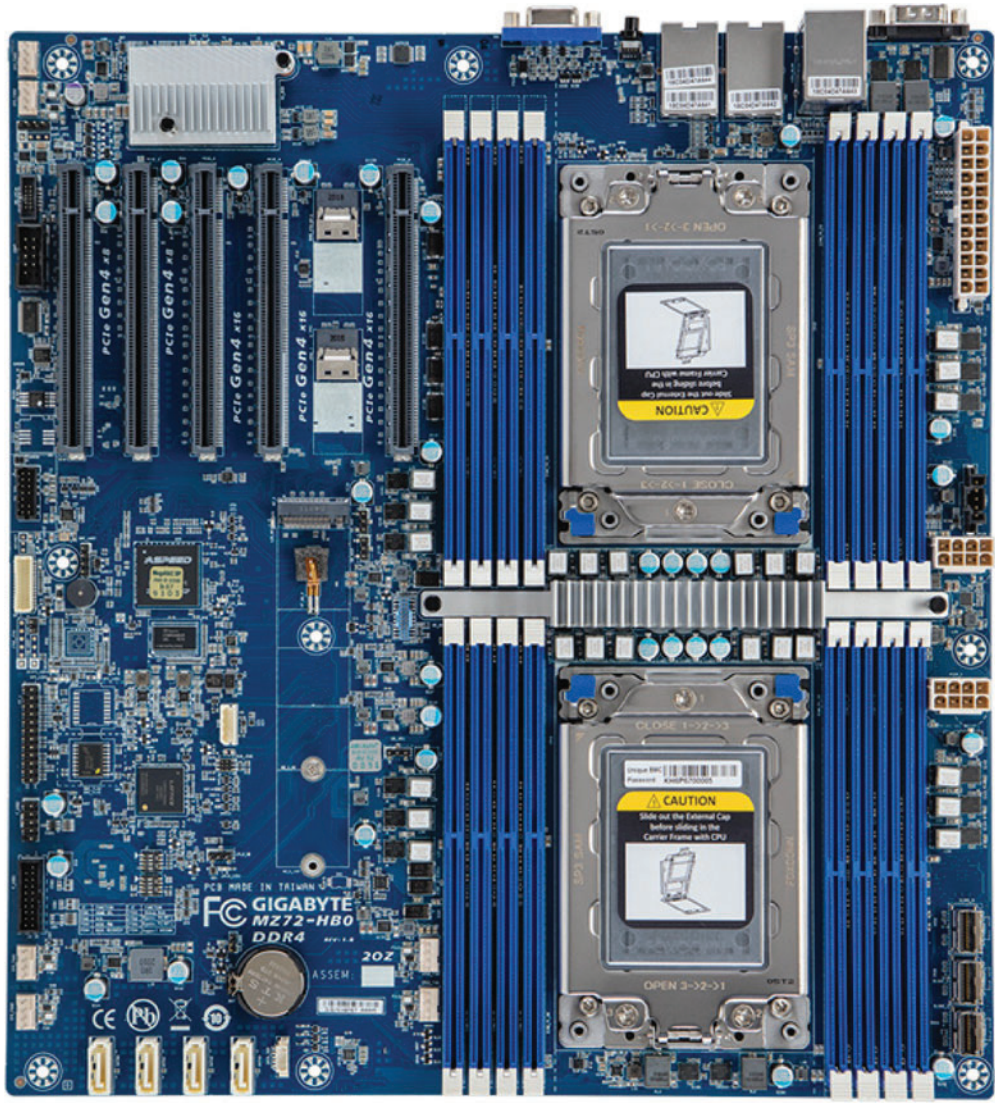
For A+ Exam Objective 3.5, be familiar with compatibility of CPU socket types (AMD and Intel) as well as multisocket motherboards.

Multisocket Motherboards

When it comes to motherboard compatibility, the two biggest things to keep in mind are the processor type and the case. If either of those are misaligned with what the motherboard supports, you're going to have problems.

Thus far, as we've talked about desktop motherboards and their CPU sockets, we have shown examples of boards that have just one socket. There are motherboards that have more than one CPU socket and conveniently, they are called *multisocket* (typically written as two words) motherboards. Figure 1.10 shows a two-socket motherboard made by GIGABYTE. The two CPU sockets are easily identifiable and note that each CPU socket has eight dedicated memory slots. Multisocket motherboards will generally use the same CPU sockets that single-socket motherboards use.

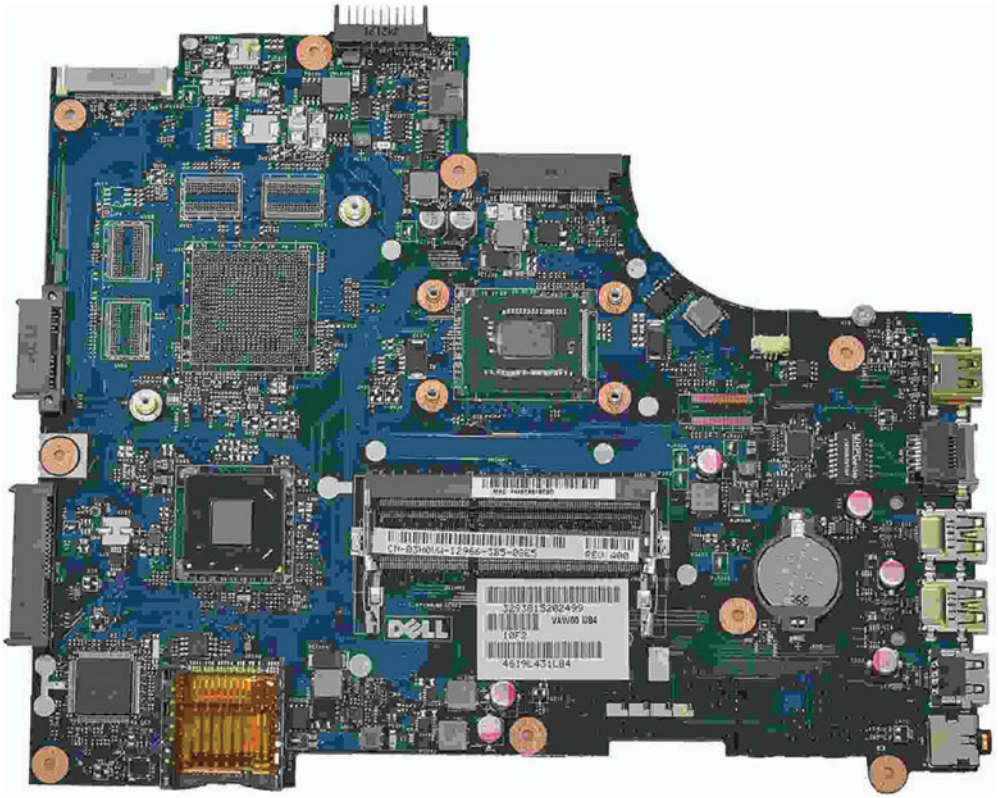
FIGURE 1.10 GIGABYTE multisocket motherboard



Mobile Motherboards

In small mobile devices, space is at a premium. Some manufacturers will use standard small-factor motherboards, but most create their own boards to fit inside specific cases. An example of an oddly shaped Dell laptop motherboard is shown in Figure 1.11. When replacing a laptop motherboard, you almost always need to use one from the exact same model; otherwise it won't fit inside the case.

FIGURE 1.11 Dell laptop motherboard

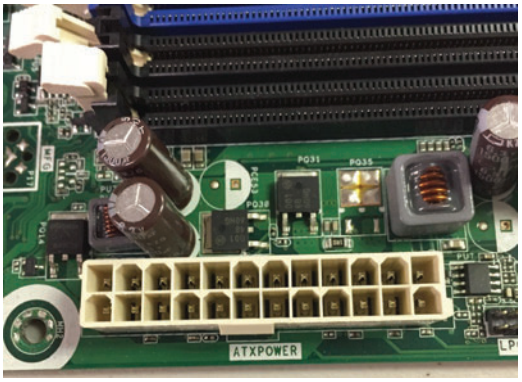


Nearly all laptop processors are soldered onto the motherboard, so you don't have to worry about CPU socket compatibility. If the CPU dies, you replace the entire motherboard. We will cover laptop components more extensively in Volume 1 Chapter 4, "Laptop and Mobile Device Hardware."

Power Connectors

In addition to these sockets and slots on the motherboard, a special connector (the 24-pin white block connector shown in Figure 1.12) allows the motherboard to be connected to the power supply to receive power. This connector is where the ATX power connector (mentioned in Volume 1 Chapter 2 in the section "Understanding Power Supplies") plugs in.

FIGURE 1.12 An ATX power connector on a motherboard



Onboard Nonvolatile Storage Connectors

Nearly all users store data, and the most widely used data storage device is a hard drive. Hard drives are great because they store data even when the device is powered off, which explains why they are sometimes referred to as nonvolatile storage. There are multiple types of hard drives, and we'll get into them in more detail in Volume 1 Chapter 2. Of course, those drives need to connect to the motherboard, and that's what we'll cover here.

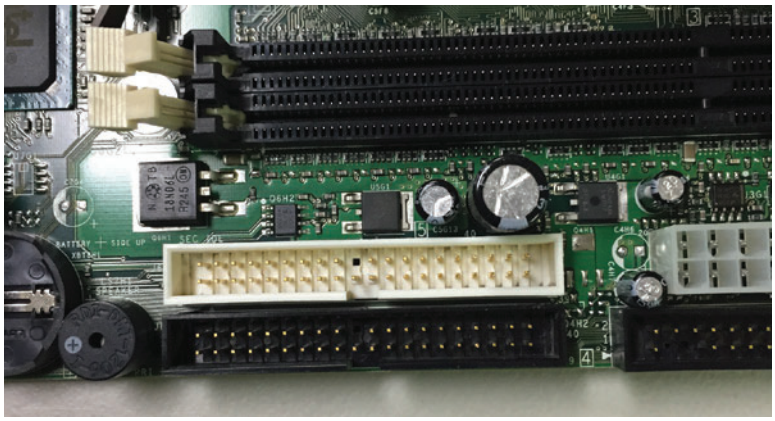
Exam Objective 3.5 lists SATA, eSATA, SAN, and M.2 motherboard connectors. Know the differences between them!

Integrated Drive Electronics/Parallel ATA

At one time, *integrated drive electronics (IDE)* drives were the most common type of hard drive found in computers. Though often thought of in relation to hard drives, IDE was much more than a hard drive interface; it was also a popular interface for many other drive types, including optical drives and tape drives. Today, IDE is referred to as Parallel ATA (PATA), and it's a legacy technology that's obsolete. We mention it to help you understand where SATA (covered next, and a test objective) came from. And believe it or not, you might still see PATA connectors and drives in use today.

Figure 1.13 shows two PATA interfaces; you can see that one pin in the center is missing (as a key) to ensure that the cable gets attached properly. The industry now favors Serial ATA instead.

FIGURE 1.13 Two PATA hard drive connectors



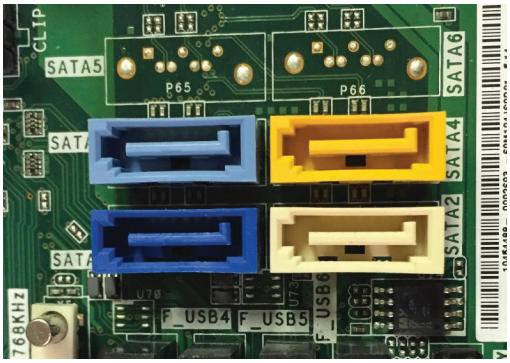
Serial ATA

Serial ATA (SATA) began as an enhancement to the original ATA specifications (IDE/PATA). Technology is proving that the orderly progression of data in a single-file path is superior to placing multiple bits of data in parallel and trying to synchronize their transmission to the point that all of the bits arrive simultaneously. Serial is simply faster than parallel in today's computer technology.

The first version of SATA, known as SATA 1.5 Gbps (and also by the less-preferred terms SATA I and SATA 150), used an 8b/10b-encoding scheme that requires 2 non-data overhead bits for every 8 data bits. The result is a loss of 20 percent of the rated bandwidth. The silver lining, however, is that the math becomes quite easy. Normally, you have to divide by 8 to convert bits to bytes. With 8b/10b encoding, you divide by 10. Therefore, the 150 Mbps throughput for which this version of SATA was nicknamed is easily derived as 1/10 of the 1.5 Gbps transfer rate. The original SATA specification also provided for hot swapping at the discretion of the motherboard and drive manufacturers.

Similar math works for SATA 3 Gbps, tagged as SATA II and SATA 300, and SATA 6 Gbps, which you might hear called SATA III or SATA 600. Note that each subsequent version doubles the throughput of the previous version. Figure 1.14 shows four SATA headers on a motherboard that will receive the data cable. Note that identifiers silkscreened onto motherboards often enumerate such headers. The resulting numbers are not related to the SATA version that the header supports. Instead, such numbers serve to differentiate headers from one another and to map to firmware identifiers, often visible within the BIOS configuration utility.

FIGURE 1.14 Four SATA headers



Another version of SATA that you will sometimes see is *external SATA (eSATA)*. As you might expect based on the name, this technology was developed for devices that reside outside of the case, not inside it. Some motherboards have an eSATA connector built in. If not, you can buy an expansion card that has eSATA ports and plugs into internal SATA connectors. Figure 1.15 shows an example of an eSATA port below two USB ports. SATA and eSATA standards are compatible. In other words, SATA 6 Gbps equals eSATA 6 Gbps.

FIGURE 1.15 eSATA port (bottom center)

SATA and eSATA ports (and PATA, for that matter) do not provide power, like USB and Thunderbolt do. Therefore, when connecting SATA and eSATA drives, a separate power connection is required in addition to the data cable. This limitation, along with slower speeds, has led to the demise of eSATA as a standard in favor of USB-C and Thunderbolt. We will cover both USB-C and Thunderbolt in more detail in Volume 1 Chapter 2.

What's in a Name?

The original *small computer system interface (SCSI)* standard was introduced in 1986 as a fast interface for hard drives and tape drives. Few motherboards had integrated SCSI controllers. If you wanted a SCSI hard drive, you needed to add an internal expansion card with a SCSI controller on it, which looked a bit like PATA but wider, and it had clips on either side. It was once popular for high-end systems, but the speed and lower cost of SATA made the original SCSI fade in popularity except for newer variants like iSCSI and *serial attached SCSI* (*SAS*; covered in Volume 1 Chapter 2), which are used in specialized applications.

As for the acronym pronunciation, the legend is that the developer wanted to call it *sexy*. After all, it was derived from the earlier *Shugart Associates System Interface (SASI)* standard which was called *sassy*. The *sexy* moniker was deemed too edgy for the computer industry at the time, and the *scuzzy* alternative took over. Who knows if the legend is true, but it makes a good story regardless.

M.2

The most recent development in expansion connections is *M.2* (pronounced "M dot 2"). So far it's primarily used for hard drives, but other types of devices, such as Wi-Fi, Bluetooth, Global Positioning System (GPS), and near-field communication (NFC) adapters are built for M.2 as well. We will cover M.2 in more depth in Volume 1 Chapter 2, thanks to how important it is to storage solutions.

It's important to call out that M.2 is a form factor, not a bus standard. The form factor supports existing SATA, USB, and PCIe buses. This means that if you hook up a SATA device to an M.2 slot (with the appropriate connector), the device speed will be regulated by SATA standards. Figure 1.16 shows an M.2 connector, highlighted and in the center pointing downward. To get to it, you will need to loosen a screw that holds a protective cover in place over the slot and remove the cover. The protective cover is shown in Figure 1.17, with the screw on the left.

M.2 is closely associated with hard drives, and the M.2 form factor is also in CompTIA A+ Exam Objective 3.4, "Compare and contrast storage devices." Because of that, we will go into more depth about M.2 and hard drive data transfer speeds in Volume 1 Chapter 2 when we discuss solid-state drives (SSDs).

Motherboard Headers

From the time of the very first personal computer, there has been a minimum expectation as to the buttons and LEDs that should be easily accessible to the user. At first, they generally appeared on the front of the case. In today's cases, buttons and LEDs have been added and placed on the top of the case or on a beveled edge between the top and the front. They have

also been left on the front or have been used in a combination of these locations. These buttons and lights, as well as other external connectors, plug into the motherboard through a series of pins known as *headers*. Examples of items that are connected using a header include the following:

- Power button
- Power light
- Reset button
- Audio jacks
- USB ports

FIGURE 1.16 M.2 connector on motherboard

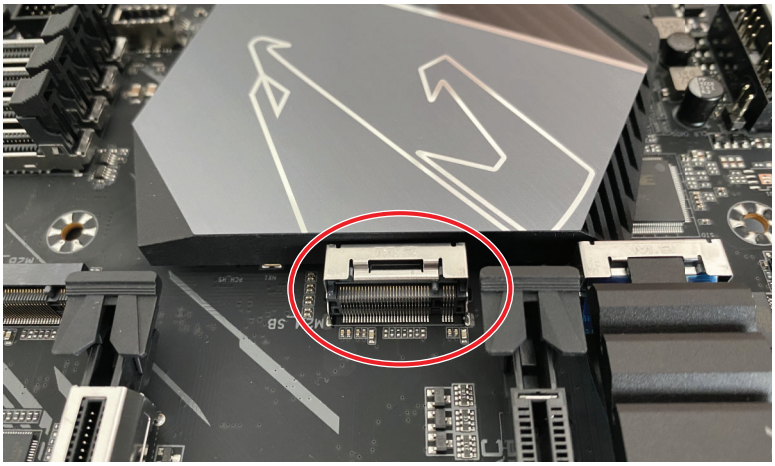
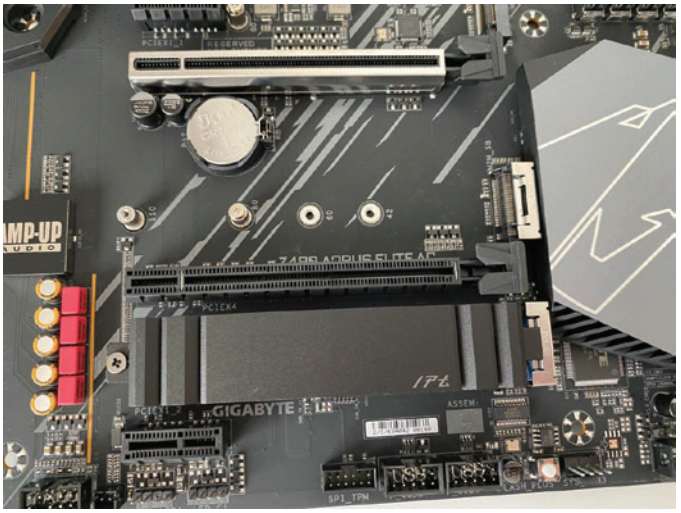
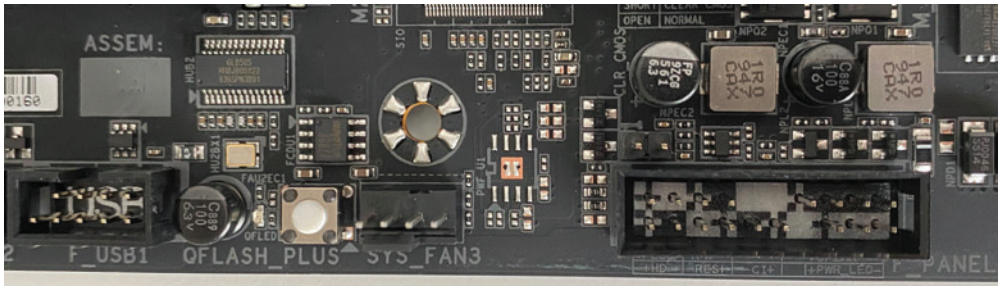


FIGURE 1.17 M.2 slot protective cover



Headers for different connections are often spread throughout different locations on the motherboard—finding the right one can sometimes be a frustrating treasure hunt. Other headers are grouped together. For example, most of the headers for the items on the front or top panel of the case are often co-located. The purpose for the header will be printed on the motherboard, and while that may tell you what should connect there, it often lacks detail in how it should be connected. The motherboard manufacturer's website is a good place to go if you need a detailed diagram or instructions. Figure 1.18 shows several headers on a motherboard. On the left is a USB header, then a system fan header in the center, and a block of front-panel headers on the right, including the hard drive light, reset button, chassis intrusion detector, and power light.

FIGURE 1.18 Motherboard headers



Exam Objective 3.5 says you need to know about motherboard headers. The exam objective doesn't list which headers you need to know, so we cover the most common ones here.

Power Button and Light

Users expect a *power button* to use to turn the computer on. (These were on the side or back of very early PCs.) The soft power feature available through the front power button, which is no more than a relay, allows access to multiple effects through the contact on the motherboard, based on how long the button is pressed. These effects can be changed through the BIOS/UEFI settings or operating system. Users also expect a *power light*, often a green LED, to assure them that the button did its job.

Reset Button

The *reset button* appeared as a way to reboot the computer from a cold startup point without removing power from the components. Keeping the machine powered tends to prolong the life of the electronics affected by power cycling. Pressing the reset button also gets around software lockups because the connection to the motherboard allows the system to restart from the hardware level. One disadvantage to power cycling is that certain circuits, such as memory chips, might need time to drain their charge for the reboot to be completely successful. This is why there is always a way to turn the computer off as well.

Audio Jacks

Early generations of optical drives had to have a special cable attached to the rear of the drive, which was then attached to the sound card if audio CDs were to be heard through the speakers attached to the sound card. Sound emanating from a CD-ROM running an application, such as a game, did not have to take the same route and could travel through the same path from the drive as general data. The first enhancement to this arrangement came in the form of a front 3.5mm jack on the drive's faceplate that was intended for headphones but could also have speakers connected to it. The audio that normally ran across the special cable was rerouted to the front jack when something was plugged into it.

Many of today's motherboards have 10-position pin headers designed to connect to standardized front-panel audio modules. Some of these modules have legacy AC'97 analog ports on them, whereas others have high-definition (HD) audio connections. Motherboards that accommodate both have a BIOS/UEFI setting that enables you to choose which header you want to activate, with the HD setting most often being the default.

USB Ports

So many temporarily attached devices feature USB connectivity, such as USB flash drives and cameras, that front-panel connectivity is a must. Finding your way to the back of the system unit for a brief connection is hardly worth the effort in some cases. For many years, motherboards have supplied one or more 10-position headers for internal connectivity of front-panel USB ports. Because this header size is popular for many connectors, only nine positions tend to have pins protruding, while the 10th position acts as a key, showing up in different spots for each connector type to discourage the connection of the wrong cable. Figure 1.19 shows USB headers on a motherboard. The labels "USB56" and "USB78" indicate that one block serves ports 5 and 6, while the other serves ports 7 and 8, all of which are arbitrary, based on the manufacturer's numbering convention. In each, the upper-left pin is "missing," which is the key.

FIGURE 1.19 Two motherboard USB headers

BIOS/UEFI

Firmware is the name given to any software that is encoded in hardware, usually a read-only memory (ROM) chip, and it can be run without extra instructions from the operating system. Most computers, large printers, routers, and devices with no operating system use firmware in some sense. The best example of firmware is a computer's *Basic Input/Output System (BIOS)*, which is burned into a chip. Also, some expansion cards, such as graphics adapters, use their own firmware utilities for setting up and configuring different functions.

The BIOS chip, also referred to as the ROM BIOS chip, is one of the most important chips on the motherboard. This special memory chip contains the BIOS system software that boots the system and allows the operating system to interact with certain hardware in the computer in lieu of requiring a more complex device driver to do so. The BIOS chip is easily identified; if you have a brand-name computer, this chip might have on it the name of the manufacturer and usually the word *BIOS*. For clones, the chip usually has a sticker or printing on it from one of the major BIOS manufacturers (AMI, Phoenix, Award, Winbond, and others).

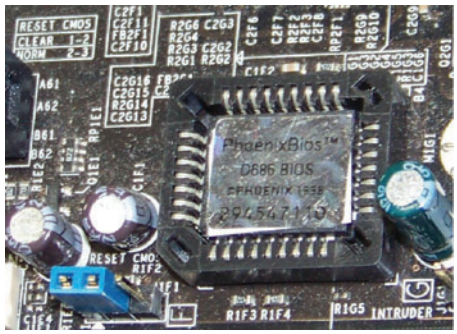
PC technology and security requirements started to outclass what the BIOS could support, and in 2002 a successor to the BIOS was released, called *Unified Extensible Firmware Interface (UEFI)*. The extensible nature of the UEFI supports key features, such as these:

- Graphical interface
- Larger hard drives (BIOS was limited to 2.2 TB)
- Better security such as Secure Boot and TPM
- Faster boot times

Even though on PCs BIOS technology is outdated, you still hear the term used quite a bit. While the technologies are different, in practice, the terms BIOS and UEFI are essentially interchangeable.

Figure 1.20 gives you an idea of what a modern BIOS/UEFI chip might look like on a motherboard. Despite the 1998 copyright on the label, which refers only to the oldest code present on the chip, this particular chip can be found on motherboards produced as late as 2009. Notice also the Reset CMOS jumper at the lower left and its configuration silkscreen at the upper left. You might use this jumper to clear the CMOS memory, discussed shortly, when an unknown password, for example, is keeping you out of the BIOS/UEFI configuration utility. The jumper in the photo is in the Reset position, not the normal operating position. System bootup is typically not possible in this state.

FIGURE 1.20 A BIOS chip on a motherboard



The BIOS/UEFI is responsible for several key functions related to the proper booting of a computer, and then it hands off control of the hardware to the operating system. Let's break some of these functions down into three groups: configuration and monitoring, security and encryption, and the power-on self-test (POST) routine.

Exam Objective 3.5 has quite a few BIOS/UEFI features to remember for the exam. They include boot options, USB permissions, TPM security features, fan considerations, Secure Boot, boot password, BIOS password, and temperature monitoring. Encryption is also part of this exam objective and covers TPM as well as HSM.

Configuration and Monitoring

The BIOS/UEFI contains basic configuration information about the CPU, RAM, and hard drives installed in a computer. It will detect the size and/or speed of these devices, and in some instances let you modify performance characteristics. Here are some features to be familiar with:

Boot Options In the BIOS/UEFI, you can set the device from which the computer boots an operating system. This can be a hard drive or an optical drive, or in some cases a flash drive. Figure 1.21 shows the boot option setting in a GIGABYTE AORUS motherboard.

FIGURE 1.21 UEFI boot option

USB Permissions USB ports can be physically disabled using the BIOS/UEFI interface. This can act as an additional security measure, because if the ports are disabled, someone can't introduce malware (such as a virus) into the computer by using a flash drive in the port.

Fan Considerations The BIOS/UEFI interface will allow you to turn system fans on or off as well as monitor the fan speeds. This is useful if you have a system that's overheating. It can also be helpful as a troubleshooting tool if you have a fan that's constantly speeding up or slowing down, or making excessive noise.

Temperature Monitoring The inside of a computer case can get hot, and the CPU is particularly good at generating excessive amounts of heat. While internal computer components are designed to withstand some heat, too much can cause them to malfunction. BIOS/UEFI utilities let you monitor temperature for the CPU, RAM, and often the ambient temperature of air inside the case as well.

Knowing how to enter the BIOS/UEFI and configure settings is an important skill for any technician. Exercise 1.1 shows you how to do it from Windows 11.

EXERCISE 1.1

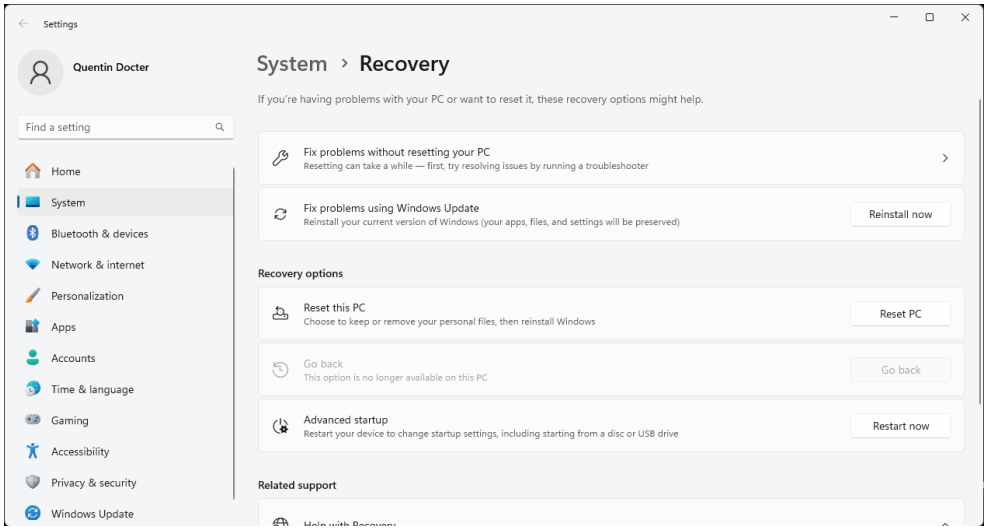
Entering the UEFI from Windows 11

Some systems may let the user enter the BIOS/UEFI by pressing a key during the boot sequence. If this is the situation, a message will appear quickly on the screen, prior to the operating system loading screen, instructing the user to press a key such as F2, F12, or Delete.

Newer computers frequently omit this option. Computers running Windows 10 and 11 allow the user to reboot into the BIOS by using the Advanced startup option within the System Recovery tool.

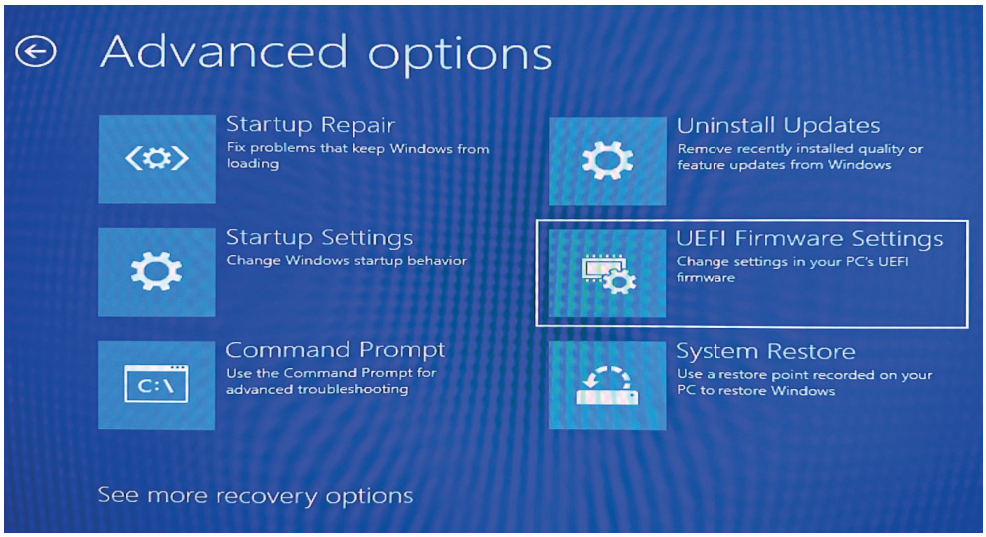
1. In the Windows search bar, type **`Advanced startup`**. The Change advanced startup options system settings app will appear in the menu. Press Enter to open it. You will see a screen similar to the one in Figure 1.22.

FIGURE 1.22 System Recovery options



2. Next to Advanced Startup, click Restart Now.
3. After the system reboots, you will see the Choose An Option screen with three options: Continue To Windows 11, Troubleshoot, and Turn Off Your PC. Click Troubleshoot.
4. On the Troubleshoot screen, click Advanced Options.
5. On the Advanced Options screen (Figure 1.23), click UEFI Firmware Settings.

FIGURE 1.23 Advanced Options



EXERCISE 1.1 *(continued)*

6. Click to confirm the reboot, and the system will reboot into the UEFI.

Explore the options; most UEFI interfaces have several tabs. Be careful, though, because changing settings in your UEFI could cause your computer to not operate properly. Don't change a setting unless you're sure you know what it does! If you do accidentally change something and you're concerned, there are two options. Either exit without saving, or reboot the computer using the power or reset button.

Security and Encryption

The BIOS/UEFI has always played a role in system security. Since the early days of the personal computer, the BIOS allowed the setting of two passwords—the user (or boot) password and the supervisor/administrator, or access, password. The *boot password* is required to leave the initial power-on screens and begin the process of booting an operating system. The administrator *BIOS password* is required before entering the BIOS/UEFI configuration utility. It is always a good idea to set the administrator password, but the boot password should not be set on public systems that need to boot on their own, in case of an unforeseen power cycle.

In more recent years, the role of the BIOS/UEFI in system security has grown substantially. BIOS/UEFI security has been extended to a point where the operating system is ready to take it over. The BIOS/UEFI was a perfect candidate to supervise security and integrity in a platform-independent way.

Most motherboards today have a dedicated security coprocessor, or cryptoprocessor, called the *Trusted Platform Module (TPM)*, installed on them. When enabled in the UEFI, UEFI works with the TPM and will boot only to an authenticated boot device. This authentication confirms that the hardware being booted to has been tied to the system containing the BIOS/UEFI and TPM, a process known as *sealing*. Sealing the devices to the system also prohibits the devices from being used after removing them from the system. For further security, the keys created can be combined with a PIN or password that unlocks their use or with a USB flash drive that must be inserted before booting.

In the AORUS UEFI, TPM is enabled in Settings ➢ Trusted Computing (Figure 1.24). The TPM settings are further shown in Figure 1.25.

FIGURE 1.24 UEFI settings

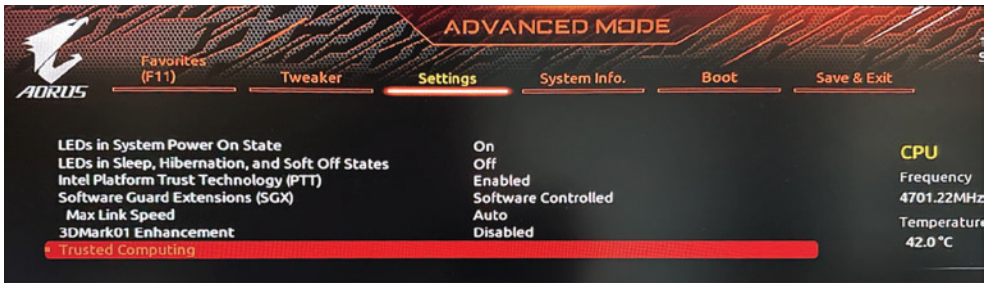


FIGURE 1.25 TPM settings

Microsoft's BitLocker uses the TPM to encrypt the entire drive. Normally, only user data can be encrypted, but BitLocker encrypts operating system files, the Registry, the hibernation file, and so on, in addition to those files and folders that file-level encryption secures. If any changes have occurred to the Windows installation, the TPM does not release the keys required to decrypt and boot to the secured volume. TPM status is shown in Windows 11 in the Device Security app. To open it, type **`tpm`** into the Windows search box and press Enter. Then click the Security Processor Details link and you will see a screen similar to the one shown in Figure 1.26.

Most motherboards come with a TPM chip installed, but if they don't, it's not possible to add one. In those situations, you can enable the same functionality by using a *hardware security module (HSM)*. An HSM is a security device that can manage, create, and securely store encryption keys—it enables users to safely encrypt and decrypt data. An HSM can take a few different forms. The simplest is a USB or PCIe device that plugs into a system. It could be set up for file encryption and decryption, required for the computer to boot, or both. For large-scale solutions, HSM-enabled servers can provide crypto services to an entire network. So while an HSM can perform the functionality of a TPM, it can have much broader application as well. They can be used together to form layers of security.

When a certain level of UEFI is used, the system firmware can also check digital signatures for each boot file it uses to confirm that it is the approved version and has not been tampered with. This technology is known as *Secure Boot*. An example of a BIOS/UEFI's

boot security screen is shown in Figure 1.27. The boot files checked include option ROMs (defined in the following section), the boot loader, and other operating system boot files. Only if the signatures are valid will the firmware load and execute the associated software.

FIGURE 1.26 Windows TPM configuration screen

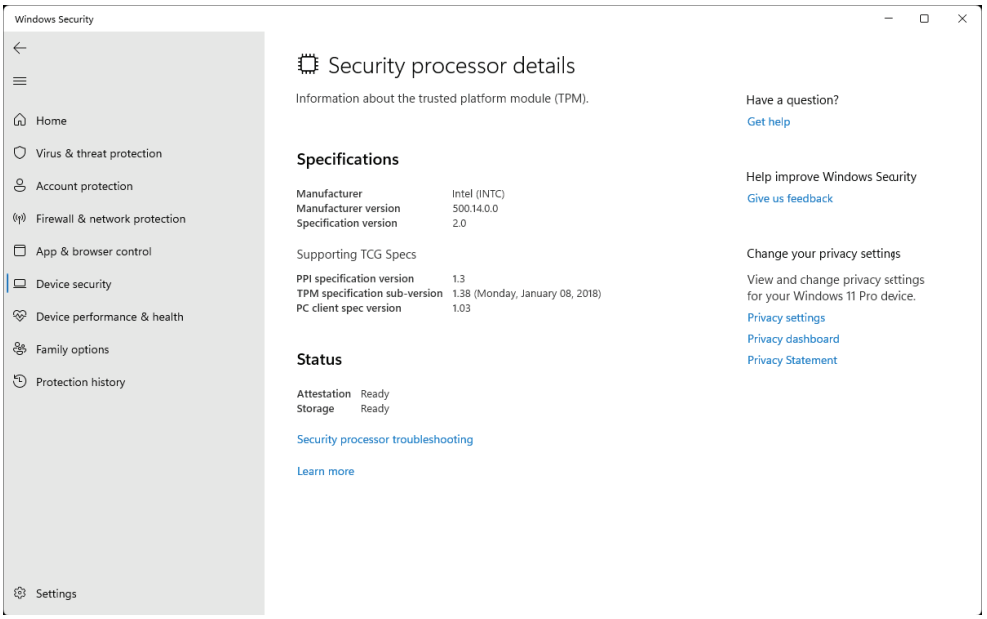
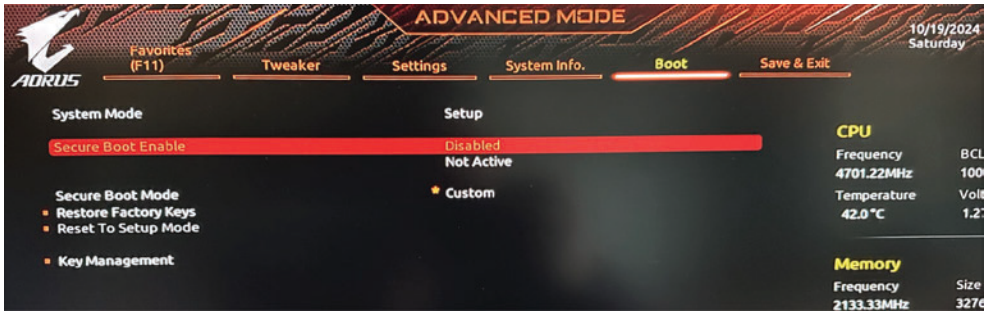


FIGURE 1.27 Secure Boot in UEFI



Some BIOS/UEFI firmware can monitor the status of a contact on the motherboard for intrusion detection. If the feature in the BIOS/UEFI is enabled and the sensor on the chassis is connected to the contact on the motherboard, the removal of the cover will be detected

and logged by the BIOS/UEFI. This can occur even if the system is off, thanks to the CMOS battery. At the next bootup, the BIOS/UEFI will notify you of the intrusion. No notification occurs over subsequent boots unless additional intrusion is detected.

POST Routine

A major function of the BIOS/UEFI is to perform a process known as a *power-on self-test (POST)*. POST is a series of system checks performed by the system BIOS/UEFI and other high-end components, such as the SCSI BIOS and the video BIOS, known collectively as *option ROMs*. Among other things, the POST routine verifies the integrity of the BIOS/UEFI itself. It also verifies and confirms the size of primary memory. During POST, the BIOS also analyzes and catalogs other forms of hardware, such as buses and boot devices, as well as manages the passing of control to the specialized BIOS/UEFI routines mentioned earlier. The BIOS/UEFI is responsible for offering the user a key sequence to enter the configuration routine as POST is beginning. Finally, once POST has completed successfully, the BIOS/UEFI selects the boot device highest in the configured boot order and executes the master boot record (MBR) or similar construct on that device so that the MBR can call its associated operating system's boot loader and continue booting up.

The POST process might end with a beep code or displayed code that indicates the issue discovered. Each BIOS/UEFI publisher has its own series of codes that can be generated. If you boot a computer and get a series of beeps and the operating system never loads, odds are there was a problem during the POST. At that point, check the motherboard manufacturer's documentation to see what the beep code is telling you.

Flashing the System BIOS/UEFI

If ever you find that a hardware upgrade to your system is not recognized, even after the latest and correct drivers (small programs that tell the OS how to interact with the hardware) have been installed, perhaps a BIOS/UEFI upgrade, also known as *flashing the BIOS*, is in order. Only certain hardware benefits from a BIOS/UEFI upgrade, such as drives and a change of CPU or RAM types. Very often, this hardware is recognized immediately by the BIOS/UEFI and has no associated driver that you must install. So, if your system doesn't recognize the new device, and there's no driver to install, the BIOS/UEFI is a logical target.

Let's be clear that we are not talking about entering the BIOS/UEFI setup utility and making changes to settings and subsequently saving your changes before exiting and rebooting. What we are referring to here is a replacement of the burned-in code within the BIOS itself. You might even notice after the upgrade that the BIOS setup utility looks different or has different pages and entries than before.

On older systems and certain newer ones, a loss of power during the upgrade results in catastrophe. The system becomes inoperable until you replace the BIOS/UEFI chip, if possible, or the motherboard itself. Most new systems, however, have a fail-safe or two. This could be a portion of the BIOS/UEFI that does not get flashed and has just enough code to

boot the system and access the upgrade image. It could be a passive section to which the upgrade is installed and switched to only if the upgrade is successful. Sometimes this is controlled onscreen. At other times, there may be a mechanism, such as a jumper, involved in the recovery of the BIOS/UEFI after a power event occurs. The safest bet is to make sure that your laptop has plenty of battery power and is connected to AC power or your desktop is connected to an uninterruptible power supply (UPS).

In all cases, if you think you need a BIOS/UEFI upgrade, do not consult the BIOS/UEFI manufacturer. Instead, go to the motherboard or system manufacturer and check its website. The motherboard or system manufacturer vendors have personalized their BIOS/UEFI code after licensing it from the BIOS/UEFI publisher. The vendor will give you access to the latest code as well as the appropriate flashing utility for its implementation.

CMOS and CMOS Battery

Your PC has to keep certain settings when it's turned off and its power cord is unplugged:

- Date
- Time
- Hard drive/optical drive configuration
- Memory
- CPU settings, such as overclocking
- Integrated ports (settings as well as enable/disable)
- Boot sequence
- Power management
- Virtualization support
- Security (passwords, Trusted Platform Module settings, etc.)

Your PC keeps these settings in a special memory chip called the *complementary metal oxide semiconductor (CMOS)* memory chip. Actually, CMOS (usually pronounced *see-moss*) is a manufacturing technology for integrated circuits. The first commonly used chip made from CMOS technology was a type of memory chip, the memory for the BIOS/UEFI. As a result, the term *CMOS* stuck and is the accepted name for this memory chip.

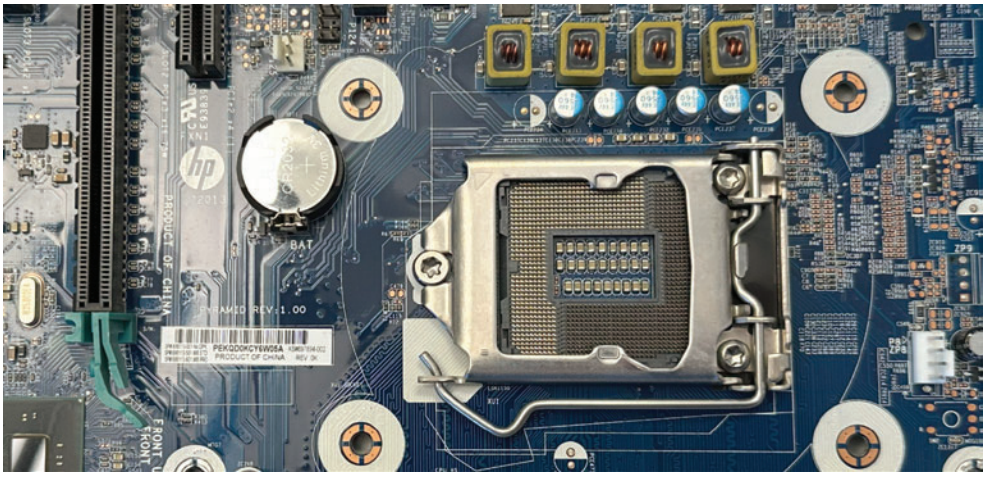
The BIOS/UEFI starts with its own default information and then reads information from the CMOS, such as which hard drive types are configured for this computer to use, which drive(s) it should search for boot sectors, and so on. Any overlapping information read from the CMOS overrides the default information from the BIOS/UEFI. A lack of corresponding information in the CMOS does not delete information that the BIOS knows natively. This process is a merge, not a write-over. CMOS memory is usually *not* upgradable in terms of its capacity and might be integrated into the BIOS/UEFI chip or the Southbridge.

Although there are technical differences, in the real world the terms BIOS/UEFI and CMOS (and BIOS/UEFI chip and CMOS chip) are used interchangeably. You will hear someone say, "Did you check the BIOS/UEFI?" or, "Did you check CMOS settings?" and they're referring to the same thing. Even though it's an oversimplification, most people feel comfortable with thinking about the BIOS/UEFI as the firmware that controls the boot process, whereas CMOS is the chip itself. It's incredibly unlikely that the A+ exam or any person is going to ask you to differentiate between the two, unless you get a job focusing specifically on those technologies.

To keep its settings, integrated circuit-based memory must have power constantly. When you shut off a computer, anything that is left in this type of memory is lost forever. The CMOS manufacturing technology produces chips with very low power requirements. As a result, today's electronic circuitry is more susceptible to damage from electrostatic discharge (ESD). Another ramification is that it doesn't take much of a power source to keep CMOS chips from losing their contents.

To prevent CMOS from losing its rather important information, motherboard manufacturers include a small battery called the *CMOS battery* to power the CMOS memory, shown to the left of the CPU socket in Figure 1.28. The batteries come in different shapes and sizes, but they all perform the same function. Most CMOS batteries look like large watch batteries or small cylindrical batteries. Today's CMOS batteries are most often of a long-life, nonrechargeable lithium chemistry.

FIGURE 1.28 CMOS battery



Understanding Processors

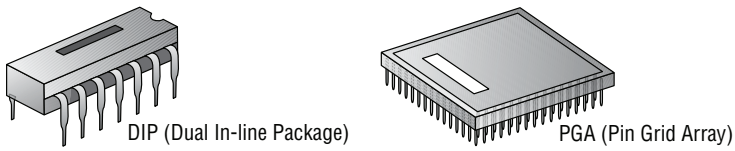
Now that you've taken a tour of the motherboard, you need to learn about the most important component on the motherboard: the CPU. The role of the CPU, or central processing unit, is to control and direct all the activities of the computer using both external and internal buses. From a technical perspective, the job of the CPU is to process, or do math, on large strings of binary numbers—0s and 1s. It is a processor chip consisting of an array of *billions* of transistors. The 13th-generation Intel Raptor Lake CPU has an estimated 12 billion.

Intel and Advanced Micro Devices, Inc. (AMD) are the two largest PC-compatible CPU manufacturers. Their chips were featured earlier in Table 1.1 during the discussion of the sockets into which they fit. From a compatibility standpoint, the most important thing to remember is that the motherboard and processor need to be made for each other. The rest of the hardware plugs into the motherboard and will be CPU brand agnostic.

The term *chip* has grown to describe the entire package that a technician might install in a socket. However, the word originally denoted the silicon wafer that is generally hidden within the carrier that you actually see. The external pins that you see are structures that can withstand insertion into a socket and are carefully threaded from the wafer's minuscule contacts. Just imagine how fragile the structures must be that you *don't* see.

Very early PC processors were manufactured in a rectangle with two rows of 20 pins known as a *dual in-line package (DIP)*—see Figure 1.29. There are still integrated circuits that use the DIP form factor; it's just no longer used for PC CPUs. Most modern CPUs use the LGA form factor although you will still see a fair number of PGA chips out there. Figure 1.9, earlier in this chapter, shows a PGA socket next to an LGA socket.

FIGURE 1.29 DIP and PGA



Intel and AMD both make extensive use of an inverted socket/processor combination of sorts. As mentioned earlier, the LGA packaging calls for the pins to be placed on the motherboard, while the mates for these pins are on the processor packaging. As with PGA, LGA is named for the landmarks on the processor, not the ones on the motherboard. As a result, the grid of metallic contact points, called *lands*, on the bottom of the CPU gives this format its name.

Figure 1.30 shows the top and bottom of an LGA CPU and a PGA CPU. Notice the lands on the left one and the pins on the right. Even though these CPUs are several years old now, the physical structure, outside of the number of pins, hasn't changed much. A decade-old CPU will look very much like a new one.

FIGURE 1.30 Top and bottom of LGA Intel and PGA AMD CPUs

Finding the CPU location in a computer is generally easy to do because it will be covered with a large heat sink, fan, or both. You likely won't be able to see the CPU itself because it's completely covered by the cooling system. The CPU is almost always located very close to the RAM to improve system speed, as shown in Figure 1.1, Figure 1.2, Figure 1.6, and Figure 1.10.

CPU Architecture

As noted in the previous section, the functional job of the processor is to do math on very large strings of 0s and 1s. How the CPU goes about doing that depends on its architecture. For commonly used processors, there are two major categories—those based on *Complex Instruction Set Computing (CISC)* and those based on *Reduced Instruction Set Computer (RISC)*.

Be familiar with the differences between x86/x64 and ARM CPUs as well as core configurations for the A+ exam, as part of Exam Objective 3.5.

x64/x86

CISC (pronounced like *disk*, but with a soft "c" like "cider") and RISC (pronounced *risk*) are examples of an instruction set architecture (ISA). Essentially, it's the set of commands that the processor can execute. Both types of chips, when combined with software, can ultimately perform all the same tasks. They just go about it differently. When programmers develop code, they develop it for a CISC or a RISC platform.

As the CISC name implies, instructions sent to the computer are relatively complex (as compared to RISC), and as such they can do multiple mathematical tasks with one instruction, and each instruction can take several clock cycles to complete. We'll talk more about CPU speeds in the "CPU Characteristics" section later, but for now, know that if a CPU is advertised as having 3.8 GHz speed, that means it can complete roughly 3.8 billion cycles in one second. The core of a processor can only do one thing at a time—it just does them very, very quickly so it looks like it's multitasking.

CISC was the original ISA for microprocessors, and the most well-known example of CISC technology is the x64/x86 platform popularized by Intel. AMD processors are CISC chips as well. So where did the terms x64 and x86 come from? First, just a bit more theory.

There is a set of data lines between the CPU and the primary memory of the system—remember the *bus*? The most common bus today is 64 bits wide, although there are still some 32-bit buses kicking around out there. In older days, buses could theoretically be as narrow as 2 bits, and 8-bit and 16-bit buses were popular for CPUs for several years. The wider the bus, the more data that can be processed per unit of time, and hence, more work can be performed. Internal registers in the CPU might be only 32 bits wide, but with a 64-bit system bus, two separate pipelines can receive information simultaneously. For true 64-bit CPUs, which have 64-bit internal registers and can run x64 versions of operating systems, the external system data bus will always be 64 bits wide or some larger multiple thereof.

In the last paragraph we snuck in the term *x64*, and by doing that we also defined it. It refers to processors that are designed to work with 64 bits of data at a time. To go along with it, the operating system must also be designed to work with x64 chips.

Contrast that with processors that can handle only 32 bits of information at once. Those are referred to as *x86* processors. You might look at that last sentence and be certain that we made a typo, but we didn't. For a long time when 32-bit processors were the fastest on the PC market, Intel was the dominant player. Their CPUs had names like 80386 (aka i386) and 80486 (i486) and were based on the older 16-bit 80286 and 8086. Since the i386 and i486 were the most popular standards, the term x86 sprung up to mean a 32-bit architecture. So even though it may seem counterintuitive due to the numbers, x64 is newer and faster than x86.

Remember that x64/x86 technology is typically found in PCs, laptops, and servers. As you'd expect, 64-bit CPUs (x64) can handle much larger amounts of data and RAM than 32-bit CPUs (x86) can.

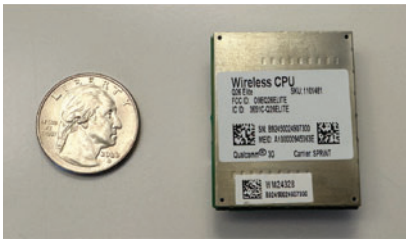
Advanced RISC Machine

Moving into the RISC architecture, the primary type of processor used today is known as an *Advanced RISC Machine (ARM)* CPU. Depending on who you talk to and which sources you prefer, there are conflicting stories about if that's actually the right acronym, as ARM is also known as an Acorn RISC Machine. Regardless of what it stands for, ARM is a competing technology to Intel and AMD x64-based CPUs.

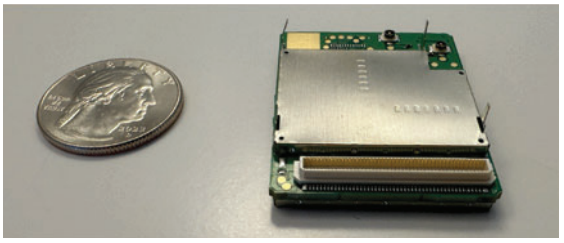
Based on the RISC acronym, one might think that the reduced set of instructions the processor can perform makes it inferior somehow, but that's not the case. Tasks just need to get executed in different ways. To use a human example, let's say that we tell you to add the number 7 to itself seven times. One way to do that is to use one step of multiplication: 7 × 7 equals 49. But what if we said you can't use multiplication? You can still get to the answer by using addition. It will just take you seven steps instead of one. Same answer, different process. That's kind of how RISC compares to CISC.

RISC processors have some advantages over their CISC counterparts. They can be made smaller than CISC chips and they produce less heat, making them ideal for mobile devices. In fact, nearly all smartphones use RISC-based chips, such as Apple's A18 and Samsung's Snapdragon and Exynos series processors. On the downside, RISC processors use more memory than CISC ones do because it takes more code to complete a task with a RISC chip. Figure 1.31 shows an older Samsung RISC processor used in a smartphone. It's about the same size as a PC CPU. Also notice that the connector isn't standard like LGA or PGA; rather, it's developed specifically for the device using it.

FIGURE 1.31 Top (a) and bottom (b) of a Samsung RISC CPU



(a)



(b)

Like x64/x86, ARM processors have evolved over time—64-bit implementations are the most current, and they are designated ARM64; 32-bit versions are known simply as ARM.

Remember that ARM chips are found mostly in smartphones, tablets, and smaller mobile devices. RISC chips handle smaller, highly optimized sets of instructions when compared to x64/x86 processors.

CPU Cores

Older processors were *single-core*, meaning that there was one set of instruction pathways through the processor. Therefore, they could process one set of tasks at a time. Designers then figured out how to speed up the computer by creating multiple cores within one processor package. Each core effectively operates as its own independent processor, provided that the operating system and applications are able to support *multicore* technology. (Nearly all do.)

Today, almost all desktop CPUs in the market are multicore. The number of cores you want may determine the processor to get. For example, the 13th-generation Intel Core i7 has 16 cores whereas the i5 has 14. Newer Intel processors will also designate two types of cores, P-cores, or performance cores, and E-cores, or efficient cores. P-cores are faster and support *hyperthreading* (the ability to manage multiple threads of an application at once), whereas E-cores are smaller and more heat-efficient. Ryzen processors will differ in core counts as well—the Ryzen 9 9950X has 16 cores whereas the Ryzen 7 9700X has 8 cores. AMD does not currently designate core types like Intel does.

As an interesting aside, throughout the 1980s and 1990s, processor clock speeds grew at very fast rates. But they haven't gotten much faster over the last 15–20 years as designers ran into technical limitations for increasing speeds. Instead, computers have gotten faster because of the introduction of multicore processors, which is basically like having four (or eight or 16) separate processors in one computer.

CPU Characteristics

When looking for a processor, you might have several decisions to make. Do you want an Intel or an AMD CPU? Which model? How fast should it be? What features does it need to support? In this section, we will take a look at some characteristics of processor performance.

Speed

The speed of the processor is generally described in clock frequency. All PC CPUs today are rated in gigahertz (GHz), or billions of cycles per second, and older ones in megahertz (MHz), or millions of cycles per second. Since the dawn of the personal computer industry, motherboards have included oscillators, which are quartz crystals shaved down to a specific geometry so that engineers know exactly how they will react when a current is run through them. The phenomenon of a quartz crystal vibrating when exposed to a current is known as the *piezoelectric effect*. The crystal (XTL) known as the *system clock* keeps the time for the flow of data on the motherboard. How the front-side bus uses the clock leads to an *effective*

clock rate known as the *FSB speed*. As discussed in the section "Types of Memory," later in this chapter, the FSB (front-side bus) speed is computed differently for different types of RAM (DDR5, DDR4, and so forth). From here, the CPU multiplies the FSB speed to produce its own internal clock rate, producing the *CPU speed* mentioned at the beginning of this paragraph.

The speed of a processor can also be tweaked by *overclocking*, or running the processor at a higher speed than the one at which the manufacturer rated it. Running at a higher speed requires more voltage and also generates more heat, which can shorten the life of the CPU. Manufacturers have historically discouraged the practice (of course, they want you to just buy a faster and more expensive CPU) because it can cause a CPU to have a shorter lifespan, and it may void any warranty. However, CPUs are sold today that specifically give you the ability to overclock them. Our official recommendation is to not do it unless the manufacturer says it's okay. If you're curious, plenty of information on how to overclock is available online.

Multithreading and Hyperthreading

The string of instructions that a CPU runs is known as a *thread*. Old processors were capable of running only one thread at a time, whereas newer ones can run multiple threads at once. This is called *multithreading*.

Intel markets their multithreading technology as *Hyper-Threading Technology (HTT)*. HTT is a form of simultaneous multithreading (SMT). SMT takes advantage of a modern CPU's superscalar architecture. Superscalar processors can have multiple instructions operating on separate data in parallel.

HTT-capable processors appear to the operating system to be two processors. As a result, the operating system can schedule two threads at the same time, as in the case of symmetric multiprocessing (SMP), where two or more processors use the same system resources. In fact, the operating system must support SMP in order to take advantage of HTT. If the current thread stalls because of missing data caused by, say, cache or branch prediction issues, the execution resources of the processor can be reallocated for a different thread that is ready to go, reducing processor downtime.

HTT manifests itself in the Windows 11 Task Manager by, for example, showing the number of logical processors as being twice as many as the number of cores the CPU has (see Figure 1.32).

For an in-market example, compare the 13th Generation Intel Core CPUs in Table 1.3. Remember that Intel E-cores (efficient cores) do not support multithreading, whereas the P-cores (performance cores) can run two threads each. So not only does the i7 have two more cores than the i5, those two cores are P-cores, so the i7 can run four more threads at one time. The i9 is even one further step up from the i7. For everyday email and Internet use, the differences won't amount to much. But for someone who is using resource-intensive apps such as online gaming or virtual reality, the differences, especially between i5 and i9 processors, can be important.

FIGURE 1.32 CPU performance in Task Manager

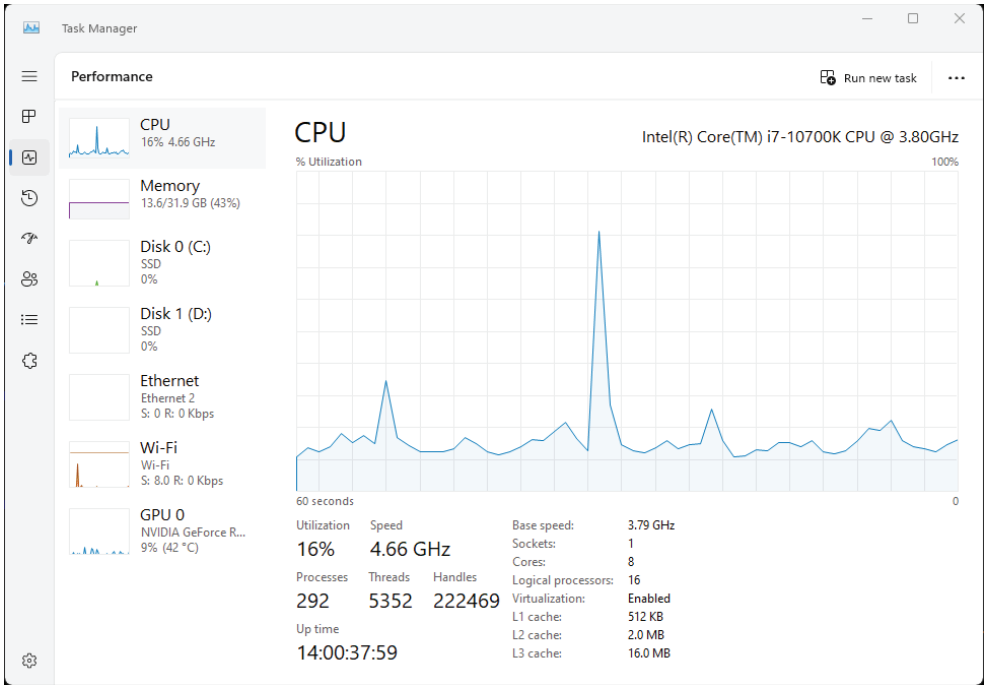


TABLE 1.3 13th Generation Intel Core CPU comparison

CPU number	P-cores	E-cores	Max CPU threads
i5-13600KF	6	8	20
i7-13700KF	8	8	24
i9-13900KF	8	16	32

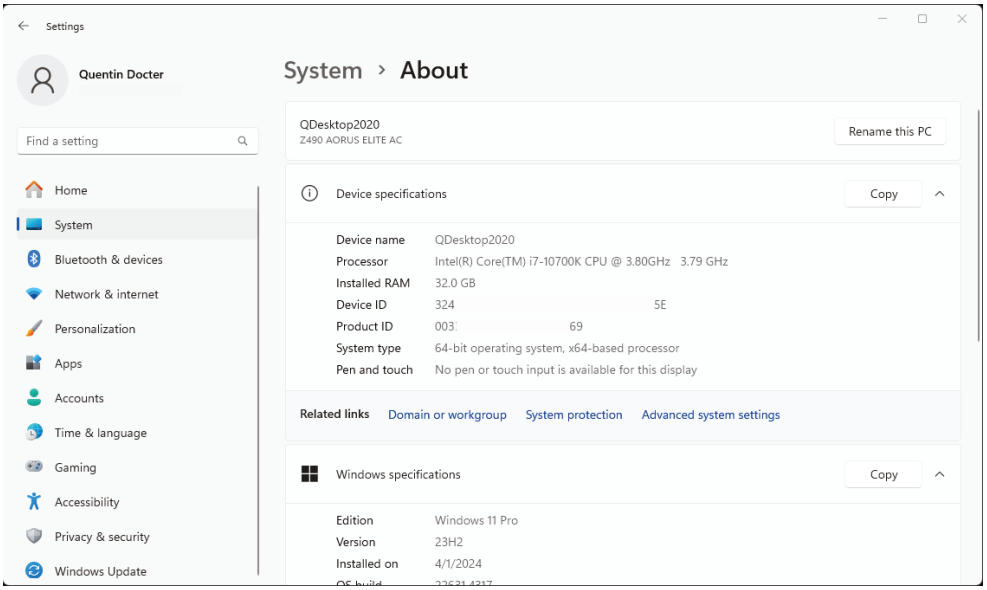
Real World Scenario

Which CPU Do You Have?

There are a few easy ways to find out which processor a computer has, without cracking the case and ripping the heat sink off the CPU itself. One is to look in Control Panel ➢ System to get the About screen, as shown in Figure 1.33. Even more detailed information can be found by running the System Information app by typing **msinfo** into the Windows search box and

pressing Enter. Of course, third-party utilities such as CPU-Z that we showed you earlier will work too. A final way to determine a computer's CPU is to save your work, exit any open programs, and restart the computer. Watch closely as the computer boots back up. You should see a notation during the POST routine that tells you what chip the system is using.

FIGURE 1.33 System information in the About screen



Virtualization Support

Hardware-based virtualization, which must be supported by the CPU, eases the burden on the system that software-based virtualization imposes. For more information on virtualization, see Volume 1 Chapter 11, "Virtualization and Cloud Computing." AMD calls its virtualization technology *AMD-V* (V for virtualization), whereas Intel calls theirs *Virtualization Technology (VT)*. Most processors made today support virtual technology, but not all. Keep in mind that the BIOS/UEFI and operating system must support it as well for virtualization to work. You may need to manually enable the virtualization support in the BIOS/UEFI before it can be used. If you have an Intel processor and would like to check its support of VT, visit the following site to download the Intel Processor Identification Utility:

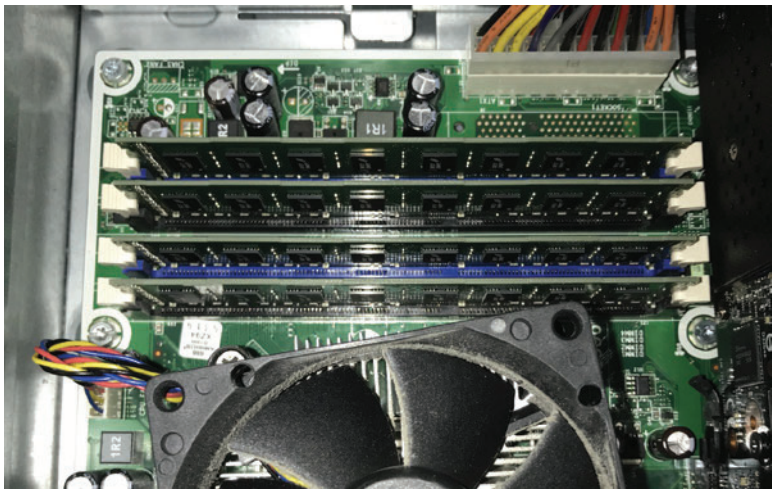
```
www.intel.com/content/www/us/en/download/12136/intel-
processor-identification-utility-windows-version.html
```

Understanding Memory

"More memory, more memory, I don't have enough memory!" Adding memory is one of the most popular, easy, and inexpensive ways to upgrade a computer. As the computer's CPU works, it stores data and instructions in the computer's memory. Contrary to what you might expect from an inexpensive solution, memory upgrades tend to afford the greatest performance increase as well, up to a point. Motherboards have memory limits; operating systems have memory limits; CPUs have memory limits.

To identify memory visually within a computer, look for several thin rows of small circuit boards sitting vertically, potentially packed tightly together near the processor. In situations where only one memory stick is installed, it will be that stick and a few empty slots that are tightly packed together. Figure 1.34 shows where memory is located in a system—in this case, all four banks are full.

FIGURE 1.34 Four memory sticks in a motherboard



Important Memory Terms

There are a few technical terms and phrases that you need to understand with regard to memory and its function:

- Parity checking
- Error-correcting code (ECC)
- Channel configurations

These terms are discussed in detail in the following sections.

ECC will sometimes be referred to as error correction code as well. As with other acronyms in computing that might have multiple ways to spell it out, focus less on what it stands for and more on what it does.

A+ Exam Objective 3.3, "Compare and contrast RAM characteristics," calls out error-correcting code (ECC) and channel configurations as topics to know.

Parity Checking

Parity checking is a rudimentary error-checking scheme offered in older RAM that offers no error correction. Parity checking works most often on a byte, or 8 bits, of data. A ninth bit is added at the transmitting end and removed at the receiving end so that it does not affect the actual data transmitted. If the receiving end does not agree with the parity that is set in a particular byte, a parity error results, and the data is deemed unusable. Systems that do not support or reserve the location required for the parity bit are said to implement *non-parity memory*.

In the early days of personal computing, almost all memory was parity-based. As quality has increased over the years, parity checking in the RAM subsystem has become more uncommon.

Error Checking and Correcting

The next step in the evolution of memory error detection is known as *error-correcting code (ECC)*. If memory supports ECC, check bits are generated and stored with the data. An algorithm is performed on the data and its check bits whenever the memory is accessed. If the result of the algorithm is all zeros, then the data is deemed valid and processing continues. ECC can detect single- and double-bit errors and actually correct single-bit errors. In other words, if a particular byte—group of 8 bits—contains errors in 2 of the 8 bits, ECC can recognize the error. If only 1 of the 8 bits is in error, ECC can correct the error.

ECC RAM is essential for environments where critical or highly sensitive data is used. Consequently, you will see ECC RAM used a lot in servers, but it's uncommon in end-user PCs. ECC RAM is also more expensive than non-ECC RAM. But as more than one server administrator has said, "If it's important enough to be on a storage server, it's important enough to use ECC RAM."

For ECC RAM to perform error checking and correcting, the motherboard it's installed on must support ECC. If ECC RAM is installed in a motherboard that does *not* support ECC, the RAM should still be recognized and function—it just won't have error checking. The opposite is not true, however. Adding non-ECC RAM to an ECC motherboard can damage the ECC functionality of the motherboard.

Channel Configurations

Standard memory controllers manage access to memory in chunks of the same size as the system bus's data width. This is considered communicating over a single channel. Most modern processors have a 64-bit system data bus. This means that a standard memory controller can transfer exactly 64 bits of information at a time. Communicating over a single channel is a bottleneck in an environment where the CPU and memory can both operate faster than the conduit between them. Up to a point, every channel added in parallel between the CPU and RAM serves to ease this constriction.

Memory controllers that support dual-channel and greater memory implementation were developed in an effort to alleviate the bottleneck between the CPU and RAM. *Dual-channel memory* is the memory controller's coordination of two memory banks to work as a synchronized set during communication with the CPU, doubling the specified system bus width from the memory's perspective. *Triple-channel memory*, then, demands the coordination of three memory modules at a time. *Quad-channel memory* is the coordination of four memory modules at once. Collectively, they are known as *multichannel memory* implementations.

Because today's processors largely have 64-bit external data buses, and because one stick of memory satisfies this bus width, there is a 1:1 ratio between banks and modules. This means that implementing multichannel memory in today's most popular computer systems requires that two, three, or four memory modules be installed at a time. Note, however, that it's the motherboard, not the memory, that implements multichannel memory (more on this in a moment). *Single-channel memory*, in contrast, is the classic memory model that dictates only that a complete bank be satisfied whenever memory is initially installed or added. One bank supplies only half the width of the effective bus created by dual-channel support, for instance, which by definition pairs two banks at a time.

In almost all cases, multichannel implementations support single-channel installation, but poorer performance should be expected. Multichannel motherboards often include slots of different colors, usually one of each color per set of slots. To use only a single channel, you populate slots of the same color, skipping neighboring slots to do so. Filling neighboring slots in a dual-channel motherboard takes advantage of its dual-channel capability.

Because of the special tricks that are played with memory subsystems to improve overall system performance, care must be taken during the installation of disparate memory modules. In the worst case, the computer will cease to function when modules of different speeds or different capacities are placed together in slots of the same channel. If all of these parameters are identical, there should be no problem with pairing modules. Nevertheless, problems could still occur when modules from two different manufacturers or certain unsupported manufacturers are installed, all other parameters being the same. Technical support or documentation from the manufacturer of your motherboard should be able to help with such issues.

Although it's not the make-up of the memory that leads to multichannel support but instead the technology on which the motherboard is based, some memory manufacturers still package and sell pairs and triplets of memory modules in an effort to give you peace

of mind when you're buying memory for a system that implements multichannel memory architecture. Keep in mind, the motherboard memory slots have the distinctive color-coding, not the memory modules.

I Can't Fill All My Memory Slots

All motherboard manufacturers document the quantity and types of modules that their equipment supports. Consult your documentation when you have questions about supported memory. Most manufacturers require that slower memory be inserted in lower-numbered memory slots. This is because such a system adapts to the first module it sees, looking at the lower-numbered slots first. Counterintuitively, however, it might be required that you install modules of larger capacity rather than smaller modules in lower-numbered slots.

Additionally, memory technology continues to advance after each generation of motherboard chipsets is announced. Don't be surprised when you attempt to install a single module of the highest available capacity in your motherboard and the system doesn't recognize the module, either by itself or with others. That capacity of module might not have been in existence when the motherboard's chipset was released. Sometimes, flashing the BIOS is all that is required. Other times it just won't work. Consult the motherboard's documentation.

Finally, know that dual- and quad-channel memory support is common today, but triple-channel is less so. As with anything else, check the motherboard's documentation to see what it supports.

Types of Memory

Memory comes in many formats. Each one has a particular set of features and characteristics, making it best suited for a particular application. Some decisions about the application of the memory type are based on suitability; others are based on affordability to consumers or marketability to computer manufacturers. The following list gives you an idea of the vast array of memory types and subtypes—and this is only a partial list of what's been in the market:

- DRAM (dynamic random access memory)
  - ADRAM (asynchronous DRAM)
  - FPM DRAM (fast page mode DRAM)
  - EDO DRAM (extended data out DRAM)
  - BEDO DRAM (burst EDO DRAM)

SDRAM (synchronous DRAM)

SDR SDRAM (single data rate SDRAM)

DDR SDRAM (double data rate SDRAM)

DDR2 SDRAM (double data rate, version 2, SDRAM)

DDR3 SDRAM (double data rate, version 3, SDRAM)

DDR4 SDRAM (double data rate, version 4, SDRAM)

DDR5 SDRAM (double data rate, version 5, SDRAM)

- SRAM (static random access memory)
- ROM (read-only memory)

Pay particular attention to all synchronous DRAM types (such as DDR4 and DDR5) as that's the most common type in use. Note that the type of memory does not dictate the packaging of the memory. Conversely, however, you might notice one particular memory packaging holding the same type of memory every time you come across it. Nevertheless, there is no requirement to this end. Let's dive into details on some of these key memory types.

DRAM

DRAM is dynamic random access memory. This is what most people are talking about when they mention RAM. When you expand the memory in a computer, you are adding DRAM chips. You use DRAM to expand the memory in the computer because it's a cheaper type of memory. Dynamic RAM chips are cheaper to manufacture than most other types because they are less complex. *Dynamic* refers to the memory chips' need for a constant update signal (also called a *refresh signal*) in order to keep the information that is written there. If this signal is not received every so often, the information will bleed off and cease to exist. Currently, the most popular implementations of DRAM are based on synchronous DRAM and include DDR4 and DDR5. Occasionally you will see some older versions of DDR in the field, and new versions periodically arrive. For example, DDR6 may show up in 2025. It will take a while before it's widely adopted, however.

Asynchronous DRAM earned its name because its speed was not tied, or synchronized, with the motherboard's FSB. This required advanced BIOS configurations to make the memory play nice with the CPU. Implementations of ADRAM included FPM, EDO, and BEDO. None of these technologies are in use today.

Synchronous DRAM (SDRAM) is named as such because it shares a common clock signal with the computer's system-bus clock, which provides the common signal that all local-bus components use for each step that they perform. This characteristic ties SDRAM to the speed of the FSB and hence the processor, eliminating the need to configure the CPU to wait for the memory to catch up.

Originally, *SDRAM* was the term used to refer to the only form of synchronous DRAM on the market. As the technology progressed, and more was being done with each clock signal on the FSB, various forms of SDRAM were developed. What was once called simply SDRAM needed a new name retroactively. Today, we use the term *single data rate SDRAM (SDR SDRAM)* to refer to this original type of SDRAM.

SDR SDRAM

SDR SDRAM is a legacy RAM technology, and it is presented here only to provide a basis for the upcoming discussion of DDR and other more advanced RAM. With SDR SDRAM, every time the system clock ticks, 1 bit of data can be transmitted per data pin, limiting the bit rate per pin of SDRAM to the corresponding numerical value of the clock's frequency. With today's processors interfacing with memory using a parallel data-bus width of 8 bytes (hence the term *64-bit processor*), a 100 MHz clock signal produces 800 MBps. That's mega*bytes* per second, not mega*bits*. Such memory modules are referred to as PC100, named for the true FSB clock rate on which they rely. PC100 was preceded by PC66 and succeeded by PC133, which used a 133 MHz clock to produce 1,066 MBps of throughput.

Note that throughput in megabytes per second is easily computed as eight times the rating in the name. This trick works for the more advanced forms of SDRAM as well. The common thread is the 8-byte system data bus. Incidentally, you can double throughput results when implementing dual-channel memory.

DDR SDRAM

Double data rate (DDR) SDRAM earns its name by doubling the transfer rate of ordinary SDRAM; it does so by *double-pumping* the data, which means transferring a bit per pin on both the rising and falling edges of the clock signal. This obtains twice the transfer rate at the same FSB clock frequency. It's the increasing clock frequency that generates heating issues with newer components, so keeping the clock the same is an advantage. The same 100 MHz clock gives a DDR SDRAM system the impression of a 200 MHz clock compared to an SDR SDRAM system. For marketing purposes, and to aid in the comparison of disparate products (DDR vs. SDR, for example), the industry has settled on the practice of using this effective clock rate as the speed of the FSB.

Module Throughput Related to FSB Speed

There is always an 8:1 module-to-chip (or module-to-FSB speed) numbering ratio because of the 8 bytes that are transferred at a time with 64-bit processors (*not* because of the ratio of 8 bits per byte). The formula in Figure 1.35 explains how this relationship works.

FIGURE 1.35 The 64-bit memory throughput formula

FSB in MHz	(~~cycles~~/second)
X 8 bytes	(bytes/~~cycle~~)
throughput	(bytes/second)

Because the actual system clock speed is rarely mentioned in marketing literature, on packaging, or on store shelves for DDR and higher, you can use this advertised FSB frequency in your computations for DDR throughput. For example, with a 100 MHz clock and two operations per cycle, motherboard makers will market their boards as having an FSB of 200 MHz. Multiplying this effective rate by 8 bytes transferred per cycle, the data rate is 1,600 MBps. Because DDR made throughput a bit trickier to compute, the industry began using this final throughput figure to name the memory modules instead of the actual frequency, which was used when naming SDR modules. This makes the result seem many times better (and much more marketable), while it's really only twice (or so) as good, or close to it.

In this example, the module is referred to as PC1600, based on a throughput of 1,600 MBps. The chips that go into making PC1600 modules are named DDR200 for the effective FSB frequency of 200 MHz. Stated differently, the industry uses DDR200 memory chips to manufacture PC1600 memory modules.

Let's make sure that you grasp the relationship between the speed of the FSB and the name for the related chips as well as the relationship between the name of the chips (or the speed of the FSB) and the name of the modules. Consider an FSB of 400 MHz, meaning an actual clock signal of 200 MHz, by the way—the FSB is double the actual clock for DDR, remember. It should be clear that this motherboard requires modules populated with DDR400 chips and that you'll find such modules marketed and sold as PC3200.

Let's try another. What do you need for a motherboard that features a 333 MHz FSB (actual clock is 166 MHz)? Well, just using the 8:1 rule mentioned earlier, you might be on the lookout for a PC2667 module. Note, however, that sometimes the numbers have to be played with a bit to come up with the industry's marketing terms. You'll have an easier time finding PC2700 modules that are designed specifically for a motherboard like yours, with an FSB of 333 MHz. The label isn't always technically accurate, but round numbers sell better, perhaps. The important concept here is that if you find PC2700 modules and PC2667 modules, there's absolutely no difference; they both have a 2667 MBps throughput rate. Go for the best deal; just make sure that the memory manufacturer is reputable.

DDR2 SDRAM

Think of the *2* in *DDR2* as yet another multiplier of 2 in the SDRAM technology, using a lower peak voltage to keep power consumption down (1.8V vs. the 2.5V of DDR). Still double-pumping, DDR2, like DDR, uses both sweeps of the clock signal for data transfer. Internally, DDR2 further splits each clock pulse in two, doubling the number of operations it can perform per FSB clock cycle. Through enhancements in the electrical interface and buffers, as well as through adding off-chip drivers, DDR2 nominally produces four times the throughput that SDR is capable of producing.

Continuing the DDR example, DDR2, using a 100 MHz actual clock, transfers data in four operations per cycle (effective 400 MHz FSB) and still 8 bytes per operation, for a total of 3,200 MBps. Just as with DDR, chips for DDR2 are named based on the perceived frequency. In this case, you would be using DDR2-400 chips. DDR2 carries on the effective FSB frequency method for naming modules but cannot simply call them PC3200 modules because those already exist in the DDR world. DDR2 calls these modules PC2-3200. (Note the dash to keep the numeric components separate.)

As another example, it should make sense that PC2-5300 modules are populated with DDR2-667 chips. Recall that you might have to play with the numbers a bit. If you multiply the well-known FSB speed of 667 MHz by 8 to figure out what modules you need, you might go searching for PC2-5333 modules. You might find someone advertising such modules, but most compatible modules will be labeled PC2-5300 for the same marketability mentioned earlier. They both support 5,333 MBps of throughput.

DDR3 SDRAM

The next generation of memory devices was designed to roughly double the performance of DDR2 products. Based on the functionality and characteristics of DDR2's proposed successor, most informed consumers and some members of the industry surely assumed the forthcoming name would be DDR4. This was not to be, however, and DDR3 was born. This naming convention proved that the *2* in DDR2 was not meant to be a multiplier but instead a revision mark of sorts. Well, if DDR2 was the second version of DDR, then DDR3 is the third. *DDR3* is a memory type, designed to be twice as fast as the DDR2 memory, that operates with the same system clock speed. Just as DDR2 was required to lower power consumption to make up for higher frequencies, DDR3 must do the same. In fact, the peak voltage for DDR3 is only 1.5V.

The most commonly found range of actual clock speeds for DDR3 tends to be from 133 MHz at the low end to less than 300 MHz. Because double-pumping continues with DDR3, and because four operations occur at each wave crest (eight operations per cycle), this frequency range translates to common FSB implementations from 1,066 MHz to more than 2,000 MHz in DDR3 systems. These memory devices are named following the conventions established earlier. Therefore, if you buy a motherboard with a 1,600 MHz FSB, you know immediately that you need a memory module populated with DDR3-1600 chips, because the chips are always named for the FSB speed. Using the 8:1 module-to-chip/FSB naming rule, the modules that you need would be called PC3-12800, supporting a 12,800 MBps throughput.

The earliest DDR3 chips, however, were based on a 100 MHz actual clock signal, so we can build on our earlier example, which was also based on an actual clock rate of 100 MHz. With eight operations per cycle, the FSB on DDR3 motherboards is rated at 800 MHz, quite a lot of efficiency while still not needing to change the original clock with which our examples began. Applying the 8:1 rule again, the resulting RAM modules for this motherboard are called PC3-6400 and support a throughput of 6,400 MBps, carrying chips called DDR3-800, again named for the FSB speed.

DDR4 SDRAM

Continuing the inevitable march of technology, *DDR4* was the next iteration of SDRAM on the market. As you would expect, the speed of DDR4 is roughly double that of DDR3. DDR3 provided data rates of approximately 800 Mbps to 2,133 Mbps, whereas DDR4 ranges between 1,600 Mbps and 3,200 Mbps. DDR also runs at a lower voltage—1.2 volts. Finally, DDR4 can support more memory per module, up to 512 GB per chip. Realistically, though, no one produces motherboards (or RAM) that support that quantity. The largest you will see are 64 GB sticks.

DDR5 SDRAM

After a long wait, *DDR5* finally hit the market at the end of 2021. Intel's Alder Lake platform was the first to support it; AMD chips could support DDR5 in early 2022 with the Zen 4 release.

DDR5 doubles the speed of DDR4 to 6.4 Gbps, as is expected for a new memory standard. Improved power efficiency means it runs at 1.1 volts. DDR5 is also the first memory module to be available in up to 128 GB modules.

Exam Objective 3.3 states that test-takers should be able to compare and contrast DDR iterations. It's highly unlikely that you will need to know intimate details of each. What's more likely to be important are a few facts:

- Each version of DDR doubles the speed of the previous version.
- You might have noticed this in the discussion on DDR or in the real world, but some RAM is labeled "PC*X*" whereas others are "DDR*X*." The *X* will be the same within a version. In other words, PC4 RAM is DDR4 RAM.
- Motherboards will only physically support one type of DDR (more on that in the upcoming "Memory Packaging" section), such as DDR4 or DDR5. The motherboard documentation will tell you which type it supports.
- Motherboards will only work with certain speeds of RAM. It used to be that there was only one speed (or maybe two speeds) of RAM a motherboard would support, such as PC3-12800, based on its chipset. Now, motherboards are more flexible and can support different RAM speeds—but it's best practice to install only one speed in a system. And of course, faster is better. Look at the motherboard documentation to see what's supported. For example, a recent AORUS ELITE motherboard supports 23 different DDR5 speeds, from DDR5 5600 (PC5-44800) through DDR5 9200 (PC5-73600).

SRAM

Static random access memory (SRAM) doesn't require a refresh signal like DRAM does. The chips are more complex and are thus more expensive. However, they are considerably faster. DRAM access times can come in at 40 nanoseconds (ns) or more; SRAM has access times faster than 10ns. SRAM is classically used for cache memory.

ROM

ROM stands for read-only memory. It is called read-only because you could not write to the original form of this memory. Once information had been etched on a silicon chip and manufactured into the ROM package, the information couldn't be changed. Some form of ROM is normally used to store the computer's BIOS because this information normally does not change often.

The system ROM in the original IBM PC contained the power-on self-test (POST), BIOS, and cassette BASIC, a simple programming language for writing and running programs. Later, IBM computers and compatibles included everything but cassette BASIC. The system ROM enables the computer to "pull itself up by its bootstraps," or *boot* (find and start the operating system).

Through the years, different forms of ROM were developed that could be altered, later ones more easily than earlier ones. The first generation was the programmable ROM (PROM), which could be written to for the first time in the field using a special programming device, but then no more. You may liken this to the burning of a DVD-R.

The erasable PROM (EPROM) followed the PROM, and it could be erased using ultraviolet light and subsequently reprogrammed using the original programming device. These days, flash memory is a form of electronically erasable PROM (EEPROM). Of course, it does not require UV light to erase its contents, but rather a slightly higher than normal electrical pulse.

Although the names of these memory devices are different, they all contain ROM. Therefore, regardless which of these technologies is used to manufacture a BIOS chip, it's never incorrect to say that the result is a ROM chip.

Memory Packaging

The memory slots on a motherboard are designed for particular module form factors or styles. RAM historically evolved from form factors no longer seen for such applications, such as dual in-line package (DIP), single in-line memory module (SIMM), and single in-line pin package (SIPP). The most popular form factors for primary memory modules today are as follows:

- DIMM (dual in-line memory module)
- SODIMM (small outline dual in-line memory module)

Desktop computers will use DIMMs. Laptops and smaller devices require SODIMMs or smaller memory packaging. Other devices may use proprietary formats. So, in addition to coordinating the speed of the components, their form factor is an issue that must be addressed.

Exam Objective 3.3 states that you should be able to compare and contrast RAM form factors: DIMM and SODIMM.

DIMM

The most popular desktop PC memory package is known as a DIMM, which stands for dual in-line memory module. DIMMs are 64-bit memory modules that are used as a package for the SDRAM family: SDR, DDR, DDR2, DDR3, DDR4, and DDR5. The term *dual* refers to

the fact that, unlike their SIMM predecessors, DIMMs differentiate the functionality of the pins on one side of the module from the corresponding pins on the other side. With 84 pins per side, this makes 168 independent pins on each standard SDR module.

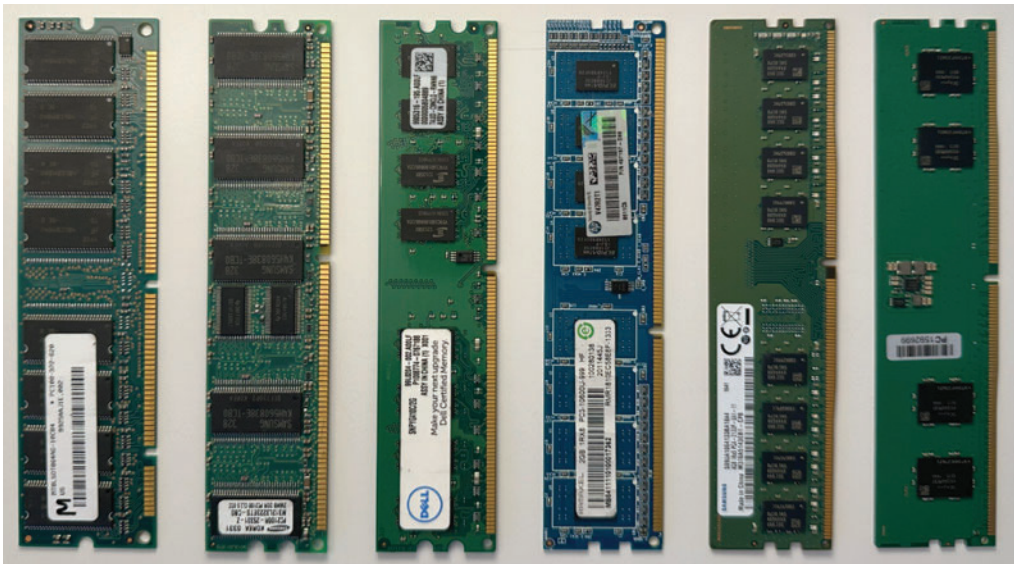
The DIMMs used for DDR memory are all the same length (5.25″ or 133.35mm), but they differ in the number of pins they have as well as where the key is in the module. Some will have built-in heat sinks, and newer modules may also have RGB lighting. Because of the position of the key, you can't put one type of DIMM into a different type of slot. Table 1.4 summarizes some key differences between the types of DDR we've introduced in this chapter.

Figure 1.36 shows SDR (on the left) and the DDR family in order from DDR to DDR5, left to right.

TABLE 1.4 DDR DIMM characteristics

Characteristic	DDR	DDR2	DDR3	DDR4	DDR5
Pins	184	240	240	288	288
Max memory	1 GB	8 GB	32 GB	64 GB	512 GB
Channels	1	1	1	1	2
Voltage	2.5 v	1.8 v	1.5 v	1.2 v	1.1 v

FIGURE 1.36 SDR and DDR DIMMs



Inserting and Removing Memory Modules

The original single in-line memory modules had to be inserted into their slots at a 45° angle. The installer then had to apply slight pressure as the module was maneuvered upright at a 90° angle to the motherboard, where a locking mechanism would grip the module and prevent it from returning to its 45° position. This procedure created a pressure that reinforced the contact of the module with its slot. Releasing the clips on either end of the module unlocked it and allowed it to return to 45°, where it could be removed.

DIMM slots, by comparison, have no spring action. DIMMs are inserted straight into the slot with the locking tabs pulled away from the module. The locking tabs are at either end of the module, and they automatically snap into place, securing the module. Pulling the tabs away from the module releases the module from the slot, allowing it to be effortlessly removed.

SODIMM

Laptop computers and other computers that require much smaller components don't use standard RAM packages, such as DIMMs. Instead, they call for a much smaller memory form factor, such as a *small outline DIMM (SODIMM)*. SODIMMs are a little more than half as wide as DIMMs, at 2.75″ (69.6mm), and have fewer pins (outlined in Table 1.5). Each version is keyed differently as well, to prevent putting the wrong chip into a motherboard. SODIMMs use the same DDR electronic specifications as DIMMs, so the number of channels, voltage, and maximum amount of memory are similar.

TABLE 1.5 SODIMM pin counts

Version	SDR	DDR	DDR2	DDR3	DDR4	DDR5
Pins	144	200	200	204	260	262

Figure 1.37 shows three SODIMMs, DDR3 through DDR5, from top to bottom. Figure 1.38 shows a DDR5 SODIMM and a DDR5 DIMM for comparison.

Understanding Cooling Systems

It's a basic concept of physics: electronic components turn electricity into work and heat. The excess heat must be dissipated or it will shorten the life of the components. In some cases (like with the CPU), the component will produce so much heat that it can destroy itself in a matter of seconds if there is not some way to remove this extra heat.

FIGURE 1.37 SODIMMs

FIGURE 1.38 DDR5 SODIMM and DIMM

Air-cooling methods are used to cool the internal components of most PCs. With air cooling, the movement of air removes the heat from the component. Sometimes, large blocks of metal called *heat sinks* are attached to a heat-producing component in order to dissipate the heat more rapidly.

The following cooling components are included in Exam Objective 3.5: fans, heat sink, thermal paste/pads, and liquid.

Case Cooling and Fans

When you turn on a computer, you will often hear lots of whirring. Most of this noise is coming from the various fans inside the computer, although older hard disk drives (HDDs) make noise too. Fans provide airflow within the computer.

Most PCs have at a minimum three types of fans:

Front Intake Fan This fan is used to bring fresh, cool air into the computer for cooling purposes. Many newer cases have two or even three front intake fans and may also have one on the top of the case as well.

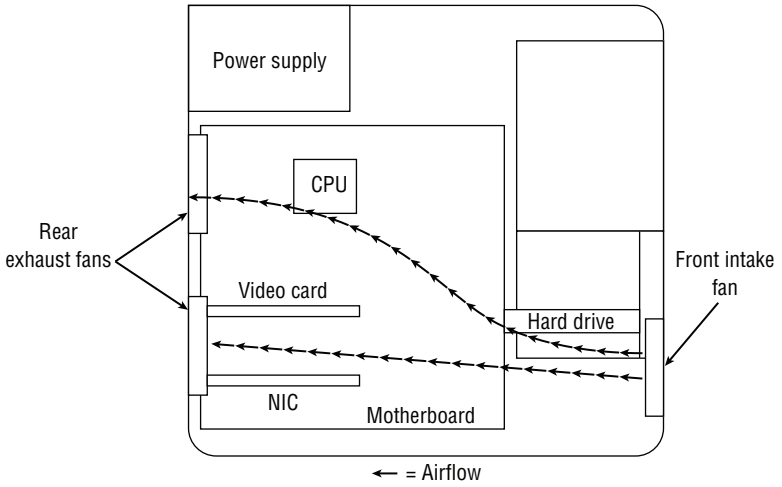
Rear Exhaust Fan This fan is used to take hot air out of the case. Newer systems may have multiple exhaust fans as well.

Power Supply Exhaust Fan This fan is usually found at the back of the power supply, and it is used to cool the power supply. In addition, this fan draws air from inside the case into vents in the power supply. This pulls hot air through the power supply so that it can be blown out of the case. The front intake fan assists with this airflow. The rear exhaust fan supplements the power supply fan to achieve the same result outside of the power supply.

Ideally, the airflow inside a computer should resemble what is shown in Figure 1.39, where the back of the chassis is shown on the left in the image.

Note that you must pay attention to the orientation of the power supply's airflow. If the power supply fan is an exhaust fan, as assumed in this discussion, the front and rear fans will match their earlier descriptions: front, intake; rear, exhaust. If you run across a power supply that has an intake fan, the orientation of the supplemental chassis fans should be reversed as well. The rear chassis fan(s) should always be installed in the same orientation the power supply fan runs to avoid creating a small airflow circuit that circumvents the cross-flow of air throughout the case. The front chassis fan and the rear fans should always be installed in reverse orientation to avoid having them fight against each other and thereby reduce the internal airflow. Reversing supplemental chassis fans is usually no harder than removing four screws and flipping the fan. Sometimes, the fan might just snap out, flip, and then snap back in, depending on the way it is rigged up.

FIGURE 1.39 System unit airflow



Motherboard Fan Power Connectors

It's important to be aware of the two main types of fan connections found on today's motherboards. One of these connectors has only three connections, while the other has four. The fan connectors and motherboard headers are interchangeable between the two pinouts, but if a chassis fan has four conductors, it's a sign that it's calling for connectivity to an extra +5VDC (volts direct current) connection that the most common 3-pin header doesn't offer. A more rare 3-pin chassis-fan connector features a +12VDC power connection for heavier-duty fans and a rotation pin used as an input to the motherboard for sensing the speed of the fan.

Four-pin CPU connections place the ground and power connections in pins 1 and 2, respectively, so that 2-pin connectors can be used to power older fans. The 4-pin header also offers a tachometer input signal from the fan on pin 3 so that the speed of the fan can be monitored by the BIOS and other utilities. Look for markings such as *CPU FAN IN* to identify this function. Pin 4 might be labeled *CPU FAN PWM* to denote the pulse-width modulation that can be used to send a signal to the fan to control its speed. This is the function lost when a 3-pin connector is placed in the correct position on a 4-pin header. Four-pin chassis-fan connectors can share the tachometer function but replace the speed control function with the extra 5V mentioned earlier.

Other power connections and types will be covered in Volume 1 Chapter 2, including the Molex connector, which can be used to power chassis and CPU fans using an adapter or the built-in connector on mostly older fans manufactured before the motherboard connectors were standardized. Figure 1.40 shows two 3-pin chassis-fan headers on a motherboard.

FIGURE 1.40 3-pin chassis-fan headers

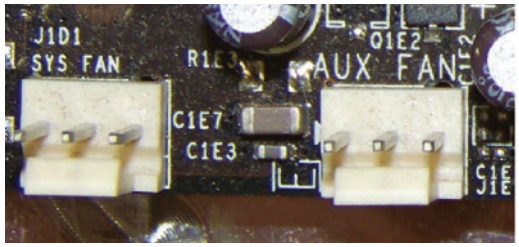
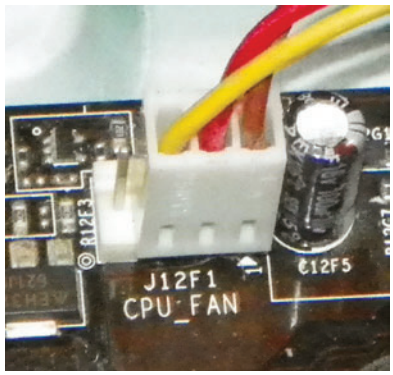


Figure 1.41 shows a 4-pin CPU fan header with an approaching 3-pin connector from the fan. Note that the keying tab is lined up with the same three pins it's lined up with in the 3-pin connectors.

FIGURE 1.41 A 4-pin CPU fan header

This physical aspect and the similar pin functions are what make these connectors interchangeable, provided that the header's function matches the role of the fan being connected. Figure 1.42 shows the resulting unused pin on the 4-pin header. Again, controlling the fan's speed is not supported in this configuration.

FIGURE 1.42 Position of a 3-pin connector on a 4-pin header



CPU Cooling

Probably the greatest challenge in cooling is the computer's CPU. It is the component that generates the most heat in a computer (aside from some pretty insane GPUs out there). As a matter of fact, if a modern processor isn't actively cooled all of the time, it will generate enough heat to burn itself up in seconds. That's why most motherboards have an internal CPU heat sensor and a CPU fan sensor, the connector for which is labeled CPU_FAN on the motherboard (as in Figure 1.41 and Figure 1.42). If no cooling fan is active, these devices will shut down the computer before damage occurs.

There are multiple CPU cooling methods, but the two most common are air cooling and liquid cooling.

Air Cooling

The parts inside most computers are cooled by air moving through the case. The CPU is no exception. However, because of the large amount of heat produced, the CPU must have (proportionately) the largest surface area exposed to the moving air in the case. Therefore, the heat sinks on the CPU are the largest of any inside the computer.

The CPU fan often blows air down through the body of the heat sink to force the heat into the ambient internal air where it can join the airflow circuit for removal from the case. However, in some cases, you might find that the heat sink extends up farther, using radiator-type fins, and the fan is placed at a right angle and to the side of the heat sink. This design moves the heat away from the heat sink immediately instead of pushing the air down through the heat sink. CPU fans can be purchased that have an adjustable rheostat to allow you to dial in as little airflow as you need, aiding in noise reduction but potentially leading to accidental overheating.

It should be noted that the highest-performing air CPU coolers use copper plates in direct contact with the CPU. They also use high-speed and high-CFM cooling fans to dissipate the heat produced by the processor. CFM is short for cubic feet per minute, an airflow measurement of the volume of air that passes by a stationary object per minute. Figure 1.43 shows a newer, large heat sink with a fan in the center. In the picture it can be tough to gauge size—this unit is about 6″ across! (And to be fair, this heat sink should have a second fan on one of the sides, but with the second fan the heatsink wouldn't fit into the author's case—the RAM was in the way.)

Many new CPU heat sinks use tubing to transfer heat away from the CPU. With any cooling system, the more surface area exposed to the cooling method, the better the cooling. Plus, the heat pipes can be used to transfer heat to a location away from the heat source before cooling. This is especially useful in cases where the form factor is small and with laptops, where open space is limited.

With advanced heat sinks and CPU cooling methods like this, it is important to improve the thermal transfer efficiency as much as possible. To that end, cooling engineers came up with a glue-like compound that helps to bridge the extremely small gaps between the CPU and the heat sink, which avoids superheated pockets of air that can lead to focal damage of the CPU. This product is known as thermal transfer compound, or simply thermal compound (alternatively, thermal grease or *thermal paste*), and it can be bought in small tubes. Single-use tubes are also available and alleviate the guesswork involved with how much you should apply. Watch out, though; this stuff makes quite a mess and doesn't want

to come off your fingers very easily. An alternative to the paste is a small *thermal pad*, which provides heat conductivity between the processor and the heat sink. Figure 1.44 shows a tube of thermal paste and a thermal pad (the pad needs to be cut to size).

FIGURE 1.43 Large heat sink and fan

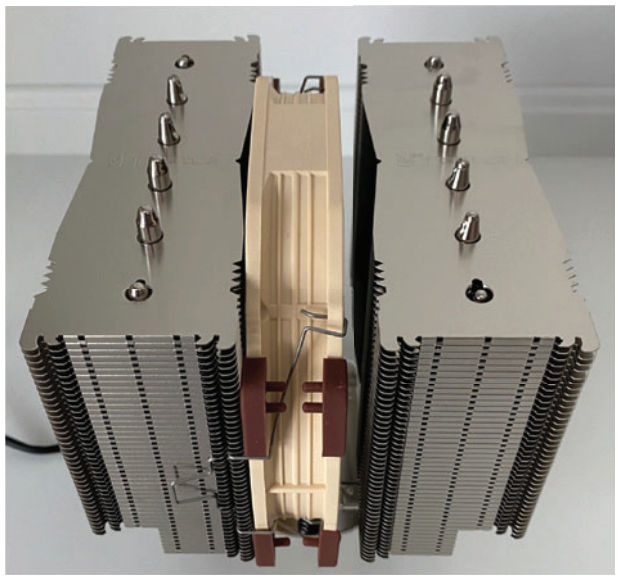
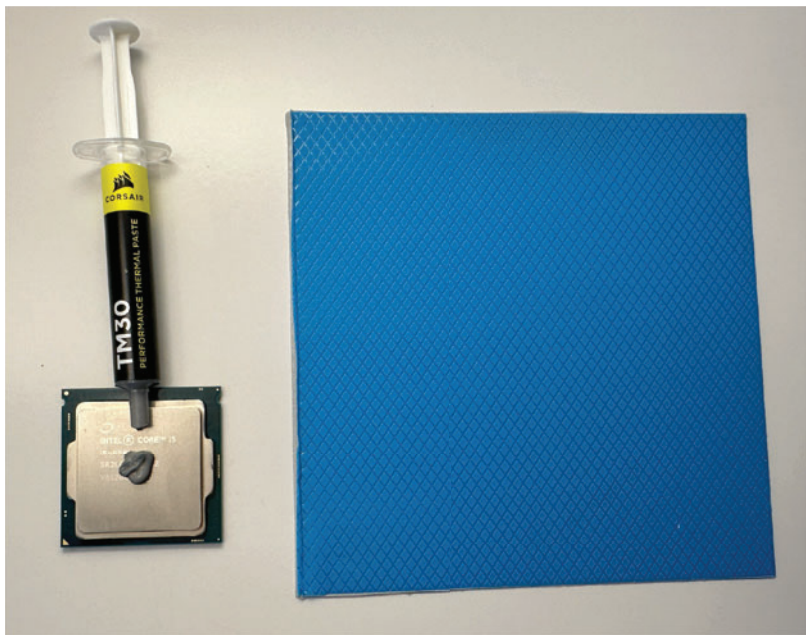


FIGURE 1.44 Thermal paste and a thermal pad



Apply the compound by placing a bead in the center of the heat sink, not on the CPU, because some heat sinks don't cover the entire CPU package. That might sound like a problem, but some CPUs don't have heat-producing components all the way out to the edges. Some CPUs even have a raised area directly over the silicon die within the packaging, resulting in a smaller contact area between the components. You should apply less than you think you need because the pressure of attaching the heat sink to the CPU will spread the compound across the entire surface in a very thin layer. It's advisable to use a clean, lint-free applicator of your choosing to spread the compound around a bit as well, just to get the spreading started. You don't need to concern yourself with spreading it too thoroughly or too neatly because the pressure applied during attachment will equalize the compound quite well. During attachment, watch for oozing compound around the edges, clean it off immediately, and use less next time.

Improving and Maintaining CPU Cooling

In addition to using thermal compound, you can enhance the cooling efficiency of a CPU heat sink by lapping the heat sink, which smoothens the mating surface using a very fine sanding element, about 1,000 grit in the finishing stage. Some vendors of the more expensive heat sinks will offer this service as an add-on.

If your CPU has been in service for an extended period of time, perhaps three years or more, it is a smart idea to remove the heat sink and old thermal compound and then apply fresh thermal compound and reattach the heat sink. Be careful, though; if your thermal paste has already turned into thermal "glue," you can wrench the processor right out of the socket, even with the release mechanism locked in place. Invariably, this damages the pins on the chip. Try running the computer for a couple of minutes to warm the paste and then try removing the heat sink again.

Counterintuitively, perhaps, you can remove a sealed heat sink from the processor by gently rotating the heat sink to break the paste's seal. Again, this can be made easier with heat. If the CPU has risen in the socket already, however, rotating the heat sink would be an extremely bad idea. Sometimes, after you realize that the CPU has risen a bit and that you need to release the mechanism holding it in to reseat it, you find that the release arm is not accessible with the heat sink in place. This is an unfortunate predicament that will present plenty of opportunity to learn.

If you've ever installed a brand-new heat sink onto a CPU, you've most likely used thermal compound or the thermal compound patch that was already applied to the heat sink for you. If your new heat sink has a patch of thermal compound pre-applied, don't add more. If you ever remove the heat sink, don't try to reuse the patch or any other form of thermal compound. Clean it all off and start fresh.

Liquid Cooling

Liquid cooling is a technology whereby a special water block is used to conduct heat away from the processor (as well as from the chipset). Water is circulated through this block to a radiator, where it is cooled.

The theory is that you could achieve better cooling performance through the use of liquid cooling. For the most part, this is true. However, with traditional cooling methods (which use air and water), the lowest temperature you can achieve is room temperature. Plus, with liquid cooling, the pump is submerged in the coolant (generally speaking), so as it works, it produces heat, which adds to the overall liquid temperature.

The main benefit to liquid cooling is silence. Only one fan is needed: the fan on the radiator to cool the water. So, a liquid-cooled system can run extremely quietly.

There are two major classifications of liquid cooling systems in use with PCs today: all-in-one (AIO) coolers and custom loop systems. AIO systems are relatively easy to install—they require about as much effort as a heat sink and fan—and are comparably priced to similarly effective air systems. Figure 1.45 shows an example from Corsair, with the pump in front and the fans behind it, attached to the radiator.

FIGURE 1.45 AIO liquid cooling system

AIO systems come in three common sizes: 120mm (with one fan, and the most common), 240mm (two fans, for overclocked components), and 360mm (three fans, for high-end multicore overclocked components). Options with RGB lighting are readily available if that's the style you want. Custom loop systems can quickly become complex and expensive, but many hardcore gamers swear by their performance. The components are essentially the same as those in an AIO system—there's a radiator, pump, fans, some tubes, and liquid. However, each part is purchased separately, and some assembly is required.

Water cooling systems often require more room inside the case than heat sinks, as well as special headers on the motherboard (or in the case) to support the fans. As always, check your documentation to be sure that your other hardware is compatible!

Cooling Other Components

As noted previously, the CPU is the most important component to cool. Most other components receive adequate cooling from the case fans, provided they are working properly and not covered in a thick layer of dust. There are, however, cooling systems for other internal components as well. Most of them involve the use of heat sinks and/or fans.

Video Card Cooling

Modern video cards run almost as hot as CPUs and will come with cooling mechanisms built-in to their packaging. Most of the time this is a fan or series of fans covering heat sinks. We will show an image of a video card with cooling fans in Volume 1 Chapter 2.

Memory Cooling

If you are going to start overclocking your computer, you will want to do everything in your power to cool all of its components, and that includes the memory.

There are two methods of cooling memory: passive and active. The passive memory cooling method just uses the ambient case airflow to cool the memory through the use of enhanced heat dissipation. For this, you can buy either heat sinks or special "for memory chips only" devices known as *heat spreaders*. These are special aluminum or copper housings that wrap around memory chips and conduct the heat away from them.

Active cooling, on the other hand, usually involves forcing some kind of cooling medium (air or water) around the RAM chips themselves or around their heat sinks. Most often, active cooling methods are just high-speed fans directing air right over a set of heat spreaders.

Hard Drive Cooling

There are both active and passive cooling devices for hard drives. Most common, however, is the active cooling bay. You install a hard drive in a special device that fits into a standard hard drive expansion bay. This device contains fans that draw in cool air over the hard drive, thus cooling it. Figure 1.46 shows an example of one of these active hard drive coolers, with two fans, mounted to a 3.5″ hard drive. As you might suspect, you can also get heat sinks for hard drives.

FIGURE 1.46 An active hard disk cooler

Summary

In this chapter, we took a tour of the key internal system components of a PC. You learned primarily about the "big three," which are the motherboard, processor, and memory. Included in the motherboard discussion were form factors, connector types, such as PCIe, M.2, and headers, motherboard and CPU compatibility, BIOS/UEFI settings, encryption, and the CMOS battery. CPU topics included features such as compatibility, architecture, multithreading, and virtualization. With RAM you learned about different types (SODIMMs and DIMMs, DDR2, DDR3, DDR4, and DDR5) and concepts such as single-, dual-, triple-, and quad-channel, error correcting, and parity.

Finally, the chapter ended with cooling systems, which keep the components from damaging themselves with excess heat. This chapter laid the foundation for the rest of the book, including the next few chapters on additional hardware components.

Exam Essentials

Know three form factors of system boards. Know the characteristics of and differences between ATX, microATX, and ITX motherboards.

Know the components of a motherboard. Be able to describe, identify, and replace (where applicable) motherboard components, such as expansion slots, memory slots, processor sockets, BIOS/UEFI (firmware), and CMOS batteries.

Be able to identify and differentiate motherboard connector types. Understand the differences between PCI, PCIe, SATA, eSATA, and M.2 connectors, as well as power connectors and headers.

Understand core concepts of motherboard compatibility. Know that Intel and AMD chips use different sockets and therefore are incompatible with each other. Also know differences between multisocket, desktop, and laptop motherboards.

Understand CPU architecture. Know the differences between x64, x86, and ARM processors, implications of single-core versus multicore CPUs, and multithreading and virtualization support.

Know what the BIOS/UEFI is responsible for. The BIOS/UEFI controls boot options, fan speeds, temperature monitoring, USB permissions, and security options such as the BIOS password, boot password, TPM, HSM, and Secure Boot.

Understand the purposes and characteristics of memory. Know about the characteristics that set the various types of memory apart from one another. This includes the actual types of memory, such as DRAM (which includes several varieties), SRAM, ROM, and CMOS, as well as memory packaging, such as DIMMs and SODIMMs. Also have a firm understanding of the different levels of cache memory as well as its purpose in general.

Know how to replace RAM, given a scenario. RAM must match the motherboard in form factor and in speed. For example, some motherboards will only accept DDR4 or DDR5 memory. The speed should be compatible with the motherboard, as indicated in the documentation.

Understand the purposes and characteristics of cooling systems. Know the different ways internal components can be cooled and how overheating can be prevented. Most cooling systems involve a fan, a heat sink, or a combination of the two, or liquid cooling and a fan. Heat sinks and fans are often attached with thermal paste or a thermal pad.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which computer component contains all the circuitry necessary for other components or devices to communicate with one another?
   - **A.** Motherboard
   - **B.** CPU
   - **C.** RAM
   - **D.** Expansion bus

2. You are told by a technician over the phone that you need to bring replacement DDR4 SDRAM memory. What type of packaging is used for DDR4 SDRAM memory?
   - **A.** 224-pin DIMM
   - **B.** 240-pin DIMM
   - **C.** 288-pin DIMM
   - **D.** 296-pin DIMM

3. You need to replace memory in a server that is using ECC RAM. Which of the following statements regarding the replacement RAM is true?
   - **A.** Installing non-ECC RAM will work fine, but the new RAM won't have error-correcting.
   - **B.** You can only install ECC RAM for it to work properly.
   - **C.** You can install non-ECC RAM and it will adapt for error-correction.
   - **D.** Installing non-ECC RAM will work fine as long as the new RAM includes a parity bit.

4. A client wants you to build a new PC for her, with a smaller case and lower power requirements. When selecting a motherboard, which form factor should you choose for a smaller size and lower power consumption?
   - **A.** ATX
   - **B.** AT
   - **C.** DTX
   - **D.** ITX

5. A motherboard failed on your desktop PC, which has an Intel Core i5-13600K processor. When you're looking for a replacement motherboard, which CPU socket does it need to have?
   - **A.** LGA 1366
   - **B.** LGA 1150
   - **C.** LGA 1200
   - **D.** LGA 1700

6. You have just replaced a processor in a computer and now need to add a cooling mechanism. What should you use to attach the cooling system to the processor?
   - **A.** Heat sink
   - **B.** Thermal paste
   - **C.** Fan
   - **D.** Superglue

7. A technician asks you how to get a computer's processor to run faster than it currently does. What is he trying to achieve?
   - **A.** Hyperthreading
   - **B.** Overclocking
   - **C.** Virtualization
   - **D.** Multicore support

8. You are assembling a new computer and are looking at new cooling systems. Which of the following cooling systems requires the use of a pump?
   - **A.** Liquid
   - **B.** Thermal paste
   - **C.** Heat sink
   - **D.** Heat sink plus fan

9. Which of the following types of processors will likely generally be preferred for mobile devices?
   - **A.** x64
   - **B.** x86
   - **C.** ARM
   - **D.** LGA

10. You press the front power button on a computer and the system boots. Later, you press it briefly and the system hibernates. When you press it again, the system resumes. You press and hold the button and the system shuts down. What is this feature called?
    - **A.** Programmable power
    - **B.** Soft power
    - **C.** Relay power
    - **D.** Hot power

11. You are training new technicians to install RAM and pointing out differences between packages. Which of the following are the numbers of pins that can be found on DIMM modules used in desktop motherboards? (Choose two.)
    - **A.** 180
    - **B.** 184

C. 200

D. 204

E. 232

F. 240

12. You are installing a new computer, and the user wants to use virtualization. Which two hardware components need to support virtual technology for this to work properly? (Choose two.)

A. RAM

B. Motherboard

C. CPU

D. BIOS

13. You find out that a disgruntled ex-employee's computer has a boot password that must be entered before the operating system is ever loaded. There is also a password preventing your access to the BIOS utility. Which of the following motherboard components can most likely be used to return the computer to a state that will allow you to boot the system without knowing the password?

A. Cable header

B. Power reset connector

C. Toggle switch

D. Jumper

14. Your Core i5 fan has a 4-pin connector, but your motherboard only has a single 3-pin header with the CPU_FAN label. Which of the following will be the easiest solution to get the necessary cooling for your CPU?

A. Plug the 4-pin connector into the 3-pin header.

B. Buy a 4-pin to 3-pin adapter.

C. Leave the plug disconnected and use only the heat sink.

D. Add an extra chassis fan.

15. You are installing a new video card into a PCIe slot. What is the combined total throughput of a PCIe 4.0 x16 slot?

A. 500 MBps

B. 1 GBps

C. 64 GBps

D. 128 GBps

16. Which of the following allows you to perform the most complete restart of the computer without removing power?

A. Start ➢ Restart

B. Start ➢ Hibernate

C. Reset button

D. Power button

17. You have just found out that a new UEFI upgrade exists for your computer, and you obtain a copy on a flash drive to install it. Which of the following is most helpful when flashing the UEFI on a desktop computer system?

A. DVD-ROM

B. Uninterruptable power supply

C. Internet connection

D. Windows administrator password

18. The 32 GB of DDR4 memory has failed in a client's desktop computer. The client requests that you upgrade the system for better performance when replacing the RAM. Which of the following is the best option?

A. Install two new 32 GB DDR4 RAM modules.

B. Install two new 128 GB DDR4 RAM modules.

C. Install two new 32 GB DDR5 RAM modules.

D. Install two new 128 GB DDR5 RAM modules.

19. You are replacing an HP laptop's motherboard. Which type of motherboard should you use?

A. ATX

B. AT

C. One for the specific model

D. mini-ITX

20. You need to install a new video card into a desktop computer. What type of expansion slot is preferred today for high-performance graphics adapters?

A. AGP

B. PCIe

C. PCI

D. SATA

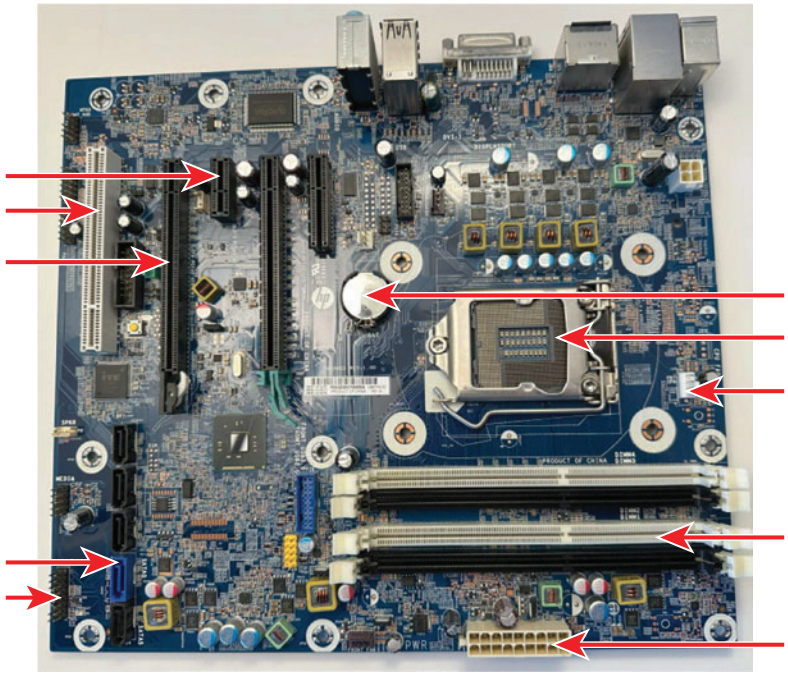
Performance-Based Question 1

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answer compares to mine, refer to Appendix B.

You have been asked to remove a dual in-line memory module and insert one with a larger capacity in its place. Describe the process for doing so.

Performance-Based Question 2

Identify the component each arrow points to in the following image of an ATX motherboard.



Chapter 2

Expansion Cards, Storage Devices, and Power Supplies

THE FOLLOWING COMPTIA A+ 220-1201 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **3.0 Hardware**

✓ **3.4 Compare and contrast storage devices.**

- Hard drives
  - Spindle speeds
  - Form factors
    - 2.5-inch
    - 3.5-inch
- Solid-state drives
  - Communications interfaces
    - Non-volatile Memory Express (NVMe)
    - SATA
    - Peripheral Component Interconnect Express (PCIe)
    - Serial Attached SCSI [Small Computer System Interface] (SAS)
  - Form factors
    - M.2
    - Mini-serial Advanced Technology Attachment (mSATA)
- Drive configurations
  - Redundant Array of Independent Disks (RAID) 0, 1, 5, 6, 10

- Removable storage
  - Flash drives
  - Memory cards
- Optical drives

✓ **3.5 Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards.**

- Expansion cards
  - Sound card
  - Video card
  - Capture card
  - Network interface card

✓ **3.6 Given a scenario, install the appropriate power supply.**

- Input 110-120 VAC vs. 220-240 VAC
- Output 3.3V vs. 5V vs. 12V
- 20+4 pin motherboard connector
- Redundant power supply
- Modular power supply
- Wattage rating
- Energy efficiency

As a PC technician, you need to know quite a bit about hardware. Given the importance and magnitude of this knowledge, the best way to approach learning about it is in sections. The first chapter introduced the topic via the primary core components, and this chapter follows up where it left off. First, we will look at adding functionality by plugging expansion cards into the motherboard. Then, we will focus on storage devices that hold data persistently—that is, they don't require power to maintain data like RAM does. Finally, we will end the chapter by looking at the simple-looking but potentially dangerous box that gives the components the energy they need—the power supply.

Installing and Configuring Expansion Cards

An *expansion card* (also known as an *adapter card*) is a circuit board that you install into a computer to increase the capabilities of that computer. Expansion cards come in varying formats for different uses, but the important thing to note is that no matter what function a card has, the card being installed must match the bus type of the motherboard into which it is being installed. For example, you can install a PCIe network card into a PCIe expansion slot only.

For today's integrated components (those built into the motherboard), you might not need an adapter to achieve the related services. Each adapter card, integrated or not, needs a driver—a software program that lets the operating system talk to the hardware—to make the device function with the operating system. Most motherboard manufacturers supply drivers with their motherboards, typically on a flash drive, that contain all the device drivers needed to get the built-in electronics recognized by the operating system. Drivers are also almost always available on the device manufacturer's website. Execution of the driver's setup program generally results in all components working properly. We'll show you how to install and update drivers in the upcoming "Adapter Configuration" section.

The following are the four most common categories of expansion cards installed today:

- Video
- Multimedia
- Network interface
- Input/output

Let's take a quick look at each of these card types, their functions, and what some of them look like.

A+ Exam Objective 3.5, "Given a scenario, install and configure motherboards, central processing units (CPUs), and add-on cards," wants you to be familiar with sound cards, video cards, capture cards, and network interface cards.

Video

A *video card* (sometimes called a *graphics card*) is the expansion card that you put into a computer to allow the computer to present information on some kind of display, typically a monitor or a projector. A video card is also responsible for converting the data sent to it by the CPU into the pixels, addresses, and other items required for display. Video cards can include dedicated chips to perform some of these functions, thus accelerating the speed of display.

You will encounter two classes of video cards: onboard cards and add-on cards. Onboard (or integrated) cards are built into the motherboard. As mentioned earlier, you need to install a device driver to get them to work properly, but those often come packaged with the motherboard itself. The upside to an integrated card is that it frees up an expansion slot. The manufacturer can either leave the slot open or design the motherboard and/or case to be smaller. One downside is that if the video card fails, you need a new motherboard, or you can install an add-on card. A second downside is that the onboard video cards aren't typically high-end. Onboard cards generally share system memory with the processor, which limits the quality of graphics one can produce. If the user wants great graphics from a powerful video card, then an add-on card is almost always the way to go. For example, serious gamers will always insist on a separate video card.

Add-on cards will have their own dedicated *graphics processing unit (GPU)*, which handles all of the video processing and frees up CPU capacity. In addition, they will have their own dedicated memory, called *video random-access memory (VRAM)*, which frees up system RAM. PCIe is the preferred expansion slot type. You might be able to find the rare, outdated motherboard that still offers a legacy AGP slot, and you might see some cheap PCI video cards, but they are uncommon. The technology on which PCIe was designed performs better for video than those on which AGP and PCI are based. Figure 2.1 shows an example of a PCIe x16 video card. The video card pictured is 10.6″ (270mm) long and takes up quite a bit of space inside the case. Most cards today have built-in fans like this one does to reduce the chance of overheating.

There is an extensive range of video cards available today on the market. For everyday usage, cards with 2–4 GB of video memory are inexpensive and will do the trick. For gamers, high-end cards with a minimum of 12–16 GB GDDR5 are recommended. Of course, over the lifespan of this book, that number is sure to increase. (As of this writing, cards with 24 GB GDDR6 are available.) High-end video cards can easily cost several thousand dollars.

FIGURE 2.1 A PCIe video expansion card

The main two standards for video cards are the NVIDIA GeForce series and the AMD Radeon (formerly ATI Radeon) line. Gamers will debate the pros and cons of each platform but know that you can get a range of performance, from good to phenomenal, from either one. When looking for a card, know how much memory is wanted or needed and how many and which types of video ports (such as HDMI or DisplayPort) are available. We will talk more about the pros and cons of several video connectors in Volume 1 Chapter 3, "Peripherals, Cables, and Connectors."

Space May Be a Challenge

Older video cards were shorter than current high-end models, often only 4–6″ long. When manufacturers started creating faster GPUs and adding memory, cards got longer. Once cards got to about 12″ long, some of them would no longer fit into standard ATX cases without running into other components.

Faster GPUs and more memory take up space and generate heat, though, which requires larger cooling systems. So instead of going longer, video card manufacturers have gone wider. The EVGA GeForce RTX 2070 Super shown in Figure 2.1 is a dual-slot card, meaning it takes up the equivalent of two back-panel slots in a standard case. Current high-end models are quad slot. Because of their width, they might preclude the use of other expansion slots on a motherboard.

When looking at buying new video card, check the size specifications and make sure it will fit with the case and motherboard you're planning to use.

Multimedia

The most basic and prolific multimedia adapter is the sound card. Video capture cards also offer multimedia experiences but are less common than sound cards.

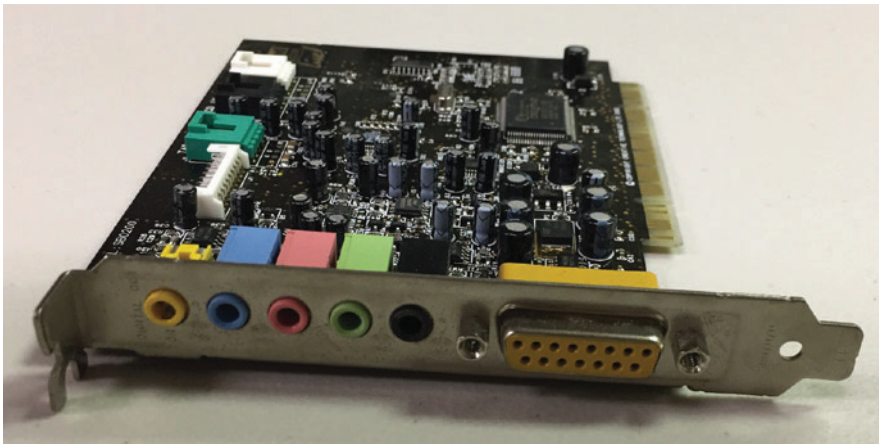
Sound Cards

Just as there are devices to convert computer signals into printouts and video information, there are devices to convert those signals into sound. These devices are known as *sound cards*. Although sound cards started out as pluggable adapters, this functionality is one of the most common integrated technologies found on motherboards today. A sound card typically has small, round 1/8″ (3.55mm) jacks on the back of it for connecting microphones, headphones, and speakers as well as other sound equipment. Older sound cards used a DA15 game port, which could be used for either joysticks or Musical Instrument Digital Interface (MIDI) controllers. Figure 2.2 shows an example of a sound card with five 3.55mm jacks and a DA15 game port. The audio jacks will be color-coded to identify their function. We know that this book isn't printed in color, so trust us when we say that from left to right, the ports are:

- **Yellow:** Digital output
- **Light blue:** Line level stereo input, such as from a sound mixer or other amplified audio equipment
- **Pink:** Microphone input
- **Lime green:** Stereo output or headphones
- **Black:** Surround sound left and right output

Some sound cards might also have an orange jack for center channel and subwoofer output, and a gray hack for surround sound rear left and right output. Other cards might have fewer ports, such as just light blue (line in), pink (microphone), and lime green (audio output).

FIGURE 2.2 A classic sound card



In our section on video cards, we noted that integrated cards have inferior performance to add-on ones, and though the same holds true for sound cards, the difference isn't quite as drastic. Many of today's motherboards come equipped with 5.1 or 7.1 analog or digital audio and support other surround sound formats as well. For everyday users and even many gamers, integrated audio is fine.

For users who need extra juice, such as those who produce movies or videos or do other audio/video (A/V) editing, a specialized add-on sound card is a must. Very good quality sound cards can be found for under $100, compared with cheaper models around $20—there isn't a huge difference in price as there is with CPUs and video cards. Look for a card with a higher sampling rate (measured in kilohertz [kHz]) and higher signal-to-noise ratio (measured in decibels [dB]). The de facto standard for sound cards is the Sound Blaster brand. Although other brands exist, they will often tout "Sound Blaster compatibility" in their advertising to show that they are legit.

Integrated motherboard audio will typically have about a 90 kHz sampling rate and 85–90 dB signal-to-noise ratio (SNR or S/N). A good add-on sound card can provide sampling of 190 kHz or more and an SNR of 115 dB or higher. The difference is noticeable, especially in professional productions that may be played at loud volumes.

In addition to audio output, many A/V editors will require the ability to input custom music from an electronic musical keyboard or other device. A term you will hear in relation to this is the MIDI standard. MIDI connections are almost always made via USB today, although you may occasionally see a round 5-pin MIDI connector. Look for the term *MIDI compatible* if attaching a digital musical device is needed.

Video Capture Cards

A *video capture card* is a stand-alone add-on card often used to save a video stream to the computer for later manipulation or sharing. This can be video from an Internet site, or video from an external device such as a digital camera or smartphone. Video-sharing sites on the Internet make video capture cards quite popular with enterprises and Internet users alike. Video capture cards need and often come with software to aid in the processing of multimedia input. While video and sound cards are internal expansion devices, capture cards can be internal (PCIe) or external (USB).

Not all video capture cards record audio signals while processing video signals. If this feature is important to you or the end user, be sure to confirm that the card supports it. Also know that capture cards work with standard video resolutions, and specific cards might be limited on the resolutions they support. Double-check the specifications to make sure the card will meet the need, and also make sure to get reviews of the software used with the device.

Because their names are similar, it's easy to confuse the purpose of video cards and video capture cards. They do completely different things.

A video card (or graphics card) creates the visual output sent to a monitor or other display device. A video capture card takes video signals from another

source, such as a gaming console, and converts them into digital data that can be recorded and saved or streamed. Be sure to understand the difference between them!

Network Interface Card

A *network interface card (NIC)* is an expansion card that connects a computer to a network so that it can communicate with other computers on that network. It translates the data from the parallel data stream used inside the computer into the serial data stream that makes up the frames used on the network. Internal cards have a connector for the type of expansion bus on the motherboard (PCIe or PCI) and external cards typically use USB. In addition to physically installing the NIC, you need to install drivers for the NIC in order for the computer to use the adapter to access the network. Figure 2.3 shows a PCIe x1 Ethernet NIC with an RJ45 port. (Network connectors are covered in more detail in Volume 1 Chapter 6, "Networking Fundamentals .")

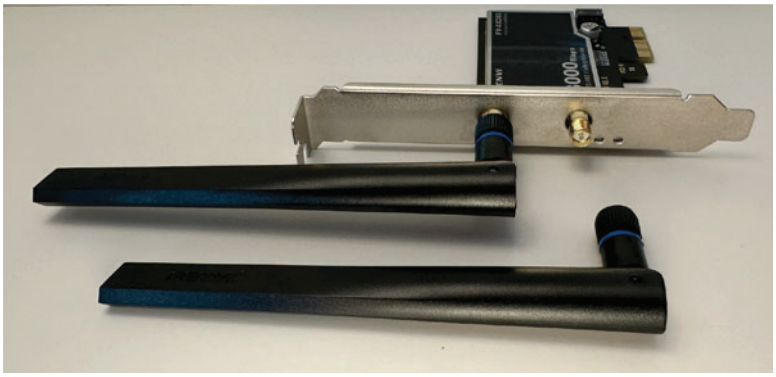
Many computers, especially mobile devices, have NIC circuitry integrated into their motherboards. Therefore, a computer with an integrated NIC wouldn't need to have a NIC expansion card installed unless it was faster or you were using the second NIC for load balancing (splitting the traffic between two cards to increase speed), security, or fault-tolerance (having a backup in case one fails) applications.

FIGURE 2.3 A network interface card

You will see two different types of network cards: wired and wireless. A wired card has an interface for the type of network it is connecting to (such as fiber connectors, Registered Jack 45 [RJ45] for unshielded twisted pair [UTP], antenna for wireless, or BNC for legacy coax). Wireless cards of course don't need to use wires, so they won't necessarily have one of these ports. Some do, just for compatibility or as a backup option.

Wireless NICs have the unique characteristic of requiring that you configure their connecting device before configuring the NIC. Wired NICs can generally create a link and begin operation just by being physically connected to a central hub or switch. Wireless (Wi-Fi) networks also use a central connectivity point, typically a wireless access point or wireless router, but that device must be configured with the network name and other parameters before any communication can occur. These concepts are explained in greater detail in Volume 1 Chapter 9, "Wireless and SOHO Networking." Figure 2.4 shows a PCIe wireless NIC for a desktop computer. Both antennae need to be attached for it to work—we took one off so you could see the connector. Other wireless antennas will have longer lead wires so the antenna can sit on top of the computer or a desk.

FIGURE 2.4 A wireless NIC



Real World Scenario

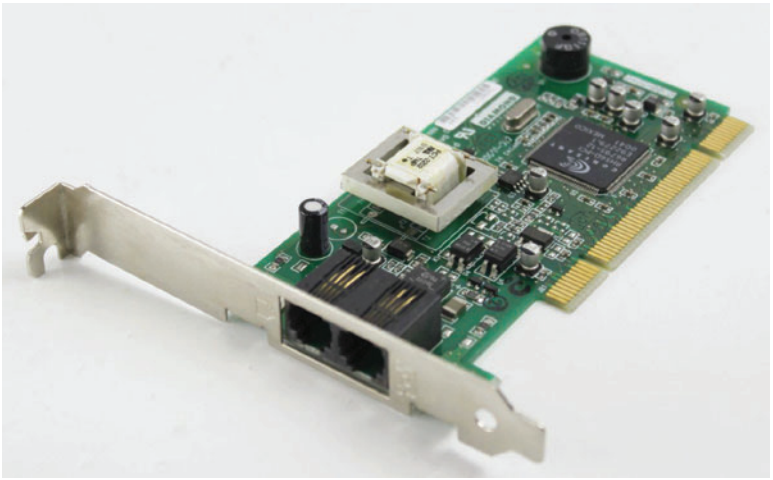
The Way of the Dinosaur. . .

Before high-speed, wireless Internet was available pretty much everywhere you went, wannabe web users needed to connect to the Internet using the plain old telephone system (POTS). Doing so required a device called a modem.

Old telephone lines were analog, and of course computers are digital. The modem got its name because it *mo*dulated and *dem*odulated analog and digital signals to enable computers to connect to each other over analog telephone lines. Modems look a bit like network cards, but the connector is smaller. Instead of the 8-pin RJ45 connector, phone lines use the 4-pin RJ11 connector. For those of you interested in history, a modem is shown in Figure 2.5. Notice it has two connectors. It's probably too hard to see in Figure 2.5, but modems with two connectors have small RJ11 connector and telephone icons etched into the faceplate. This tells you which connector to use to plug into the wall, and which one to

plug a telephone into. If the modem was in use, the telephone could not be used for voice calls. Only one connection at a time!

FIGURE 2.5 An internal analog modem



Input/Output

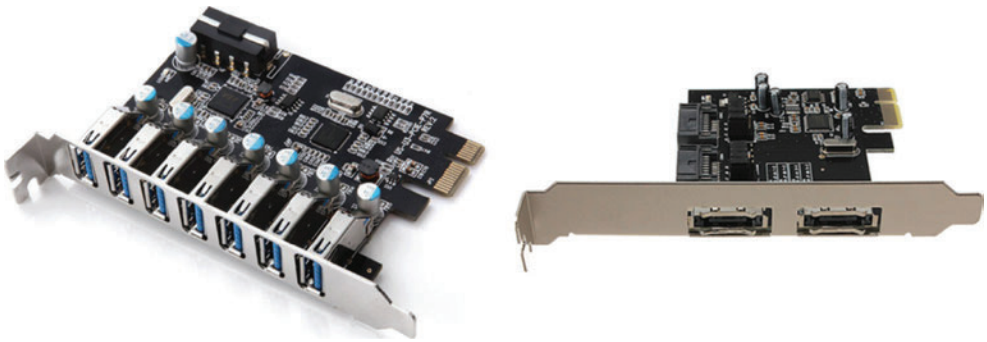
An *input/output card* is often used as a catchall phrase for any expansion card that enhances the system, allowing it to interface with devices that offer input to the system, output from the system, or both. The following are common examples of modern I/O cards:

- USB cards
- Storage cards, such as eSATA
- Thunderbolt cards

Figure 2.6 shows a 7-port PCIe x1 USB expansion card (left) next to an eSATA card (right). USB cards commonly come in 2-, 4-, and 7-port configurations, whereas eSATA cards often have one or two external ports. This eSATA card also has two internal SATA connectors on the top (left, in the picture) of the card. You'll also find cards that have multiple port types, such as eSATA and USB.

These cards are to be installed in a compatible slot on the motherboard. Their configuration is minimal, and it is usually completed through the operating system's Plug and Play (PnP) process. Nevertheless, check the BIOS/UEFI settings after installation for new entries in the menu structure. It's the job of the BIOS/UEFI to track all the hardware in the system and supply resources as needed. For example, a new Thunderbolt expansion card might allow you to configure whether attached Thunderbolt devices should be allowed to wake the system, how long a delay should be observed before waking the system, and various settings for how to use memory and other resources.

FIGURE 2.6 USB and eSATA expansion cards



Adapter Configuration

Most adapter cards will be automatically recognized by a PnP operating system such as Windows or macOS. On the off chance that the device isn't recognized or isn't installed properly, a driver might need to be manually installed. In addition, some adapters require additional configuration of features. We will look at drivers and adapter configuration here.

Installing and Updating Device Drivers

As noted previously, when you install a new device, a PnP OS should recognize it and install a driver for you. This goes for both internal and external devices. While this process works flawlessly the vast majority of the time, it's not perfect. In addition, sometimes the default driver supplied by the OS is older or doesn't support all of the features of a device like a newer driver will. It's good practice to always have the most updated driver for every device. For some devices such as video cards, driver updates might appear every six months. For others, such as basic keyboards or mice, it might be every decade.

Drivers in Windows are updated through the Device Manager utility, shown in Figure 2.7. If a device is not functioning properly, the device icon will have a yellow exclamation point or a red X over it. If you see devices like this listed, updating the driver is a good step to resolving the problem.

Device Manager has options to see driver details, update the driver, roll back the driver (this is useful if a driver update causes problems with the device working), disable the device, and uninstall the device. To get to these options, right-click the device in the Device Manager list and click Properties. Then open the Driver tab, as shown in Figure 2.8.

Exercise 2.1 shows you how to update a device driver in Windows 11. As pre-steps for the exercise, you should do the following:

1. Go to the manufacturer's website and download the latest driver. Save it in a place that's easy to find.
2. Double-click the downloaded driver file. It will do one or both of the following things:
    a. Extract the driver files to a separate location. Again, make that location easy to find.
    b. Start installing the driver (if so, go ahead and complete the process and review Exercise 2.1 for reference).

FIGURE 2.7 Windows 11 Device Manager

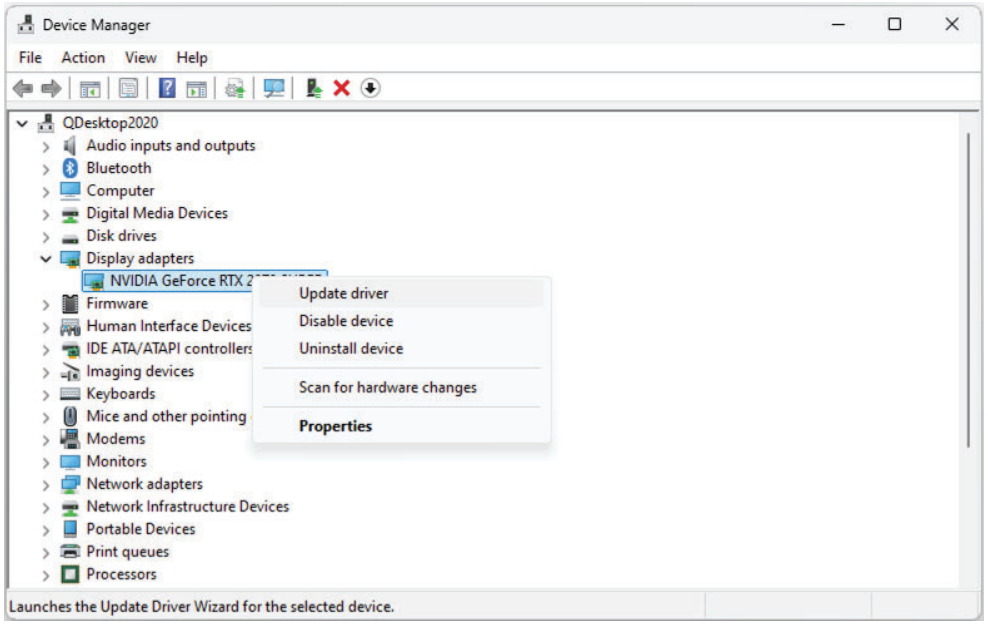
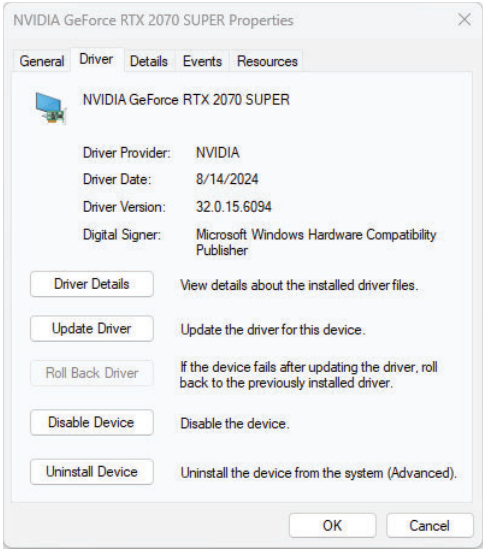


FIGURE 2.8 Video card properties



The video driver we downloaded for Exercise 2.1 does the extraction routine and then immediately starts the installation. The installation program is shown as an example in Figure 2.9. The installation will vary by driver, but this gives you an idea of what the program will look like. Again, if you choose to install or update a driver this way, that's completely fine; you won't need to complete the steps in Exercise 2.1, though.

FIGURE 2.9 NVIDIA driver installation

EXERCISE 2.1

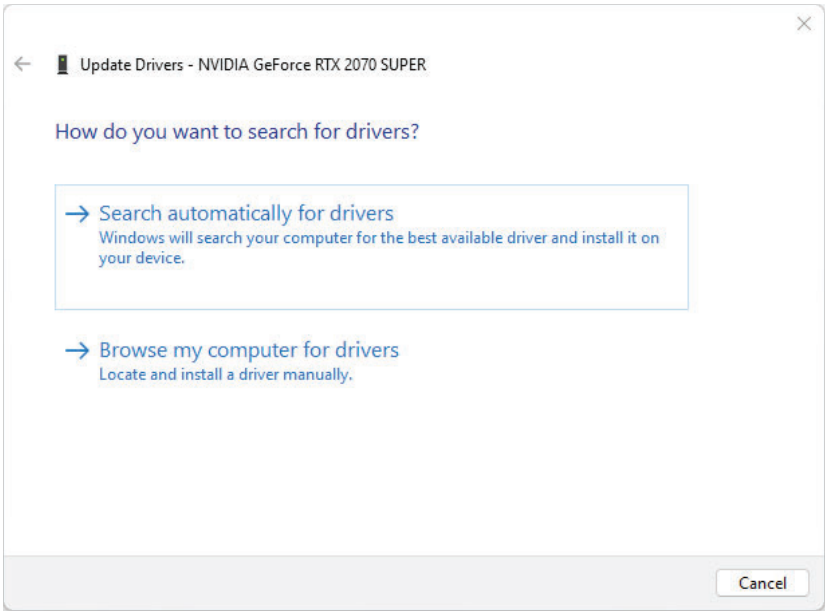
Updating a Device Driver in Windows 11

This exercise uses a video card as an example, but the process is similar for all devices. It will also work for new driver installation (provided the device is detected in Device Manager) or driver updates.

1. Open Device Manager by typing **`device manager`** into the Windows 11 search box and pressing Enter.
2. Right-click the device. You will see a menu similar to the one in Figure 2.7.
3. Click Update Driver. That will begin the driver update process shown in Figure 2.10.

EXERCISE 2.1 *(continued)*

FIGURE 2.10 Update driver in Windows



4. Have Windows search for the driver, or specify the location where you saved the extracted driver files.
5. Follow the prompts to complete the installation.

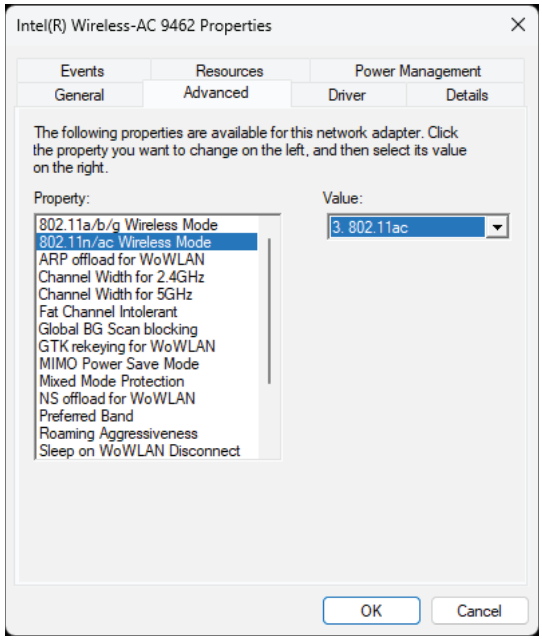
Configuring Devices

Most devices come preconfigured with the most popular options. Some devices might not have configuration options at all, whereas others have quite a few. Some devices might have software settings configured through an app provided by the device manufacturer. If there is no app, try opening Device Manager, right-clicking the device, and choosing Properties. Again, options will vary by device. Here are some examples:

- Video cards might require you to set the resolution or enable/disable other advanced features.
- Most sound cards tend to work with no specific configuration, but advanced features will need to be implemented through the operating system or through utilities that came with the adapter.

- The functions of video capture cards are sometimes not native to the operating system and therefore come with advanced utilities that must be learned and configured before the adapters will work as expected.
- Wireless network adapters often require the installation of a screw-on antenna, which should be postponed until after the card is fully inserted and physically secured in the system. Software configuration that allows these cards to communicate with a wireless access point can be challenging for the novice. To give you an example of what a wireless NIC configuration screen may look like, take a look at Figure 2.11. We will cover some of these and others in Volume 1 Chapter 9. (Most of them you will never need to set in your entire career.)

FIGURE 2.11 Wireless NIC properties



Wired network adapters tend to be easier to configure than wireless ones. Nevertheless, even wired NICs might require manual configuration of certain protocol settings, such as IP addressing, duplex, and speed, in order for them to be productive. We will cover these topics in later networking chapters. In general, installation and configuration steps for expansion cards can be summarized as follows:

1. Ensure that the computer is powered off and that you are properly grounded.
2. Install the adapter into an open slot.
3. Connect power, if needed.

   This most often applies to video cards.

4. After booting up the computer, install the driver.

   Again, Plug and Play may take care of this automatically for you.
5. If the card isn't recognized or providing the expected functionality, check the BIOS for configuration settings.
6. Check Device Manager to see if the device is recognized and if there are any error badges.
7. For other configuration options, use the utility provided by the manufacturer, if applicable.

In any event, consult the documentation provided with your adapter or the manufacturer's website for additional configuration requirements or options. The more specialized the adapter, the more likely it will come with specialty-configuration utilities.

Understanding Storage Devices

What good is a computer without a place to put everything? Storage media hold the files that the operating system needs to operate and the data that users need to save. What about saving to the cloud? The computers that make up the cloud, rather than the local computer, hold the storage media. The many different types of storage media differ in terms of their capacity (how much they can store), access time (how fast the computer can access the information), and the physical type of media used. The following types of storage devices are covered here:

- Conventional hard disk drives
- Solid-state drives
- RAID
- Removable storage, including optical drives

After that, we'll finish this section by looking at installing, removing, and configuring storage devices.

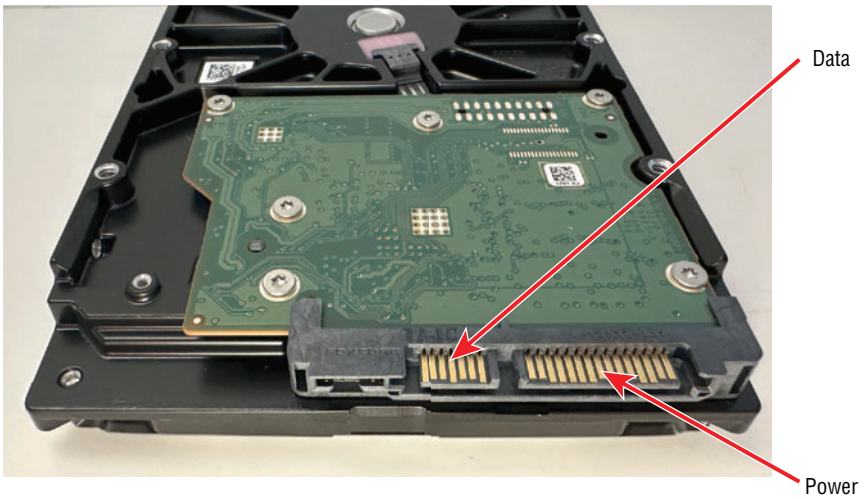
Conventional Hard Disk Drives

Hard disk drive (HDD) systems (or *hard drives* for short) are used for permanent, nonvolatile data storage and quick access. Hard drives typically reside inside the computer, where they are semi-permanently mounted with no external access (although there are external and removable hard drives) and can hold more information than other forms of storage. Conventional hard drives use a *magnetic storage* medium on rotating metal platters.

Hard drives, both conventional and solid-state (covered next), are nonvolatile storage. This means that unlike RAM or cache, they retain their data when power is shut off.

Hard drives often connect to the motherboard's SATA interface, which you learned about in Volume 1 Chapter 1, "Motherboards, Processors, and Memory." The back of the hard drive will have data and power connectors. Figure 2.12 offers a bottom view of a SATA hard drive, and shows the data and power connectors.

FIGURE 2.12 SATA hard drive connectors



Putting Ancient Technology Aside

The predecessor to SATA for storage connections was Parallel Advanced Technology Attachment (PATA), which we briefly introduced in Volume 1 Chapter 1. PATA was used for hard drives, optical drives, and magnetic tape drives. Today, PATA hard drives are essentially obsolete because of their slow transfer speeds. Most PATA hard drives followed the ATA/100 standard, which had a maximum transfer speed of 100 MBps.

You might still run across a PATA drive, so Figure 2.13 shows you the data and power connectors for one. The data connector on the right is 40 pins and uses a flat ribbon cable. There is a missing pin, or key, in the center of the top row of pins. The 4-pin power connector on the left is called Molex (which is still an exam objective). We show the *Molex connector* from the power supply later in this chapter in Figure 2.32 and again in Volume 1

Chapter 3. In between them is the jumper block, which is used to configure the drive as primary or secondary (there can only be two PATA drives on one PATA controller).

FIGURE 2.13 PATA hard drive connectors

Anatomy of a Hard Drive

A hard drive is constructed in a cleanroom to avoid the introduction of contaminants into the hermetically sealed drive casing. Once the casing is sealed, most manufacturers seal one or more of the screws with a sticker warning that removal of or damage to the seal will result in voiding the drive's warranty. Even some of the smallest contaminants can damage the precision components if allowed inside the hard drive's external shell.

Inside the sealed case of the hard drive lie one or more platters, where the actual data is stored by the read/write heads. The heads are mounted on a mechanism that moves them in tandem across both surfaces of all platters. Older drives used a stepper motor to position the heads at discrete points along the surface of the platters, which spin at thousands of revolutions per minute on a spindle mounted to a hub. Newer drives use voice coils for a more analog movement, resulting in reduced data loss because the circuitry can sense where the data is located through a servo scheme, even if the data shifts due to changes in physical disc geometry. Figure 2.14 shows the internal components of a conventional hard drive.

Before a hard drive can store data, it must be prepared. The drive is separated into logical areas with a process called partitioning. After that, the partitions can be formatted with a filesystem and then it's ready to store data.

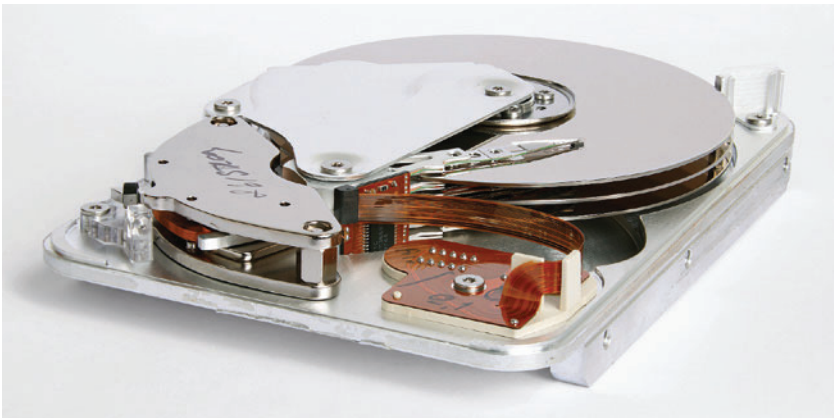
We will talk more about filesystems such as FAT32, NTFS, and others in Volume 2 Chapter 1, "Operating System Basics."

HDD Spindle Speeds

As the electronics within hard drives got faster, they were capable of requesting data at higher and higher rates. If the platters were spinning at a constant rate, however, the information could be accessed only as fast as a given fixed rate. To make information

available to the electronics more quickly, manufacturers increased the speed at which the platters spin from one generation of drives to the next, with multiple speeds coexisting in the marketplace for an unpredictable period, at least until the demand died down for one or more speeds.

FIGURE 2.14 Anatomy of a hard drive



By Eric Gaba, Wikimedia Commons user Sting, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=11278668

Exam Objective 3.4, "Compare and contrast storage devices," calls out spindle speeds and form factors as topics to know for the exam.

The following *spindle speeds* (or spin rates) have been used in the industry for the platters in conventional magnetic hard disk drives:

- 5,400 rpm
- 7,200 rpm
- 10,000 rpm
- 12,000 rpm
- 15,000 rpm

While it is true that a higher revolutions per minute (rpm) rating results in the ability to move data more quickly, many applications do not benefit from increased disk-access speeds. As a result, you should choose only faster drives, which are also usually more expensive per byte of capacity, when you have an application for this type of performance, such as for housing the partition where the operating system resides or for very disk-intensive programs. For comparison, a 7,200 rpm SATA hard drive can sustain data read speeds of about 100 MBps, which is about the same as a PATA ATA/100 7,200 rpm drive. A 10,000 rpm (also known as 10k) SATA drive can top out around 200 MBps.

Most conventional hard drives sold are 5,400 rpm or 7,200 rpm. Faster drives were usually reserved for specialized installations.

Higher speeds also consume more energy and produce more heat. The lower speeds can be ideal in laptops, where heat production and battery usage can be issues with higher-speed drives. Even the fastest conventional hard drives are slower than solid-state drives are at transferring data.

HDD Form Factors

Physically, the most common hard drive *form factors* (sizes) are 3.5″ and 2.5″. Desktops traditionally use 3.5″ drives, whereas the 2.5″ drives are made for laptops—although most laptops today avoid using conventional HDDs. Converter kits are available to mount a 2.5″ drive into a 3.5″ desktop hard drive bay. Figure 2.15 shows the two drives together. As you can see, the 2.5″ drive is significantly smaller in all three dimensions, but it does have the same storage capacity and connectors as its bigger cousin.

FIGURE 2.15 A 3.5″ and a 2.5″ hard drive

Reading Is Like Breathing In, and Writing Is Like Breathing Out

There are two categories of conventional hard drives that you will find in the marketplace, and they differ in how they read and write data. The first is conventional magnetic recording (CMR), which is the "traditional" hard drive. Data is written in tracks (think of concentric circles on the platter), with a small gap between each track. A newer technology is called shingled magnetic recording (SMR), which overlaps tracks to increase storage capacity.

CMR drives are better for situations where data will be frequently read from and written to a drive. SMR drives are less expensive per platter and more energy efficient, but slower at transferring data. Therefore, they are best suited for things like data backups or archives.

Neither CMR nor SMR is an exam objective, but knowing details like this can help you make a better decision if you're in the market for a new drive.

Solid-State Drives

Unlike conventional hard drives, *solid-state drives (SSDs)* have no moving parts—they use the same solid-state memory technology found in the other forms of flash memory. You can think of them as big versions of the flash drives that are so common.

Because they have no moving parts, SSDs are capable of transferring data much more quickly than HDDs could ever dream of doing. Recall from the "HDD Spindle Speeds" section that a 10k SATA HDD tops out at about 200 MBps. Even the slowest SSDs will run circles around that. The true speed of an SSD will be determined, of course, by the drive itself, but also the interface to which it's attached.

And because there's no need for spinning platters and read/write heads, SSDs can be made much smaller than HDDs, making them better for laptops and portable devices. SSDs have several other advantages over their mechanical counterparts as well, including the following:

- Faster start-up and read times
- Less power consumption and heat produced
- Silent operation
- Generally more reliable because of a lack of moving parts
- Less susceptible to damage from physical shock and heat production
- Higher data density per square centimeter

The disadvantages of SSDs are as follows:

- The technology to build an SSD is more expensive per byte.
- All solid-state memory is limited to a finite number of write (including erase) operations. Lack of longevity could be an issue. This doesn't seem to be a problem with reputable drive manufacturers, but it could become a problem as a device ages.

You will find that SSDs in the market generally have lower overall capacity than HDDs. For example, it's common to find HDDs over 8 TB in size, with 24 TB drives pacing the market. Conversely, the biggest commercially available SSD (as of this writing) has been 8 TB for a few years now. As for cost, HDDs run about 2–3 cents per GB and low-end SATA SSDs are about three to four times as expensive. Faster SSDs such as NVMe drives (which we'll get to in a minute) can be from four to 10 times as expensive. Of course, prices are subject to (and it's guaranteed they will) change!

When used as a replacement for traditional HDDs, SSDs are expected to behave in a similar fashion, mainly by persistently storing data—retaining contents across a power cycle. With SSD, you can also expect to exceed the read/write speed of the HDD. SSDs can be made faster still by including a small amount of DRAM as a cache.

SSDs come in various shapes and sizes and have a few different interfaces and form factors. We will cover those in the upcoming "SSD Communication Interfaces" and "SSD Form Factors" sections.

SSDs are in Exam Objective 3.4. Be able to compare and contrast the NVMe, SATA, PCIe, and SAS communications interfaces, and the M.2 and mSATA form factors.

SSD Communication Interfaces

The advent of the SSD was a major advancement for the computer industry. Solid-state drives are basically made from the same circuitry that RAM is, and they are really, really fast. When they were first on the market, the limitation in the system was the SATA controller that most hard drives were plugged into. So as enterprising computer engineers are known to do, some started looking into ways to overcome this barrier. The result is that there have been more interfaces designed for storage devices, with much faster speeds. We will look at four of them here: SATA, PCIe, NVMe, and SAS.

SATA

At this point, SATA is a technology you should be somewhat familiar with. We covered it in Volume 1 Chapter 1, and the hard drive interface is shown earlier in this chapter in Figure 2.12. SATA is a bit unique among the other subjects in this SSD section because it can support mechanical hard drives as well as SSDs.

SSDs came onto the market in the mid-1990s before SATA was even a thing, but they were limited in popularity because of the major bottleneck—the PATA communications interface. Once SATA came along in the early 2000s, the two technologies felt like glorious companions. SATA 1.x could transfer data at 150 MBps, which was a lot faster than the conventional hard drives at the time. (The most common conventional standard was ATA/100, which maxed out at 100 MBps.) Then, of course, came SATA 2.x (300 MBps) and eventually SATA 3.x (600 MBps). Comparatively, conventional hard drives appear to be painfully slow. And they are. Table 2.1 summarizes the standards and speeds.

TABLE 2.1 SATA standards and speeds

Version	Nickname	Date	Speed
1.x	SATA-150	2003	150 MBps
2.x	SATA-300	2004	300 MBps
3.x	SATA-600	2009	600 MBps

Keep in mind that of all the SSD technologies we discuss in this chapter, the SATA interface is the slowest of them. So while SATA SSDs are about 6 times faster than conventional HDDs, there is a lot of performance upside. SATA SSDs are still popular today because they are plentiful and cheap (compared to other SSDs), and motherboards usually have more SATA connectors than any other type of hard drive connector.

PCIe

Peripheral Component Interconnect Express (PCIe) is another technology that was covered in Volume 1 Chapter 1 and is used for SSDs. PCIe was first introduced in 2002, technically a year before SATA, and both technologies took a little while to get widely adopted. Like SATA, PCIe has gone through several revisions, with each version being faster than the previous one. Table 2.2 shows the throughput of PCIe versions.

TABLE 2.2 PCIe standards and transfer rates

Version	Date	Transfer rate	Throughput per lane (one direction)	Total x16 throughput (bidirectional)
1.0	2003	2.5 GT/s	250 MBps	8 GBps
2.0	2006	5.0 GT/s	500 MBps	16 GBps
3.0	2010	8.0 GT/s	1 GBps	32 GBps
4.0	2017	16.0 GT/s	2 GBps	64 GBps
5.0	2019	32.0 GT/s	4 GBps	128 GBps
6.0	2022	64.0 GT/s	8 GBps	256 GBps

PCIe 6.0 was released in 2022, and PCIe 7.0 is expected sometime in late 2025. As you might expect, it will double the speed of PCIe 6.0 and be compatible with older PCIe standards. Devices supporting the newest PCIe specifications typically take two to three years to hit the market following a new version release.

Looking at Table 2.2, something might immediately jump out at you. The transfer rate is specified in gigatransfers per second (GT/s). This unit of measure isn't very commonly used, and most people talk about PCIe in terms of its data throughput. Before moving on, though, let's take a quick look at what GT/s means. (It's highly unlikely you will be tested on this, but you might be curious.)

PCIe is a serial bus that embeds clock data into the data stream to help the sender and receiver keep track of the order of transmissions. Two clock bits are used for every 8 bits of data. Therefore, PCIe uses what's called "8b/10b" encoding—8 bits of data are sent in a

10-bit bundle, which is decoded at the receiving end. Gigatransfers per second refers to the total number of bits sent, whereas the throughput data you might be more used to seeing refers to data only and not the clock.

A little math using PCIe 1.0 might help illustrate the transfer rate versus data throughput. PCIe 1.0 has a transfer rate of 2.5 billion (giga) bits per second. Because it uses 8b/10b encoding, you then multiply by 0.8 (8/10) to get to bits of data, which is 2 trillion. But that's bits. Divide that by 8 to turn it into bytes, and you are left with 250 million bytes, or 250 MBps. Again, don't expect to see this on the test, but we know that some of you like these technical details!

Also remember that PCIe cards can have different numbers of channels—x1, x2, x4, x8, and x16. So, for example, a PCIe 3.0 x4 card will have total data throughput of 8 GBps (1 GBps for one lane in one direction, times two for bidirectional, times four for the four channels).

What does all of this mean for SSDs? First, take a look at a picture of a PCIe SSD in Figure 2.16. This is an Intel SSD 750, and here are some of its specs:

- **Interface:** PCIe 3.0 x4 – NVMe
- **Capacity:** 1.2 TB
- **Sequential read:** 2.4 GBps
- **Sequential write:** 1.2 GBps
- **Endurance:** 70 GB of writes per day for five years

FIGURE 2.16 Intel PCIe SSD

This drive is a few years old, but it's perfect for our discussion. Based on Table 2.2, PCIe 3.0 x4 has a maximum throughput of 8 GBps. This drive doesn't get close to that level, but it still reads about four times faster than the best SATA drives. This is an example of the difference between theoretical maximums of standards versus practical realities of creating hardware. Also note its rated endurance. Recall that SSDs have a finite lifespan, although it will take your average user quite some time to hit this mark.

Remember that, unlike SATA, which is designed for storage devices such as hard drives and optical drives, PCIe is more of a universal connector. Video cards, sound cards, network cards, and many other devices use PCIe slots as well. Before attempting to install a PCIe hard drive into a computer, make sure that there's an open PCIe slot of the appropriate type first!

Also remember that PCIe cards can get power from the slot itself. Devices that require a lot of power such as video cards will have their own power connector from the power supply, but PCIe hard drives don't need anything other than what the PCIe slot provides.

NVMe

Created by a consortium of manufacturers, including Intel, Samsung, Dell, SanDisk, and Seagate, and released in 2011, *Non-Volatile Memory Express (NVMe)* is an open standard designed to optimize the speed of data transfers. Unlike SATA and PCIe, NVMe isn't related to a specific type of connector. Said another way, there is no NVMe connector—think of it as a nonvolatile memory chip that can be used in SATA, PCIe, or M.2 (which we will cover in the next section) slots. Figure 2.17 shows a 1 TB Western Digital M.2 NVMe SSD.

FIGURE 2.17 M.2 NVMe SSD

NVMe drives are incredibly fast—current NVMe SSDs can support data reads of up to 3.5 GBps, provided that the interface they are plugged into supports it as well, of course. An NVMe SATA 3 SSD will still be limited to 600 MBps; older PCIe versions in theory might have limitations, but you're not going to find any PCIe 1.0 or 2.0 NVMe SSDs, so it's not a problem.

One potential issue you might see with NVMe hard drives is that in order to be a boot drive, the motherboard must support it. If the motherboard has a built-in M.2 slot, odds are that the BIOS will support booting from an NVMe drive. If you are adding it using a PCIe slot, the BIOS might not be able to boot from that drive. Always check the motherboard documentation to ensure it supports what you're trying to do.

SAS

When the Small Computer System Interface (SCSI) standard came out in 1986, it was among the fastest storage transfer technologies available, delivering 5 MBps throughput. In the early-to-mid 1990s, it was found mostly in servers. If someone ran SCSI in their home system, you knew that they were serious about high performance. Over time, revisions boosted the speed to 320 MBps (a later implementation could hit 640 MBps, but it was not widely adopted). Faster speeds were still needed though, and thanks to advancements in USB, serial technologies were showing more promise for long-term success.

Switching over to a serial standard, SCSI developers released *Serial Attached SCSI (SAS)* in 2003. It featured a solid 3 Gbps bandwidth, which is 375 MBps (see the "Converting the Numbers" sidebar in this section). At the time, the primary competing storage standards were PATA (commonly 100 MBps, with some 133 and 167 MBps implementations) and SATA 1.x (150 MBps). So SAS was very fast at the time. Subsequent versions doubled the speed (where have you heard that before?); Table 2.3 shows versions, release dates, and throughput. SAS-5 is supposedly under development, offering 45 Gbps throughput, but there is no official release date set.

TABLE 2.3 SAS standards

Version	Released	Throughput
SAS-1	2003	3 Gbps (375 MBps)
SAS-2	2009	6 Gbps (750 MBps)
SAS-3	2013	12 Gbps (1500 MBps or 1.5 GBps)
SAS-4	2017	24 Gbps (3000 MBps or 3.0 GBps)

Converting the Numbers

Notice that SAS speeds are quoted in gigabits per second (Gbps), whereas a lot of other standards are quoted in gigabytes per second (GBps) or even megabytes per second (MBps). We don't throw around these terms to confuse you—rather, we use what the standards' developers use. And yes, SCSI is in MBps and SAS is Gbps even though they were developed by the same group.

Here are some conversions, for reference:

- 125 MBps equals 1 Gbps.
- 8 Gbps equals 1 GBps.
- 1000 MBps equals 1 GBps.

Today, SAS is found mostly in servers and dedicated storage devices. You can purchase PCIe SAS expansion cards for desktop PCs, but with M.2 drives commonly available, there's not much point.

SAS hard drives come in 2.5″ and 3.5″ form factors, and here's the sometimes tricky part—the connector on the hard drive is nearly identical to a SATA connector. The only difference is that a SAS drive has a plastic block in between the data and power connectors, whereas SATA has a gap between them. SAS and SATA are also electronically compatible. SATA drives, which are less expensive than SAS drives, can be connected to a SAS controller and usually work properly, albeit at slower SATA speeds. SAS drives, however, cannot be connected to a SATA controller. Additionally, you should never mix SAS and SATA drives in the same hard drive array (logical grouping of hard drives) lest you want to deal with performance inconsistencies.

SSD Form Factors

Whereas a *communications interface* is the method the device uses to communicate with other components, a *form factor* describes the shape and size of a device. The two SSD form factors you need to know for the A+ exam are mSATA and M.2.

mSATA

The Serial ATA International Organization has developed several specifications—you've already been introduced to SATA and eSATA. Next on the list is a form factor specifically designed for portable devices such as laptops and smaller—*mini-Serial ATA (mSATA)*. mSATA was announced in 2009 as part of SATA version 3.1 and hit the market in 2010.

mSATA uses the same physical layout as the Mini PCI Express (mPCIe) standard, and both have a 30mm-wide 52-pin connector. The wiring and communications interfaces between the two standards are different though. mSATA uses SATA technology, whereas mPCIe uses PCIe. In addition, mPCIe card types are as varied as their larger PCIe cousins are, including video, network, cellular, and other devices. mSATA, on the other hand, is dedicated to storage devices based on SATA bus standards. mSATA cards come in 30mm × 50.95mm full-size and 30mm × 26.8mm half-size cards. Figure 2.18 shows a full-sized mSATA SSD on top of a 2.5″ SATA SSD.

The wiring differences between mSATA and mPCIe can pose some interesting challenges. Both types of cards will fit into the same slot, but it depends on the motherboard as to which is supported. You might have heard us say this before, but as always, check the motherboard's documentation to be sure.

FIGURE 2.18 mSATA SSD and a 2.5″ SATA SSD

Source: Vladsinger / Wikimedia Commons / CC BY SA 3.0

M.2

Originally developed under the name Next Generation Form Factor (NGFF), *M.2* (pronounced "M dot 2") was born from the desire to standardize small form factor SSD hard drives. We touched briefly on M.2 in Volume 1 Chapter 1, and we mentioned that although M.2 is primarily used for hard drives, it supports other types of cards such as Wi-Fi, Bluetooth, Global Positioning System (GPS), and near-field communication (NFC) connectivity, as well as PCIe and SATA connections. It's a form factor designed to replace the mSATA standard for ultra-small expansion components in laptops and smaller devices. Whereas mSATA uses a 30mm 52-pin connector, M.2 uses a 66-pin connector, most commonly 22mm wide.

One interesting connectivity feature of M.2 is that the slots and cards are keyed such that only a specific type of card can fit into a certain slot. The keys are given letter names to distinguish them from each other, starting with the letter *A* and moving up the alphabet as the location of the key moves across the expansion card. Table 2.4 explains the slot names, some interface types supported, and common uses.

TABLE 2.4 M.2 keying characteristics

Key	Common interfaces	Uses
A	PCIe x2, USB 2.0	Wi-Fi, Bluetooth, and cellular cards
B	PCIe x2, SATA, USB 2.0, USB 3.0, audio	SATA and PCIe x2 SSDs

Key	Common interfaces	Uses
E	PCIe x2, USB 2.0	Wi-Fi, Bluetooth, and cellular cards
M	PCIe x4, SATA	PCIe x4 SSDs

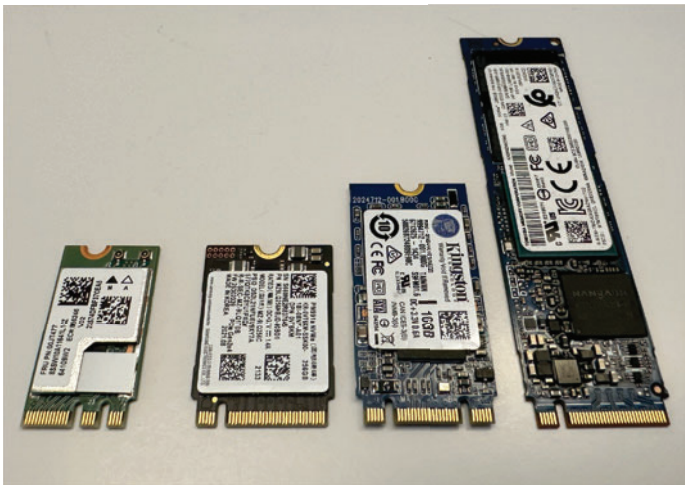
Another interesting feature of the cards is that they are also named based on their size. For example, you will see card designations such as 1630, 2242, 2280, 22110, or 3042. The first two numbers refer to the width, and the rest to the length (in millimeters) of the card.

Let's look at some examples. Figure 2.19 shows four different M.2 cards. From left to right, they are as follows:

- A- and E-keyed 1630 Wi-Fi and Bluetooth adapter
- M-keyed 2230 SSD
- B- and M-keyed 2242 SSD
- M-keyed 2280 SSD

Of the four, only an M-keyed SSD can get the fastest speeds, because it supports PCIe x4. SSDs on the market are keyed B, M, or B+M. A B-keyed or M-keyed SSD won't fit in a B+M socket. A B+M keyed drive will fit into a B socket or an M socket, however.

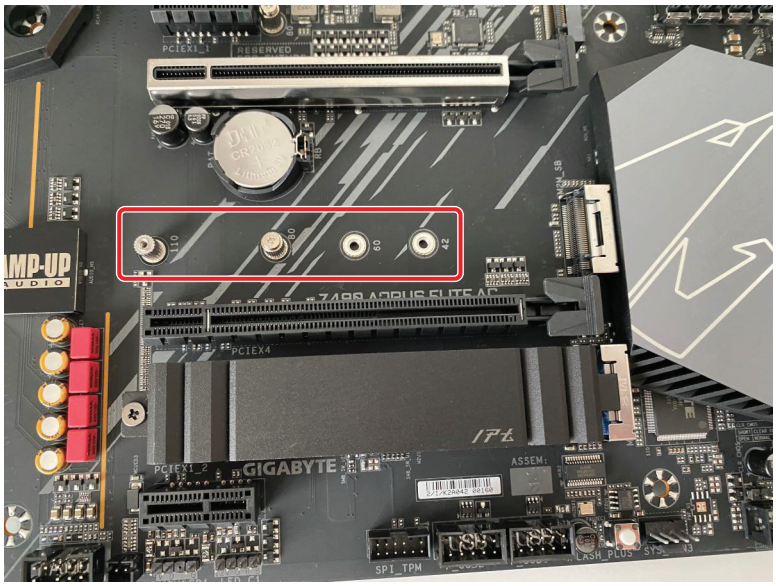
FIGURE 2.19 Four M.2 cards



Maximum speeds are always increasing. For reference, at the time of writing, some of the fastest 4 TB NVMe M.2 SSDs support PCIe 5.0 x4 and have read speeds up to 14.1 GBps and write speeds up to 12.6 GBps.

Figure 2.20 shows a motherboard with two M.2 slots. The bottom one is hidden by its long, narrow metal cover. These function as protection and heat spreaders and get replaced over a card when it's added to the system. Be sure to take any plastic cover off the bottom of the spreader before putting it back over an M.2 card! The top one is an uncovered M-keyed slot. We put a highlighted box around the numbered post holes for 42mm, 60mm, 80mm, and 110mm cards. Use only the post hole needed for the length of card you're installing (in Figure 2.20, the post is screwed into the 80mm hole), and then the 110 post for the heat spreader. Some high-end SSDs come with their own heatsinks—in those cases, it's not necessary to reinstall the original cover. Like PCIe, M.2 devices derive power from the motherboard connector.

FIGURE 2.20 M.2 slots



As mentioned earlier, M.2 is a form factor, not a bus standard. M.2 supports SATA, USB, and PCIe buses. What does that mean for M.2 hard drives? It means that if you purchase an M.2 SATA hard drive, it will have the same speed limitation as SATA III, or about 600 MBps. That's not terrible, but it means that the primary advantage of an M.2 SATA drive versus a conventional SATA SSD is size. An M.2 PCIe hard drive is an entirely different story. PCIe, you will recall, is much faster than SATA. A PCIe 2.0 x1 bus supports one-way data transfers of 500 MBps. That is close to SATA III speed, and it's only a single lane for an older standard. NVMe M.2 drives kick up the speed even further. If you want the gold standard for hard drive speed, NVMe M.2 is the way to go.

RAID

Multiple hard drives can work together as one system, or array, providing increased performance (faster disk reads and writes), fault tolerance (protection against one or more disks failing), or both. Such systems are called *Redundant Array of Independent (or Inexpensive) Disks (RAID)*. RAID can be implemented in software, such as through the operating system, or in hardware, such as through the motherboard BIOS or a RAID hardware controller. Hardware RAID is more efficient and offers higher performance but at an increased cost.

Exam Objective 3.4 mentions RAID. You need to be able to compare and contrast RAID 0, 1, 5, 6, and 10.

There are several types of RAID. The following are the most commonly used RAID levels:

RAID 0 *RAID 0* is also known as *disk striping*, where a striped set of equal space from at least two drives creates a larger volume. This is in contrast to unequal space on multiple disks being used to create a simple *volume set*, which is not RAID 0. RAID 0 doesn't provide the fault tolerance implied by the *redundant* component of the name. Data is written across multiple drives, so one drive can be reading or writing while another drive is doing the same. This makes for faster data access. If any one of the drives fails, however, all content is lost. Some form of redundancy or fault tolerance should be used in concert with RAID 0.

RAID 1 Also known as *disk mirroring*, *RAID 1* is a method of producing fault tolerance by writing all data simultaneously to two separate drives. If one drive fails, the other contains all of the data, and it will become the primary drive. Disk mirroring doesn't help access speed, however, and the cost is double that of a single drive. If a separate host adapter is used for the second drive, the term *duplexing* is attributed to RAID 1. Only two drives can be used in a RAID 1 array.

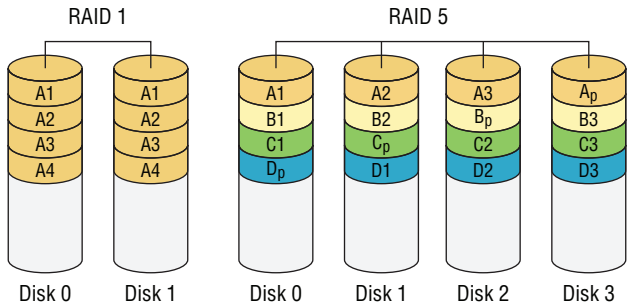
RAID 5 *RAID 5* combines the benefits of both RAID 0 and RAID 1, creating a redundant striped volume set. Sometimes you will hear it called a *stripe set with parity*. Unlike RAID 1, however, RAID 5 does not employ mirroring for redundancy. Each stripe places data on *n*-1 disks, and parity computed from the data is placed on the remaining disk. The parity is interleaved across all the drives in the array so that neighboring stripes have parity on different disks. If one drive fails, the parity information for the stripes that lost data can be used with the remaining data from the working drives to derive what was on the failed drive and to rebuild the set once the drive is replaced.

The same process is used to continue to serve client requests until the drive can be replaced. This process can result in a noticeable performance decrease, one that is predictable because all drives contain the same amount of data and parity. Furthermore,

the loss of an additional drive results in a catastrophic loss of all data in the array. Note that while live requests are served before the array is rebuilt, nothing needs to be computed for stripes that lost their parity. Recomputing parity for these stripes is required only when rebuilding the array. A minimum of three drives is required for RAID 5. The equivalent space of one drive is lost for redundancy. The more drives in the array, the less of a percentage this single disk represents. Because of the parity stripe, the total capacity of a RAID 5 array is the sum of the capacities of all disks in the array, minus one disk. If more than one drive fails, the RAID 5 array will fail as well.

Figure 2.21 illustrates RAID 1 and RAID 5.

FIGURE 2.21 RAID 1 and RAID 5



RAID 6 *RAID 6*, also called *double-parity RAID*, is similar to RAID 5. The biggest difference is that RAID 6 uses a second parity stripe, meaning that it can recover from multiple disk failures. Because of this, it's considered more fault-tolerant than RAID 5 and is a better choice for mission-critical data. It will be a little slower than RAID 5, though, because two parity stripes need to be calculated. A minimum of four hard disks is required to implement RAID 6. The total storage capacity of the array will be the sum of all drive capacities minus two drives, to account for the double-parity stripes.

RAID 10 Also known as *RAID 1+0, RAID 10* adds fault tolerance to RAID 0 through the RAID 1 mirroring of each disk in the RAID 0 striped set. Its inverse, known as *RAID 0+1*, mirrors a complete striped set to another striped set just like it. Both of these implementations require a minimum of four drives and, because of the RAID 1 component, use half of your purchased storage space for mirroring.

Table 2.5 summarizes the key data for RAID systems.

TABLE 2.5 RAID characteristics

Version	Nickname	Fault tolerant?	Min. drives	Can recover from X drive failures	Capacity
RAID 0	Disk striping	No	2	0	Add total space from all drives.

Version	Nickname	Fault tolerant?	Min. drives	Can recover from X drive failures	Capacity
RAID 1	Disk mirroring	Yes	2	1	Add total space from all drives, divide in half.
RAID 5	Stripe set with parity	Yes	3	1	All drives minus one drive.
RAID 6	Double parity	Yes	4	2	All drives minus two drives.
RAID 10	RAID 1+0	Yes	4	2*	Add total space from all drives, divide in half.

* The number of drive failures a RAID 10 system can sustain depends on a few factors. If both mirrored or both original drives fail, the data can be recovered. But if Disk 1 of both the original set and the mirrored set fail, the data is not recoverable. Always have a data backup!

Removable Storage and Media

Thus far we've focused on storage media that is internal to a PC, but external and removable storage options exist as well. Among the other types of storage available are flash drives, memory cards, optical drives, and external hard drives. The following sections present the details about removable storage solutions.

You need to know about removable storage such as flash drives and memory cards, as well as optical drives, as part of Exam Objective 3.4.

Flash Memory

Once used only for primary memory, the same components that sit on your motherboard as RAM can be found in various physical sizes and quantities among today's solid-state storage solutions. These include older removable and nonremovable flash memory mechanisms, Secure Digital (SD) and other memory cards, and USB flash drives. Each of these technologies has the potential to store reliably a staggering amount of information in a minute form factor. Manufacturers are using innovative packaging for some of these products to provide convenient transport options (such as keychain attachments) to users. Additionally, recall the SSD alternatives to magnetic hard drives mentioned earlier in this chapter.

For many years, modules known as flash memory have offered low- to mid-capacity storage for devices. The name comes from the concept of easily being able to use electricity to alter the contents of the memory instantly. The original flash memory is still used in devices that require a nonvolatile means of storing critical data and code often used in booting the device, such as routers and switches.

For example, Cisco Systems uses flash memory in various forms to store its Internetwork Operating System (IOS), which is accessed from flash during bootup and, in certain cases, throughout operation uptime and therefore during an administrator's configuration sessions. Lesser models store the IOS in compressed form in the flash memory device and then decompress the IOS into RAM, where it is used during configuration and operation. In this case, the flash memory is not accessed again after the boot-up process is complete, unless its contents are being changed, as in an IOS upgrade.

The following sections explain a bit more about today's most popular forms of flash memory: USB flash drives and memory cards.

USB Flash Drives

USB *flash drives* are incredibly versatile and convenient devices that enable you to store large quantities of information in a very small form factor. Many such devices are merely extensions of the host's USB connector, extending out from the interface but adding little to its width, making them easy to transport, whether in a pocket or a laptop bag. Figure 2.22 shows a few flash drives.

FIGURE 2.22 USB flash drives

USB flash drives capitalize on the versatility of the USB interface, taking advantage of Windows' Plug and Play, AutoPlay, and Safely Remove Hardware features and the physical connector strength. Upon insertion, these devices announce themselves to Windows File Explorer as removable drives, and they show up in the Explorer window with a drive letter. This software interface allows for drag-and-drop copying and most of the other Explorer functions performed on standard drives.

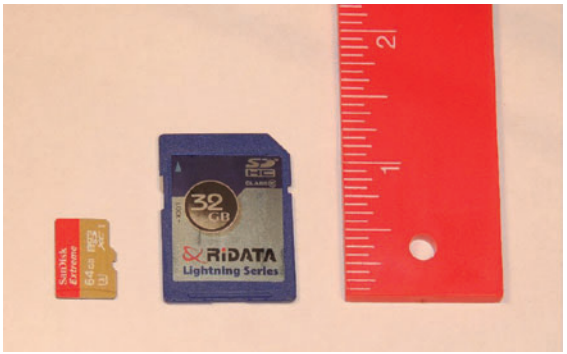
Flash drives exist in a variety of capacities and price points. You can find 32 GB to 64 GB drives for under $5, while drives in the 512 GB to 1 TB range might cost $50 or more. USB flash drives may be Type-A (the old rectangular connecter) or Type-C (the newer oval-shaped one).

SD and Other Memory Cards

Today's smaller devices require some form of removable solid-state memory that can be used for temporary and permanent storage of digital information. Modern electronics, as well as most contemporary digital still cameras, use some form of removable *memory card* to store still images permanently or until they can be copied off or printed out. Of these, the *Secure Digital (SD)* format has emerged as the preeminent leader of the pack, which includes the older MultiMediaCard (MMC) format on which SD is based. Both of these cards measure 32mm × 24mm, and slots that receive them are often marked for both. The SD card is slightly thicker than the MMC and has a write-protect notch (and often a switch to open and close the notch), unlike MMC.

Even smaller devices, such as mobile phones, have an SD solution for them. One of these products, known as *miniSD*, is slightly thinner than SD and measures 21.5mm × 20mm. The other, *microSD*, is thinner yet and only 15mm × 11mm. Both of these reduced formats have adapters that allow them to be used in standard SD slots. Figure 2.23 shows an SD card and a microSD card next to a ruler based on inches. There are other formats as well, such as CompactFlash (CF), which is used in digital cameras, but they are far less common than SD cards.

FIGURE 2.23 Typical SD cards



Some printers and laptops have built-in memory card readers. If not, users can purchase a USB-attached card reader like the one shown in Figure 2.24. The one in Figure 2.24 is simple, with slots for SD and microSD cards, as well as a USB-A port for flash drives.

Hot-Swappable Devices

Many of the removable storage devices mentioned are *hot-swappable*. This means that you can insert and remove the device with the system powered on. Most USB-attached devices without a filesystem fall into this category. Non–hot-swappable devices, in contrast, either cannot have the system's power applied when they are inserted or removed or have some sort of additional conditions for their insertion or removal. One subset is occasionally

referred to as cold-swappable, the other as warm-swappable. The system power must be off before you can insert or remove cold-swappable devices. An example of a cold-swappable device is anything connected to a SATA connector on the motherboard.

FIGURE 2.24 A USB-attached SD card reader

Warm-swappable devices include USB flash drives and external drives that have a filesystem. Windows and other operating systems tend to leave files open while accessing them and write cached changes to them at a later time, based on the algorithm in use by the software. Removing such a device without using the Safely Remove Hardware and Eject Media utility can result in data loss. However, after stopping the device with the utility, you can remove it without powering down the system—hence, the *warm* component of the category's name. These are officially hot-swappable devices.

Hardware-based RAID systems benefit from devices and bays with a single connector that contains both power and data connections instead of two separate connectors. This is known as *Single Connector Attachment (SCA)*. SCA interfaces have ground leads that are longer than the power leads so that they make contact first and lose contact last. SATA power connectors are designed in a similar fashion for the same purpose. This arrangement ensures that no power leads make contact without their singular ground leads, which would often result in damage to the drive. Drives based on SCA are hot-swappable. RAID systems that have to be taken offline before drives are changed out, but the system power can remain on, are examples of warm-swappable systems.

Optical Drives

The final category of storage devices we will look at is optical drives. They get their name because instead of storing data using magnetic fields like conventional HDDs, they read and write data with the use of a laser. The laser scans the surface of a spinning plastic disc, with data encoded as small pits and bumps in the track of the disc.

Optical storage devices began earnestly replacing floppy diskette drives in the late 1990s. While they had far higher capacity than floppies, they were never intended to replace HDDs, because HDDs greatly exceed the capacity and read/write performance of optical drives. For about 20 years, optical drives were practically required components, but today they are far less common. The popularity of high-speed Internet access, streaming services, and

high-capacity USB drives that don't require a separate hardware reader have relegated them to low-priority peripherals.

The most advanced optical storage technology used is the *Blu-ray Disc (BD)* drive. It replaced the *digital versatile disc (DVD)*, also called *digital video disc drive*, which in turn replaced the *compact disc (CD)* drive. Each type of optical drive can also be expected to support discs of the technologies that came before it. Figure 2.25 shows an example of an internal 5¼″ DVD-ROM drive, which also accepts CD-ROM discs. Modern optical drives are indistinguishable from older ones, aside from obvious markings concerning their capabilities. External drives that connect via USB are more popular (and portable!) than their internal cousins.

FIGURE 2.25 A DVD-ROM drive

The original compact disc was intended to be read-only. Data was burned, or written, to the disc once and that was it. Based on consumer demand, manufacturers started creating re-recordable discs. The basic CD-ROM that could both read and write was called a *burner*, a nickname that stuck with future generations of optical writers as well.

There's a lot of interesting history behind the development of each iteration of optical discs, complete with competing standards, intrigue, and infighting. Because optical discs are becoming more and more obscure, we'll cover just some of the key points to know:

- CDs could only be written to one side. Eventually that became known as single-sided (SS).
- DVDs used better technology that allowed the ability to write on both sides (dual-sided [DS]), and have two layers of data on one side (dual-layer [DL]).
- Dual-sided DVDs doubled the capacity of single-sided ones. But in dual-layer DVDs, accommodations needed to be made to avoid overlapping tracks, resulting in about a 10 percent reduction in capacity.
- Further technology enhancements, such as using a violet laser instead of a red one and improved optics, allowed BDs to have much greater capacity. (The name Blu-ray is a bit of creative license from the change in laser color.)

The last piece to cover is transfer rates. Each standard has its own base data transfer speed, which is its 1x speed. Devices in the market will read and write at several multiples of the base speed, which we call the common transfer rate—it reflects the fastest devices typically available. Also note that when burning discs, the transfer rate will be much slower than what's listed here. For example, you might find a BD burner that reads at 16x but burns at 6x. Table 2.6 outlines some basic specifications of the different optical technologies.

TABLE 2.6 Optical disc and drive specifications

Technology	Format	Standard capacity	1x transfer rate	Common transfer rate
CD	SS	700 MB	150 Kbps	7.8 Mbps (52x)
DVD	SS, SL	4.7 GB	1.4 Mbps	33.6 Mbps (24x)
DVD	SS, DL	8.5 GB		
DVD	DS, SL	9.4 GB		
BD	SS, SL	25 GB	4.5 Mbps	72 Mbps (16x)
BD	SS, DL	50 GB		
BD	DS, DL	100 GB		

SS = single-sided; DS = double-sided; SL = single-layer; DL = dual-layer

Installing, Removing, and Configuring Storage Devices

The removal and installation of storage devices, such as hard drives and optical drives, is pretty straightforward. There really isn't any deviation in the process of installing or exchanging the hardware. Fortunately, with today's operating systems, little to no configuration is required for such devices. The Plug and Play BIOS and operating system work together to recognize the devices. However, you still have to partition and format out-of-the-box hard drives before they will allow the installation of the operating system. Nevertheless, today's operating systems allow for a pain-free partition/format/setup experience by handling the entire process, if you let them.

Removing Storage Devices

Removing any component is frequently easier than installing the same part. Consider the fact that most people could destroy a house, perhaps not safely enough to ensure their well-being, but they don't have to know the intricacies of construction to start smashing away. On the other hand, very few people are capable of building a house. Similarly, many could

figure out how to remove a storage device, as long as they can get into the case to begin with, but only a few could start from scratch and successfully install one without tutelage.

This section details the removal of internal storage devices, and the section "Installing Storage Devices" details their installation. The focus here is on internal devices as opposed to external ones. Today's external storage devices are eSATA-, USB-, and perhaps Thunderbolt-attached devices, making them completely Plug and Play. Only the software preparation of external hard drives is a consideration, but the same procedure for the software preparation of internal devices works for external devices as well.

In Exercise 2.2, you'll remove an internal storage device.

EXERCISE 2.2

Removing an Internal Storage Device

1. With the power source removed from the system, ground yourself and the computer to the same source of ground.

   The most reliable way to ground yourself is to turn the computer off but leave it plugged into an electrical outlet. Then, wear an antistatic wrist strap and attach the alligator clip end to an unpainted metal portion of the case you're working in. This will effectively drain static electricity from your body before you touch the sensitive electronic components. If you don't have an antistatic strap, touch and keep touching (or frequently touch) an unpainted metal portion of the case to remove excess static electricity.

2. Remove the cover from the system, exposing the internal components.
3. Unplug all connections from the storage device you wish to remove. These include data and power connections as well as any others, such as audio connections to the sound card or motherboard (for an optical drive). Always be sure to grip the connector, not the wires.
4. Gather the appropriate antistatic packaging for all static-sensitive components that will be reused in the future, including any adapter cards that the storage device plugs into.
5. Remove any obstructions that might hinder device removal, such as component cables attached to adapter cards or the adapter cards themselves, storing them in antistatic packaging so that they can be reused.
6. Remove related adapter cards from the motherboard, storing them in antistatic packaging so that they can be reused.
7. Remove the machine screws holding the storage device to the chassis. These could be on the side of the device or on the bottom.

EXERCISE 2.2 *(continued)*

8. Some devices, especially hard drives because they have no front access from the case, pull out of the chassis toward the rear of the case, whereas others, such as optical drives, generally pull out from the front. A gentle nudge from the rear of the device starts it on its way out the front. Go ahead and remove the device from the case. If you discover other components that obstruct the storage device's removal, repeat step 5.

Installing Storage Devices

An obvious difference among storage devices is their *form factor*. This is the term used to describe the physical dimensions of a storage device. Form factors commonly have the following characteristics:

- 3½″ wide vs. 5¼″ wide
- Half height vs. full height vs. 1″ high and more
- Any of the laptop specialty form factors

You will need to determine whether you have an open bay in the chassis to accommodate the form factor of the storage device that you want to install. Adapters exist that allow a device of small size to fit into a larger bay. For obvious reasons, the converse is not also true.

In Exercise 2.3, you'll install an internal storage device.

EXERCISE 2.3

Installing an Internal Storage Device

1. With the power source removed from the system, ground yourself and the computer to the same source of ground.
2. Remove the cover from the system, exposing the internal components.
3. Locate an available bay for your component, paying attention to your device's need for front access. If you do not see one, look around; some cases provide fastening points near the power supply or other open areas of the case. If you still do not see one, investigate the possibility of sacrificing a rarely or never used device to make room.
4. Remove any obstructions that might hinder device installation, such as component cables attached to adapter cards or the adapter cards themselves, storing them in antistatic packaging to be reused.
5. Find the proper screws for the storage device, and set any jumpers on the drive while it is in hand. Then insert the device into the bay. Keep in mind that some insert from the rear of the bay and some from the front.
6. Line up the screw holes in the device with the holes in the bay. Note that many devices rarely insert as far as they can before lining up with the chassis's holes, so don't be

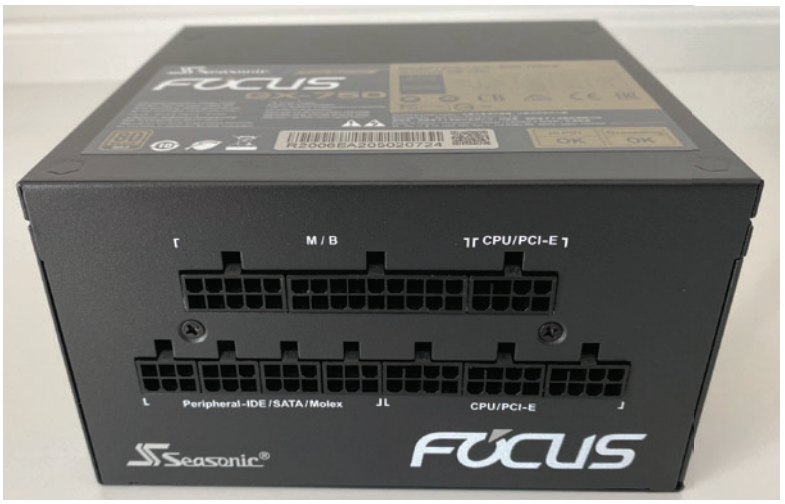
surprised when pushing the device all the way into the bay results in misalignment. Other devices that require front access stop themselves flush with the front of the case, and still others require you to secure them while holding them flush.

7. Use at least two screws on one side of the device. This keeps the device from sliding in the bay as well as from rotating, which happens when you use only one screw or one screw on each side. If the opposite side is accessible, go ahead and put at least one screw in the other side. Most devices allow for as many as four screws per side, but eight screws are not necessary in the vast majority of situations.
8. Connect the data cable from the device to the adapter card or motherboard header.
9. Attach a power connector from the power supply to the device, if needed (it is for PATA, SATA, and SAS). Be sure to insert the connector completely.
10. Once the drive is attached, unground yourself, and turn the computer on to verify that the drive is functional.
11. If the drive is working properly, replace the computer cover.

Understanding Power Supplies

The computer's components would not be able to operate without power. The device in the computer that provides this power is the *power supply* (see Figure 2.26). A power supply converts 100–240V AC current into the DC voltages that a computer needs to operate. These are +3.3VDC, +5VDC, +12VDC, and –12VDC. The jacket on the leads carrying each type of voltage has a different industry-standard color-coding for faster recognition.

FIGURE 2.26 A desktop power supply



You will see the term PSU in reference to a *power supply unit*.

Throughout this section, you will see us use the terms *watts*, *volts*, and *amps*. If you're working with electricity a lot, you might also see the term *ohms*. To help understand what these terms mean, let's use an analogy of water flowing through a pipe. Amps would be the amount of water flowing through the pipe; voltage would be the water pressure; and watts would be the power that the water could provide. (Watts mathematically are volts × amps.) If there were a filter or other barrier in the pipe, that would provide resistance, which is measured in ohms. In non-analogous terms, amps are the unit of current flow; volts are the unit of force; watts are the unit for power (watts = volts × amps); and ohms are resistance.

The abbreviation *VDC* stands for *volts DC. DC* is short for *direct current*. Unlike alternating current (AC), DC does not alter the direction in which the electrons flow. AC for standard power distribution does so 50 or 60 times per second (50 or 60 Hz, respectively). Sometimes you will see people abbreviate a current such as positive 5 volts as +5V, and other times as +5VDC. It's really a matter of preference.

Be aware that DC voltage is not safer than AC voltage, despite its common use in batteries and lower-power components. Direct current is more likely to cause a prolonged clamping of the muscles than AC, which is more likely to fibrillate the heart, which results in a deadly loss of coordination of the various cardiac muscles. Furthermore, power supplies contain transformers and capacitors that can discharge *lethal* amounts of current even when disconnected from the wall outlet for long periods. They are not meant to be serviced, especially by untrained personnel. Do *not* attempt to open them or do any work on them. Simply replace and recycle them when they go bad.

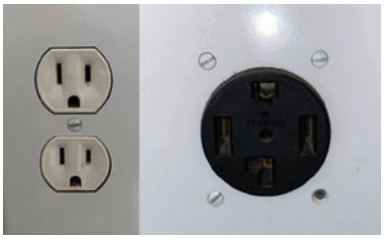
A+ Exam Objective 3.6, "Given a scenario, install the appropriate power supply," highlights input of 110-120VAC and 220-240VAC, as well as output of 3.3V vs. 5V vs. 12V. Understand the voltages power supplies require and produce, including wattage ratings.

Power Supply Input

Computer power supplies need to get their power from somewhere, and that is typically a wall outlet. There may be an intermediary battery backup device in-between called an uninterruptible power supply (UPS), which we will talk about later in the "Battery Backup Systems" section, but the point is that the power supply doesn't just generate its own power. It converts AC power from the wall into DC power that components use.

Countries have differing standards on the voltage provided by wall outlets. In the United States, it's typically 110V and 220V. The 110V outlets are the "normal" outlets that most electronics, including computers, are plugged into. The 220V outlets are for high-energy devices such as electric ranges and clothes dryers. Fortunately, the two plugs are completely different (as shown in Figure 2.27) to help us avoid plugging the wrong thing into the wrong place and frying the component. As noted, though, other countries have different standards, and power supply manufacturers want to ensure their devices work in different countries.

FIGURE 2.27 110V (left) and 220V (right) wall outlets



Therefore, some power supplies have a recessed, two-position slider switch, often a red one, on the rear that is exposed through the case. *Dual-voltage options* on such power supplies read 110 and 220, 115 and 230, or 120 and 240. This selector switch is used to adjust for the voltage level used in the country where the computer is in service. Newer PSUs will autodetect the incoming power level and adjust accordingly—they are referred to as *universal voltage* power supplies.

As noted earlier, in the United States, the power grid supplies anywhere from 110V to 120V. However, in Europe, for instance, the voltage supplied is double, ranging from 220V to 240V. Although the voltage is the same as what is used in the United States to power high-voltage appliances, the amperage is much lower. The point is, if the PSU has a switch, it's not there to allow multiple types of outlets to be used in the same country. If the wrong voltage is chosen in the United States, the power supply will expect more voltage than it receives and might not power up at all. If the wrong voltage is selected in Europe, however, the power supply will receive more voltage than it is set for. The result could be disastrous for the entire computer and could result in sparking or starting a fire. Always check the switch before powering up a new or recently relocated computer. In the United States and other countries that use the same voltage, check the setting of this switch if the computer fails to power up. Or, simply use universal voltage power supplies.

Power Supply Output and Wattage Ratings

Power supplies all provide the same voltages to a system, such as +3.3V, +5V, +12V, and –12V. Each of these can be referred to as a *rail*, because each one comes from a specific tap (or rail) within the power supply. Some power supplies provide multiple 12V rails in an effort to supply more power overall to components that require 12V. For instance,

in *dual-rail power supplies*, one rail might be dedicated to the CPU, whereas the other is used to supply power to all of the other components that need 12V.

The problem that can arise in high-powered systems is that although the collective power supplied by all rails is greater than that supplied by power supplies with a single rail, each rail provides less power on its own. As a result, it is easier to overdraw one of the multiple rails in such a system, causing a protective shutdown of the power supply. Care must be taken to balance the load on each of the rails if a total amperage greater than any one rail is to be supplied to attached components. Otherwise, if the total power required is less than any single rail can provide, there is no danger in overloading any one rail.

Power supplies are rated in watts. A watt is a unit of power. The higher the number, the more power your computer can draw from the power supply. Each power supply comes with a *wattage rating*—think of this rating as the "capacity" of the device to supply power. Most computers require power supplies in the 500- to 600-watt range. Higher wattage power supplies, say 800- to 1300-watt, might be required for more advanced systems that employ power-hungry graphics cards or multiple disk drives, for instance. As of this writing, power supplies of up to 2,000 watts were available for desktop machines. It is important to consider the draw that the various components and subcomponents of your computer place on the power supply before choosing one or its replacement.

Real World Scenario

How Much Power Do You Really Need?

Having a power supply that doesn't provide sufficient wattage to system components will result in brownouts, intermittent reboots, or even the system failing to power up. Having too much power is a little inefficient, but you won't have to worry about failures such as those. The question is, how much do you really need?

Most power-hungry devices (and we're looking at you, video cards) will give you a minimum power supply requirement. Generally speaking, the video card is the biggest power user, so if you meet these requirements, you should be fine. The manufacturer assumes you have otherwise standard gear in the system and adds a bit of a buffer.

Another option, if you are building a system, is to use the System Builder feature on a website we love called `PCPartPicker.com`. As you add components to the system, the site automatically calculates the estimated wattage drawn and displays it on the page. Pick a PSU that supplies more than that amount of power and you should be in the clear.

Energy Efficiency

As you've learned, power supplies take AC current from the wall and convert it into DC power for PC components. The conversion isn't 100 percent efficient, and heat generation is a side effect of the lack of efficiency. Of course, excess heat is an enemy of sensitive electronic

components. In addition, wasted energy conversion means that more energy is needed to produce the same output, which will increase electric bills.

Energy efficiency is part of Exam Objective 3.6. Understand how it works and what it means when selecting a power supply.

In 2004, the 80 PLUS certification program was launched to rate power supplies on their *energy efficiency*—that is, how much of the input energy they convert into output energy. PSUs that have 80 percent or greater efficiency at 20 percent, 50 percent, and 100 percent of their rated loads can qualify for 80 PLUS certification. There isn't a law or rule that requires manufacturers to certify their PSUs, nor are there requirements for users or companies to purchase certified PSUs. The certification is designed to help users make better choices for their computing situations, and potentially save money and wear-and-tear on their components in the process.

There are currently six standards of 80 PLUS certification. The higher the level, the more efficient the PSU is. Table 2.7 shows the levels and minimum efficiencies required to obtain the certification. It's highly unlikely that you will be tested on the efficiency thresholds, but it would be a good idea to know the relative rankings, just in case.

TABLE 2.7 80 PLUS rating minimum efficiency

Efficiency at	80 PLUS	80 PLUS Bronze	80 PLUS Silver	80 PLUS Gold	80 PLUS Platinum	80 PLUS Titanium
20% load	80%	82%	85%	87%	90%	90%
50% load	80%	85%	88%	90%	92%	92%
100% load	80%	82%	85%	87%	89%	94%

At a minimum, you should look for 80 PLUS–certified PSUs. Going higher than that depends on the circumstances. Users running powerful, high-end systems with energy-hungry components can save several dollars a month on energy bills by going with a 80 PLUS Gold or higher PSU. Over time, a more efficient PSU can save a user money, not to mention potentially extend the lifetime of the computer's components.

Power Connectors

The connectors coming from the power supply are quite varied these days. Some PSUs will have connectors permanently attached, whereas other PSUs give you the ability to attach and detach power connectors as needed, based on the devices installed in the system. The following sections detail and illustrate the most common power connectors in use today.

For Exam Objective 3.6, be familiar with the 20+4 pin motherboard power connector.

ATX and ATX12V Connectors

ATX motherboards use a single block connector from the power supply. When ATX boards were first introduced, this connector was enough to power all the motherboard, CPU, memory, and all expansion slots. The original ATX system connector provides the six voltages required, plus it delivers them all through one connector: an easy-to-use single 20-pin connector. Figure 2.28 shows an example of an ATX system connector.

FIGURE 2.28 20-pin ATX power connector

The development of PCIe was great for the computer industry but challenging from a power standpoint because PCIe devices require more power than PCI ones did. For ATX motherboards with PCIe slots, the 20-pin system connector proved inadequate. This led to the ATX12V 2.0 standard and the even higher-end EPS12V standard for servers. These specifications call for a 24-pin connector that adds further positive voltage leads directly to the system connector. The 24-pin connector looks like a larger version of the 20-pin connector. The corresponding pins of the 24-pin motherboard header are actually keyed to accept the 20-pin connector. Adapters are available if you find yourself with the wrong combination of motherboard and power supply. Some power supplies feature a 20-pin connector that snaps together with a separate 4-pin portion for flexibility, called a 20+4 connector, which can be seen in Figure 2.29. Otherwise, it will just have a 24-pin connector.

The adapter mentioned in the previous paragraph is called a *20-pin to 24-pin motherboard adapter*.

ATX12V 2.1 introduced a new 6-pin connector (Figure 2.30), which was specifically designed to give additional dedicated power to the PCIe adapters that required it. It provided a 75W power source to such devices. ATX12V 2.2 replaced the 75W 6-pin connector with a 150W 8-pin connector. The connector in Figure 2.30 is modular. It comes standard as a 6-pin connector, but also has a 2-pin additional attachment to make it into an 8-pin one.

FIGURE 2.29 A 24-pin ATX12V 2.x connector, in two parts

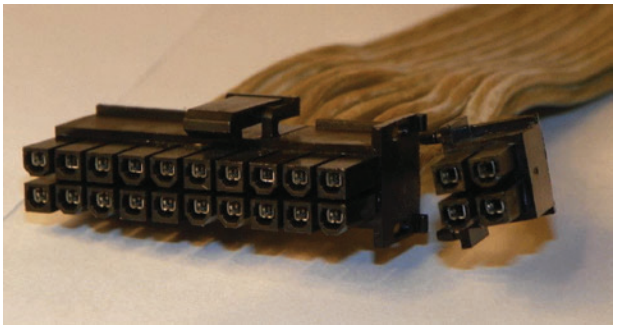


FIGURE 2.30 6-pin ATX12V 2.1 PCIe connector with extender for an 8-pin ATX12V 2.2 PCIe connection

SATA Power Connectors

SATA drives arrived on the market with their own power requirements in addition to their new data interfaces. (Refer back to Figure 2.12 to see the SATA data and power connectors.) For SATA drives, use the 15-pin SATA power connector, a variant of which is shown in Figure 2.31. The fully pinned connector is made up of three +3.3V, three +5V, and three +12V leads interleaved with two sets of three ground leads. Each of the five sets of three common pins is supplied by one of five single conductors coming from the power supply. When the optional 3.3V lead is supplied, it is common to see it delivered on an orange conductor.

Older PATA hard drives and optical drives used a thicker 4-pin power connector called a Molex connector. An example is shown in Figure 2.32. The connector is keyed with two beveled corners, making it nearly impossible to install incorrectly.

FIGURE 2.31 SATA power connector

FIGURE 2.32 Molex power connector

Modular Power Supplies

On older PSUs, all power connectors were hardwired into the power supply itself. This so-called *fixed cable* PSU had a number of interesting side effects. One was that no matter how many or how few internal devices were present, there were a fixed number of connectors. Power supply manufacturers generally provided enough so most users wouldn't run short of connectors. The flip side of this was that there were often four to six unused connectors, but the cables were still taking up space inside the case. Zip ties and thick rubber bands helped maintain the chaos.

Modular power supplies are part of Exam Objective 3.6. Know the difference between them and fixed cable PSUs, and how to install them.

As the variety of internal components became more complex, the need arose to have more flexibility in terms of the connectors provided. Out of this need rose an elegant solution—the *modular power supply*. From a functional standpoint, it works just as a non-modular power

supply does. The difference is that none of the power cables are permanently attached—only the ones that are needed are connected. The PSU shown back in Figure 2.26 shows the side of a fully modular power supply. The top row has connectors for the motherboard (left and center) and CPU or PCIe device. On the bottom row, you can see four 6-pin peripheral connectors and three 8-pin ones to power the CPU or PCIe devices. Nearly all PSUs sold in the market today are modular.

You will also occasionally see semi-modular PSUs on the market. Generally, the motherboard and CPU connectors will be hardwired, whereas the peripheral connectors can be added as needed. There are two potential disadvantages to using a fully modular or semi-modular power supply. First, some PSU manufacturers use proprietary connectors. Always be sure to keep the extra power connectors around (many come with a bag to store unused cables) just in case they are needed. Second, modular PSUs can take up a little more room in the case. Plugging the power connectors into the PSU can take up to an extra half inch. Usually this isn't an issue, but it can be in smaller cases.

Redundant Power Supplies

Nearly every computer you will work with has one and only one power supply—is that enough? If the PSU supplies the right amount of wattage to safely power all components, then the answer is nearly always yes. There are some instances, though, where power redundancy is helpful or even critical. Within the realm of power redundancy, there are two paths you can take: redundant power supplies within a system or battery backups. Let's look at both.

Redundant power supplies are included in Exam Objective 3.6. Understand how they work and how to install one.

Multiple PSUs

It's almost unheard of to see two power supplies installed in a desktop computer. There's generally no need for such a setup, and it would just be a waste of money. And for laptops and mobile devices, it's simply not an option. For servers, though, having a *redundant power supply (RPS)*, meaning a second PSU installed in the system, might make sense. The sole reason to have two power supplies is in case one fails, the other can take over. The transition between the two is designed to be seamless and service will not be disrupted.

Based on its name and our description so far, it might seem as though this means installing two full-sized PSUs into a computer case. Given the limited amount inside a case, you can imagine how problematic this could be. Fortunately, though, PSU manufacturers make devices that have two identical PSUs in one enclosure. One such example is shown in Figure 2.33. The total device is designed to fit into ATX cases and is compliant with ATX12V and EPS12V standards. If one fails, the other automatically takes over. They are hot-swappable, so the failed unit can be replaced without powering the system down.

FIGURE 2.33 Hot-swappable redundant PSUs

Source: Rainer Knäpper / Wimimedia Commons / CC BY-SA 2.0 DE

Although an RPS can help in the event of a PSU failure, it can't keep the system up and running if there is a power outage.

Battery Backup Systems

The second type of power redundancy is a battery backup system that the computer plugs into. This is commonly referred to as an *uninterruptible power supply (UPS)*.

These devices range in size from as small as a brick to as large as a room. Some just have a few indicator lights, whereas others have LCD displays that show status and menus and come with their own management software. The UPS will have several power plugs. It might divide the plugs such that a few of them provide surge protection only, whereas others provide surge protection as well as backup power. Figure 2.34 shows a small UPS that could be used in a home or small office to protect a few machines, and Figure 2.35 shows a room-sized one that can support dozens of servers in the event of a power failure. It's approximately 10 feet long, 6 feet wide, and 6 feet tall.

FIGURE 2.34 An uninterruptible power supply front (a) and back (b)

(a) (b)

FIGURE 2.35 Large UPS

Inside the UPS are one or more batteries and fuses. Much like a surge suppressor, a UPS is designed to protect everything that's plugged into it from power surges. UPSs are also

designed to protect against power sags and even power outages. Energy is stored in the batteries, and if the power fails, the batteries can power the computer for a period of time so that the administrator can then safely power it down. Many UPSs and operating systems will also work together to safely power down automatically a system that gets switched to UPS power. These types of devices may be overkill for Uncle Bob's machine at home, but they're critically important fixtures in server rooms.

The UPS should be checked periodically to make sure that its battery is operational. Most UPSs have a test button that you can press to simulate a power outage. You will find that batteries wear out over time, and you should replace the battery in the UPS every couple of years to keep the UPS dependable.

UPSs all have a limit as to how many devices they can handle at once. There are three numbers to pay attention to when looking for a UPS:

- Volt-ampere (VA) rating
- Power factor (PF)
- Watts

The VA rating shows the maximum load the UPS can support. The higher the number, the bigger the load. A UPS will also have a power factor, which is a measure of efficiency, somewhere between 0 and 1.0. Again, the higher, the better. Finally, the wattage supported is calculated by taking the VA × PF. For example, a 1,000 VA load UPS with a 0.8 PF will support a maximum of 800 watts. Always choose a UPS with a VA of at least 1.2 times the anticipated need. UPS power limitations should be strictly observed. If overloaded, it can cause a short, which could potentially result in fire.

Replacing Power Supplies

Sometimes power supplies fail. Sometimes you grow out of your power supply and require more wattage than it can provide. Often, it is just as cost effective to buy a whole new case with the power supply included rather than dealing with the power supply alone. However, when you consider the fact that you must move everything from the old case to the new one, replacing the power supply becomes an attractive proposition. Doing so is not a difficult task.

Regardless of which path you choose, you must make sure the power connection of the power supply matches that of the motherboard to be used. Additionally, the physical size of the power supply should factor into your purchasing decision. If you buy a standard ATX-compatible power supply, it might not fit in the petite case you matched up to your micro ATX motherboard. In that scenario, you should be on the lookout for a smaller form factor power supply to fit the smaller case. Odds are that the offerings you find will tend to be a little lighter in the wattage department as well.

Exercise 2.4 details the process to remove an existing power supply. Use the reverse of this process to install the new power supply. Just keep in mind that you might need to procure the appropriate adapter if a power supply that matches your motherboard can no longer be found. There is no post-installation configuration for the power supply, so there is nothing to cover along those lines. Many power supply manufacturers have utilities on their websites that allow you to perform a presale configuration so that you are assured of obtaining the most appropriate power supply for your power requirements.

EXERCISE 2.4

Removing a Power Supply

1. With the power source removed from the system, ground yourself and the computer to the same source of ground.
2. Remove the cover from the system, exposing the internal components.
3. After locating the power supply, which can come in a variety of formats and appear on the left or right side of the case, follow all wiring harnesses from the power supply to their termini, disconnecting each one.
4. Remove any obstructions that appear as if they might hinder the removal of the power supply.
5. Using the dimensions of the power supply, detectable from the inside of the case, note which machine screws on the outside of the case correspond to the power supply. There are often four such screws in a non-square pattern. If your case has two side panels, and you removed only one, there will likely be one or more screws holding the other panel on that appear to be for the power supply. These do not need to be removed. If all case screws have been removed, pay attention to their location and do not use these holes when securing the new power supply.
6. Remove the screws that you identified as those that hold the power supply in place. Be aware that the power supply is not lightweight, so you should support it as you remove the final couple of screws.
7. Maneuver the power supply past any obstructions that did not have to be removed, and pull the power supply out of the case.

AC Adapters as Power Supplies

Just as the power supply in a desktop computer converts AC voltages to DC for the internal components to run on, the AC adapter of a laptop computer converts AC voltages to DC for the laptop's internal components. And AC adapters are rated in watts and selected for use with a specific voltage just as power supplies are rated. One difference is that AC adapters

are also rated in terms of DC volts out to the laptop or other device, such as certain brands and models of printer.

Because both power supplies and AC adapters go bad on occasion, you should replace them both and not attempt to repair them yourself. When replacing an AC adapter, be sure to match or exceed the wattage requirements for the device. In addition, while new AC adapters use USB-C, older AC adapters may have proprietary connectors and either positive or negative polarity. You can read more on this subject in Volume 1 Chapter 4, "Laptop and Mobile Device Hardware."

Summary

In this chapter, you learned about three classes of personal computer components that finish our tour of the inside of the case—expansion cards, storage devices, and power supplies.

Expansion cards add helpful capabilities, such as video, audio, network connections, and additional ports for devices and peripherals. Storage devices provide long-term data capacity. Examples include conventional spinning hard drives and SSDs. RAID arrays can help provide performance increases and additional data protection. Other removable storage devices include flash drives, memory cards, and optical drives.

Storage communications' interfaces you learned about included NVMe, SATA, PCIe, and SAS. Form factors covered included M.2 and mSATA.

Finally, we discussed power supply safety as well as the various requirements and connectors. You learned about how power supplies take AC current and convert it into DC power for internal components in 2.3V, 5V, and 12V. You also learned about redundant power supplies, modular power supplies, wattage ratings, and efficiency ratings. Finally, you learned how to remove, install, and configure storage devices and how to replace power supplies.

Exam Essentials

Know how to install and configure expansion cards to provide needed functionality. Understand the functionality that video cards, sound cards, network cards, and capture cards provide. Know where to install them and broadly how to configure them.

Be familiar with the components of a conventional hard drive and the anatomy of a hard drive. Conventional hard drives store data magnetically on rotating metal platters. Be familiar with magnetic hard drive speeds, including 5,400, 7,200, 10,000, and 15,000 rpm. Form factors are 2.5″ and 3.5″.

Understand the advantages that solid-state drives have over conventional drives. SSDs are much faster than magnetic hard drives, produce less heat, and can be made much smaller physically. They are also less susceptible to shock from drops.

Know the differences between three SSD communications interfaces and two form factors. The SSD communications interfaces are NVMe, SATA, PCIe, and SAS. The two form factors to know are M.2 and mSATA.

Understand the details surrounding optical storage. From capacities to speeds, you should know what the varieties of optical storage offer. Blu-ray is the newest, fastest, and largest capacity, followed by DVDs and finally CDs.

Understand the different flash drive and memory card options available. Know the differences between SD cards, CompactFlash, microSD, and miniSD. Be able to identify which cards can fit into specific types of slots natively or with an adapter.

Understand the characteristics of five types of RAID configurations. You need to know RAID 0, RAID 1, RAID 5, RAID 6, and RAID 10. RAID 0 is disk striping, which can improve speed but does not provide fault tolerance. RAID 1 is disk mirroring, which gives fault tolerance but no performance increase. RAID 5 is striping with parity, which can give some performance boost along with fault tolerance. RAID 6 is double-parity RAID. It's similar to RAID 5, with a second parity stripe. RAID 10, also known as RAID 1+0, adds mirroring to a striped set. Understand what hot-swappable means.

Know about power supplies and their connectors. Power supplies are commonly made in ATX and other, smaller form factors. Regardless of their type, power supplies must offer connectors for motherboards and internal devices. Know the differences among the connectors and why you might need a 20-pin to 24-pin motherboard adapter. Also understand why AC adapters are related to power supplies.

Understand power supply characteristics that determine performance. Power supplies can take AC input from 110V to 240V and may have a switch on the back to determine which source to expect. Output to internal components will be 3.3V, 5V, and 12V. Capacity is measured in watts and expressed as a wattage rating. The amount of AC energy converted to DC power is called energy efficiency—the higher, the better. There are also redundant and modular power supplies.

Know how to remove, install, and configure storage devices. Know the difference between the data and power connectors used on storage devices. Know what it means to partition and format a hard drive. Be aware of the physical differences in storage device form factors.

Know how to remove, install, and configure power supplies. Know the difference between the modern motherboard power headers and know when an adapter might be required. Be familiar with how to fasten power supplies to the chassis as well as how to unfasten them.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. You are installing a new graphics adapter in a Windows 11 system. Which of the following expansion slots is designed for high-speed, 3D graphics adapters?
   - **A.** USB
   - **B.** NVMe
   - **C.** PCI
   - **D.** PCIe

2. You need to install a new M.2 hard drive into a desktop computer. The motherboard has an open M.2 slot that is M-keyed. Which types of M.2 hard drives will you be able to install into that motherboard?
   - **A.** B-key
   - **B.** M-key
   - **C.** B+M-key
   - **D.** A, B, and C
   - **E.** B and C

3. You install a new NIC for a user, and he asks what it does. Which of the following best characterizes the functions of this device?
   - **A.** Multimedia
   - **B.** Communications
   - **C.** Input/Output
   - **D.** Storage

4. A client wants to ensure that their server does not lose power if their power supply fails. Which of the following best describes the type of power supply needed in this situation?
   - **A.** A power supply that provides twice as much wattage than what is needed to power system components
   - **B.** A power supply that provides a voltage output selector
   - **C.** A modular power supply
   - **D.** A redundant power supply

5. A client is looking for a desktop storage technology that provides the fastest possible data transfers. Which technology should you recommend?
   - **A.** SATA
   - **B.** NVMe
   - **C.** 10,000 rpm
   - **D.** 15,000 rpm

6. You are choosing an optical drive format for a client. The client needs to store video files which are about 20 GB in size. Which technology should they use?
   - **A.** Double-sided, double-layer DVD+R
   - **B.** Single-sided, single-layer Blu-ray disc
   - **C.** Double-sided, single-layer DVD-R
   - **D.** Double-sided, single-layer DVD+R

7. Which of the following best describes the concept of hot-swappable devices?
   - **A.** Power does not need to be turned off before the device is inserted or removed.
   - **B.** The device can be removed with power applied after it is properly stopped in the operating system.
   - **C.** Care must be taken when swapping the device because it can be hot to the touch.
   - **D.** The device can be swapped while still hot, immediately after powering down the system.

8. Of the following voltage pairings, which one accurately represents the input and output, respectively, of power supplies and AC adapters?
   - **A.** AC in, AC out
   - **B.** DC in, DC out
   - **C.** AC in, DC out
   - **D.** DC in, AC out

9. What are the output voltages that are commonly produced by PC power supplies? (Choose all that apply.)
   - **A.** +3.3VDC
   - **B.** +4.3VDC
   - **C.** +5VDC
   - **D.** +8VDC
   - **E.** +12VDC
   - **F.** –12VDC
   - **G.** +110VDC
   - **H.** –110VDC

10. You are installing a new power supply in a desktop computer. Which of the following statements about power supplies is true?
    - **A.** You must make sure that the voltage selector switch on the back of the power supply is switched to the lower setting if the computer is going to be used in Europe.
    - **B.** SATA hard drives most often use the same type of power connector that PATA hard drives use.
    - **C.** Power supplies require a 20-pin to 24-pin motherboard adapter to supply power to ATX-based motherboards.
    - **D.** Power supplies convert AC input to DC output.

11. You have been asked to install a new hard drive for a client. Which of the following is not a consideration when installing an internal storage device?

- **A.** You should match the form factor of the drive or adapt it to an available drive bay or slot.
- **B.** You should secure the drive with at least two screws on one side and preferably two on each side.
- **C.** Due to the high revolutions at which modern hard drives spin, you must secure an external power source because the internal power supplies do not have the capacity.
- **D.** You need to be sure that the routing of the drive's data cable, if applicable, does not obstruct the engineered flow of air across internal components.

12. A client just bought a new digital camera for his photography business. Which of the following is specifically designed as an internal storage device for cameras?

- **A.** Memory cards
- **B.** Optical discs
- **C.** Flash drives
- **D.** mSATA

13. You are installing a second hard drive into a desktop computer. The hard drive is an NVMe PCIe card. Which of the following steps will you need to take during installation?

- **A.** Install the hard drive into an open PCIe slot.
- **B.** Install the hard drive into an open PCIe slot. Connect the NVMe PCIe data cable.
- **C.** Install the hard drive into an open PCIe slot. Connect the NVMe PCIe power connector.
- **D.** Install the hard drive into an open PCIe slot. Connect the NVMe PCIe data cable and power connector.
- **E.** Install the hard drive into an open NVMe slot.
- **F.** Install the hard drive into an open NVMe slot. Connect the NVMe PCIe data cable.
- **G.** Install the hard drive into an open NVMe slot. Connect the NVMe PCIe power connector.
- **H.** Install the hard drive into an open NVMe slot. Connect the NVMe PCIe data cable and power connector.

14. You are purchasing a new video card in a desktop computer. For the best performance, which type of video card should you purchase?

- **A.** PCI x16
- **B.** PCI x128
- **C.** AGP
- **D.** PCIe x128
- **E.** PCIe x16

15. Which of the following is not a consideration when upgrading power supplies?

- **A.** You might find that you do not have a matching motherboard connector on your new power supply.
- **B.** You might find that your case has a nonremovable power supply.
- **C.** You might find that your power rating is not adequate on the new power supply.
- **D.** You might find that you do not have enough of the appropriate connectors coming from the power supply for the devices that you have installed.

16. You are choosing a flash standard for a client. They want the smallest flash cards possible. Which standard should you recommend?

- **A.** SD
- **B.** CompactFlash
- **C.** microSD
- **D.** miniSD

17. Your goal is to build the fastest computer possible. Which of the following hard drive technologies should you choose for the fastest read performance?

- **A.** M.2 PCIe
- **B.** M.2 SATA
- **C.** NVMe M.2
- **D.** NVMe SATA

18. You have been asked to configure a RAID 5 system for a client. Which of the following statements about RAID 5 is true?

- **A.** It provides fault tolerance but no performance improvement.
- **B.** It provides performance improvement but no fault tolerance.
- **C.** If multiple drives in the array fail, the data can be saved provided that one drive is still operational.
- **D.** It requires at least three drives to implement.

19. You have a motherboard with integrated video, and the video fails. You install a video add-on card. Which of the following statements are the most accurate? (Choose two.)

- **A.** The add-on card will not function properly because of the integrated video.
- **B.** The add-on card will function if it is bridged to the integrated video chipset.
- **C.** The add-on card will be detected automatically and function properly if it's PCIe.
- **D.** For the card to work properly, you might need to disable the integrated video in the BIOS.
- **E.** The add-on card will work properly by using the integrated card's video driver.

20. When replacing a power supply, which of the following tends to vary among power supplies and must be chosen properly to support all connected devices?

A. Wattage

B. Voltage

C. Amperage

D. Resistance

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answer compares to the authors', refer to Appendix B.

Detail the process for removing a power supply from a computer chassis.

Chapter 3

Peripherals, Cables, and Connectors

THE FOLLOWING COMPTIA A+ 220-1201 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

- ✓ **3.0 Hardware**
- ✓ **3.1 Compare and contrast display components and attributes.**
  - Types
    - Liquid crystal display (LCD)
      - In-plane switching (IPS)
      - Twisted nematic (TN)
      - Vertical alignment (VA)
    - Organic light-emitting diode (OLED)
    - Mini light-emitting diode (Mini-LED)
  - Touch screen/digitizer
  - Inverter
  - Attributes
    - Pixel density
    - Refresh rates
    - Screen resolution
    - Color gamut
- ✓ **3.2 Summarize basic cable types and their connectors, features, and purposes.**
  - Peripheral cables
    - USB 2.0
    - USB 3.0

- Serial
- Thunderbolt

- Video cables
  - High-definition Multimedia Interface (HDMI)
  - DisplayPort
  - Digital Visal Interface (DVI)
  - Video Graphics Array (VGA)
  - USB-C
- Hard drive cables
  - Serial Advanced Technology Attachment (SATA)
  - External SATA (eSATA)
- Adapters
- Connector types
  - MicroUSB
  - MiniUSB
  - USB-C
  - Molex
  - Lightning
  - DB9

Thus far, our discussion of computer components has focused primarily on those inside the case. With knowledge of the key internal components under your belt, it is time to turn our attention to the outside of the computer. Dozens of external devices are available to enhance a computer's functionality. We'll cover a variety of them that add video, audio, input, output, and storage capabilities.

Of course, to connect external peripherals, we need some sort of cable and connector. Not everything is wireless yet! Consequently, we will also discuss the interfaces and cables associated with common peripherals. With that, it's now time to think outside the box.

Understanding Peripheral Devices

Peripheral devices add much needed functionality to computers, beyond the core components. Having a fast processor and terabytes of hard drive space is great, but it doesn't complete the picture. Users need the ability to input data and easily see and use the output that the processor generates. Of course, the types of devices that can input or receive output are quite varied. In the following sections, we are going to break peripheral devices into four categories:

- Video
- Audio
- Input and output
- Storage

We realize that video and audio are indeed input or output devices, but because they are more specialized, we will cover them separately. After this section, you will have a good understanding of purposes of and uses for several common peripheral devices, as well as how they connect to a PC.

Knowledge of a variety of peripheral devices was previously tested on A+ exams. With the newest version, the objectives covering peripherals were removed. Even though you might not see these items on the exam, we feel it's important to cover them, as having this knowledge will make you a more complete technician. Besides, it will help you understand what all the cables and connectors are for!

Video Devices

The primary method of getting information out of a computer is to use a computer video display. Display systems convert computer signals into text and pictures and display them on a TV-like screen. As a matter of fact, early personal computers used television screens because it was simpler to use an existing display technology than to develop a new one. The most common video device used is a monitor.

Monitors

Most display systems work the same way. First, the computer sends a signal to a device called the *video card*—an expansion board installed in an expansion bus slot or the equivalent circuitry integrated into the motherboard—telling it to display a particular graphic or character. The video card then renders the character for the display—that is, it converts the single instruction into several instructions that tell the display device how to draw the graphic and sends the instructions to the display device based on the connection technology between the two. The primary differences after that are in the type of video card you are using (digital or analog) and the type of display (LCD, LED, IPS, and so forth).

Types of Monitors

PC monitors today are generally based on some form of *liquid crystal display (LCD)* technology. First used with portable computers and then adapted to desktop monitors, LCDs are based on the concept that when an electrical current is passed through a semi-crystalline liquid, the crystals align themselves with the current. When transistors are combined with these liquid crystals, patterns can be formed. Patterns can then be combined to represent numbers or letters. LCDs are relatively lightweight and don't consume much power.

A+ Exam Objective 3.1, "Compare and contrast display components and attributes," expects you to know the difference between display types LCD, IPS, TN, VA, OLED, and Mini-LED. The LCD inverter is also listed as an objective.

Liquid crystals produce no light, so LCD monitors need a lighting source to display an image. Traditional LCDs use a fluorescent bulb, typically about 8″ long and just slimmer than a pencil, called a *backlight* to produce light. Backlights require a high-voltage, high-frequency energy that's not provided directly by a power supply, and in LCD monitors the energy is provided by a separate component called the *inverter*. The inverter is a small circuit board installed behind the LCD panel that takes DC current and inverts it to AC for the backlight. If you are having problems with flickering screens or dimness, it's more likely that the inverter is the problem, not the backlight itself.

There are two things to keep in mind if you are going to replace an inverter. First, they store and convert energy, which means they have the potential to discharge that energy. To an inexperienced technician, they can be dangerous. Second, make sure the replacement inverter was made to work with the LCD backlight that you have. If they weren't made for each other, you might have problems with a dim screen or poor display quality.

Inverters can discharge energy, which can cause severe injury to you. Be careful when working with them!

There are three popular variants of LCD monitors in use today: *twisted nematic (TN)*, *in-plane switching (IPS)*, and *vertical alignment (VA)*. The three differ in the ways they align the liquid crystals in the display. Let's take a look at each:

Twisted Nematic TN is the oldest of the three technologies. It has somewhat restricted viewing angles and not the best color reproduction. Colors may appear washed out or too blended together. On the bright side, they have very little lag and can handle high refresh rates such as 240 Hz with ease. (We will talk about refresh rates in the upcoming "Adjusting Display Settings" section.) That makes them a popular choice for competitive gamers. They're not the best choice for high-end video needs, but they are inexpensive, which also makes them good for everyday office use.

In-Plane Switching IPS followed TN into the market, and they have the best viewing angles and color reproduction of all LCD monitors. They usually have a bit more lag than TN monitors do, but the difference is minimal in higher-end models. IPS monitors are ideal for those seeking the best color experience, such as graphic designers and video artists. They are also the best choice for someone who wants to mount them vertically, like a lot of programmers do.

Vertical Alignment VA has the best contrast ratios of the three, meaning the difference between the dark and bright colors is the best. Color reproduction tends to be good but not quite as good as IPS, and VA monitors do have a bit more lag (but we're talking milliseconds here). It's more a jack-of-all-trades technology than the other two options.

Most LCDs today use a panel of *light-emitting diodes (LEDs)* instead, which consume less energy, run cooler, and live longer than fluorescent bulbs. Therefore, when you see a monitor advertised as an LED monitor, it's really an LCD monitor with LED backlighting.

Recent advancements in LED technology have enabled manufacturers to make smaller LED backlights and pack many more together. The resulting technology is called *Mini light-emitting diode (Mini-LED)*, and it produces better contrast ratios (darker darks and brighter brights) than a standard LED. This can be helpful in bright environments. They will also be more expensive than LED displays. Compared to OLEDs (covered next), they will have better brightness (but not as dark of darks), similar resolution, slightly inferior viewing angles and color saturation, but also be a little less expensive.

The final type of LED monitor we'll cover is an *organic light-emitting diode (OLED)* display. An organic light-emitting compound forms the heart of the OLED, and it is placed between a positive anode and a negative cathode, which produce a current that runs through the electroluminescent compound, causing it to emit light. An OLED, then, is the combination of the compound and the electrodes on each side of it. The electrode in the back of the OLED cell is usually opaque, allowing a rich black display when the OLED cell is not lit. The front electrode should be transparent, to allow the emission of light from the OLED.

If thin-film electrodes and a flexible compound are used to produce the OLEDs, an OLED display can be made flexible, which is not only cool, but it allows it to function in places where other display technologies could never work.

Because OLEDs create the image in an OLED display and supply the light source, there is no need for a backlight, so power consumption is less than it is in LCD or LED panels. Additionally, the contrast ratio of OLED displays exceeds that of LCD/LED panels, meaning that in darker surroundings, OLED displays produce better images than LCD/LED panels produce. The viewing angles are also the best. The only performance downside of OLEDs is that they can be susceptible to *burn-in*, which is a permanent discoloration or artifact on a monitor screen that happens after an image has been displayed for a long time. That said, OLED monitors are generally some of the highest-quality monitors you will find on the market today, and usually more expensive than LEDs or Mini-LEDs. OLED is commonly found in smaller devices such as smartphones as well.

Nearly all monitors in use today are flat-screen monitors and measure no more than a few inches deep. Occasionally, though, you will run into a monitor that's not flat and can be over a foot deep. These relics are based on old television technology called the cathode ray tube (CRT). CRT monitors are bulky and heavy, and almost certainly use an analog signal (and VGA connector) instead of digital.

Using Displays for Input

If you have ever used a smartphone, and it's likely you have, you're already familiar with the concept of using a display for input. The screen of a smartphone acts as both a display and input device, and such a component is referred to as a *touchscreen*. Of course, touchscreens aren't just for small mobile devices. Laptops commonly include them, and many desktop monitors have touchscreen capabilities as well.

Touchpads and digitizers are part of Exam Objective 3.1. Understand how they work!

There are two types of touchscreens. *Resistive touchscreens* use pressure from a stylus or finger to register the contact. *Capacitive touchscreens* use electrical conductivity, which makes the human finger the perfect stylus. Resistive touchscreens allow for more granularity because they can account for more sensors per square inch. Capacitive touchscreens are more responsive and better suited to multitouch applications (which can register multiple touches at the same time), but they generally don't work if the user is wearing gloves, which can block the finger's electrical current. If you need help remembering the difference, think of resistive screens as providing resistance or needing pressure. And for capacitive screens, think of a capacitor, which stores electricity (and capacitive touchscreens utilize conductivity).

Touchscreens are said to digitize (or make digital) physical input. It follows, then, that touchscreens are a form of *digitizer*, or a device that can be written or drawn on, and the content will be converted from analog input to digital images on the computer. When built into the display, they might be the glass of the display itself, or they might be implemented

as an overlay for the display. Other types of digitizers are not built into the screen and may be USB devices that look like notepads, or pads on the palm rest or next to the touchpad of a laptop.

Adjusting Display Settings

Although most monitors are automatically detected by the operating system and configured for the best quality that they and the graphics adapter support, sometimes manually changing display settings, such as for a new monitor or when adding a new adapter, becomes necessary. Let's start by defining a few important terms:

- Refresh rate
- Screen resolution
- Multiple displays

Each of these terms relates to settings available through the operating system by way of display option settings.

Refresh Rate The *refresh rate*—technically, the vertical scan frequency—specifies how many times in one second the image on the screen can be completely redrawn, if necessary. Refresh rate is specified in hertz (Hz), which indicates how many times per second the screen is refreshed.

For LCD screens, the refresh rate may or may not be adjustable. The lowest standard refresh rate is 60 Hz, but higher-end monitors will be in the 360 Hz to 480 Hz range.

Higher refresh rates translate to more fluid video motion. Think of the refresh rate as how often a check is made to see if each pixel has been altered by the source. If a pixel should change before the next refresh, the monitor is unable to display the change in that pixel. Therefore, for gaming and home-theater systems, higher refresh rates are an advantage.

While the refresh rate is selected for the monitor, the refresh rate you select must be supported by both your graphics adapter and your monitor because the adapter drives the monitor. If a monitor supports only one refresh rate, it does not matter how many different rates your adapter supports—without overriding the defaults, you will be able to choose only the one common refresh rate. It is important to note that on some monitor and video card combinations, as the resolution you select increases, the higher supported refresh rates begin to disappear from the selection menu. If you want a higher refresh rate, you might have to compromise by choosing a lower resolution. Exercise 3.1 shows you where to change the refresh rate in Windows 11.

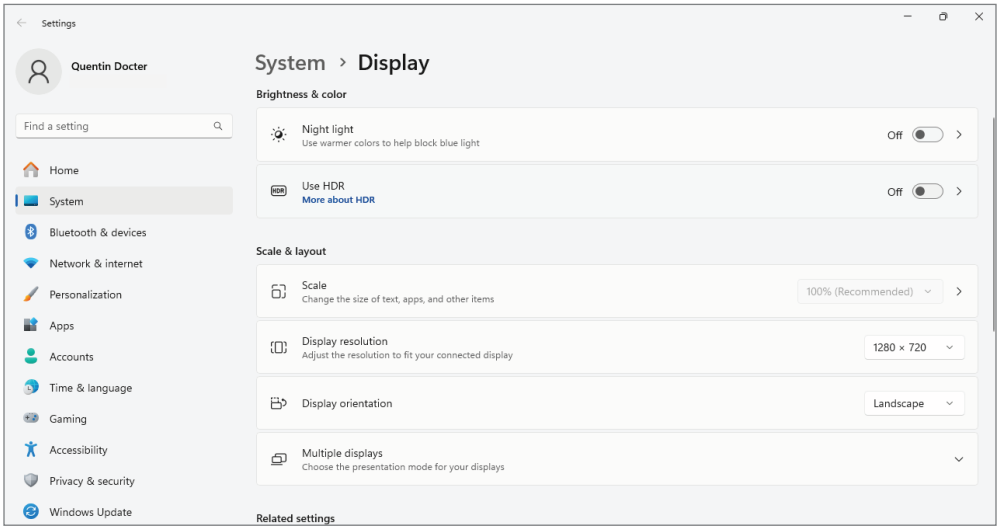
EXERCISE 3.1

Changing the Refresh Rate in Windows 11

1. Right-click a blank portion of the desktop.
2. Click Display Settings to open the System ➢ Display app, as shown in Figure 3.1.

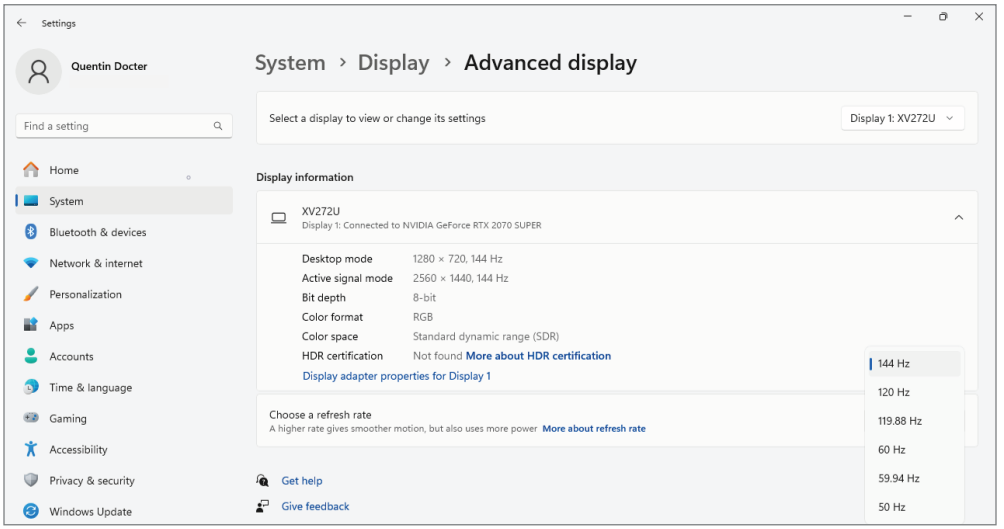
EXERCISE 3.1 *(continued)*

FIGURE 3.1 Display settings



3. Near the bottom of the window in the Related Settings section, click Advanced Display. You will get the Advanced Display Settings window. For the computer in this example, only one monitor is attached. If multiple displays are present, the refresh rate can be independently set. The display choice is made using the drop-down menu at the top of the Advanced Display app (you can see it in Figure 3.2).

FIGURE 3.2 Advanced display refresh rates



4. In the Choose A Refresh Rate box, the current refresh rate will be shown on the right side. Click the down arrow next to the refresh rate, and you will be able to see other refresh rate options, like in Figure 3.2. Note that many monitors will only have one available refresh rate.

   For this monitor running at 1280 × 720, there are six options. Note that if the resolution is set to the default 2560 × 1440, there are only three: 144 Hz, 120 Hz, and 59.95 Hz.

5. Choose the refresh rate you want for this selected monitor.

6. The display will go dark for a few seconds, and then your screen will reappear. You will have a pop-up window asking if you want to keep the changes or revert to previous settings. The default is to revert.

   Chances are you won't notice a difference after selecting a new refresh rate. The good news is that if it does mess up the image on the screen, you can wait about 15 seconds and the computer will revert back to the previous selection.

7. Close the Advanced Display window.

Eyestrain and Headaches

Some users (including this author) may complain about getting headaches when using displays with refresh rates set at 60 Hz or lower. The display screen flickers whenever it's refreshed, and it's almost always imperceptible.

As an aside, if you ever watch a live news broadcast and they are interviewing someone with a display behind them, you will often see the monitor flashing or what appears to be a rolling horizontal bar on the image on their screen. You're seeing the monitor refresh. Television and broadcast cameras are typically set to 30 Hz or 60 Hz (movie cameras usually operate at 24 Hz). If this is different than the display it's filming, it will cause the image to flicker or appear to roll.

Lower-frequency refresh rates such as 60 Hz can cause eyestrain, leading to headaches. In instances like this, invest in a monitor and video card that support higher refresh rates.

Screen Resolution *Resolution* is defined by how many software picture elements (pixels) are used to draw the screen. An advantage of higher resolutions is that more information can be displayed in the same screen area. A disadvantage is that the same objects and text displayed at a higher resolution appear smaller and might be harder to see. Up to a point, the added crispness of higher resolutions displayed on high-quality monitors compensates for the negative aspects.

The resolution is described in terms of the visible image's dimensions, which indicate how many rows and columns of pixels are used to draw the screen. For example, a resolution of 2560 × 1440 means 2560 pixels across (columns) and 1440 pixels down (rows) were used to draw the pixel matrix. The video technology in this example would use 2560 × 1440 = 3,686,400 pixels to draw the screen. Resolution is a software setting that is common among CRTs, LCDs, and projection systems, as well as other display devices.

Setting the resolution for your monitor is fairly straightforward. If you are using an LCD, for best results you should use the monitor's *native resolution*, which comes from the placement of the transistors in the hardware display matrix of the monitor. For a native resolution of 1920 × 1080, for example, there are 2,073,600 transistors (LCDs) or cells (OLED) arranged in a grid of 1920 columns and 1080 rows. Trying to display a resolution other than 1920 × 1080 through the operating system tends to result in the monitor interpolating the resolution to fit the differing number of software pixels to the 2,073,600 transistors, often resulting in a distortion of the image on the screen.

Some systems will scale the image to avoid distortion, but others will try to fill the screen with the image, resulting in distortion. On occasion, you might find that increasing the resolution beyond the native resolution results in the need to scroll the desktop in order to view other portions of it. In such instances, you cannot see the entire desktop all at the same time. The monitor has the last word in how the signal it receives from the adapter is displayed. Adjusting your display settings to those that are recommended for your monitor can alleviate this scrolling effect.

To change the resolution in Windows 11, right-click the desktop and choose Display Settings (as in Exercise 3.1). There is a pull-down menu for Display Resolution (it can be seen back in Figure 3.1). Click it and choose the resolution you want. Just like when you changed the refresh rate, changing the resolution will bring up a pop-up window asking if you want to keep the changes or revert back to the previous setting. If the change messes up the image and makes it unreadable, wait about 15 seconds and it will revert back to the previous setting.

Understanding Aspect Ratios

The term *aspect ratio* refers to the relationship between the horizontal and vertical pixel counts that a monitor can display. For example, old square-ish CRTs were shaped to support a display that conformed to 4:3 ratios, such as 800 × 600 or 1024 × 768. If you divide the first number by 4 and multiply the result by 3, the product is equal to the second number. Additionally, if you divide the first number by the second number, the result is approximately 1.3, the same as 4 ÷ 3. Displays with a 16:10 aspect ratio have measurements that result in a dividend of 16 ÷10 = 1.6.

When LCD monitors first became popular, they had wider screens and most supported a 16:10 ratio. Because the ATSC (Advanced Television Systems Committee) standard for widescreen television aspect ratios is 16:9 (1.778), computer monitors are trending more toward this same aspect ratio. As a result, the once popular 1920 × 1200, 16:10 resolution is uncommon and the 1920 × 1080, 16:9 resolution is the widescreen default. If you have a monitor that supports one and you try to set it to the other, the image may look squished or stretched, or the monitor may not display at all.

Multiple Displays Whether regularly or just on occasion, you may find yourself in a position where you need to use two or more monitors on the same computer simultaneously. For example, you may need to work in multiple spreadsheets at the same time and having two monitors makes it much easier. Or, if you are giving a presentation and would like to have a presenter's view on your laptop's LCD but need to project a slideshow onto a screen, you might need to connect an external projector to the laptop. Simply connecting an external display device does not guarantee that it will be recognized and work automatically. You might need to change the settings to recognize the external device or adjust options such as the resolution or the device's virtual orientation with respect to the built-in display. Exercise 3.2 guides you through this process.

When you have dual displays, you have the option to extend your desktop onto a second monitor or to clone your desktop on the second monitor. To change the settings for multiple monitors in Windows 11, follow the steps in Exercise 3.2, after ensuring that you have a second monitor attached.

EXERCISE 3.2

Changing the Settings for Multiple Monitors

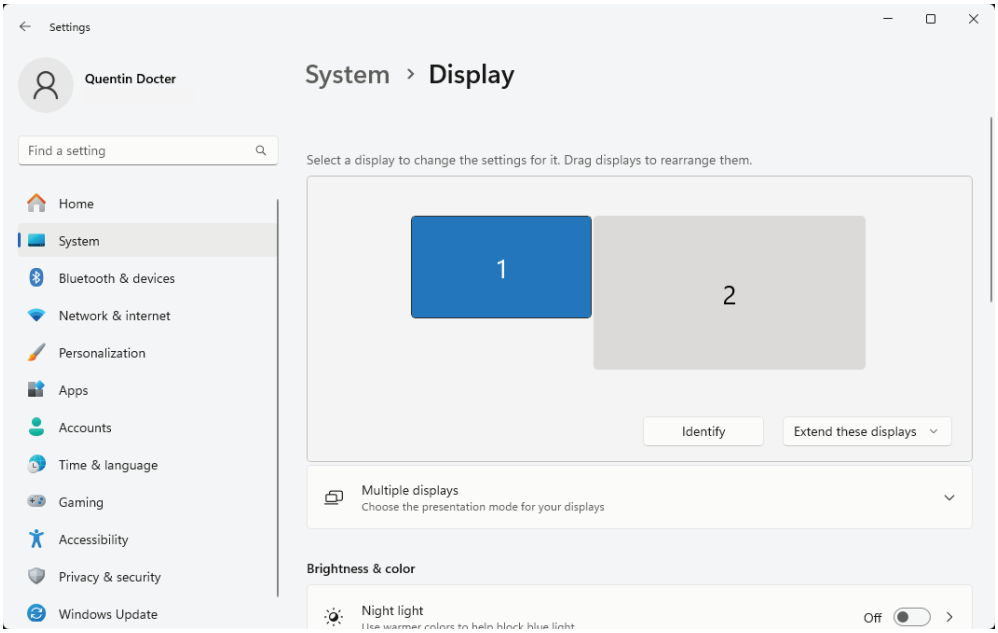
1. Right-click a blank portion of the desktop.
2. Click Display Settings to open the Display Settings window.

   If a second monitor has been detected, you will see a screen similar to the one shown in Figure 3.3. Otherwise, you will need to click the Detect button in the Multiple Displays section.

   Notice that the first monitor is highlighted. If you were to change settings such as scale, resolution, or orientation, it would affect the monitor that's highlighted.

EXERCISE 3.2 *(continued)*

FIGURE 3.3 Multiple displays detected



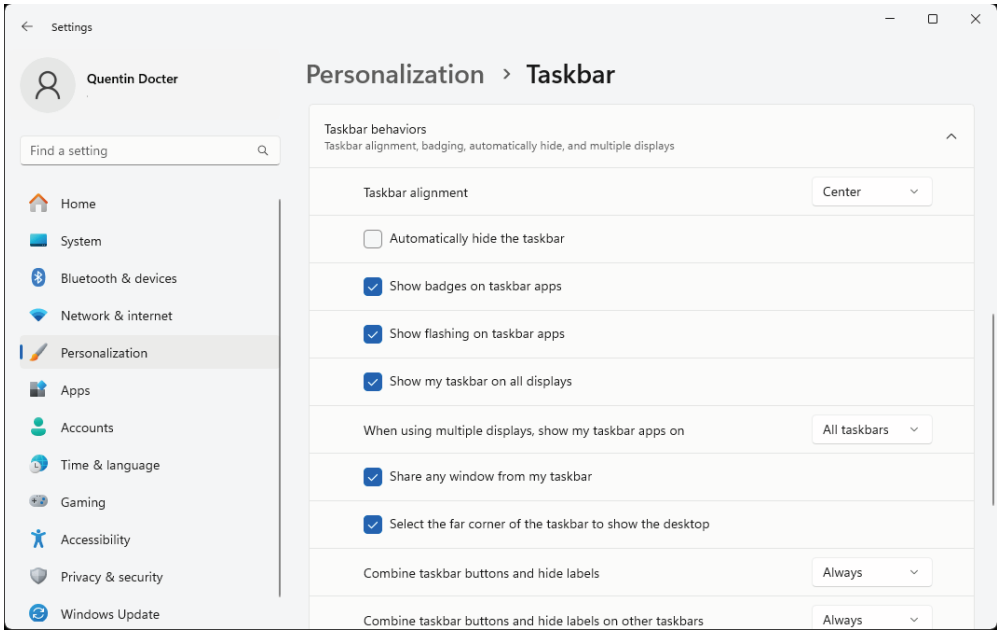
3. In the window showing the two displays, you have options to show an image on only one of the monitors, duplicate the displays, or extend the displays. In Figure 3.3, look where it says Extend These Displays, and click on the drop-down arrow. If your computer is not on Extend, choose that option now.
4. In the area showing two monitors, click and hold the second monitor and drag it around. Notice that you can place it above, below, left, or right of the first monitor. This will affect some display features, including where you need to move the mouse cursor to get it to appear on the second monitor.
5. Move the second monitor to be above the first monitor, and close Display Settings.
6. Move your mouse until you get the cursor to appear on the second screen.

   The rest of the steps are optional.
7. Open Display Settings and configure the second monitor to be in the position you want it to be relative to the first monitor.

   Does the second monitor (the one that's extended) show the Windows taskbar as well? If so, let's remove it. If not, let's add it.
8. Open the Taskbar settings app by right-clicking the taskbar and choosing Taskbar settings.

9. Scroll down to the Taskbar Behaviors section and click the drop-down arrow. In there, you will see the option Automatically Hide The Taskbar, as shown in Figure 3.4. If it's selected, deselect the box. If not, then select the box.

FIGURE 3.4 Taskbar settings



10. Configure the taskbar how you would like it, and then close the Taskbar app.

Selecting a Monitor

If you go to your favorite electronics retailer and search for monitors, the number of choices can be overwhelming. Here are a few tips to help narrow the field to a manageable number of options.

Exam Objective 3.1 lists four display attributes you need to know: pixel density, refresh rates, screen resolution, and color gamut.

Determine what the monitor will be used for. If it's a general-purpose monitor, it doesn't need to be the newest and fanciest technology. If it will be for a gamer, though, the minimum performance specifications just went up. For example, the resolution will need to be QHD or better and the refresh rate at least 144 Hz.

Choose a size. This depends on how far the user will be sitting from the screen and how much desk space is available. For a regular office or home office setup, somewhere

between 24″and 27″ is the most popular today, with some people preferring the slightly larger 32″. Anything bigger than that could be too large or not fit on the desk. If it will be viewed from across the room, then go as large as makes sense.

Pick a resolution. The resolutions available may be somewhat dependent on the size of monitor you choose. Remember that higher resolutions result in sharper images. Here are the three common options today:

1080p/Full High Definition (FHD) (1920 × 1080) This is the basic resolution used today. It's tenable for monitors up to 24″, but for larger monitors a higher resolution is needed; otherwise, icons and images appear oversized.

1440p/Quad High Definition (QHD) (2560 × 1440) For 27″ and larger displays, this is the base performance you want to look for. Sometimes you will hear this referred to as 2k resolution.

2160p/Ultra High Definition (UHD) (3840 × 2160) Commonly referred to as 4k, it's the resolution of choice for serious gamers and others who want ultra-sharp and crisp pictures. It's also great for those wanting to watch 4k movies on a monitor.

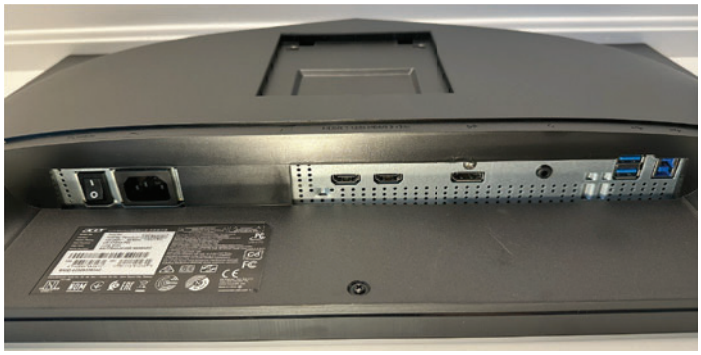
There are other resolutions available, but the three we listed are the most common. The highest available right now is 8k (UHD-2), which is 7680 × 4320. While you can buy 8k monitors, it may be too much of a good thing right now because of limited content available in 8k, and there are healthy debates on how much better it really is than 4k on screens smaller than 55″ due to limitations in human eyesight.

If you get an ultrawide monitor (and many are curved), you may see resolutions such as Wide QHD (WQHD [3440 × 1440]) or 4k offshoots like 7680 × 2160.

Choose a refresh rate. Basic, run-of-the-mill LCD screens will have a 60 Hz refresh rate; slightly nicer models will boast a 75 Hz rate. You probably won't notice too much of a difference between them. Gamers or those using their computer monitor to watch live-action sports or movies will want at least 144 Hz as a minimum. Hard-core gamers and ultimate performance seekers will want to go to at least 240 Hz. That kind of refresh rate is overkill for people staring at office applications and email all day.

Weigh optional features. Monitors today come with a wide variety of bells and whistles to enhance the user experience. Options to consider include curved or ultrawide vs. widescreen, number of I/O ports, integrated speakers and/or webcam and microphone, and voice assistant availability. Figure 3.5 shows the ports on a typical 27″ monitor. In the center are two HDMI ports, and moving to the right there is a DisplayPort, headphone jack, two USB-A ports, and a USB-B port. On the side of the monitor (not shown) are two additional USB-A ports.

FIGURE 3.5 Monitor ports



In addition, depending on the users' needs you might want to consider looking at *pixel density* or *color gamut* (both are test objectives, by the way). Pixel density refers to the number of pixels in an inch, or pixels per inch (PPI). The higher the number, the more pixels are in an inch, which can result in crisper and sharper images. Note that this is not the same as resolution. Resolution measures how many pixels wide and tall an image is. For example, imagine a QHD (2560 × 1440) image on a 24″monitor versus a 32″ monitor. Each one would use 2560 pixels wide, but the smaller monitor would need to have a greater PPI to squeeze them in.

Color gamut is the range of colors the display can produce. Most displays today can easily produce over 16 million colors, whereas the human eye can only detect around 1 million. So what's the big deal? A display with a higher color gamut can be important for visual designers and others who need high-quality color reproduction, because it will produce smoother and more vibrant images. Lower color gamut displays may produce washed-out looking images.

There are several color gamut standards. They include sRGB, which is standard for computer monitors and televisions; Adobe RGB, used for professional color imaging; NTSC, created for televisions; DCI-P3, used in movie theaters; and Rec. 709, used for high-definition TVs and monitors. It's unlikely that you'll need to memorize these standards any time soon, but do know that the higher the color gamut, the more colors the display can produce.

Consider the price. Last but not least is price. The higher the specs, the more it will cost. Balance the user's needs with their budget to determine the right monitor to select.

Projection Systems

Another major category of display device is the video projection system, or *projector*. A portable projector can be thought of as a condensed video display with a lighting system that projects the image onto a screen or other flat surface for group viewing. Interactive whiteboards have become popular over the past decade to allow presenters to project an image

onto the board as they use virtual markers to draw electronically on the displayed image. Remote participants can see the slide on their system as well as the markups made by the presenter. The presenter can see the same markups because the board transmits them to the computer to which the projector is attached, causing them to be displayed by the projector in real time.

To accommodate using portable units at variable distances from the projection surface, a focusing mechanism is included on the lens. Other adjustments, such as keystone, trapezoid, and pincushion, are provided through a menu system on many models as well as a way to rotate the image 180 degrees for ceiling-mount applications.

The key characteristics of projectors are resolution and brightness. Resolutions are similar to those of computer monitors. Brightness is measured in lumens. A *lumen (lm)* is a unit of measure for the total amount of visible light that the projector gives off, based solely on what the human eye can perceive and not on invisible wavelengths. Sometimes the brightness is even more of a selling point than the maximum resolution that the system supports because of the chosen environment in which it operates. For example, it takes a lot more to display a visible image in a well-lit office than it does in a darkened theater.

If you are able to completely control the lighting in the room where the projection system is used, producing little to no ambient light, a projector producing as little as 1,300 lumens is adequate in a home theater environment, while you would need one producing around 2,500 lumens in the office. However, if you can only get rid of most of the ambient light, such as by closing blinds and dimming overhead lights, the system should be able to produce 1,500 to 3,500 lumens in the home theater and 3,000 to 4,500 lumens in the office. If you have no control over a very well-lit area, you'll need 4,000 to 4,500 lumens in the home theater and 5,000 to 6,000 lumens in the business setting.

By way of comparison, a 60W standard light bulb produces about 800 lumens. Output is not linear, however, because a 100W light bulb produces over double, at 1,700 lm. Nevertheless, you couldn't get away with using a standard 100W incandescent bulb in a projector. The color production is not pure enough and constantly changes throughout its operation due to deposits of soot from the burning of its tungsten filament during the production of light. High-intensity discharge (HID) lamps, like the ones found in projection systems, do more with less by using a smaller electrical discharge to produce far more visible light. Expect to pay considerably more for projector bulbs than for standard bulbs of a comparable wattage.

Real World Scenario

Burning Up

Although it doesn't take long for the fan to stop running and the projector to cool down, this phase should never be skipped to save time. With projector bulbs being one of the priciest consumables in the world of technology, doing so may cost you more than a change in your travel arrangements.

A fellow instructor relayed the story of carrying his own portable projector with him on the road. At the end of a week's class, he would power down the projector and get his laptop

and other goodies packed away. Just before running out of the door, he would unplug the projector and pack it up. As with many instructors, this gentleman's presentations increased in density and length as he became more and more comfortable with the material.

Over time, his presentation had been running later and later each Friday afternoon, edging him ever closer to his airline departure time. He admitted that he had gotten into the habit of yanking the power plug for his projector from the wall and quickly stuffing the unit into the carrying case before darting out the door. It's no wonder then that a few months after he started this behavior, his projector failed catastrophically. Replacing the bulb was not the solution.

One caveat with projectors is that you must never pull the electrical plug from the outlet until you hear the internal fan cut off. There is enough residual heat generated by the projector bulb that damage to the electronics or the bulb itself (discoloration or outright failure) can occur if the fan is not allowed to remove enough heat before it stops running. Without a connection to an electrical outlet, the fan stops immediately. The electronics have the appropriate level of heat shielding so that the fan removes enough heat during normal operation to avoid damage to the shielded components.

Audio Devices

Audio devices, true to their name, produce sound by plugging into a sound card. Many sound cards today are integrated into a device's motherboard, but some computers still have separate audio expansion cards. Audio devices can provide output, such as through speakers or headphones, or input with a microphone.

Speakers and headphones generally connect with a 1/8″ (3.5mm) audio connector, as shown in Figure 3.6. You may recall seeing the ports for these in Volume 1 Chapter 2, "Expansion Cards, Storage Devices, and Power Supplies," when we covered sound cards. Most audio connectors have two thin black bands engraved on them, which separates the connector into three parts: the tip, ring, and sleeve. Because of this, sometimes you will see these connectors referred to as *TRS connectors*. The tip provides left audio, the first band above the black groove (the ring) provides right audio, and the sleeve is the ground. You'll notice that the audio connector in Figure 3.6 has three black bands, providing four connections and making it a TRRS connector. The fourth one is for the microphone.

FIGURE 3.6 1/8″ audio connector

Headsets that provide audio and a microphone are popular for audio conferencing calls and video gaming. A sample headset is shown in Figure 3.7. This model connects via USB, as do most headsets. Volume controls and a microphone mute are located on the right earpiece.

FIGURE 3.7 A USB headset

Although discussed throughout this chapter, the *microphone* has yet to be formally defined, a definition that is at once technical and simple. Microphones convert sound waves into varying electrical signals. The result can be recorded, transmitted, or altered in a variety of ways, including amplification.

When installing a microphone, you must match its connector with an available one on the computer. Modern choices include the classic analog pink TRS connector and USB. Wireless versions also exist, but their receiver might still be connected to a standard I/O port. Alternatively, the microphone could be paired with a built-in Bluetooth transceiver, headphones, or headset.

Configuring a microphone on a PC can be done in a few places in Windows, depending on what you need to do. For microphone configuration options such as input sensitivity, look for the Recording tab of the Sound applet in Control Panel. Options here include setting the levels and choosing enhancements, such as noise suppression and echo cancellation. To select which apps can use the microphone, go to Start ➢ Settings ➢ Privacy & Security ➢ Microphone. Specialized applications may also have internal configuration for the microphone, passing most details of the configuration back to the operating system.

Input and Output Devices

An *input device* is one that transfers information from outside the computer system to an internal storage location, such as system RAM, video RAM, flash memory, or disk storage. Without input devices, computers would be unable to change from their default boot-up state. An *output device* does the opposite of an input device—it takes information that's stored in RAM or another location and spits it back out somewhere for the user to do something with it. We've already covered monitors, which are the most common output device. The other major type of output device is a printer. Volume 1 Chapter 5, "Printers and Multifunction Devices," is dedicated to them. Further, some devices are capable of managing both input and output.

Frequently, when discussed together, input and output are abbreviated as I/O.

Keyboard

The *keyboard* is easily the most popular PC input device, so much so that it's more of a necessity. Very few users would even think of beginning a computing session without a working keyboard. Fewer still would even know how. The U.S. English keyboard places keys in the same orientation as the QWERTY typewriter keyboards, which were developed in the 1860s. Wired keyboards are almost always attached via USB. Wireless keyboards will often have a USB dongle that is attached to the computer, but they can also use Bluetooth.

Keyboards have also added separate number pads to the side and function keys (not to be confused with the common laptop key labeled *Fn*), placed in a row across the top of the keyboard above the numerical row. Key functionality can be modified by using one or more combinations of the Ctrl, Alt, Shift, and laptop Fn keys along with the normal QWERTY keys.

In addition to the layout for a standard keyboard, other keyboard layouts exist—some not nearly as popular. For example, without changing the order of the keys, an ergonomic keyboard is designed to feel more comfortable to users as they type. The typical human's hands do not rest with the fingers straight down. Ergonomic keyboards, therefore, should not place keys flat and along the same plane. To accomplish that goal, manufacturers split the keyboard down the middle, angling keys on each side downward from the center. Doing so fits the keys to the fingers of the hands when they are in a relaxed state. Figure 3.8 shows an example of an ergonomic keyboard. Even more exotic-looking ergonomic keyboards exist and may provide better relief for users who suffer from repetitive-use problems such as carpal tunnel.

Here are two places to go for keyboard-related settings in Windows 11:

- Keyboard properties, by typing **keyboard** into the Windows search box and pressing Enter
- Typing settings, by clicking Start ➢ Settings ➢ Time & Language ➢ Typing

FIGURE 3.8 An ergonomic keyboard

Mouse

Although the computer *mouse* was born in the 1970s at Xerox's Palo Alto Research Center (PARC), it was in 1984 that Apple made the mouse an integral part of the personal computer with the introduction of the Macintosh. In its most basic form, the mouse is a hand-fitting device that uses some form of motion-detection mechanism to translate its own physical two-dimensional movement into onscreen cursor motion. Many variations of the mouse exist, including trackballs, tablets, touchpads, and pointing sticks. Figure 3.9 illustrates a basic wireless optical mouse.

FIGURE 3.9 A computer mouse

The motion-detection mechanism of the original Apple mouse was a simple ball that protruded from the bottom of the device so that when the bottom was placed against a flat surface that offered a slight amount of friction, the mouse would glide over the surface but the ball would roll, actuating two rollers that mapped the linear movement to a Cartesian plane and transmitted the results to the software interface. This method of motion detection has been replaced by optical receptors to catch LED light reflected from the surface the mouse is used on. Note that most optical mice will have problems working on a transparent glass surface because of the lack of reflectivity.

The mouse today can be wired to the computer system or connected wirelessly. A wired mouse typically uses a USB port, which also provides power. Wireless versions will have a USB dongle or connect via Bluetooth. They are powered with batteries, and the optical varieties deplete these batteries more quickly than their mechanical counterparts.

The final topic is one that is relevant for any mouse: buttons. The number of buttons that you need your mouse to have depends on the software interfaces you use. For the Macintosh, one button has always been sufficient, but for a Windows-based computer, at least two are recommended—hence, the term *right-click*. Today, the mouse is commonly found to have a wheel on top to aid in scrolling and other specialty movement. The wheel has even developed a click in many models, sort of an additional button underneath the wheel. Buttons on the side of the mouse that can be programmed for whatever the user desires are common today as well.

There are several variants on pointer devices, such as trackballs. A *trackball* is like an inverted mouse. Both devices place the buttons on the top, which is where your fingers will be. A mouse places its tracking mechanism on the bottom, requiring that you move the entire assembly as an analogue for how you want the cursor on the screen to move. In contrast, a trackball places the tracking mechanism, usually a ball that is about one inch in diameter, on the top with the buttons. You then have a device that need not be moved around on the desktop and can work in tight spaces and on surfaces that would be incompatible with the use of a mouse. The better trackballs place the ball and buttons in such a configuration that your hand rests ergonomically on the device, allowing effortless control of the onscreen cursor.

Other input devices you may encounter are touchpads (especially on laptops), signature pads, game controllers, barcode and QR code scanners, magnetic or chip readers, and near-field communication (NFC) devices such as tap-to-pay devices. They almost always connect to the computer using a USB port.

KVM Switches

Some users will need multiple computers in the same location, but they don't want to hassle with multiple sets of input and output devices. For them, a KVM switch may be a good solution.

A KVM switch isn't an input or output device per se, but it allows you to switch between sets of input and output devices. The *KVM switch* is named after the devices among which it allows you to switch. The initials stand for *keyboard*, *video*, and *mouse*. KVM switches come in a variety of models, with different connector types available.

The purpose of the switch is to allow you to have multiple computers attached to the same keyboard, monitor, and mouse. You can use these three devices with only one system at a time. Some switches have a dial that you turn to select which system attaches to the components, while others feature buttons for each system connected. Common uses of KVM switches include using the same components alternately for a desktop computer and a laptop docking station or having a server room with multiple servers but no need to interface with them simultaneously.

Storage Devices

We spent quite a lot of time in Volume 1 Chapter 2 discussing storage options, such as hard drives and optical drives. These devices are frequently internal to the case, but external options are available as well.

Take optical drives, for instance. In order to save space and reduce costs, manufacturers usually don't include internal optical drives. If users want to play a Blu-ray or DVD movie, they will need to attach an external optical drive. External optical drives can be used for data backups as well. These external drives will most likely connect via USB or eSATA.

External storage drives can greatly enhance the storage capacity of a computer, or they can provide networked storage for several users. A plethora of options is available, from single drives to multidrive systems with several terabytes of capacity. Figure 3.10 shows an external network-attached storage (NAS) device.

Looking at Figure 3.10, you can see that this is a self-enclosed unit that can hold up to five hard drives, which are externally accessible—a nice feature. Some hold more; some hold fewer. This NAS also has status lights next to the drives to indicate proper operation or failure as well as a small status display. Nicer NAS systems enable you to hot-swap hard drives and have built-in fault tolerance as well.

In addition to the hardware, the NAS device contains its own operating system, meaning that it acts like its own file server. In most cases, you can plug it in, do some very minor configuration, and have instant storage space on your network. As far as connectivity goes, NAS systems will connect to a PC through a USB or eSATA port, but that is primarily so you can use that PC to run the configuration software for the NAS. The NAS also connects to the network, and that is how all network users access the storage space.

Understanding Cables and Connectors

Peripheral devices used with a computer need to attach to the motherboard somehow. They do so through the use of ports and cables. A *port* is a generic name for any connector on a computer or peripheral into which a cable can be plugged. A *cable* is simply a way of

connecting a peripheral or other device to a computer using multiple copper or fiber-optic conductors inside a common wrapping or sheath. Typically, cables connect two ports: one on the computer and one on some other device.

FIGURE 3.10 A network-attached storage device

Source: Bin im Garten / Wikimedia Commons / CC BY SA 3.0

The A+ exam objectives break cables and connectors into two different subobjectives, but really they need to be discussed together. After all, a cable without a connector doesn't do much good, and neither does a connector without a cable. In the following sections, we'll look at three different classifications of cables and the connectors that go with them: peripheral, video, and hard drive.

For the A+ exam, you will also need to be familiar with network cables and connectors. We will cover those in depth in Volume 1 Chapter 6, "Networking Fundamentals."

Peripheral Cables and Connectors

Some cables are for specific types of devices only. For example, SATA is for storage devices. And HDMI can transmit audio as well as video, but for the most part, though, we associate HDMI with video.

Unlike HDMI and SATA, the cables and connectors in this section are specifically designed to connect a variety of devices. For example, someone may have a USB hub with a wireless mouse, network card, Lightning cable (to charge an iPhone), and flash drive all attached at the same time. Those four devices serve very different purposes, but they all share the USB connection in common. We'll start with the highly popular USB and then discuss Lightning ports, Thunderbolt cables, and serial cables.

Exam Objective 3.2, "Summarize basic cable types and their connectors, features, and purposes," lists four peripheral cable types you should know: USB 2.0, USB 3.0, Serial, and Thunderbolt.

Universal Serial Bus

It seems like we've mentioned USB a thousand times already in this book, because it's everywhere and it's like everyone knows what it is. But we have yet to define or explain it. *Universal Serial Bus (USB)* is a standard used to connect a wide variety of peripherals, such as keyboards, mice, digital cameras, printers, scanners, hard drives, and network cards, to computers. USB was designed by several companies, including Intel, Microsoft, and IBM, and is currently maintained by the USB Implementers Forum (USB-IF).

USB technology is fairly straightforward. Essentially, it is designed to be Plug and Play—just plug in the peripheral and it should work, provided that the software is installed to support it. Many standard devices have drivers that are built into the common operating systems or automatically downloaded during installation. More complex devices come with drivers to be installed before the component is connected.

USB host controllers can support up to 127 devices, which is accomplished through the use of a 7-bit identifier. (A bit has two values, 0 and 1. Therefore, a string of 7 consecutive bits has 2^7 possible unique values, or 128 total.) The 128th identifier, the highest address, is used for broadcasting to all endpoints. Realistically speaking, you'll probably never get close to this maximum. Even if you wanted to try, you won't find any computers with 127 USB ports. Instead, you would plug in a device known as a *USB hub* (shown in Figure 3.11) into one of your computer's USB ports, which will give you several more USB ports from one original port. Understand that a hub counts as a device for addressing purposes. Hubs can be connected to each other, but interconnection of host controllers is not allowed; each one and its connected devices are isolated from other host controllers and their devices. As a result, USB ports are not considered networkable ports. Consult your system's documentation to find out if your USB ports operate on the same host controller.

Another nice feature of USB is that devices can draw their power from the USB cable, so you may not need to plug in a separate power cord. This isn't universally true, though, as some peripherals still require external power.

USB Standards

Even though USB was released in 1996, the first widely used standard was USB 1.1, which was released in 1998. It was pretty slow—only 12 Mbps at full speed and 1.5 Mbps at low speed—so it was only used for keyboards, mice, and printers. When USB 2.0 came out in 2000 with a faster transfer rate of 480 MBps (called *Hi-Speed*), video devices were possible.

The newer USB 3.x, USB4, and USB4 v2 standards have increased throughput even further. Table 3.1 lays out the specifications and speeds for you.

FIGURE 3.11 A 4-port USB hub

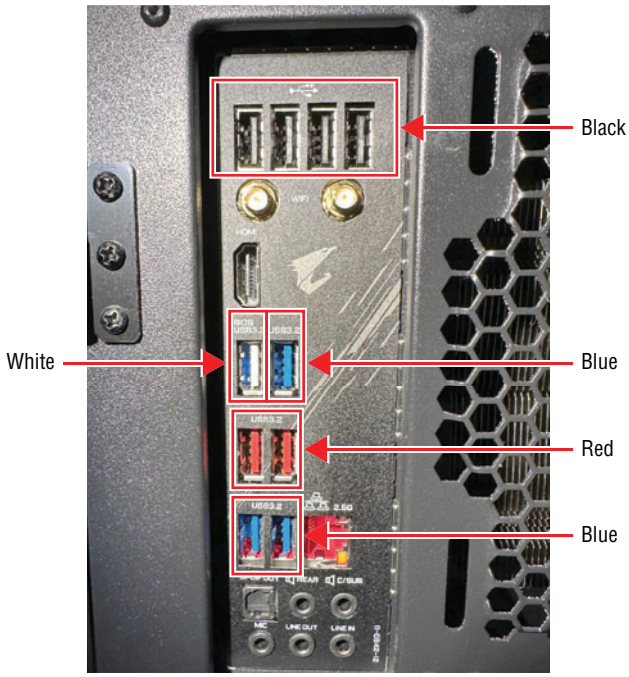
TABLE 3.1 USB specifications

Specification	Release year	Maximum speed	Trade name	Color
USB 1.1	1998	12 Mbps	Full-Speed	White
USB 2.0	2000	480 Mbps	Hi-Speed	Black
USB 3.0	2008	5 Gbps	SuperSpeed	Blue
USB 3.1	2013	10 Gbps	SuperSpeed+	Teal
USB 3.2	2017	20 Gbps	SuperSpeed+	Red
USB4	2019	40 Gbps	USB4 40 Gbps	n/a
USB4 2.0	2022	80 Gbps	USB4 80 Gbps or USB4 v2	n/a

The USB 1.x and 2.x specifications didn't recommend a specific color for the ports, but when USB 3.0 was released, the USB Implementers Forum suggested that the ports and cable connectors be colored blue, to signify that they were capable of handling higher speeds. Device manufacturers are not required to follow the color-coding scheme, so you may see some inconsistency. For example, if you look at image 3.12, you will see 10 USB ports on the back of an ASUS motherboard. We know the book prints in black-and-white, so bear with us as we describe the scene. The top row of 4 ports are all black, so you can assume these

are USB 2.0. Below the HDMI port, there are two USB ports, white and blue, and both are labeled USB 3.2. The third row of USB ports are also labeled 3.2 and are red, and the fourth row are also labeled USB 3.2 but are blue. One can assume in this case that the bottom six USB ports all support USB 3.2, regardless of their color.

FIGURE 3.12 USB ports on a motherboard



A yellow USB port is "always on," meaning it's capable of charging a connected device even if the PC is sleeping or shut down. USB4 v2 is the newest standard, and in many ways is comparable to Thunderbolt 5 (discussed later). USB4 v2 is typically implemented over USB-C cables.

USB Power

As mentioned previously, USB ports provide power to devices plugged into them. Typical power for attached USB devices is 5V. The maximum current (amps) and wattage will depend on the connected device and USB standard being used.

All USB ports are also capable of functioning as charging ports for devices such as tablets, smartphones, and smart watches. The charging standard, called *USB Battery Charging*, was released in 2007. USB Power Delivery (PD) was developed in 2012. Technically, they are different standards, but in practice, USB ports are capable of supporting both standards at the same time. Table 3.2 outlines some of the versions and the maximum power that they provide. The newest version, USB PD 3.1, requires the use of a USB-C cable.

TABLE 3.2 USB power standards

Standard	Year	Maximum power
USB Battery Charging 1.0	2007	5V, 1.5A (7.5W)
USB Battery Charging 1.2	2010	5V, 5A (20W)
USB Power Delivery 1.0	2012	20V, 5A (100W)
USB Power Delivery 2.0 (specified use of Type-C connectors but only up to 15W)	2014	5V, 3A (15W)
USB Power Delivery 3.0	2015	20V, 5A (100W)
USB Power Delivery 3.1	2021	48V, 5A (240W)

A smartphone or tablet typically needs a minimum of about 7.5 watts to charge properly. The Battery Charging 1.0 standard was good enough, but not for larger devices. For example, about 20 watts is required to power a small laptop computer, and standard 15-inch laptops can require 60 watts or more. With USB PD, one USB port can now provide enough power for a laptop as well as a small printer.

Because of the capabilities of USB PD, it's becoming common to see devices up to laptop size lose their standard AC power ports and adapters—they may just have a USB-C port instead. To get the full capabilities of USB PD, you need to use a USB-C port and cable.

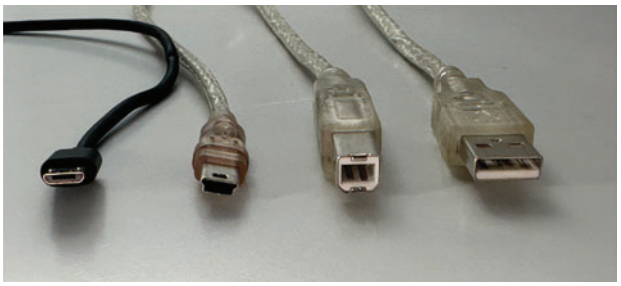
USB Cables and Connectors

In order to achieve the full speed of the specification that a device supports, the USB cable needs to meet that specification as well. In other words, USB 1.x cables cannot provide USB 2.0 and 3.x performance, and USB 2.0 cables cannot provide USB 3.x performance. Otherwise, the connected device will have to fall back to the maximum version supported by the cable. This is usually not an issue, except for the lost performance, but some high-performance devices will refuse to operate at reduced levels. Note that all specifications are capable of Low Speed, which is a 1.5 Mbps performance standard that has existed since the beginning of USB time.

Throughout most of its history, USB has relied on a small suite of standard connectors. The two broad classifications of connectors are designated Type-A and Type-B connectors, and there are micro and mini versions of each. A standard USB cable has some form of Type-A connector on the end that plugs into the computer or hub, and some form of Type-B or proprietary connector on the device end. Figure 3.13 shows four classic USB 1.x/2.0 cable connectors. From left to right, they are as follows:

- Micro-USB
- Mini-USB
- Type-B
- Type-A male

FIGURE 3.13 Standard USB connectors



Small form factor devices, including many smartphones and smaller digital cameras, use a *micro-USB* or *mini-USB* connector, unless the manufacturer has developed its own proprietary connector. Micro-USB connectors (and modified ones) are popular with many Android phone manufacturers.

In 2014, a new connector named *USB Type-C* (or simply *USB-C*) was developed. USB-C is designed to replace Type-A and Type-B, and, unlike its predecessors, it's reversible. That means no more flipping the connector over several times to figure out which way it connects. Type-C cables will also be able to provide more power to devices than classic cables were. Figure 3.14 shows a Type-C connector and a Type-A connector. You can see that while the Type-A connector is rectangular-shaped, the Type-C connector has rounded corners and looks more like an elongated oval.

FIGURE 3.14 USB Type-C (top) and Type-A (bottom)

One point of confusion for many is dissociating the connector type from the standard. Because USB 3.1 and USB-C were both released around the same time, people often think that they are one and the same—but they're not. USB 3.1 can be implemented using classic A and B connectors, and USB 2.0 can work over a Type-C connector.

USB was designed to be a short-distance technology. Because of this, USB cables are limited in length. USB 1.x and 2.0 can use cables up to 5 meters long, whereas USB 3.x can use cables up to 3 meters long. The maximum length of a USB4 cable is even shorter yet at 80 centimeters (0.8 meters). In addition, if you use hubs, you should never use more than five hubs between the system and any component.

Despite the seemingly locked-up logic of USB connectivity, it is occasionally necessary to alter the interface type at one end of a USB cable. For that reason, there are a variety of simple, passive converters on the market with a USB interface on one side and a USB or different interface on the other. Along with adapters that convert USB Type-A to USB Type-B, there are adapters that will convert a male connector to a female one. In addition, you can convert USB to a lot of other connector types, such as USB to Ethernet (shown in Figure 3.15), USB to SATA, USB to eSATA, USB to PS/2, USB to serial, and a variety of others.

FIGURE 3.15 Kensington USB-to-Ethernet adapter

For more information on USB, check out www.usb.org.

Lightning

Introduced in 2012 with the iPhone 5, the *Lightning connector* was Apple's proprietary connector for iPads until the third-generation iPad Pro in 2018 and iPhones up until the iPhone 15. It's an 8-pin connector that replaced Apple's previous 30-pin dock connector. A standard Lightning cable has a USB Type-A or USB-C connector on one end and the Lightning connector on the other, as shown in Figure 3.16. It's not keyed, meaning that you can put it in with either edge up.

FIGURE 3.16 Lightning cable

Lightning cables support USB 2.0. You will find cables that are USB-C to Lightning, as well as various Lightning adapters, such as those to HDMI, DisplayPort, audio, and Lightning to female USB Type-A (so you can plug a USB device into an iPad or iPhone).

Thunderbolt

Where there's lightning, there's thunder, right? Bad joke attempts aside, in computer circles Lightning connectors don't have anything to do with Thunder(bolt). *Thunderbolt*, created in collaboration between Intel and Apple and released in 2011, combines PCI Express 2.0 x4 with the DisplayPort 1.x technology. While it's primarily used for video (to replace DisplayPort), the connection itself can support multiple types of peripherals, much like USB does.

Thunderbolt Standards

For most of their histories, Thunderbolt and USB have been competing standards. Thunderbolt was designed more for video applications and USB was the slower "jack of all trades" port, but in reality, they could be used for almost the exact same list of peripherals. It just depended on what your computer supported. But as we pointed out in the USB section, the new USB4 version is based on Thunderbolt 3, providing the same speed and using the same connectors. Table 3.3 shows the five Thunderbolt versions and some key characteristics.

TABLE 3.3 Thunderbolt standards

Version	Year	Maximum throughput	Connector	Other new features
Thunderbolt 1	2011	10 Gbps	Mini DisplayPort	
Thunderbolt 2	2013	20 Gbps	Mini DisplayPort	DisplayPort 1.2 (can send video to a 4k display)
Thunderbolt 3	2015	40 Gbps	USB-C	10 Gbps Ethernet support
Thunderbolt 4	2020	40 Gbps	USB-C	Can support two 4k displays or one 8k display; 32 Gbps PCIe
Thunderbolt 5	2024	80/120 Gbps	USB-C	Can support multiple 8k displays and 540 Hz refresh rates, or 3 displays at 144 Hz

Thunderbolt 3 was released in 2015 and doubled the bandwidth to 40 Gbps. It supports PCIe 3.0 and DisplayPort 1.2, meaning that it can support dual 4K displays at 60 Hz or a single 4K display at 120 Hz. It also provides up to 100 watts of power to a device.

Thunderbolt 4 was released in 2020. Perhaps the most interesting thing about the release is what it didn't do, which is increase data transfer rates versus Thunderbolt 3. It still had a maximum bandwidth of 40 Gbps. And the maximum of 100 watts of power to attached devices didn't change either. The big advantages of Thunderbolt 4 included support for two 4k displays or one 8k display and the requirement to support 32 Gbps data transfers via PCIe, up from 16 Gbps in version 3. Version 5, released in 2024, doubles the speed of version 4.0 to 80 Gbps, and can be boosted to 120 Gbps for limited amounts of time. It also supports multiple 8k displays at once, and it can provide 240 watts of charging power.

Thunderbolt Cables and Connectors

The most common Thunderbolt cable is a copper, powered active cable extending as far as 3 meters, which was designed to be less expensive than an active version of a DisplayPort cable of the same length. There are also optical cables in the specification that can reach as far as 60 meters. Copper cables can provide power to attached devices, but optical cables can't.

Additionally, and as is the case with USB, Thunderbolt devices can be daisy-chained and connected via hubs. Daisy chains can extend six levels deep for each controller interface, and each interface can optionally drive a separate monitor, which should be placed alone on the controller's interface or at the end of a chain of components attached to the interface.

As noted in Table 3.3, Thunderbolt changed connectors between versions 2 and 3. Figure 3.17 shows two Thunderbolt 2 interfaces next to a USB port on an Apple MacBook Pro. Note the standard lightning-bolt insignia by the port. Despite its diminutive size, the Thunderbolt port has 20 pins around its connector bar, like its larger DisplayPort cousin. Of course, the functions of all the pins do not directly correspond between the two interface types, because Thunderbolt adds PCIe functionality.

FIGURE 3.17 Two Thunderbolt 2 interfaces

Starting with Thunderbolt 3, the connector was changed to standard USB-C connectors, as shown in Figure 3.18. Notice that the lightning bolt icon remained the same.

Converters are available that connect Thunderbolt connectors to VGA, HDMI, and DVI monitors. Active converters that contain chips to perform the conversion are necessary in situations such as when the technology is not directly pin-compatible with Thunderbolt—as with VGA and DVI-A analog monitor inputs, for example. Active converters are only slightly more expensive than their passive counterparts but still only a fraction of the cost of Thunderbolt hubs. One other advantage of active connectors is that they can support resolutions of 4k (3840 × 2160) and higher.

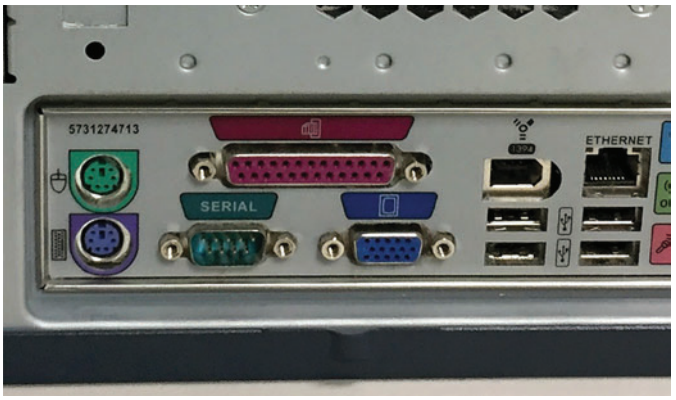
FIGURE 3.18 Two Thunderbolt 3 interfaces

Source: Amin / Wikimedia Commons / CC BY SA 4.0

Serial Ports

Before USB came along in 1998, *serial ports* were considered slow and inferior to parallel ports. Still, serial enjoyed use among peripherals that didn't need to transfer information at high speeds, such as mice, modems, network management devices, and even printers. Figure 3.19 shows a 9-pin serial port. It's the one marked "Serial," and it's also the only male connector on the back of the PC.

FIGURE 3.19 Several peripheral ports



As you might expect, a serial cable attaches to the serial port. Figure 3.20 shows a female DB-9 serial connector. To make things more confusing, sometimes you will hear people refer to the image in Figure 3.20 as an RS-232 cable or connector. Even though the terms are often used interchangeably, there is a technical difference.

FIGURE 3.20 DB-9 serial connector

DB-9 refers to a specific type of D-sub connector that has 9 pins. *RS-232*, on the other hand, is a communications standard for serial transmission. In other words, systems may communicate with each other using RS-232 over a DB-9 connection. But RS-232 can be used on other types of serial cables as well, such as DB-15 or DB-25. Generally speaking, if someone asks for an RS-232 serial cable, they *mean* a DB-9 cable with female connectors. But it's always best to confirm.

RS-232 did have a few advantages over USB—namely, longer cable length (15 meters vs. 3–5 meters) and a better resistance to electromagnetic interference (EMI). Still, USB has made old-school serial ports nearly obsolete. About the only time they are used today is for management devices that connect to servers or network routers with no keyboard and monitor installed.

Other Keyboard and Mouse Connectors

At one time, the most popular connectors for keyboards and mice were round connectors called *Personal System/2 (PS/2)* connectors. The PS/2 connector (there are two on the left in Figure 3.19) is a smaller 6-pin mini-DIN connector. Many PCs included a PS/2 keyboard connector as well as a PS/2 mouse connector right above it on the motherboard. The keyboard connector was colored purple, and the mouse one green. The ends of the keyboard and mouse cables would be purple and green as well. Today, the PS/2-style connector has been replaced by the USB port.

If you do run into a keyboard or mouse with a PS/2 connector, or a motherboard with a PS/2 port, you can buy a PS/2-to-USB adapter to make them work with more current hardware.

Video Cables and Connectors

Computer displays are ubiquitous—they're easily the most widely used peripheral. Different standards exist to connect displays to the computer, and you need to be familiar with five of them for the exam. We will start with the older technologies and work toward the present.

Exam Objective 3.2 lists VGA, DVI, HDMI, DisplayPort, and USB-C as video cables and connectors to know.

Video Graphics Array Connector

The *Video Graphics Array (VGA) connector* was the de facto video standard for computers for years and is still in use today. First introduced in 1987 by IBM, it was quickly adopted by other PC manufacturers. The term *VGA* is often used interchangeably to refer to generic analog video, the 15-pin video connector, or a 640 × 480 screen resolution (even though the VGA standard can support much higher resolutions). Figure 3.21 shows a VGA port, as well as the male connector that plugs into the port. Nearly all VGA connectors are blue.

FIGURE 3.21 VGA connector and port

Understanding D-Sub Ports and Connectors

The VGA connector is an example of a *D-subminiature connector*, also known as a *D-sub connector*. For a number of years, D-sub was the most common style of connector found on computers. Their names are typically designated with D*X-n*, where the letter *X* is replaced by a letter from *A* to *E*, which refers to the size of the connector, and the letter *n* is replaced by the number of pins or sockets in the connector.

D-sub connectors are usually shaped like a trapezoid and have at least two rows of pins with no other keying structure or landmark. Several were shown back in Figure 3.19. At the bottom center in Figure 3.19 is a DE-15F 15-pin display-connector port, which may also be referred to as an HD-15 or DB-15 port. The top one is a classic DB-25 parallel port, and the

bottom left is a *DB-9* serial port. The IEEE 1394 (FireWire—now obsolete) and two dusty USB ports are shown for a size comparison.

The "D" shape ensures that only one orientation is possible. If you try to connect them upside down or try to connect a male connector to another male connector, they just won't go together and the connection can't be made. By the way, male interfaces have pins, while female interfaces have sockets. Be on the lookout for the casual use of *DB* to represent any D-sub connector. This is very common and is accepted as an unwritten de facto standard, even if some are technically DE- or DA- connectors. Also note that you will see them written without the hyphen or with a space, such as DB15 or DB 15.

VGA technology is the only one on the objectives list that is purely analog. It has been superseded by newer digital standards, such as DVI, HDMI, and DisplayPort, and it was supposed to be phased out starting in 2013. A technology this widely used will be around for quite a while, though, and you'll still see it occasionally in the wild (or still in use).

All the video connector types introduced from here on are digital standards.

Digital Visual Interface

The analog VGA standard ruled the roost for well over a decade but it had a lot of shortcomings. Digital video can be transmitted farther and at higher quality than analog, so development of digital video standards kicked off in earnest. The first commercially available one was a series of connectors known collectively as *Digital Visual Interface (DVI)* and was released in 1999.

At first glance, the DVI connector might look like a standard D-sub connector. On closer inspection, however, it begins to look somewhat different. For one thing, it has quite a few pins, and for another, the pins it has are asymmetrical in their placement on the connector. The DVI connector is usually white and about an inch long. Figure 3.22 shows what the connector looks like coming from the monitor.

FIGURE 3.22 DVI connector

Source: Evan Amos / Wikimedia Commons / Public domain

There are three main categories of DVI connectors:

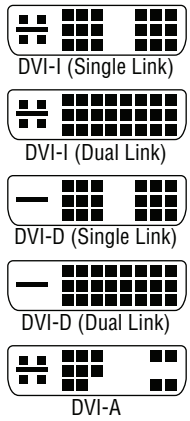
DVI-A *DVI-A* is an analog-only connector. The source must produce analog output, and the monitor must understand analog input.

DVI-D *DVI-D* is a digital-only connector. The source must produce digital output, and the monitor must understand digital input.

DVI-I *DVI-I* is a combination analog/digital connector. The source and monitor must both support the same technology, but this cable works with either a digital or an analog signal.

The DVI-D and DVI-I connectors come in two varieties: single-link and dual-link. The dual-link options have more conductors—taking into account the six center conductors—than their single-link counterparts; therefore, the dual-link connectors accommodate higher speed and signal quality. The additional link can be used to increase screen resolution for devices that support it. Figure 3.23 illustrates the five types of connectors that the DVI standard specifies.

FIGURE 3.23 Types of DVI connectors



DVI-A and DVI-I analog quality is superior to that of VGA, but it's still analog, meaning that it is more susceptible to noise. However, the DVI analog signal will travel farther than the VGA signal before degrading beyond usability. Nevertheless, the DVI-A and VGA interfaces are pin-compatible, meaning that a simple passive *DVI-to-VGA adapter*, as shown in Figure 3.24, is all that is necessary to convert between the two. As you can see, the analog portion of the connector, if it exists, comprises the four separate color and sync pins and the horizontal blade that they surround, which happens to be the analog ground lead that acts as a ground and physical support mechanism even for DVI-D connectors.

FIGURE 3.24 DVI-to-VGA adapter

It's important to note that DVI-I cables and interfaces are designed to interconnect two analog or two digital devices; they cannot convert between analog and digital. DVI cables must support a signal of at least 4.5 meters, but better cable assemblies, stronger transmitters, and active boosters result in signals extending over longer distances.

One thing to note about analog versus digital display technologies is that all graphics' adapters and all monitors deal with digital information. It is only the connectors and cabling that can be made to support analog transmission. Before DVI and HDMI encoding technologies were developed, consumer digital video display connectors could not afford the space to accommodate the number of pins that would have been required to transmit 16 or more bits of color information per pixel. For this reason, the relatively few conductors of the inferior analog signaling in VGA were appealing.

High-Definition Multimedia Interface

High-Definition Multimedia Interface (HDMI) is an all-digital technology that advances the work of DVI to include the same dual-link resolutions using a standard HDMI cable but with higher motion-picture frame rates and digital audio right on the same connector. HDMI was introduced in 2002, which makes it seem kind of old in technology years, but it's a great, fast, reliable connector that will probably be around for several years to come. HDMI cabling also supports an optional Consumer Electronics Control (CEC) feature that allows transmission of signals from a remote-control unit to control multiple devices without separate cabling to carry infrared signals.

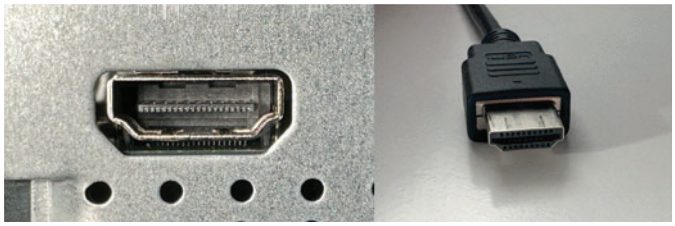
HDMI cables, known as Standard and High Speed, exist today in the consumer space. Standard cables are rated for 720p resolution as well as 1080i, but not 1080p. High Speed cables are capable of supporting not only 1080p, but also the newer 4k and 8k technologies. Figure 3.25 shows an HDMI port and cable.

In June 2006, revision 1.3 of the HDMI specification was released to support the bit rates necessary for HD DVD and Blu-ray Disc. This version also introduced support for "deep color," or color depths of at least one billion colors, including 30-, 36-, and 48-bit color. However, not until version 1.4, which was released in May 2009, was the High Speed HDMI cable initially required.

With version 1.4 came HDMI capability for the controlling system—the television, for instance—to relay Ethernet frames between its connected components and the Internet, alleviating the need for each and every component to find its own access to the LAN for Internet

access. Both Standard and High Speed cables are available with this Ethernet channel. Each device connected by such a cable must also support the HDMI Ethernet Channel specification, however.

FIGURE 3.25 HDMI port and cable



Additional advances that were first seen in version 1.4 were 3D support, 4K resolution (but only at a 30 Hz refresh rate), an increased 120 Hz refresh rate for the 1080 resolutions, and an Audio Return Channel (ARC) for televisions with built-in tuners to send audio back to an A/V receiver without using a separate output cable. Version 1.4 also introduced the anti-vibration Type-E locking connector for the automotive-video industry and cables that can also withstand vibration as well as the hot/cold extremes that are common in the automotive world.

Version 2.0 of HDMI (2013) introduced no new cable requirements. In other words, the existing High Speed HDMI cable is fully capable of supporting all new version 2 enhancements. These enhancements include increasing the 4K refresh rate to 60 Hz, a 21:9 theatrical widescreen aspect ratio, and 32-channel audio. Note that 7.1 surround sound comprises only eight channels, supporting the more lifelike Rec. 2020 color space and multiple video and audio streams to the same output device for multiple users. Version 2.0a, released in 2015, primarily added high dynamic range (HDR) video, but it does not require any new cables or connectors.

The most recent version (as of this writing) is HDMI 2.1, released in November 2017. Version 2.1 specifies a new cable type called 48G, which provides for 48 Gbps bandwidth. 48G cables are backward compatible with older HDMI versions. You can also use older cables with 48G-capable devices, but you just won't get the full 48 Gbps bandwidth. HDMI 2.1 also provides for 120 Hz refresh rates for 4k, 8k, and 10k video, and it supports enhanced Audio Return Channel (eARC), which is needed for object-based audio formats, such as DTS:X and Dolby Atmos.

Even though the HDMI connector is not the same as the one used for DVI, the two technologies are electrically compatible. HDMI is compatible with DVI-D and DVI-I interfaces through proper adapters, but HDMI's audio and remote-control pass-through features are lost. Additionally, 3D video sources work only with HDMI. Figure 3.26 shows a *DVI-to-HDMI adapter* between DVI-D and the Type-A 19-pin HDMI interface. Compare the DVI-D interface in Figure 3.26 to the DVI-I interface in Figure 3.24, and note that the ground blade

on the DVI-D connector is narrower than that of the DVI-A and DVI-I connectors. The DVI-D receptacle does not accept the other two plugs for this reason; in addition, the four analog pins around the blade have no sockets in the DVI-D receptacle.

FIGURE 3.26 DVI-to-HDMI adapter

Unlike DVI-D—and, by extension DVI-I—DVI-A and VGA devices cannot be driven passively by HDMI ports directly. An *HDMI-to-VGA adapter* must be active in nature, powered either externally or through the HDMI interface itself.

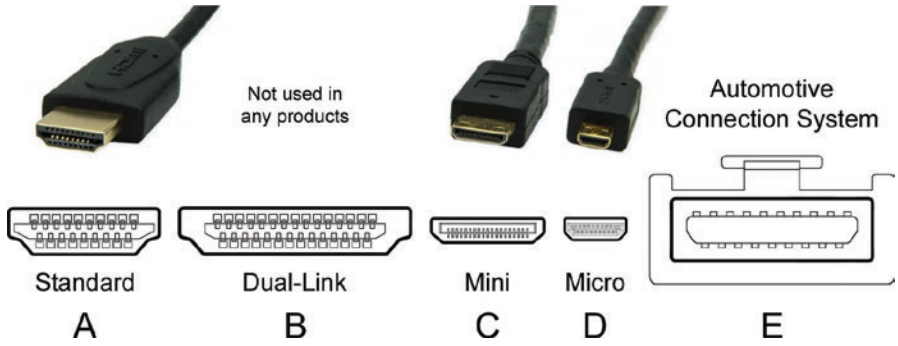
HDMI cables should meet the signal requirements of the latest specification. As a result, and as with DVI, the maximum cable length is somewhat variable. For HDMI, cable length depends heavily on the materials used to construct the cable. Passive cables tend to extend no farther than 15 meters, while adding electronics within the cable to create an active version results in lengths as long as 30 meters.

Smaller HDMI connectors

Multiple versions of HDMI connectors are available in the marketplace. The standard connector that you're probably used to seeing, and the ones shown in Figure 3.25 and Figure 3.26, are the 19-pin Type-A connector. The Type-A connector and the 29-pin Type-B connector were specified in HDMI version 1.0 and haven't changed much since then. Type-B connectors were intended for higher-resolution products but are not used in the market today.

HDMI version 1.3 specified a smaller 19-pin Type-C connector for portable devices. The Type-C connector, also referred to as a *mini-HDMI connector*, is compatible with the Type-A connector, but it still requires an adapter due to its smaller size. HDMI version 1.4 specified two more interfaces: Type-D and Type-E. If Type-C is a mini-HDMI interface, then you might refer to the Type-D connector as micro-HDMI. Figure 3.27 shows all five HDMI connectors. Also compatible with Type-A interfaces because they have the same 19 pins, Type-D interfaces require just a simple adapter for conversion.

FIGURE 3.27 HDMI connector types



Source: C0nanPayne / Wikimedia Commons / Public domain

The mini-HDMI and micro-HDMI connectors are most often used on smaller portable devices, such as tablets, smartphones, and digital cameras. As mentioned previously, the Type-E connector has a locking mechanism and is intended for use in automobiles or other environments that are susceptible to vibration, which could cause a connector and cable to become disconnected.

DisplayPort

DisplayPort is a royalty-free digital display interface from the Video Electronics Standards Association (VESA) that uses less power than other digital interfaces and VGA. Introduced in 2008, it's designed to replace VGA and DVI. To help ease the transition, it's backward compatible with both standards, using an adapter. In addition, an adapter allows HDMI and DVI voltages to be lowered to those required by DisplayPort because it is functionally similar to HDMI and DVI. DisplayPort cables can extend 3 meters, unless an active cable powers the run, in which case the cable can extend to 33 meters. DisplayPort is intended primarily for video, but, like HDMI, it can transmit audio and video simultaneously.

Figure 3.28 shows a DisplayPort port as well as a connector. Note the beveled keying at the bottom-left corner of the port. The DisplayPort connector latches itself to the receptacle with two tiny hooks. A push-button mechanism serves to release the hooks for removal of the connector from the receptacle—you can see the button is on the top of the connector on the right in Figure 3.28.

The DisplayPort standard also specifies a smaller connector, known as the *Mini DisplayPort (mDP) connector*. The mDP is electrically equivalent to the full-size DP connector and features a beveled keying structure, but it lacks the latching mechanism present in the DP connector. Figure 3.29 shows an mDP to DisplayPort cable, so you can see both connectors at the same time.

FIGURE 3.28 A DisplayPort port and cable

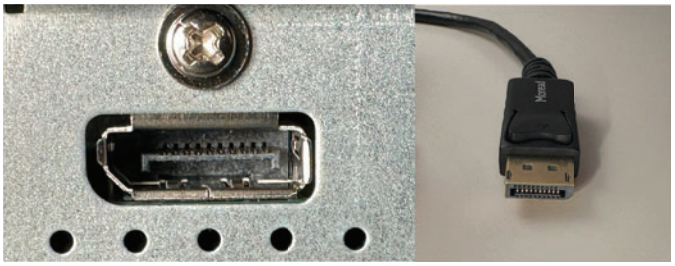


FIGURE 3.29 mDP and DisplayPort connectors

The final video cable connector to know for the exam is USB-C. Because we've already covered USB-C in depth earlier in this chapter, we won't repeat the information here. Do be aware that USB-C can be used for video connections as well as other peripherals.

Hard Drive Cables and Connectors

At the beginning of this chapter, we said that we were going to move outside the box and talk about external peripherals, cables, and connectors. For the most part that's true, but here we need to take a small digression to talk about connecting hard drives, most of which are internal. Some of this you already learned in Volume 1 Chapter 2, so this could feel like a review. Of course, there are SATA and PATA connectors, but we'll also throw in two new ones—SCSI and eSATA.

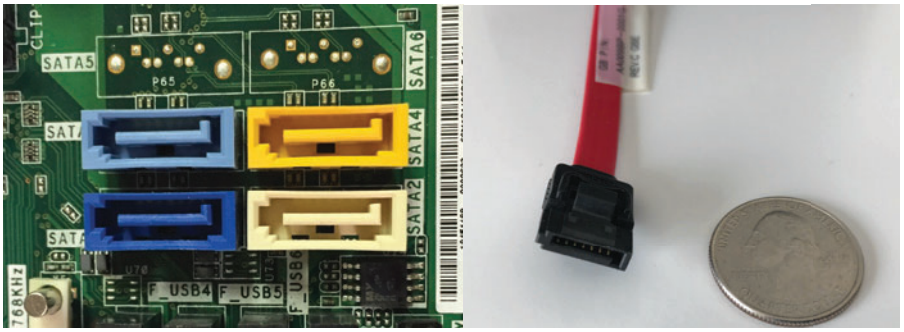
Optical drives use the same connectors as hard drives.

Remember that all drives need some form of connection to the motherboard so that the computer can "talk" to the disk drive. Regardless of whether the connection is built into the motherboard (*onboard*) or on an adapter card (*off-board*), internal or external, the standard for the attachment is based on the drive's requirements. These connections are known as *drive interfaces*. The interfaces consist of circuitry and a port, or *header*.

Serial Advanced Technology Attachment

The most common hard drive connector used today is *Serial Advanced Technology Attachment (SATA)*. Figure 3.30 shows SATA headers, which you have seen before, and a SATA cable. Note that the SATA cable is flat, and the connector is keyed to fit into the motherboard header in only one way. SATA data cables have a 7-pin connector. SATA power cables have 15 pins and are wider than the data connector.

FIGURE 3.30 SATA connectors and cable



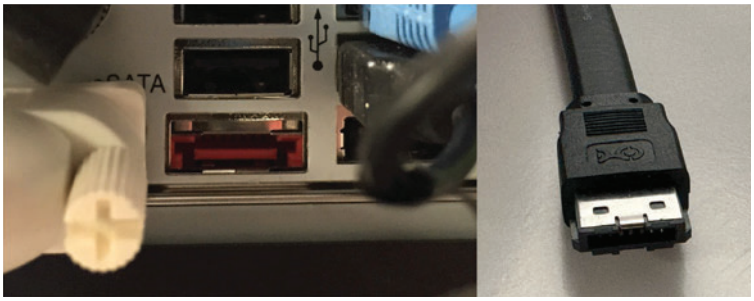
The SATA we've discussed so far is internal, but there's an external version as well, appropriately named *external SATA (eSATA)*. It uses the same technology, only in an external connection. The port at the bottom center of the left image in Figure 3.31 is eSATA, and the image on the right is an eSATA cable. It entered the market in 2003, is mostly intended for hard drive use, and can support up to 15 devices on a single bus.

Table 3.4 shows some of the eSATA specifications.

TABLE 3.4 eSATA specifications

Version	Year	Speed	Names
Revision 1.0	2003	1.5 Gbps	SATA I, SATA 1.5 Gb/s
Revision 2.0	2005	3.0 Gbps	SATA II, SATA 3 Gb/s
Revision 3.0	2009	6.0 Gbps	SATA III, SATA 6 Gb/s

FIGURE 3.31 eSATA port and cable



You will commonly see the third generation of eSATA (and SATA) referred to as SATA 6 or SATA 6 Gb/s. This is because if they called it SATA 3, there would be confusion with the second generation, which had transfer speeds of 3.0 Gbps.

An interesting fact about eSATA is that the interface does not provide power, which is a big negative compared to its contemporary high-speed serial counterparts. To overcome this limitation, there is another eSATA port that you might see, called *Power over eSATA*, eSATA+, eSATAp, or eSATA/USB. It's essentially a combination eSATA and USB port. Since the port is a combination of two others, neither sanctioning body officially recognizes it (which is probably why there are so many names—other companies call it what they want to). Figure 3.32 shows this port.

FIGURE 3.32 USB over eSATA

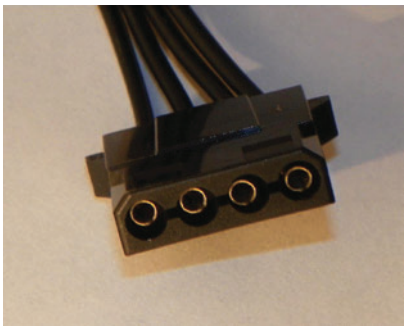
You might have noticed that the most recent eSATA specification was released over a decade ago. eSATA is basically dead, because newer standards such as USB4 and Thunderbolt 3 (and newer) provide not only power but faster data transmission speeds as well.

You can see that this port is slightly different from the one in Figure 3.31, and it's also marked with a USB icon next to the eSATA one. On the market, you can purchase cables that go from this port to an eSATA device and provide it with power via the eSATAp port.

A Curtain Call for PATA

We mentioned PATA in Volume 1 Chapter 2 and noted that it's essentially obsolete now, but you still might run into it on occasion. In addition, the power connector for PATA drives (and older optical drives) is the *Molex connector*, which still does happen to be an exam objective. So for the sake of completing our review of hard drive cables and connector types covered in the exam objectives, Figure 3.33 shows a 4-pin Molex connector. Historically, most were white or opaque plastic, but there's no standard color.

FIGURE 3.33 Molex connector



Summary

In this chapter, you first learned about peripheral types. We broke them into four categories: video, audio, input/output, and storage. Video peripherals include monitors and projectors. There aren't many audio connectors, but most use the TRS connector you learned about. Input and output devices are plentiful, and we concentrated on keyboards and mice. Storage devices and optical drives can be external, and an example is an external network-attached storage (NAS) device.

In the second section of the chapter, you learned about various cable and connection types, and the purposes and uses of peripheral types. First, you learned about peripheral cables and connectors, such as USB and its variants, Lightning, Thunderbolt, and serial. Then we moved on to video cables. Topics included the analog VGA standard, as well as the digital standards DVI, HDMI, and DisplayPort. Then, we covered hard drive connections and cables related to SATA and eSATA, with a quick nod to the PATA-era Molex power connector.

Exam Essentials

Understand the differences between display types. You should be able to know the advantages and disadvantages of using IPS, TN, and VA LCD displays, as well as OLEDs and mini-LEDs.

Know what touch screens, digitizers, and inverters are used for. Touchscreens and digitizers have the same function, which is to take physical input from a finger or stylus and convert that to digital signals for the computing device to use. An inverter is a small circuit board that converts DC power to AC power for an LCD backlight.

Know four different display attributes. Be familiar with pixel density (the number of pixels per inch), refresh rates (how often the screen is redrawn), screen resolution (the number of pixels across and down on a display), and color gamut (the range of colors a display can produce).

Recognize and understand different peripheral connectors and adapters. Expansion cards and motherboards have external connectivity interfaces. The interfaces have connectors that adhere to some sort of standard for interconnecting with a cable or external device. Knowing these specific characteristics can help you differentiate among the capabilities of the interfaces available to you. Understanding when to use an adapter to convert one connector to another is crucial to achieving connectivity among differing interfaces. Adapters you should know are DVI to HDMI, USB to Ethernet, and DVI to VGA.

Recognize and be able to describe display connectors specifically. Although a type of peripheral connector, display connectors are in a class all their own. Types include VGA, various versions of DVI, HDMI, mini-HDMI, DisplayPort, and USB-C.

Recognize and understand the purpose of hard drive cables and connectors. The connectors you should recognize are SATA and eSATA. Molex connectors are used to power older PATA devices. They each connect hard drives or optical drives.

Know the various peripheral cables and their connectors. Multipurpose cables include USB, Lightning, Thunderbolt, and serial. USB has the largest variety of connectors, including USB-A and USB-B and their mini- and micro- versions, as well as the newer USB-C. USB cables also can have proprietary connectors, such as Apple's Lightning connector. Thunderbolt can use a proprietary connector or USB-C. Serial cables have a DB-9 connector.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which of the following could be used with an analog monitor? (Choose two.)
   - **A.** HDMI
   - **B.** DVI-A
   - **C.** DVI-D
   - **D.** DisplayPort
   - **E.** VGA

2. You are attaching an external hard drive to a computer with USB 2.0 ports. The user is concerned about data transfer speed. What is the maximum speed of USB 2.0?
   - **A.** 480 Mbps
   - **B.** 5 Gbps
   - **C.** 10 Gbps
   - **D.** 20 Gbps

3. You got a new desktop computer for a designer at your company. On the front panel is a rectangular yellow USB port. What does that mean?
   - **A.** It's USB 2.0.
   - **B.** It's USB 3.0.
   - **C.** It's a port capable of charging devices even with the system powered off.
   - **D.** It's a port to be used with USB-C compatible devices.

4. You are installing a replacement monitor on a desktop computer. Which of the following is true regarding a monitor's refresh rate?
   - **A.** As long as the graphics adapter can refresh the image at a particular rate, the attached monitor can accommodate that refresh rate.
   - **B.** The refresh rate is normally expressed in MHz.
   - **C.** The refresh rate is normally selected by using the controls on the front panel of the monitor.
   - **D.** As you lower the resolution, the maximum refresh rate allowed tends to increase.

5. Which of the following connector types is typically not associated with hard drive use?
   - **A.** USB-C
   - **B.** eSATA
   - **C.** DB-9
   - **D.** SATA

6. The company photographer is interested in buying a new digital camera. He wants to be able to plug it directly into his laptop to transfer pictures for editing. What type of connector will most likely be on his camera?
   - **A.** Mini-USB
   - **B.** Thunderbolt
   - **C.** Molex
   - **D.** Lightning

7. You are installing a new video card that has an HDMI port. Which of the following are true about HDMI cables? (Choose all that apply.)
   - **A.** They can transmit analog or digital signals.
   - **B.** They can transmit audio as well as video.
   - **C.** They are not used to connect to projectors.
   - **D.** They can use HDMI or mini-HDMI connectors.

8. Which of the following display interfaces is equivalent to DisplayPort with PCIe added in?
   - **A.** Thunderbolt
   - **B.** HDMI
   - **C.** DVI
   - **D.** VGA

9. Which of the two standards use the same connector? (Choose two.)
   - **A.** eSATA
   - **B.** Thunderbolt 3
   - **C.** USB4
   - **D.** USB 2.0
   - **E.** Mini-USB

10. How do you connect a DVI-A interface on a peripheral to a DVI-D interface on the computer?
    - **A.** With a DVI-I cable.
    - **B.** With a cable that is terminated on one end with a DVI-A connector and on the other end with a DVD-D connector.
    - **C.** You wouldn't interconnect those two interfaces.
    - **D.** With a standard DVI cable.

11. A technician says he needs to provide power to a PATA hard drive and a DVD-ROM in a desktop. The computer now has a new fully modular power supply, and the technician wants to know what type of power connectors he should take. What do you tell him?
    - **A.** SATA
    - **B.** SCSI
    - **C.** Serial
    - **D.** Molex

12. Which of the following connectors is oval shaped?

A. Micro-USB

B. Mini-USB

C. Thunderbolt 2

D. Thunderbolt 3

13. Which of the following connector types are capable of conducting power and transmitting data? (Choose all that apply.)

A. Lightning

B. Molex

C. USB-C

D. DB-9

E. Micro-USB

14. Which of the following statements is true about eSATA cables and connectors?

A. eSATA cables provide power and data to devices.

B. eSATA connectors are smaller than USB-C connectors.

C. eSATA is faster than USB4 and Thunderbolt 3.

D. eSATA maximum throughput is 6 Gbps.

15. Which of the following is not an example of a connector for a standard peripheral input device?

A. 1/8″ jack

B. Molex

C. D-sub

D. USB

16. You are asked to hook up a projector in a classroom. What type of cable are you most likely to need?

A. USB Type A

B. eSATA

C. HDMI

D. DB-9

17. A user with an iPhone 14 needs to charge her device and asks you to bring a cable. Which of the following connector types should you ensure that the cable has?

A. Thunderbolt

B. Lightning

C. Mini-USB

D. Micro-USB

18. Which of the following types of cables will transfer data at the slowest rate?

A. USB4

B. Serial

C. Thunderbolt 3

D. HDMI

19. What is the maximum data transfer rate of USB High Speed?

A. 10 Gbps

B. 5 Gbps

C. 480 Mbps

D. 12 Mbps

20. VGA cables use what type of signal between the adapter and monitor?

A. Digital

B. Analog

C. Composite

D. Compressed

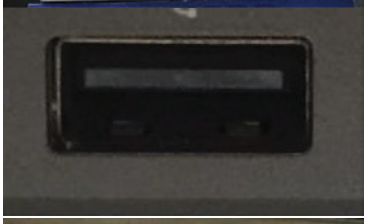
Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

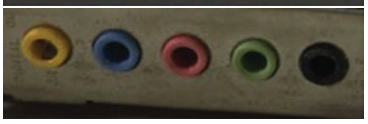
Looking at the back of a computer, you see the interfaces shown in the following graphic. Which type of cables do you need to plug into each one?

A:

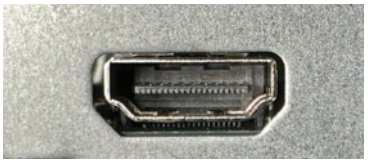
B:



C:

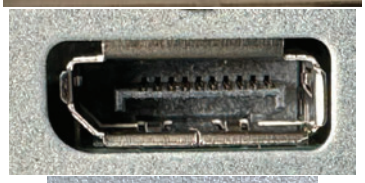


D:



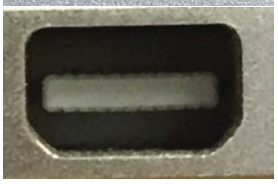
E:

F:



G:

H:



Chapter

4

Laptop and Mobile Device Hardware

THE FOLLOWING COMPTIA A+ 220-1201 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

- ✓ **1.0 Mobile Devices**
- ✓ **1.1 Given a scenario, monitor mobile device hardware and use appropriate replacement techniques.**
  - Battery
  - Keyboard/keys
  - Random-access memory (RAM)
  - Hard disk drive (HDD)/solid-state drive (SSD)
  - Wireless cards
  - Physical privacy and security components
    - Biometrics
    - Near-field scanner features
  - Wi-Fi antenna connector/placement
  - Camera/webcam
  - Microphone
- ✓ **1.2 Compare and contrast accessories and connectivity options for mobile devices.**
  - Connection methods
    - Universal Serial Bus (USB)/USB-C/microUSB/miniUSB
    - Lightning
    - Near-field communication (NFC)
    - Bluetooth
    - Tethering/hotspot

- Accessories
  - Stylus
  - Headsets
  - Speakers
  - Webcam
- Docking station
- Port replicator
- Trackpad/drawing pad/track points

In 1943, the president of IBM, Thomas Watson, was quoted as saying, "I think there is a world market for maybe five computers." Somewhat more recently, in 1977, Ken Olsen, the founder of one-time computer industry giant Digital Equipment Corporation (DEC), stated, "There is no reason anyone would want a computer in their home." Four years later the personal computer was introduced, ironically by IBM. It's about 80 years past the first quote and about 50 years past the second, but looking back at the history of computers, it's amazing to see how far the industry has come.

As recently as the 1990s, portable computers were luxuries that were affordable to only the wealthy or the select few businesspeople who traveled extensively. As with all other technologies, though, portable systems became smaller, lighter (more portable), more powerful, and less expensive. Because the technology and price disparity between the two platforms has decreased significantly, laptops have outsold desktops since the mid-2000s. For about 20 years, laptop computers enjoyed a status as the most popular mobile computing device available. Desktop computers still were more powerful and cheaper, but with a laptop, users weren't tethered to a desk.

Technology continued its inevitable progression, and smaller mobile devices came into play. Tablets, smartphones, and even wearable devices obtained enough power and features to be considered computers. Laptop sales have been flat since about 2012, whereas smartphones and wearables have continued to increase in market share. Further, smartphones overtook laptops as the most popular device to access the Internet as of 2015. In stark contrast to the computers of yesteryear, today's most popular computing devices can be so small that the likelihood of misplacing them is a real and ongoing concern.

Every indication is that the movement toward mobile computing will continue, so you definitely need to be well versed in portable technologies, which contain both nifty features and frustrating quirks. In this chapter, you will learn about the features and quirks of laptop and mobile device hardware. There are some differences—for example, the majority of hardware in a laptop is more similar to that of a desktop than to that of a smartphone, but there are similarities as well. The display of a laptop (and to be fair, most desktop-sized monitors) is a lot like those in smaller devices. Throughout the chapter, we will cover laptops and smaller mobile devices as though they are similar, and call out specific differences where appropriate. We will start with installing and configuring laptop hardware and components, including details on display technologies. Then we'll finish with setting up and configuring accessories and ports on mobile devices.

Working with Laptop and Mobile Device Hardware

Hardware in all computing devices needs to perform the same tasks. There are devices that control input and output, processing, short-term and long-term storage, displaying information, and connecting to other computers. This is true regardless of the size of the device. Granted, smaller devices are space-constrained, so the hardware components that do these tasks need to be smaller and consume far less energy.

Working with hardware, then, should be similar across devices, and for the most part it is. Maybe instead of having a physical keyboard you'll have a virtual one (and you can't exactly replace it, per se), but at least you know how to input to the device. Or maybe the display is smaller, but it and the video card still perform the same tasks that their larger counterparts do. In the following sections, we will look at the specifics regarding working with hardware in laptops and smaller devices.

Understanding the Differences Between Device Types

The first personal computers developed were more similar to today's desktop computers than they were to laptops or mobile devices. The smaller devices do trace their ancestry back to the desktop, though, so it makes sense to compare them to their bigger counterparts. Here, we'll take a high-level look at what makes laptops and mobile devices unique.

Laptops vs. Desktops

Laptops are similar to desktop computers in architecture in that they contain many parts that perform similar functions. However, the parts that make up a laptop are completely different from those in desktop computers. The obvious major difference is size; laptops are space challenged. Another primary concern is heat. Restricted space means less airflow, meaning parts can heat up and overheat faster.

To overcome space limitations, laptop parts are physically much smaller and lighter, and they must fit into the compact space of a laptop's case. It might not sound like much, but there really is a major difference between a 4-pound laptop and a 5-pound laptop if you're hauling it around in its carrying case all day. Also, laptop parts are designed to consume less power and to shut themselves off when not being used, although many desktops also have components such as video circuitry that go into a low-power state when not active. Finally, most laptop components are proprietary—not just to the manufacturer but even to the specific model. The motherboard is especially proprietary, and the *liquid crystal display (LCD)* screen from one laptop will not necessarily fit on another.

Manufacturers have also pushed out smaller and smaller portables that are laptop adjacent. For example, in 2007 the first netbooks were introduced. A *netbook* is an extremely small laptop computer that is lighter in weight and more scaled down in features than a standard laptop. The term *netbook* is rarely used today, but Chromebooks are an example of that type of technology. Users are attracted to Chromebooks because of their enhanced portability and affordability. The features that remain are ideal for Internet access and emailing.

However, many users would find them insufficient for mainstream business usage. Tablets are even smaller yet, but they are typically characterized as a mobile device, so we'll hold off on talking about them just yet.

If you've shopped for a laptop, you have no doubt noticed that the prices of desktop PCs are often quite a bit lower than those for laptop computers, yet the desktops are usually faster and more powerful. If you've ever wondered what makes a laptop so much different from a PC, here are the primary differences between laptops and desktops:

Portability This is probably the most obvious difference. Laptops are designed to be portable. They run on batteries, so you aren't tied to one spot at home or at the office. Networking options are available that allow you to connect to a network wirelessly and do work from just about anywhere, including airports, coffee shops, and so on. As anyone who has tried to bring their full-tower PC to a LAN party can tell you, desktops just aren't that portable.

Cost Laptops tend to cost more than desktop computers with similar features. The primary reason is that portability requires smallest components and unique proprietary designs so that those components fit into the smallest size necessary. Miniature versions of components cost more money than standard-sized (desktop) versions. The cost discrepancy between desktops and laptops has shrunk considerably in the last several years, but it still exists.

Performance By and large, laptops are always going to lose out somewhere in the performance department. Compromises must often be made between performance and portability, and considering that portability is the major feature of a laptop, performance is what usually suffers. While it is possible to have a laptop and a desktop with comparable performance, the amount of money one would have to spend for a "desktop replacement" laptop is considerable. This is not to say that a laptop can't outperform a desktop—it's just that the "bang for the buck" factor is higher in a desktop.

Expandability Because desktop computers were designed to be modular, their capabilities can be upgraded quite easily. It is next to impossible to upgrade the processor or motherboard on most laptops. Other than memory and hard drives, most laptop upgrades consist of adding an external device through one of the laptop's ports, such as a USB port.

Quality of Construction Considering how much abuse laptops get, it is much more important that the materials used to construct the laptop case and other components be extremely durable. Durability is important in a desktop too, but it won't be tested as much as in a laptop.

Building Your Own

This anecdote comes from one of the authors: "During an A+ course, I gave the class the assignment to go out on the web and put together the most powerful and complete computer they could for under a thousand dollars. The class was for non-degree-seeking adults,

so nothing was graded; it was simply to provide experience with spec'ing out and pricing the parts that go into making a complete system.

"One of the students had her eye on a new laptop for personal use. Because she noticed the trend toward being able to build a desktop computer for less than she could buy one, the student assumed the same about laptops. Unfortunately, I had not specifically mentioned the fact that there are no standards for building complete laptop clones, unlike with desktops.

"You can't reliably build your own laptop. Because laptop components are designed to exacting specifications to fit properly inside one manufacturer's case, there generally are no universal motherboards, video boards, and so on for laptops. Memory and hard drives are the exception. You can get different brands of memory and hard drives for laptops, but you can't buy a motherboard from one company and the video circuitry from another."

Characteristics of Mobile Devices

If you were asked to define the primary characteristic of mobile devices, you would probably answer, "They are small," and you wouldn't be wrong. Three overarching characteristics of mobile devices make working with them unique versus working with laptops or desktops: field servicing and upgrading, input methods, and secondary storage. We'll discuss each one in turn.

Field Servicing and Upgrading

Ever since the dawn of the portable computer, manufacturers and service providers have based a percentage of their success on warranties and "house calls" to repair devices on the fritz. It's a fact that quasi-permanent components, such as displays and motherboards, are widely considered replaceable only with identical components in laptops and smaller devices. However, technically minded users could take it upon themselves to expand the capabilities of their own system by, for instance, upgrading the hard drive, increasing RAM, using expansion cards and flash devices, and attaching wired peripherals.

Although the ability to repair and expand the functionality of portable devices in the field has become all but obviated, it has been shown with current and past generations of mobile devices that users are not averse to giving up expandability and replaceable parts as long as functionality and convenience outshine the loss.

Although many Android and other non-Apple devices allow the replacement of batteries and the use of removable memory cards as primary storage, even this basic level of access is removed in Apple's mobile devices, including its iPad line of tablet computers. In an effort to produce a sleeker mobile phone, even Android devices have been developed without user access to the battery. For Apple, however, in addition to producing a nice compact package, it is all part of keeping the technology as closed to adulteration as possible. Supporters of this practice recognize the resulting long-term quality. Detractors lament the lack of options.

To service closed mobile devices of any size, you may have to seek out an authorized repair facility and take or send your device to them for service. Attempting your own

repairs can void any remaining warranty, and it can possibly render the device unusable (also known as "bricking" a phone, because you basically limited its functionality to that of a brick). For example, a special screwdriver-like tool is required to open Apple's devices. You cannot simply dig between the seams of the case to pop the device open. Even if you get such a device to open, there is no standard consumer pipeline for parts, whether for repair or upgrading. If you want to try the repair yourself, you could be on your own. You may be able to find helpful videos on YouTube or `www.ifixit.com` to provide some guidance, though.

Anyone who has been around the business for more than just a few years has likely seen their fair share of components and systems with no user-serviceable parts. For these situations, an authorized technician can be dispatched to your location, home or work, with the appropriate tools, parts, and skills to field-service the system for you. On a slightly different, perhaps subtler note, the bottom line here is that many of today's mobile devices, including some of the larger tablet-style devices, have no field-serviceable parts inside, let alone user-serviceable parts. In some extremes, special work environments similar to the original clean manufacturing environment have to be established for servicing.

Input Methods

With decreased size comes increased interaction difficulties. Human interfaces can become only so small without the use of projection or virtualization. In other words, a computer the size of a postage stamp is fine as long as it can project a full-sized keyboard and a 60″ display, for example. Using microscopic real interfaces would not sell much product. Thus, the conundrum is that users want smaller devices, but they do not want to have to wear a jeweler's loupe or big-screen virtualization glasses to interact with their petite devices.

As long as the size of the devices remains within the realm of human visibility and interaction, modern technology allows for some pretty convenient methods of user input. Nearly all devices, from tablet size down, are equipped with touchscreens, supplying onscreen keyboards and other virtual input interfaces. On top of that, more and more of the screens are developing the capability to detect more than one contact point.

We discuss touchscreens more in the section "Input Devices," later in this chapter.

Generically, this technology is referred to in the industry as multitouch, and it is available on all Apple devices with touch input, including the touchpads of the Apple laptops. Apple, through its acquisition of a company called FingerWorks, holds patents for the capacitive multitouch technology featured on its products. Today, multitouch is more about functionality than novelty. Nevertheless, the markets for both business and pleasure exist for multitouch.

Certainly, touchscreens with the capability to sense hundreds of separate points of contact can allow large-scale collaboration or fun at parties. Imagine a coffee table that can allow you to pull out a jigsaw puzzle with the touch of an icon, remembering where you and three friends left off. Imagine all of you being able to manipulate pieces independently and simultaneously and being able to send the puzzle away again as quickly as you brought it out so that you can watch the game on the same surface. This technology exists, and it is for

sale today. Early examples were built on Microsoft's PixelSense technology, including the Samsung SUR40. Companies like Ideum build multitouch platform tables, including a monster 86″ 4K ultra-high-definition (UHD) display with 100 touchpoints, allowing up to eight people to use it simultaneously.

On a smaller scale, our mobile devices allow us to pinch and stretch images on the screen by placing multiple fingers on that screen at the same time. Even touchpads on laptops can be made to differentiate any number of fingers being used at the same time, each producing a different result, including pointing and clicking, scrolling and right-clicking, and dragging—all one-handed with no need to press a key or mouse button while gesturing.

HTC created an early touchscreen software interface called TouchFLO that has matured into HTC Sense, and it is still in use today on its Android-based mobile devices. TouchFLO is not multitouch capable, nor does it specify the physical technology behind the touchscreen, only the software application for it. Theoretically, then, TouchFLO and multitouch could be combined.

The primary contribution of TouchFLO was the introduction of an interface that the user perceives as multiple screens, each of which is accessible by an intuitive finger gesture on the screen to spin around to a subsequent page. On various devices using this concept, neighboring pages have been constructed side by side or above and below one another. Apple's mobile devices employ gestures owing to the contributions of TouchFLO, bringing the potential of combining TouchFLO-like technology and multitouch to bear.

Users of early HTC devices with resistive touchscreen technology met with difficulty and discord when flowing to another screen. The matte texture of the early resistive screens was not conducive to smooth gesturing. Capacitive touchscreen technology is a welcome addition to such a user interface, making gestures smooth and even more intuitive than ever.

Recall from Volume 1 Chapter 3, "Peripherals, Cables, and Connectors" that resistive touchscreens use pressure and capacitive touchscreens use a change in electrical conductivity to detect a touch.

Secondary Storage

Computers of all sizes and capabilities use similar forms of RAM for primary storage—the storage location for currently running instructions and data. Secondary storage—the usually nonvolatile location where these instructions and data are stored on a more permanent basis—is another story.

The primary concern with smaller devices is the shock they tend to take as the user makes their way through a typical day. Simply strapping a phone to your hip and taking the metro to work presents a multitude of opportunities for a spinning disk to meet with catastrophe. The result would be the frequent loss of user information from a device counted on more and more as technology advances.

Just as most telephony subscribers have migrated from a home landline that stays put to a mobile phone that follows them everywhere, many casual consumers are content to use their mobile device as their primary or only computing system, taking it wherever they go. As a result, the data must survive conditions more brutal than most laptops because laptops are most often shut down before being transported.

The most popular solution is to equip mobile devices with very small solid-state drives (SSDs) in place of larger magnetic or solid-state drives. There are no moving parts, the drive stays cooler and resists higher temperature extremes, and SSDs require less power to run than their conventional counterparts.

Now that we've illustrated the primary differences between laptops, mobile devices, and desktops, let's examine some principles for taking laptops apart and putting them back together.

Disassembling and Reassembling Laptops

Desktop computers often have a lot of empty space inside their cases. This lets air circulate and also gives the technician some room to maneuver when troubleshooting internal hardware. Space is at a premium in laptops, and rarely is any wasted. With a desktop computer, if you end up having an extra screw left over after putting it back together, it might not be a big deal. With laptops, every screw matters, and you'll sometimes find yourself trying to identify visually miniscule differences between screws to make sure that you get them back into the right places.

Even though repairing a laptop poses unique issues, most of the general troubleshooting and safety tips that you use when troubleshooting a desktop still apply. For example, always make sure that you have a clean and well-lit workspace and be cautious of electrostatic discharge (ESD). General safety tips and ESD prevention are covered in Volume 2 Chapter 9, "Safety and Environmental Concerns." For now, our general advice is to use antistatic mats or wrist straps if they're available.

Throughout this section, we'll use the word *laptop* almost exclusively. The principles covered here apply to nearly all portable devices, though, even tablets and smartphones.

One of the key principles for working with laptops is using the right tools to tear the thing apart. It's doubtful that any technician goes into a job thinking, "Hey, I'm going to use the wrong tools just to see what happens." With laptops, though, it's especially important to ensure that you have exactly the tools you'll need for the job. The two main camps of materials you need are the manufacturer's documentation and the correct hand tools. We'll also emphasize the importance of organization.

Using the Manufacturer's Documentation

Most technicians won't bat an eye at whipping out their cordless screwdriver and getting into a desktop's case. The biggest difference between most desktops is how you get inside the case. Once it's opened, everything inside is pretty standard fare.

Laptops are a different story. Even experienced technicians will tell you to not remove a single screw until you have the documentation handy and have taken pictures of the pre-surgery state (if you're allowed to) unless you're incredibly familiar with that particular

laptop. Most laptop manufacturers give you access to repair manuals on their websites. Table 4.1 lists the service and support websites for some of the top laptop manufacturers.

TABLE 4.1 Laptop manufacturers' service and support websites

Company	URL
Apple	`https://support.apple.com/mac`
Asus	`https://www.asus.com/support`
Dell	`https://www.dell.com/support`
HP	`https://support.hp.com`
Lenovo	`https://support.lenovo.com`
Sony	`https://www.sony.com/electronics/support`

If the site you need isn't listed, a quick Google search should do the trick. Once you are at the right website, search for the manual using the laptop's model number.

Some laptop manufacturers have a policy that if you open the case of a laptop, the warranty is voided. Be sure to understand your warranty status and implications of cracking the case open before you do it.

Using the Proper Hand Tools

Once you have the manual in hand or on your screen, you must gather the proper hand tools for the job. For some laptops, you only need the basics, such as small Phillips-head and flat-head screwdrivers. For others, you may need a Torx driver. Gather the tools you need and prepare to open the case. A small flashlight might also come in handy. Small PC technician toolkits are readily available online or from your favorite electronics retailer. They may have a few different sizes of screwdrivers, hex drivers, Torx drivers, tweezers, a screw grabber, and a few other assorted goodies, all in a convenient carrying case. An example is shown in Figure 4.1. Find one you like and never leave home without it.

The Consequences of Using the Wrong Tools

It's been said once, but it's important to say it again: always use the right tool for the job when repairing laptops. If the documentation says that you need a T-10 Torx driver, make sure that you have a T-10 Torx driver.

Not using the right tools can result in the stripping of the screw head. If you strip a screw head in a desktop, you might have alternative methods of removing the screw. Laptops are far less forgiving. If you strip a screw head and are unable to turn the screw, you may never be able to remove it. That could result in needing to scrap the device.

FIGURE 4.1 PC technician toolkit

Organization and Documentation

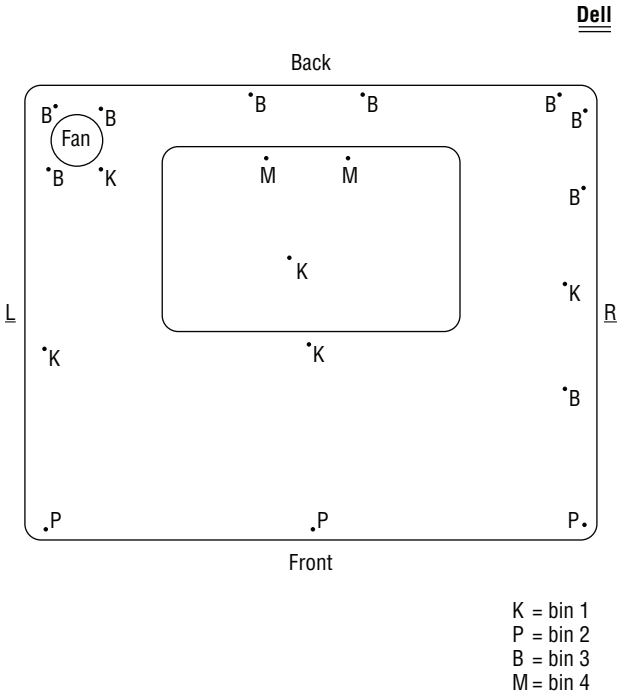
Before you crack open the case of your laptop, have an organization and documentation plan in place. Know where you are going to put the parts. Have a container set aside for the screws. You can purchase small plastic containers that have several compartments in them with lids that snap tightly shut, into which you can place screws. You can also use containers designed to organize prescription pills or fishing tackle. The bottom of an egg carton works well too, provided that you don't need to transport the screws from place to place. You don't want the screws falling out and getting lost!

For documentation, many technicians find it handy to draw a map of the computer they're getting into, such as the one shown in Figure 4.2. It can be as complex as you want it to be, as long as it makes sense to you. Taking pictures with your phone is also a smart move, provided that you're allowed to use your phone and don't violate any security or privacy policies.

The drawing in Figure 4.2 shows the locations of the screws, and it also calls out where the screws should be placed once they're removed. Again, this type of documentation can be as simple or complex as you want it to be, as long as it makes sense and helps you stay organized.

Now that we've covered some key principles, let's take a look at specific components, technologies involved, and how to install and configure them.

FIGURE 4.2 Laptop repair road map



Installing and Configuring Laptop Hardware

In the following sections, you will learn about the various components that make up laptops and how they differ from desktop computer components. These sections deal specifically with laptops, because smaller devices generally don't have field-replaceable components (or you can get specialized training on how to repair them). If you don't remember exactly what each component does, it may help you to refer back to earlier hardware chapters occasionally as you read this chapter.

Laptop Case

A typical laptop case is made up of three main parts:

- The display—usually an LCD, light-emitting diode (LED), or organic light-emitting diode (OLED) display
- The case structure, which is the metal reinforcement inside the laptop that provides rigidity and strength and to which most components are mounted
- The case, or the plastic or aluminum cover that surrounds the components and provides protection from the elements

Some laptops have cases made of a strong, lightweight metal, such as aluminum or titanium. However, the majority of laptop cases are made of plastic, usually acrylonitrile butadiene styrene (ABS) or an ABS composite.

Laptop cases are made in what is known as a *clamshell design*. In a clamshell design, the laptop has two halves, hinged together at the back. The display portion, called the top half, often includes a webcam, microphone, and Wi-Fi antenna. All other components, including the motherboard, memory, storage, keyboard, battery, cooling fan, and speakers, are in the bottom half.

Occasionally, part of the laptop's case or the device's frame will crack and need to be replaced. However, you usually can't just replace the cracked section. Most often, you must remove every component from inside the laptop's case and swap the components over to the new one. This is a labor-intensive process because the screws in laptops are often very small and hard to reach.

Often, repairing a cracked case may cost several hundred dollars in labor alone. Most times, people who have cracked laptop cases wait until something else needs to be repaired before having the case fixed. Or, they just wait until it's time to upgrade to a new system. The decision on when to repair or replace the laptop boils down to a few factors. The primary one is if the user can live with the damage. While they can be annoying, most case problems don't inhibit the operation of the machine. The secondary factor is money. The user (or company) needs to decide if it's really worth spending the money needed to fix the issue immediately.

One of the components you may need to service is the speakers, which are generally built into the bottom part of the case. Exercise 4.1 will provide an example of removing speakers from a Dell Inspiron 13 7000 laptop. We'll use that model for an example throughout this chapter. We realize that this specific model is a few years old and you might not have access to one, but the exercises can help you understand the process. Besides, there are two key points regarding these exercises:

- The basic clamshell laptop design hasn't changed in over three decades. Parts have gotten smaller but they are likely in similar places to where their predecessors were.
- It would be impractical to list the specific replacement steps for all makes and models out there. If you have a different model, download its service manual and perform the necessary steps.

To get the most out of the exercises in this chapter, it's best if you're able to perform them on a laptop. However, we realize that not everyone has access to a laptop they can tear apart. If you don't have a laptop you can perform surgery on, here are a few options you can try:

- Download a service manual for the laptop you have (or another model that seems interesting). Look for directions on how to remove or replace the components we describe so that you can get a feel for how you would perform the tasks.
- Visit a local computer repair shop and ask if they have any old laptops that they are going to take to recycling. Explain that you are studying for the A+ exam and would like to get experience taking apart a laptop. See if they will let you take one apart at their facility. (You never know until you ask!)
- Visit YouTube or `www.ifixit.com` and look up videos for how to replace components on your laptop or another model that seems interesting.

Before we get to Exercise 4.1, we want to remind you of a few safety steps to take before working on a laptop. We're not going to repeat these instructions before every exercise, but always perform these steps before beginning digging into a laptop (or a desktop, for that matter):

1. Turn off the computer.
2. Disconnect all external peripherals and cables, such as keyboards, mice, monitors, and network cables.
3. Unplug it from the power source.
4. Ensure that your workspace is free of clutter and well lit.
5. Have the manual or help videos handy, just in case.
6. Make sure you have the proper tools readily available. (This might include a phone to take pictures with.)
7. Ground yourself with an ESD wrist strap or other antistatic protection.

EXERCISE 4.1

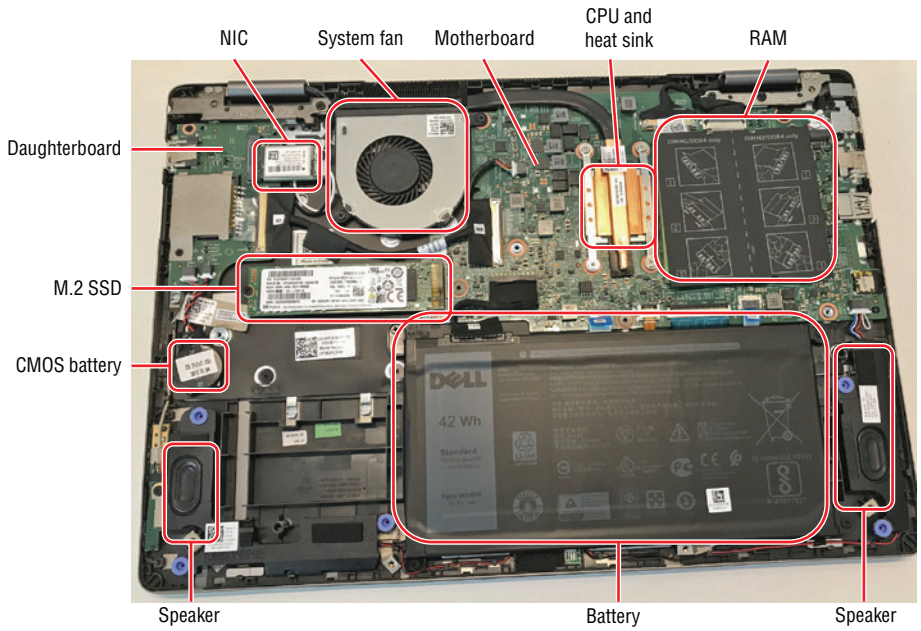
Removing Speakers from a Laptop

1. Remove the bottom cover. As shown in Figure 4.3, this model requires removing nine screws to take the cover off.

FIGURE 4.3 Removing nine screws to take off the bottom cover

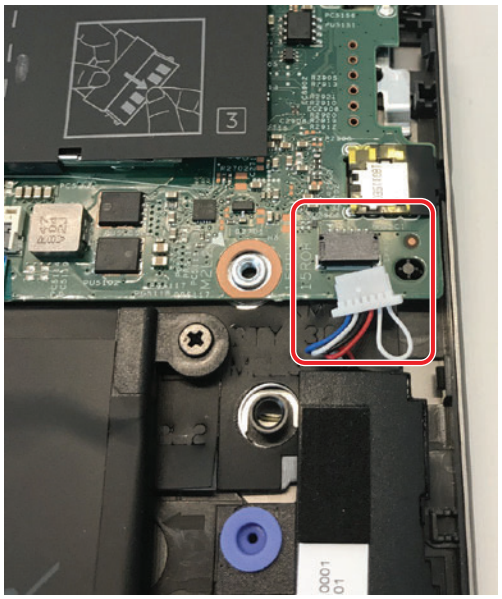
2. Make a mental note of the key components and their location, or better yet, take a picture with your phone like the one in Figure 4.4.

FIGURE 4.4 Inside the laptop



3. Disconnect the speaker cable from the motherboard, as shown in Figure 4.5.

FIGURE 4.5 Disconnecting the speaker cable



EXERCISE 4.1 *(continued)*

4. Remove the speaker cable from the wire routing guides on the base assembly.

   Figure 4.6 shows the speaker wire that travels from the left speaker to the right speaker.

FIGURE 4.6 The speaker wire

5. Remove the rubber grommets that hold the speakers in.

   Figure 4.6 shows the two from the right speaker. The left speaker has three.

6. Carefully remove both speakers from the base assembly.

To replace the speakers, follow the steps in reverse order. Insert the new speakers, replace the grommets, run the speaker wire through the guides, and connect the speaker to the motherboard.

Removing the speakers from the Dell Inspiron 13 7000 was relatively easy. On some laptops, the speakers or speaker wire are completely buried underneath several other components, and you need to remove those first. As we move through this chapter, you will see situations where you need to remove several components before you can get to the one you want.

Laptop Displays

The display system is the primary component in the top half of the clamshell case. (The wireless antenna, webcam, and microphone often reside here too, and we'll get to them in just a bit.) Much like all other laptop components, the display is more or less a smaller version of its desktop counterpart. What is unique to laptop displays, though, is that for some

time, the technology used in them was actually more advanced than what was commonly used in desktops. This is due to LCD technology.

Before LCD technology, computer displays used cathode-ray tube (CRT) technology (like old-school televisions) and were big, bulky, and hardly mobile. We introduced LCD, LED, and OLED concepts in Volume 1 Chapter 3, but we'll do a quick refresher in the "Screen" section after we talk about video cards. Let's quickly cover the different components that are required to make these types of displays work on a laptop.

Some laptops feature screens that rotate orientation depending on how they are held (much like smartphones and tablets), or the screens themselves will rotate horizontally or be entirely removable. Rotating and removable screens cost more money, so consider your needs or the needs of the user when recommending such features.

Video Card

The video card in a laptop or desktop with a flat-screen (LCD, LED, OLED, or Mini-LED) monitor does the same thing, regardless of what type of machine it's in. It's responsible for generating and managing the image sent to the screen. In this section we will bundle all of those types of monitors together as LCDs for the sake of brevity. LCD monitors are digital, so laptop video cards generate a digital image. Laptop manufacturers put video cards that are compatible with the given display in a laptop, and most laptop manufacturers choose to integrate the LCD circuitry on the motherboard to save space.

Odds are that the laptop you're using has an integrated video card. If the video card fails, you're likely looking at a motherboard replacement. Very few laptops have a replaceable video card. For those that do, replacing or upgrading it will probably resemble replacing system memory.

Real World Scenario

Video Memory Sharing

If your video card is built into your motherboard, it likely won't have its own memory but will share system memory with the processor. Note that there is nothing wrong with this type of setup; in fact, it often brings the cost of the laptop down. It's just that instead of having 16 GB of RAM and 4 GB of video RAM (for example), you would have only 16 GB total. So if your video card were using 4 GB, the system would be left with only 12 GB. That will slow everything down.

How much of a difference does all of this make? Well, it depends on what you're doing with your laptop. If you're using it for the Internet and light work, probably not much difference.

If you're working with more video-intensive applications, using a computer with shared memory might slow things down to an unacceptable level. This usually brings up two questions: What's the optimal balance? And where do I change this?

To answer the first question, again, it depends on what you are doing. If you perform more video-intensive operations (or if you're gaming), then you might want to set aside more memory for the video card. If you're not as concerned with rapid pixilation, then less is fine. Which brings us to the second question: Where do you set it? Shared memory is configured in the system BIOS/UEFI. Each BIOS/UEFI is different, so be sure to consult your owner's manual if you have any questions. Keep in mind that some BIOSs/UEFIs will allow you to set aside only a certain amount of memory—say, 512 MB to 2 GB—for video memory.

How does this affect your computer when you upgrade the memory? First, keep in mind that some of your memory will be taken by the video card, so you might want to upgrade to more memory than you originally had planned, if the laptop supports it. Second, after upgrading the memory, you will need to go into the BIOS/UEFI and reconfigure how much memory you want allocated to the video card.

Screen

The screen, as you would expect, shows the image just as it does on a full-sized, stand-alone display unit. It will function the exact same way as an LCD, LED, OLED, touchscreen, or whatever display technology you want to talk about. We won't review each of the different types of display technologies here—refer back to Volume 1 Chapter 3 if you need a refresher. What we will do, however, is provide a quick reminder of key components that make the screen work.

Backlight The *backlight* provides lighting for LCD screens. Recall that OLED screens don't need backlights.

Inverter For systems with a backlight, the *inverter* is needed to provide AC power to it. Inverters often look like a long, thin rod. Keep in mind that they can store energy, which can be shocking to the unwary technician.

Digitizer Many laptop displays are touchscreen, and the digitizer is the component that translates the touch into electrical signals for the computer. Most of the time on laptops, the glass part of the display contains the digitizer, webcam, and microphone.

Webcams are nearly universal on laptops today. The most common placement is right above the display on the laptop, although some are below the screen. Most laptops with webcams will also have a smaller circle to the left of the lens, which has a light that turns on to illuminate the user when the webcam is on. Some people are a bit paranoid about their webcams, and they will put something over the camera to block it when it's not in use. Many laptops will have a built-in, slidable shutter for this purpose.

Exam Objective 1.1, "Given a scenario, monitor mobile device hardware and use appropriate replacement techniques," highlights three components in the display you should know how to replace: Wi-Fi antenna connector, camera/webcam, and microphone.

If you're going to use a webcam to conduct a video call with someone, it helps if they can hear you too. That's where the *microphone* comes into play. Microphones are often built into the display unit as well. The webcam shown in Figure 4.7 has the illumination light to the left and the microphone inputs on both sides of the lens. Generally speaking, the illumination light will only be on when the webcam is active—it can be a good signal to the user as to whether they are "live" or not. Microphones can also be built into the bottom half of the clamshell, either above the keyboard or somewhere on the front bezel.

FIGURE 4.7 Webcam and microphone

Wi-Fi Antenna

Practically all laptops produced today include built-in Wi-Fi capabilities. Considering how popular wireless networking is today, it only makes sense to include 802.11 functionality without needing to use an expansion card. With laptops that include built-in Wi-Fi, the *Wi-Fi antenna* is generally run through the upper half of the clamshell case. This is to get the antenna higher up and improve signal reception. The wiring will run down the side of the display and through the hinge of the laptop case and plug in somewhere on the motherboard.

The Wi-Fi antenna won't affect what you see on the screen, but if you start digging around in the display, know that you'll likely be messing with your wireless capabilities as well.

It's worth repeating again that the Wi-Fi antenna wiring usually runs through one of the hinges of the laptop case. Damage to a hinge can cause Wi-Fi problems. Later in the "Other Internal Components" section, we have an exercise on removing the Wi-Fi card, in which you will disconnect the antenna wires.

Replacing Components Within a Laptop Display

While going through the exercises in this chapter, you've probably noticed that the inside of a laptop is tight and complex, and navigating it can be a challenge. Laptop displays are the

same way. There are not as many components as compared to the bottom half of a laptop, but space is even tighter. Within a display, the components you might need to replace include the screen (or touchscreen/digitizer, which some manufacturers call the display panel), Wi-Fi antenna, webcam, microphone, and inverter.

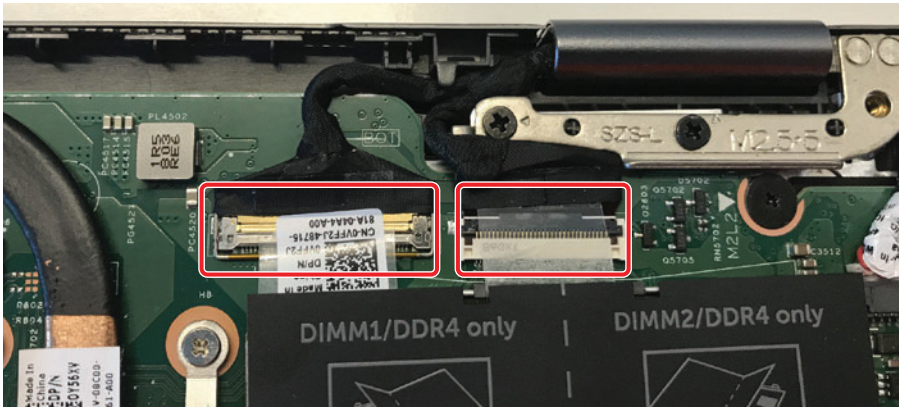
In most cases, servicing components within the display means removing the display assembly from the base assembly. Exercise 4.2 shows you how to do that.

EXERCISE 4.2

Removing the Display Assembly

1. Remove the bottom cover.
2. Remove the wireless NIC from the base assembly.
3. Open the latch for the display cable and disconnect it from the motherboard. It's the left connector shown in Figure 4.8.

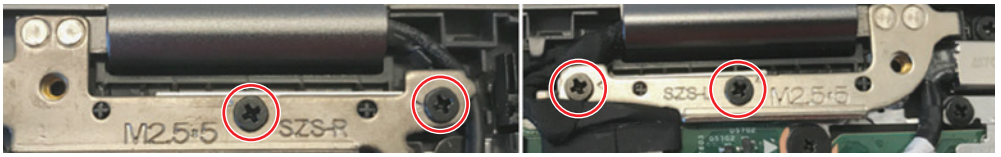
FIGURE 4.8 Display cable (left) and touchscreen board cable connectors (right)



4. Disconnect the touchscreen board cable from the motherboard. It's the right connector shown in Figure 4.8.
5. Open the computer so that the display assembly is parallel with the base assembly, and lay it screen side down on a flat surface.

   We recommend that you place it on a soft surface to avoid scratching the screen.
6. Remove the screws that secure the display assembly to the base assembly. There are two screws for each hinge, and they are highlighted in Figure 4.9.

FIGURE 4.9 Display assembly hinge screws



7. Lift the display assembly off of the base assembly.

   Note that the NIC antenna wires, display cable and connector, and touchscreen board cable and connector will go with the display assembly.

If you're replacing the entire display unit, all you need to do is get the new unit and reverse the steps you followed in Exercise 4.2. If you're replacing a component within the display unit, you need to go further. To get to the Wi-Fi antenna, webcam, microphone, and inverter, you must also remove the display panel. Exercise 4.3 gives you the general steps needed to accomplish this.

EXERCISE 4.3

Removing the Display Panel

1. Remove the bottom cover.
2. Remove the display assembly.
3. Remove the display cable from the routing guides inside the hinge covers.
4. Use a plastic scribe to pry apart the back of the display assembly and the display panel.

   Step 4 is a tricky step. There's not much of a gap between the display panel and the back of the display assembly. It's generally best to try to start prying it apart near the center of one of the sides of the panel—that's where you're likely to get the most separation to get the scribe in there.

   Also, we highly recommend using a plastic scribe. You can use a very thin flat screwdriver, but the metal is more likely to mark up or damage the case than the plastic scribe. Once you start getting separation, gently work your way around the edge of the display panel to separate the whole thing.

5. Identify the component you need to replace, and remove it.

EXERCISE 4.3 *(continued)*

Figure 4.10 shows a Lenovo ThinkPad with the display bezel removed. You can see wires running along the side and top. Some of the wires go to the webcam, and others are for the wireless antenna. As always, consult the laptop's documentation to identify the proper components.

FIGURE 4.10 Wires inside a laptop display

Motherboards and Processors

As with desktop computers, the motherboard of a laptop is the backbone structure to which all internal components connect. However, with a laptop, almost all components must be integrated onto the motherboard, including onboard circuitry for the USB, video, expansion, and network ports of the laptop. With desktop systems, the option remains to not integrate such components. Because of the similarities between laptop and desktop components, some material in the next few sections will be familiar to you if you have read Volume 1 Chapter 1, "Motherboards, Processors, and Memory."

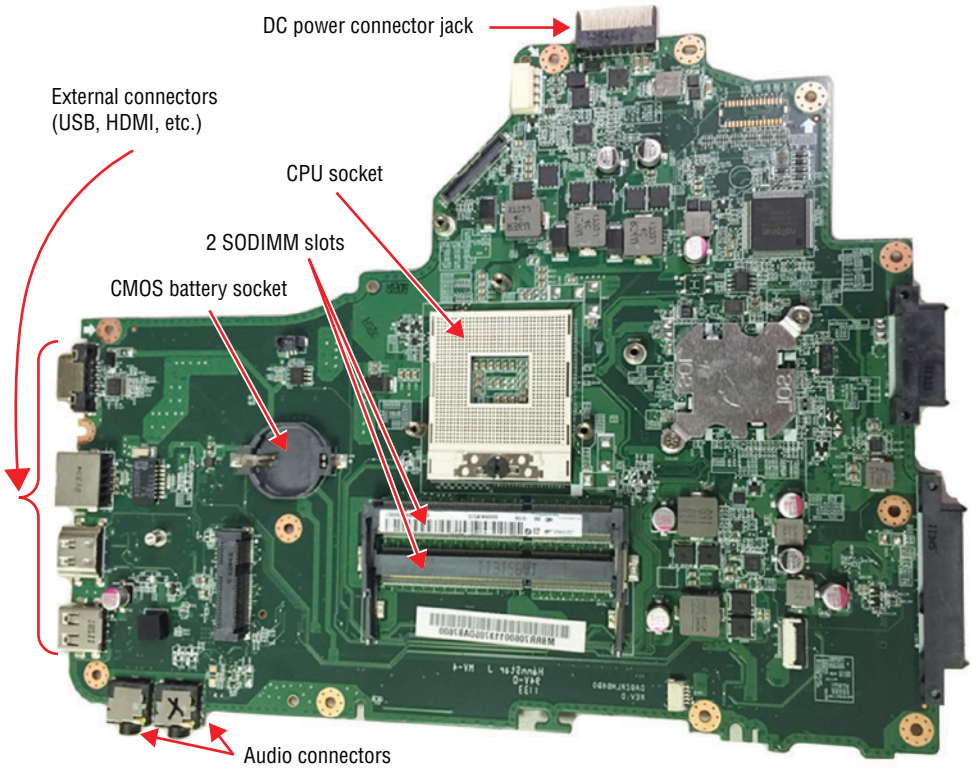
Laptop Motherboards

The primary differences between a laptop motherboard and a desktop motherboard are the lack of standards and the much smaller form factor. As mentioned earlier, most motherboards are designed along with the laptop case so that all the components will fit inside. Therefore, the motherboard is nearly always proprietary, and that's what we mean by "lack of standards." They use the technologies you're used to, such as USB and Wi-Fi, but it's very unlikely that you're going to be able to swap a motherboard from one laptop to another, even if both laptops are from the same manufacturer. Figure 4.11 shows an example of a laptop motherboard. Its unusual shape is designed to help it fit into a specific style of case with the other necessary components.

To save space, components of the video circuitry (and possibly other circuits as well) may be placed on a thin circuit board that connects directly to the motherboard. This circuit board is often known as a *riser card* or a *daughterboard*; an example is shown in

Figure 4.12. We also labeled them in the Dell in Figure 4.4. They're harder to see when they're in the case with other components, but you may be able to tell that they have very different shapes from those in Figure 4.11 and Figure 4.12.

FIGURE 4.11 A laptop motherboard



Dell happens to call this daughterboard an input/output board, or I/O board. This model also has another daughterboard. If you were ordering parts, you'd want to specify I/O board or daughterboard, but for general purposes, the term daughterboard is applicable.

Having components performing different functions (such as video, audio, and networking) integrated on the same board is a mixed bag. On one hand, it saves a lot of space. On the other hand, if one part goes bad, you have to replace the entire board, which is more expensive than just replacing one expansion card. Exercise 4.4 walks you through the steps to remove the motherboard from the Dell Inspiron 13 7000. As you'll see, you need to remove several components and disconnect several connectors before you can get the motherboard out.

FIGURE 4.12 A laptop daughterboard

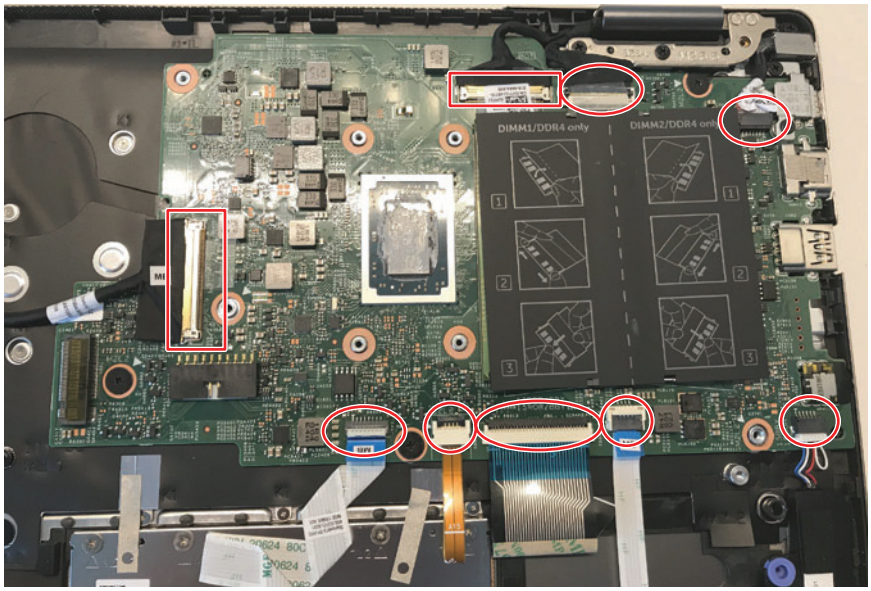
EXERCISE 4.4

Removing the Motherboard from a Laptop

1. Remove the bottom cover.
2. Remove the battery.
3. Remove the fan.
4. Remove the heat sink.
5. Remove the solid-state drive.
6. Disconnect all the connectors plugged into the motherboard.

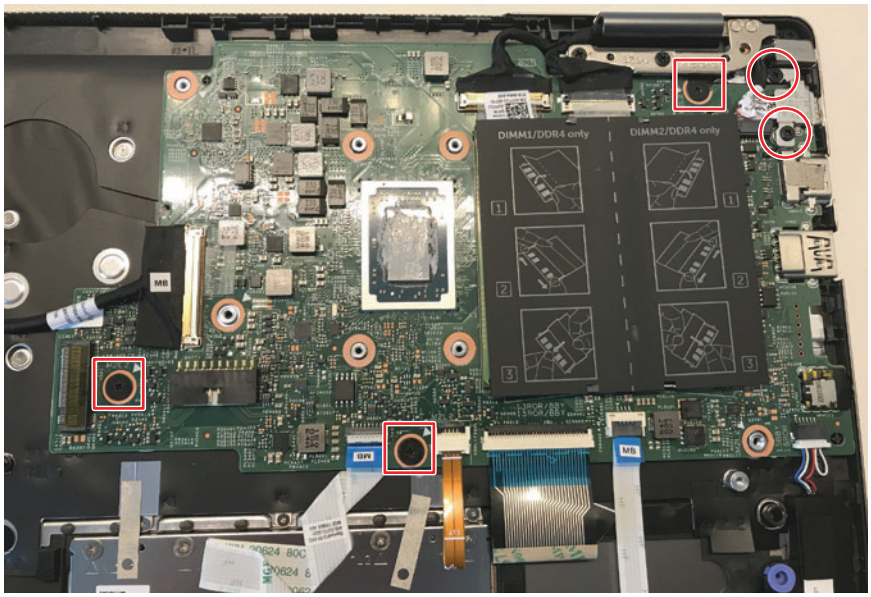
   Be careful, because two of them (for the I/O cable and the display cable) have a latch that needs to be released first. By our count, there are nine connectors. We've highlighted them in Figure 4.13. The two with latches have rectangles around them, and the rest have ovals.

FIGURE 4.13 Disconnecting the motherboard connectors



7. Remove the two screws from the system board bracket. They're shown in Figure 4.14 with circles around them.

FIGURE 4.14 Removing the screws



EXERCISE 4.4 *(continued)*

8. Remove the system board bracket.
9. Remove the three screws that hold the motherboard to the base assembly. In Figure 4.14, they are highlighted with squares around them.
10. Lift the motherboard off the base assembly.

To replace the motherboard, follow the steps in reverse order.

Laptop Processors

Just as with desktop computers, the central processing unit (CPU) is the brain of the laptop computer. And just like everything else, compared to desktop hardware devices, laptop hardware devices are smaller and not quite as powerful. The spread between the speed of and number of cores in a laptop CPU and that of a desktop model can be significant. Fortunately, the gap has closed over the years, and laptop processors today are pretty fast—most people think they perform just fine. It's up to the user to determine if the difference in speed hurts their usage experience.

Laptops have less space than desktops, and therefore the CPU is usually soldered onto the motherboard and is not upgradable. You can see the processor of the Dell we've been working on in Figure 4.14—it's the small silver square to the left of the RAM. Within confined computing spaces, heat is a major concern. Add to that the fact that the processor is the hottest-running component, and you can see where cooling can be an issue. To help combat this heat problem, laptop processors are engineered with the following features:

Streamlined Connection to the Motherboard Nearly all desktop processors mount using pin connectors, whether on the CPU or on the motherboard (as is the case with LGA sockets). Pins and sockets are big and bulky, meaning they're not a laptop's friends. Laptop processors are generally either soldered directly to the motherboard or attached using the Micro-FCBGA (Flip Chip Ball Grid Array) standard, which uses balls instead of pins. In most cases, this means that the processor cannot be removed, meaning no processor upgrades are possible.

Lower Voltages and Clock Speeds To combat heat, you can either slow the processor down (run it at a lower speed) or give it less juice (run it at a lower voltage). Again, performance will suffer compared to a desktop processor, but lowering heat is the goal here.

Active Sleep and Slowdown Modes Most laptops will run the processor in a lower power state when on battery power to extend the life of the battery. This is known as *processor throttling*. The motherboard works closely with the operating system to determine if the processor really needs to run at full speed. If it doesn't, it's slowed down to save energy and to reduce heat. When more processing power is needed, the CPU is throttled back up.

While many portable computers will have processors that have just as many features as their desktop counterparts, others will simply use stripped-down versions of desktop processors. Although there's nothing wrong with this, it makes sense that components specifically designed for laptops fit the application better than components that have been retrofitted for laptop use. Consider an analogy to the automobile industry: it's better to design a convertible from the ground up than simply to cut the top off an existing coupe or sedan.

Memory

Laptops don't use standard desktop computer memory chips, because they're too big. In fact, for most of the history of laptops, there were no standard types of memory chips. If you wanted to add memory to your laptop, you had to order it from the laptop manufacturer. Of course, because you could get memory from only one supplier, you got the privilege of paying a premium over and above a similar-sized desktop memory chip.

Exam Objective 1.1 includes RAM as a component you need to monitor and replace.

Fortunately, the industry seems to have settled on one form factor, which is the *small outline dual inline memory module (SODIMM)*, which we introduced in Volume 1 Chapter 1. Recall that they're much smaller than standard DIMMs, measuring 67.6 millimeters (2.6″) long and 32 millimeters (1.25″) tall. SODIMMs are available in a variety of configurations, including older 32-bit (72-pin and 100-pin) and 64-bit (144-pin SDRAM, 200-pin DDR/DDR2, 204-pin DDR3, 260-pin DDR4, and 262-pin DDR5) options. Different standards of DDR SODIMMs are only a few millimeters longer or shorter than other versions. You probably won't be able to tell the difference unless they are right next to each other, or unless you try to install them and they don't fit in the socket. (You should never have the latter problem, though, because you will check the documentation first!) Figure 4.15 shows a laptop DDR5 SODIMM above a desktop DDR5 DIMM for a size comparison.

FIGURE 4.15 Laptop SODIMM and Desktop DIMM

Laptop manufacturers may still opt to go the proprietary route due to design considerations that favor a custom solution. In fact, many ultra-thin laptop models will have the RAM soldered onto the motherboard to save space. In such systems, the memory is not upgradable. To see what kind of memory your laptop uses, check either the manual or the manufacturer's website. You can also check third-party memory producers' websites (such as `www.crucial.com`).

Just as with desktop computers, make sure the SODIMM you want to put into the laptop is compatible with the motherboard. The same standards that apply to desktop memory compatibility apply to laptops. This means that you can find DDR, DDR2, DDR3, DDR4, and DDR5 SODIMMs for laptops. DDR has topped out at 1 GB per module, whereas DDR2 and DDR3 SODIMM modules can be purchased in sizes up to 8 GB and DDR4 up to 32 GB and DDR5 up to 64 GB (at the time this book was being written). Exercise 4.5 shows you how to replace SODIMMs in a laptop.

EXERCISE 4.5

Replacing Laptop Memory

1. Remove the bottom cover.
2. Remove the battery.
3. Locate the system memory.

   In the Dell Inspiron 13 7000, the two RAM modules are under Mylar flaps.
4. Pull a Mylar flap back to reveal the RAM, as shown with one of the SODIMMs in Figure 4.16.

FIGURE 4.16 SODIMM in a laptop

5. Carefully spread apart the metal securing clips on each end of the memory module. It will pop up, as shown in Figure 4.17.

FIGURE 4.17 Removing a SODIMM

6. Lift the memory module out of the slot.

When replacing the SODIMM, be sure to align the notch, and insert the module at the approximate angle it was at when it popped up after being released. Slowly press down on the module until it clicks into place. If it doesn't click into place, it's not inserted properly.

For several years, a laptop memory form factor called MicroDIMM enjoyed some popularity. The MicroDIMM is over 50 percent smaller than a SODIMM—only about 45.5mm (about 1.75″) long and 30mm (about 1.2″) wide. Another major difference is that the MicroDIMM does not have any keying notches on the bottom. Figure 4.18 shows a 172-pin MicroDIMM compared to a DDR5 SODIMM; notice that it's a lot closer to being square-shaped than the SODIMM is. It was designed for the ultra-light and portable subnotebook style of computer. Common MicroDIMM form factors included 64-bit modules with 172 or 214 pins for DDR2/DDR3. DDR4 and DDR5 MicroDIMMs have not been produced. Replacing a MicroDIMM is just like replacing a SODIMM.

FIGURE 4.18 172-pin MicroDIMM and a SODIMM

Storage

Storage is important for every computer made. If you can't retrieve important files when you need them, the computer isn't very useful. While the trend is moving toward storing more data online (in the cloud), there's still considerable need for built-in storage.

Exam Objective 1.1 lists hard drives (HDDs) and solid-state drives (SSDs) as devices to monitor and know how to replace.

Laptop Hard Drives

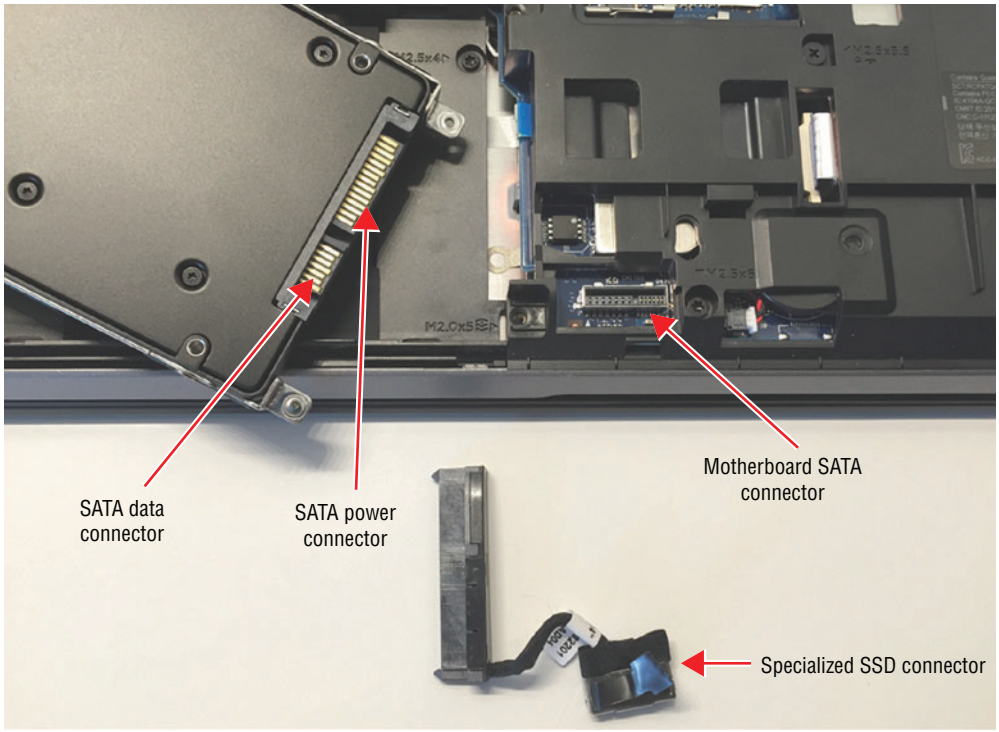
Laptops don't have the room for the full-sized 3.5″ hard drives that desktop computers use. Smaller form factor drives at 2.5″ or 1.8″ that are less than ½″ thick are more appropriate. These drives share the same controller technologies as desktop computers; however, they use smaller connectors. Figure 4.19 shows an example of a standard 3.5″ hard drive compared to a 2.5″ laptop hard drive.

FIGURE 4.19 A 3.5″ hard drive (left) compared to a 2.5″ hard drive (right)

To save space and heat, almost all laptops today use a *solid-state drive (SSD)*, which we introduced in Volume 1 Chapter 2, "Expansion Cards, Storage Devices, and Power Supplies." Recall that, unlike conventional magnetic hard drives, which use spinning platters, SSDs have no moving parts. They use the same solid-state memory technology found in the other forms of flash memory. Otherwise, they perform just like a traditional magnetic HDD, except they're a lot faster.

Connecting a regular SSD in a desktop is usually just like connecting a regular HDD; they have the same Serial Advanced Technology Attachment (SATA) and power connectors. Laptops often have a specialized connector and a single cable that handles both data and power, as shown in Figure 4.20. Most manufacturers also make them in the same physical dimensions as traditional hard drives, even though they could be made much smaller, like removable flash drives. (This is probably to preserve the "look" of a hard drive so as to not confuse consumers or technicians.)

FIGURE 4.20 2.5″ SSD, motherboard connector, and cable



Newer SSDs may come in the even smaller M.2 form factor. In fact, the Dell Inspiron we've been working on in this chapter has an M.2 SSD. Figure 4.21 shows the SSD and the M.2 connector it plugs into. Exercise 4.6 walks you through removing an M.2 SSD from a laptop.

FIGURE 4.21 M.2 SSD and M.2 connector

EXERCISE 4.6

Removing an M.2 SSD from a Laptop

1. Remove the bottom cover.
2. Remove the screw that holds the SSD to the base assembly.

   The drive will pop up slightly. The screw is highlighted in Figure 4.22.
3. Slide the drive directly out of the connector.

FIGURE 4.22 Removing an M.2 SSD

Removing 2.5″ or 1.8″ SSDs from a laptop will require a few more steps than Exercise 4.6 did. First, disconnect the drive cable from the motherboard (refer to Figure 4.20), and then remove the two to four screws that hold the drive or its mounting bracket in place. Figure 4.23 shows an example of an SSD in a Lenovo ThinkPad that uses three screws—we've already removed them but highlighted the holes for you. Then lift out the drive (and bracket as needed). We show the disconnected drive next to the laptop in Figure 4.24.

FIGURE 4.23 2.5″ SATA SSD

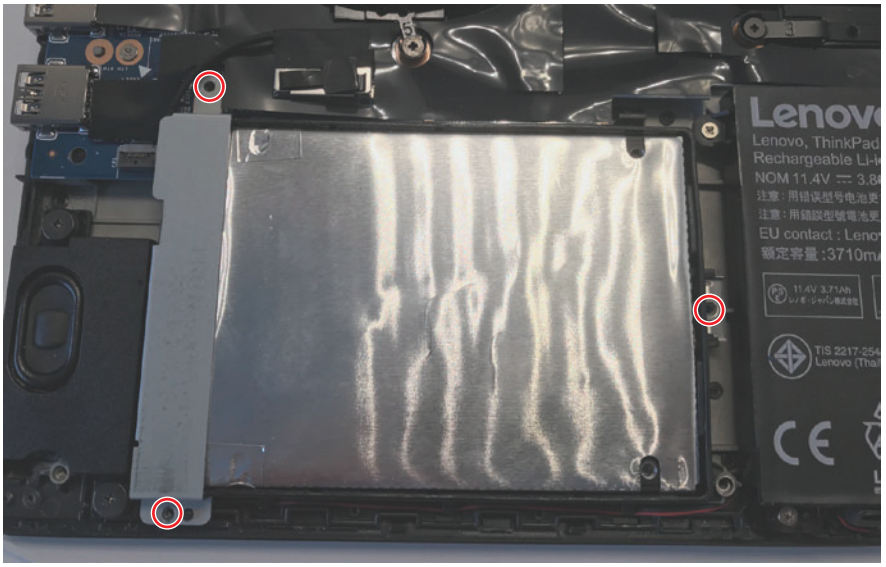
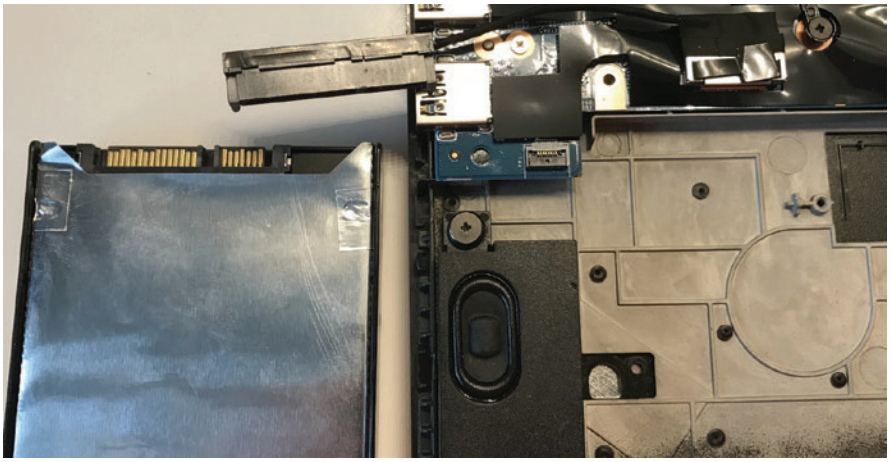


FIGURE 4.24 Disconnected SATA SSD



SSDs are perfect for laptops because they are faster, produce less heat, and are more shock resistant than conventional hard drives. They are more expensive per gigabyte, but that's the trade-off you have to make for better performance.

Hard Drive Data Migration

After replacing a hard drive or upgrading a laptop to a newer model, it may be necessary to get all of the user's data from the old drive (or laptop) onto the new one. Doing so is often referred to as data *migration* or hard drive migration. When setting up for a migration, there are two key questions to ask. The first is, what needs to be migrated? There's a big difference between moving someone's data and moving operating system settings and configurations. The second is, will the old drive be accessible when the new one is up and running? If so, there are more options to perform a migration. The answers to these will help determine which of the migration methods are preferable—either manual file copying or using migration software.

Copying Files Manually If all that's needed is to get user data from the old device to the new one, then manually copying files is usually an easy option. If the old drive will be inaccessible after the replacement, then files can be copied from the old drive to the cloud or to an external hard drive first. The drive replacement can be made, and then the files copied from the cloud or external drive to the new hard drive. If the old drive will still be accessible, then files can be copied across a network (if conditions permit) or copied from computer to computer with a transfer cable.

The downside to this method is that it doesn't transfer any settings or configurations (such as a user's desktop, color scheme, installed printers, etc.), and it normally doesn't work for apps. Apps will need to be reinstalled on the new drive, and the user will have to reconfigure their settings.

Using Migration Software Migration software can move files, settings, configurations, and apps from one drive to another. In most cases for this to work, both drives need to be accessible. For example, the old drive needs to be in a different operational computer, in a second expansion slot on the new computer, or otherwise connected, such as through a USB port using a USB-to-SATA adapter. If the connection is available, the migration software can work its magic.

There are several different migration apps available on the market. For example, Laplink PCmover (`https://go.laplink.com`) works with Windows computers. It has several versions, including the Home edition, which is good for normal users, and a Professional version that can copy multiple user profiles. Macrium Reflect X (`www.macrium.com`) is another option for Windows-based systems, and SuperDuper (`https://shirt-pocket.com`) works for macOS. All of these apps feature a graphical interface, where you choose what you want to migrate and the software takes it from there.

Laptop Optical Drives

Nearly all laptops have a hard drive, but rarely does a laptop made today have an internal optical drive. There just isn't room for one. If you need one, you can attach an external optical drive via an expansion port such as USB. It might be a bit slower than an internal drive, but it's better than not having one at all.

Input Devices

Because of the small size of laptops, getting data into them presents unique challenges to designers. They must design a keyboard that fits within the case of the laptop. They must also design some sort of pointing device that can be used with graphical interfaces like Windows. The primary challenge in both cases is to design these peripherals so that they fit within the design constraints of the laptop (low power and small form factor) while remaining usable.

Exam Objective 1.1 includes keyboard and keys as laptop hardware you should know how to replace.

Keyboards

A standard-sized desktop keyboard wasn't designed to be portable. It wouldn't fit well with the portable nature of a laptop. That usually means laptop keys are not normal size; they must be smaller and packed together more tightly. People who learned to type on a typewriter or a full-sized keyboard often have a difficult time adjusting to a laptop keyboard.

Keyboards may need to be replaced if keys are missing or are stuck and won't function. If a user has spilled a beverage onto their keyboard, it could cause problems that necessitate a replacement. On most laptop keyboards, you can't replace just one key; the whole board must be replaced to fix a malfunctioning key. If needed, an external USB keyboard can be added as a solution until the faulty built-in one is repaired.

Laptop keyboards are built into the lower portion of the clamshell. Sometimes, they can be removed easily to access peripherals below them, like memory and hard drives, as in the Lenovo ThinkPad series. Other times, removing the keyboard is one of the most challenging tasks, because nearly all other internal components need to be removed first. Exercise 4.7 illustrates the joy of removing the keyboard from the Dell Inspiron 13 7000.

EXERCISE 4.7

Removing a Laptop Keyboard

1. Remove the bottom cover.
2. Remove the battery.
3. Remove the SSD.
4. Remove the wireless card.

EXERCISE 4.7 *(continued)*

5. Remove the fan.
6. Remove the heat sink.
7. Remove the display assembly.
8. Remove the I/O board.
9. Remove the power adapter port.
10. Remove the motherboard.
11. Remove the speakers.
12. Remove the 14 screws that hold the keyboard shield to the base assembly.
13. Lift the keyboard from the base assembly.

As you can see, it was necessary to remove 11 components before we got to the keyboard. With a laptop like this, replacing the keyboard is not a quick-service item. This is also a great time to remember that you should always take pictures (if allowed) and document screw types and locations when tearing laptops apart. After removing 11 components, you might be dealing with several screw types and things can get confusing!

Pointing Devices

In addition to using the keyboard, you must have a method of controlling the onscreen pointer in the Windows (or other graphical) interface. Most laptops today include multiple USB ports for connecting a mouse, and you can choose from a wide range of wired or wireless full-sized or smaller mice. There are several additional methods for managing the Windows pointer. Here are some of the more common ones:

- Trackball
- Touchpad
- Point stick
- Touchscreen

Because of different pointing-device preferences, some laptops include multiple pointing devices to appeal to a wider variety of people.

Trackball Many early laptops used trackballs as pointing devices. A *trackball* is essentially the same as a mouse turned upside down. The onscreen pointer moves in the same direction and at the same speed that you move the trackball with your thumb or fingers.

Trackballs are cheap to produce. However, the primary problem with trackballs is that they do not last as long as other types of pointing devices; a trackball picks up dirt and oil from operators' fingers, and those substances clog the rollers on the trackball and prevent it from functioning properly.

Trackpad/Touchpad To overcome the problems of trackballs, a newer technology that has become known as the trackpad was developed. Many people call the trackpad a touchpad, but Touchpad is actually the trademarked name of Hewlett-Packard (HP) trackpad devices. They are one and the same, though.

A *trackpad* is a device that has a pad of touch-sensitive material. The user draws with their finger on the trackpad, and the onscreen pointer follows the finger motions. Included with the trackpad are two buttons for left- or right-clicking (although with some trackpads, you can perform the functions of the left-click by tapping on the trackpad, and Macs have one button). Figure 4.25 shows a trackpad on a laptop.

FIGURE 4.25 Laptop trackpad

Yet another name for trackpad-type devices is *drawing pad*. Typically, people only refer to it as a drawing pad if it's larger than a typical laptop trackpad and attached to a computer via USB. Drawing pads are often the size of a sheet of paper.

One problem people have with a trackpad is the location. You'll notice that the trackpad is conveniently placed right below the laptop keyboard, which happens to be where your palms rest when you type. Sometimes this will cause problems, because you can inadvertently cause your mouse cursor to do random things like jump across the screen. Most trackpads have settings to allow you to control the sensitivity, and they will also differentiate between a palm touching them and a finger. If you have a sensitive trackpad that is giving you trouble, you can disable it altogether. Exercise 4.8 shows you how to do that in Windows 11. The specific steps to disable the trackpad will differ by manufacturer—you will almost always be able to disable it through the operating system, and some laptops have a function key to disable it as well. The steps in Exercise 4.8 were performed on an HP ZBook laptop. Consult the laptop documentation if you are unable to locate the setting.

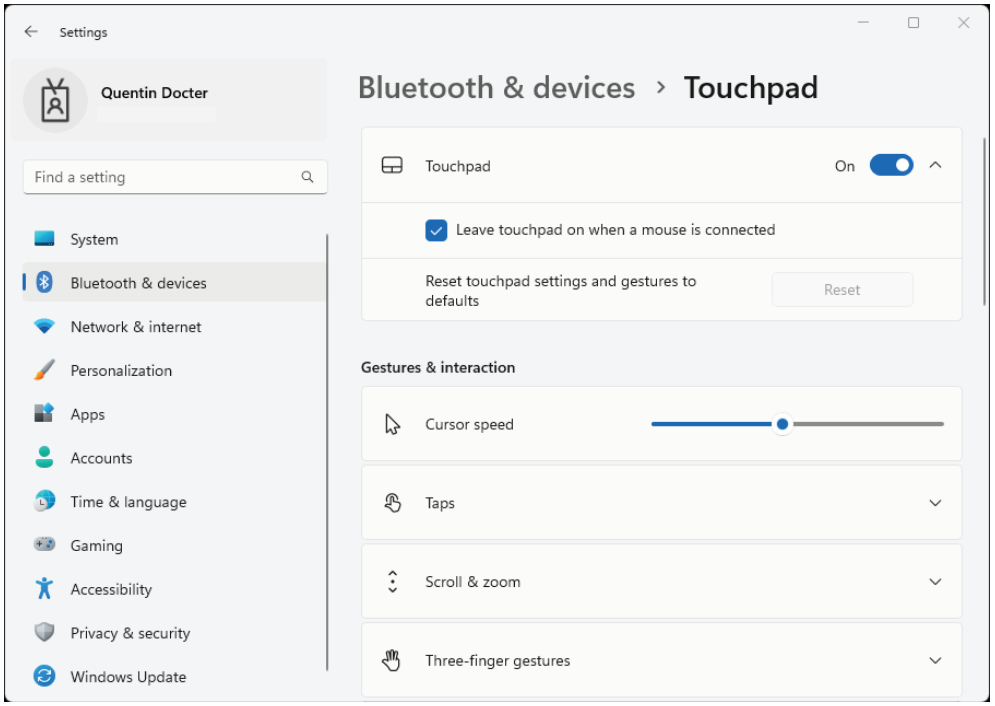
EXERCISE 4.8

Disabling a Touchpad in Windows 11

1. Open the Touchpad settings app by typing **touchpad** in the Windows search box and pressing Enter. You will see a screen similar to the one in Figure 4.26.

EXERCISE 4.8 *(continued)*

FIGURE 4.26 Touchpad settings



Remember that Touchpad is an HP trademark. On other laptops, trackpad settings are often in the Mouse Settings app.

2. Turn the Touchpad off by sliding the toggle bar to the off position. Notice there is also an option to leave the touchpad on (or turn it off) when a mouse is connected.
3. (Optional) Explore the other Touchpad settings, such as how to manage taps, scrolling and zooming, and multi-finger gestures.
4. Close the Touchpad window when finished.

The trackpad can be replaced if it becomes defective, just like most other laptop components. Before you replace it, first make sure that it's not just disabled in the operating system.

Track Point/Point Stick With the introduction of the ThinkPad series of laptops, IBM introduced a new feature known as the *Touchpoint*, generically known as a *point stick* or a *track point*. (ThinkPads are now produced by Lenovo.) The track point is a pointing device that uses a small rubber-tipped stick. When you push the track point in a particular direction, the onscreen pointer goes in the same direction. The harder you push, the faster the onscreen pointer moves. The point allows fingertip control of the onscreen pointer, without the reliability problems associated with trackballs. Figure 4.27 shows a track point.

FIGURE 4.27 Track point

Track points can have their own problems, however. The stick may not return to center properly, causing the pointer to drift when not in use. You might also notice the rubber cover for the stick becoming a bit gummy with extended use. Most manufacturers supply replacement covers of varying textures with new systems. Exercise 4.9 shows you how to disable and manage track points.

EXERCISE 4.9

Disabling a Track Point in Windows 11

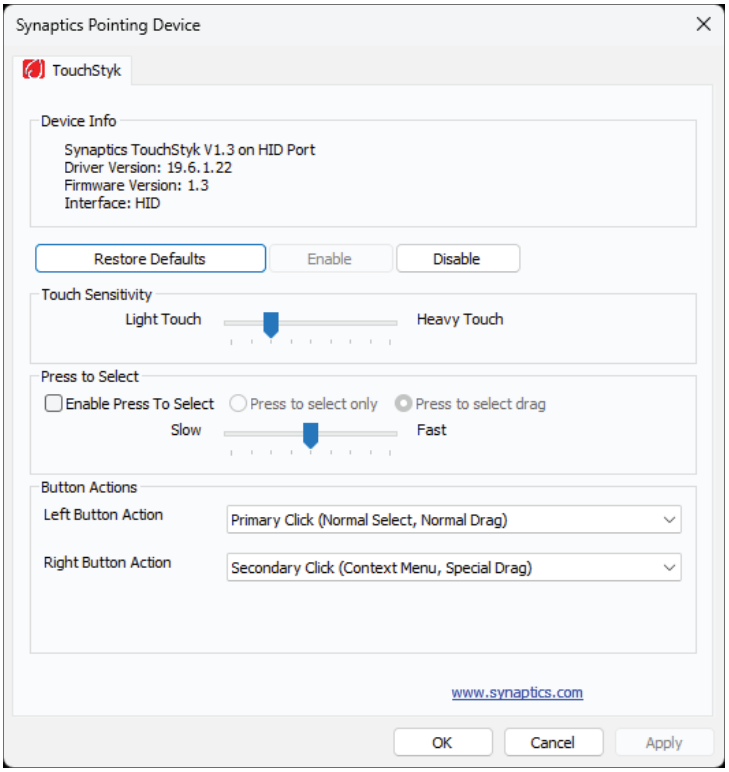
1. Open the Mouse settings app by typing **`mouse`** in the Windows search bar and pressing Enter.

   You may be able to type **`pointstick`** in the search bar and press Enter to open Mouse Settings as well.

2. In Mouse Settings, scroll down to the PointStick Settings area and click menu arrow on the right side. You will get a Properties screen like the one in Figure 4.28.

EXERCISE 4.9 *(continued)*

FIGURE 4.28 Point stick properties



3. Click Disable to disable the track point.
4. (Optional) Explore other configuration options for the track point.
5. Once you're satisfied with the configuration, click OK to close the Properties window.

Touchscreen *Touchscreens* are standard fare for smartphones and tablets and many laptop computers as well. We've already introduced the technology as a display device, so here we'll cover it as an input device. The idea is pretty simple: it looks like any other display device, but you can touch the screen and the system senses it. It can be as simple as registering a click, like a mouse, or it can be more advanced, such as capturing handwriting and saving it as a digital note.

Although the technical details of how touchscreens work are beyond the scope of this chapter, there are a few things you should know:

- Some touchscreens work with any object touching them, whereas others require a conductive input, such as your finger. iPhones are a great example of this, to

which anyone who lives in cold climates and wears gloves can attest.

- Some touchscreens are coated with a film that is sensitive to touch. Cleaning these screens with regular glass cleaner can ruin the touchscreen nature of the device. It's best to clean those devices only with a damp cloth, as needed.
- Some touchscreens need to be calibrated in order to properly interpret the user's input. See the manufacturer's documentation for details.

User options to configure touchscreens are often limited to calibrating the screen for touch input or pen input. If the touchscreen seems to be sensing input incorrectly or not detecting input from the edges of the screen, it might be time to recalibrate it. In Windows, open Control Panel ➢ Tablet PC Settings, and then click Calibrate. Windows will ask the user to specify pen or touch input. Once that's specified, the user will be able to draw on the screen with their desired input tool, and the system will sense it. (In most cases, there will be a crosshairs on the screen and the user needs to touch where it is.) At the end, the user needs to save the calibration data for it to take effect.

Many laptops sold today are advertised as 2-in-1 devices, meaning that the top portion of the clamshell that holds the display folds completely over, effectively turning the laptop into a glorified tablet. These devices use touchscreens, and when they are in tablet mode, the keyboard is disabled. Some will also automatically change the screen orientation to landscape or portrait, depending on how it's being held, just like a smartphone or tablet.

Internal Expansion

Although laptop computers are less expandable than their desktop counterparts, many can be expanded to some extent. The two primary forms of internal expansion used in laptops today are Mini PCIe and M.2.

Exam Objective 1.1 lists wireless cards as devices you should know how to replace in a laptop. To be safe, you should understand how to replace both Mini PCIe and M.2 expansion cards.

Mini PCIe

Since around 2005, *Mini PCIe* has been the most common slot for laptop expansion cards. We introduced PCIe in Volume 1 Chapter 2, and mini PCIe is just like the full version except the connectors are smaller. These cards reside inside the case of the laptop and are connected via a 52-pin card edge connector. Mini PCIe cards come in two sizes. The full-sized cards are 30mm wide and 51mm long. Half-sized cards (one is shown in Figure 4.29, with the connector at the bottom) are 30mm wide and 27mm long. Mini PCIe cards support USB and PCIe x1 functionality, and at the same speeds. Additionally, Mini PCIe cards have the 1.5V and 3.3V power options.

FIGURE 4.29 Mini PCIe card in a laptop

M.2

We also introduced M.2 in Volume 1 Chapter 2, so we won't go into a great amount of detail here. Figure 4.21 earlier in this chapter shows an M.2 hard drive and slot. For purposes of comparing it to Mini PCIe, know that it uses a narrower connector (22mm vs. 30mm) that has more pins (66-pin vs. 52-pin). M.2 supports USB 2.0 and newer. The slowest M.2 slots support PCIe x2, and M-keyed slots support PCIe x4, making it much faster than Mini PCIe. Most M.2 expansion cards focus on communications or storage. Common types of cards you will see in the market include the following:

- Wi-Fi network cards
- Bluetooth cards
- Cellular cards
- SSDs

Many laptops don't come with any free internal expansion slots, either M.2 or Mini PCIe. For example, the Dell we've been working on has two M.2 slots, but both are filled. One has

the SSD, and the other is used by the Wi-Fi card. Be sure to check the documentation before you buy an expansion card for a laptop, to see what it supports.

Batteries and Power Adapters

Because portable computers have unique characteristics as a result of their portability, they have unique power systems as well. Portable computers can use either of two power sources: batteries or adapted power from an AC or DC source. Regardless of the source of their power, laptops utilize DC power to energize their internal components. Therefore, any AC power source needs to be rectified (converted) to DC. Most laptop display backlights, on the other hand, require high-voltage, low-amperage AC power. To avoid a separate external AC input, an inverter is used to convert the DC power that is supplied for the rest of the system to AC for the backlight. In case it's not obvious, converters and inverters perform opposite functions, more or less.

Understand how to replace batteries in a laptop, per Exam Objective 1.1.

Batteries

There are many different battery chemistries that come in various sizes and shapes. Nickel cadmium (NiCd), lithium-ion (Li-ion), and nickel-metal hydride (NiMH) have been the most popular chemistries for laptop batteries. A newer battery chemistry, lithium-polymer (Li-poly), has been gaining in prominence over recent years for smaller devices.

Laptop batteries may be easily removable without cracking the case. Figure 4.30 shows a removable Li-ion battery for an HP laptop that just pops off the back side of the case.

FIGURE 4.30 A removable laptop Li-ion battery

A removable battery is very easy to replace in the event of a battery failure. However, most laptops today make use of an internal battery, such as the one shown in Figure 4.31. This particular battery is very thin—only 5mm (⅕″) thick. Exercise 4.10 shows you how to remove an internal laptop battery.

EXERCISE 4.10

Removing an Internal Laptop Battery

If a laptop won't power on while on battery power but it works while plugged in and the battery shows it has a charge, a simple re-seating of the battery might solve the problem. Unplug it, let the laptop sit for a few seconds, and then plug it back in.

1. Remove the bottom cover.
2. Disconnect the battery from the motherboard.

   Its connector is highlighted with a rectangle in Figure 4.31.
3. Remove the four screws holding the battery in place.

   In Figure 4.31, the screws are highlighted with circles.
4. Lift the battery out of the base assembly.

FIGURE 4.31 Internal laptop battery

Even though replacing an internal battery requires slightly more effort than replacing an external battery, it's still generally a simple operation.

Battery chemistries can be compared by energy density and power density. Energy density measures how much energy a battery can hold. Power density measures how quickly the stored energy can be accessed, focusing on access in bursts, not prolonged runtime. An analogy to the storage and distribution of liquids might help solidify these concepts. A gallon bucket has a higher "energy density" and "power density" than a pint bottle; the bucket holds more and can pour its contents more quickly. Another common metric for battery comparison is rate of self-discharge, or how fast an unused battery reduces its stored charge.

Real World Scenario

Is That Battery Really Dead?

Some batteries, such as nickel cadmium (NiCd) batteries, suffer from a performance-affecting chemical memory loss. Others, such as lithium-ion (Li-ion), don't suffer from this affliction but do suffer from so-called digital memory loss that plagues the built-in gauges that monitor the charge left in the battery. This effect can be observed in software gauges that read the battery's charge level. The digital memory effect manifests itself as a sudden loss of power when the gauges register, say, 30 percent remaining capacity. The fix, much like the fix for chemical memory in NiCd batteries, is to allow a full discharge once a month or so. This is called *battery calibration* and can be performed right in the device while it's using the battery.

Other than this occasional full discharge, Li-ion batteries last longer when you partially discharge them and then recharge them, making them ideal for laptops and personal handheld devices, such as smartphones, that tend to get used sporadically on battery power before being plugged back in to charge.

Power Adapters

Most laptop computers can also use AC power with a special adapter (called an *AC adapter*) that converts AC power input to DC output. The adapter can be integrated into the laptop, but more often it's a separate "brick" with two cords: one that plugs into the back of the laptop and another that plugs into a wall outlet. Figure 4.32 is a photo of the latter. Most AC adapters use a USB-C connection, which has been popular since 2014. Some laptop manufacturers will still roll out proprietary connectors, though. In those cases, only an AC adapter that matches the connector will work.

Use caution when selecting a replacement AC adapter for your laptop. You should choose one rated for the same or higher wattage than the original. Regarding the input voltage of the adapter, care must also be taken to match the adapter to the power grid of the surrounding region. Most adapters are autoswitching (universal voltage), but some adapters may have a fixed AC input requirement. Purchasing the wrong unit can result in a lack of functionality or damage to the laptop. If or when working internationally, you should always ascertain whether some sort of converter is required, even for autoswitching adapters.

FIGURE 4.32 A laptop AC adapter

Other Internal Components

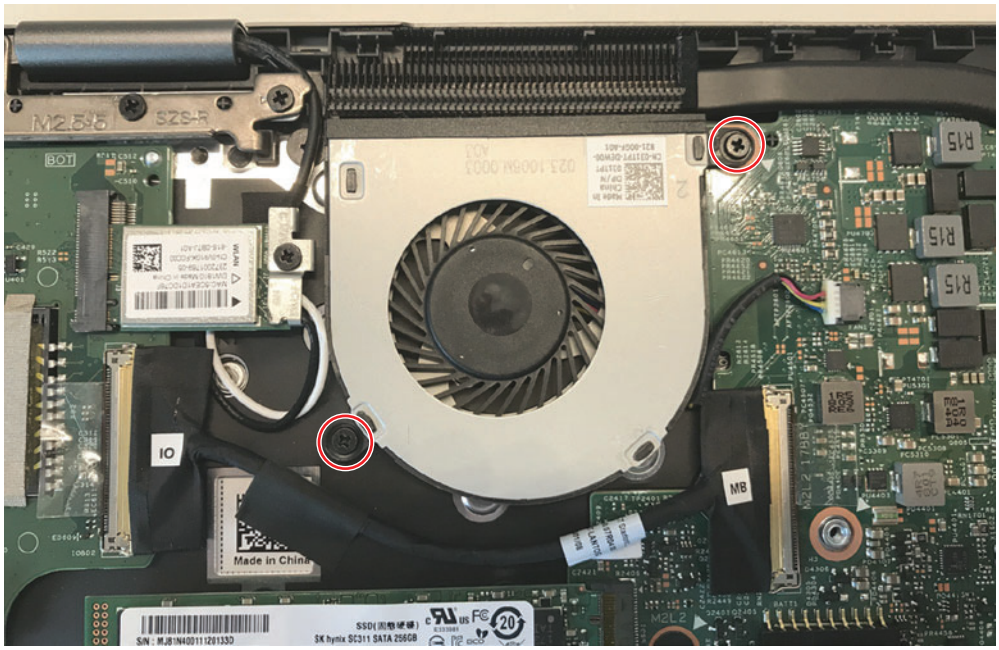
There are a few internal components we've referenced in chapter exercises, but we haven't given explicit details on how to remove them. That's what this section is for. Note that these components aren't currently on the exam objectives, but it helps to know how to remove them anyway. The four components we'll examine are the fan, heat sink, wireless NIC, and CMOS battery. And since we'll be talking about CMOS, which is related to the BIOS/UEFI, we'll look at how to upgrade (or flash) the BIOS/UEFI as well. Exercise 4.11 shows you how to remove the system fan.

EXERCISE 4.11

Removing the System Fan

1. Remove the bottom cover.
2. Remove the two screws that hold the system fan in place. They are highlighted in Figure 4.33.
3. Lift the fan away from the base assembly.

FIGURE 4.33 Removing the system fan



Now that the fan is removed, Exercise 4.12 shows you how to remove the CPU heat sink.

EXERCISE 4.12

Removing the CPU Heat Sink

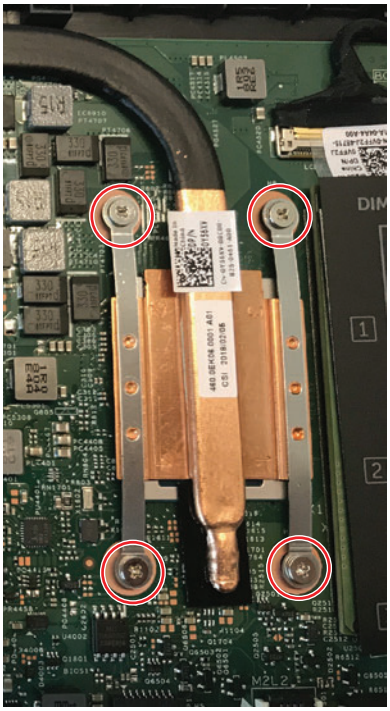
1. Remove the bottom cover.
2. Remove the system fan.
3. Loosen the four screws holding the heat sink in place, as highlighted in Figure 4.34.

   The screws will not completely come out of the heat sink bracket.
4. Lift the heat sink straight up off the processor.

   It might stick a bit, since most are adhered with a thermal compound.

EXERCISE 4.12 *(continued)*

FIGURE 4.34 Removing the heat sink



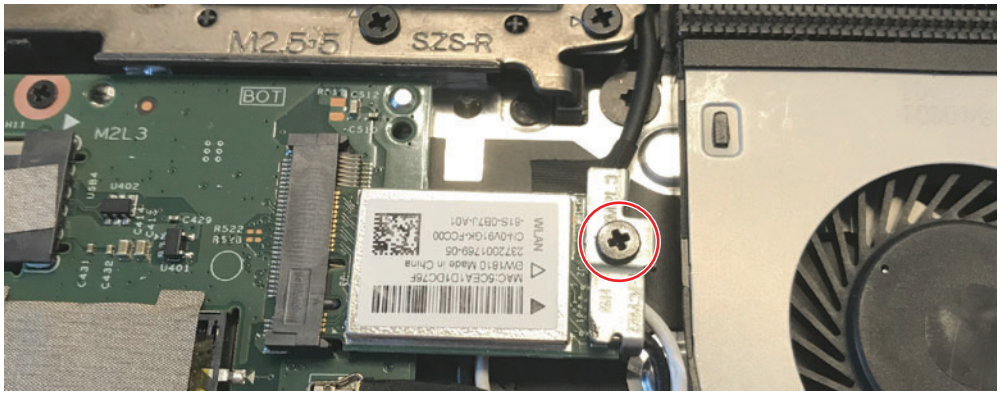
Exercise 4.13 shows you how to remove the wireless NIC. Perhaps ironically, it actually does have wires, but those are for the antenna that runs up into the display.

EXERCISE 4.13

Removing the Wireless NIC

1. Remove the bottom cover.
2. Remove the screw holding the wireless card bracket in place. It's circled in Figure 4.35.

FIGURE 4.35 Removing the wireless NIC



3. The wireless card will pop up, as shown in Figure 4.36.

FIGURE 4.36 Disconnecting the two antenna wires

4. Disconnect the two antenna wires that are attached to the end of the card.
5. Pull the wireless NIC straight out of the M.2 socket.

When you're reconnecting the wireless card, the white antenna cable will go on the main post, which is indicated by a white triangle. The black antenna cable attaches to the auxiliary connector, which is marked with a black triangle.

Exercise 4.14 shows how to remove the CMOS battery. In most laptops, the CMOS battery is covered by a black rubber coating.

EXERCISE 4.14

Removing the CMOS Battery

1. Remove the bottom cover.
2. Disconnect the connector from the motherboard. It's highlighted with a box in Figure 4.37.

FIGURE 4.37 Disconnect the CMOS battery

3. Peel the battery off the base assembly.

   Odds are good that it's been glued into place, so you might have to pull hard.

Replacement batteries often come with a small amount of adhesive to secure the new battery in place.

Flashing the system BIOS/UEFI is usually a pretty straightforward process. You can download a BIOS/UEFI update from the manufacturer's website and then run the program. Exercise 4.15 shows sample steps for flashing the BIOS on a Dell laptop.

EXERCISE 4.15

Flashing the System BIOS

1. Turn on the computer.
2. Visit `https://www.dell.com/support`.
3. Navigate to Drivers & Downloads.
4. Select the operating system installed on your laptop.
5. Find the BIOS section.
6. Select the file and click Download to download the BIOS.
7. After the download is complete, locate the file on your laptop.
8. Double-click the file to run it and begin the BIOS update process. Follow the onscreen instructions.

Physical Privacy and Security Components

Mobile devices are mostly self-contained units, which aids in their portability. One downside to the portability is that mobile devices can easily be carried away by someone other than their rightful owner. And, if a user is doing work on a laptop in a public place, others could shoulder-surf and see things they're not supposed to. The use of physical privacy and security components can help deter these unwanted behaviors.

Exam Objective 1.1 directs you to know about two physical privacy and security components: biometrics and near-field scanners.

Biometrics

Many mobile devices allow you to log in or unlock the screen through *biometrics*, or the use of a body part. For example, you can unlock your smartphone by enabling the facial recognition feature, or some older models have a fingerprint scanner for the same purpose. Other options for higher-end security systems include voice recognition and retinal scanning.

Laptops may have a fingerprint scanner built into the keyboard, such as the square one shown to the right of the Intel sticker in Figure 4.38. Others may be a rectangle or a circle. If a laptop doesn't have a biometric scanner and you want to add one, there are many USB options available.

The use of biometrics can increase device security. Someone may be able to guess your password or see you type it in and can hack you that way. But fingerprints and other biometric features are unique. You may have seen movies where a super-secret spy ring replicates someone's fingerprint to gain access to a system, but in real life that type of thing is virtually unheard of.

FIGURE 4.38 Laptop fingerprint scanner

Near-Field Scanner Features

In Volume 1 Chapter 9, "Wireless and SOHO Networking," we will cover the technical details of *near-field communication (NFC)*. For now, know that NFC is used extensively for mobile payment systems because it's fast and convenient. The payment system is enabled through the use of a *near-field scanner*.

NFC is a wireless communications method, so the signals sent to and from NFC devices could be intercepted by a malicious third device. Because the maximum distance for NFC is about 10 centimeters (about 4″), it would be pretty difficult for someone to intercept NFC data without the sender and receiver knowing it, but it's not impossible. Confidentiality, denial of service (DoS) and on-path attacks are potential security concerns with NFC. The key thing to remember is, if using NFC for payment, be aware of your surroundings and keep an eye out for potentially suspicious-looking electronic devices.

Cable Locks

One way that you can help to physically secure your laptop is through the use of a physical laptop lock, also known as a *cable lock*. Essentially, a cable lock anchors your device to a physical structure, making it nearly impossible for someone to walk off with it. Figure 4.39 shows a cable with a keyed lock. With others, a small combination lock is used. If you grew up using a bicycle lock, these will look really familiar.

FIGURE 4.39 A cable lock

Sometimes a cable lock is called a *Kensington lock* (named after a company that makes them) or a *K-lock*.

Here's how it works. First, find a secure structure, such as the permanent metal supports of your workstation at work. Then, wrap the lock cord around the structure, putting the lock through the loop at the other end. Finally, secure the lock into your cable lock hole on the back or side of your laptop (Figure 4.40), and you're secure. If you forget your combination or lose your key, you're most likely going to have to cut through the cord, which will require a large cable cutter or a hacksaw.

FIGURE 4.40 Cable lock in a laptop

If someone wants your laptop bad enough, they can break the case and dislodge your lock. Having the lock in place will deter most people looking to make off with it, though.

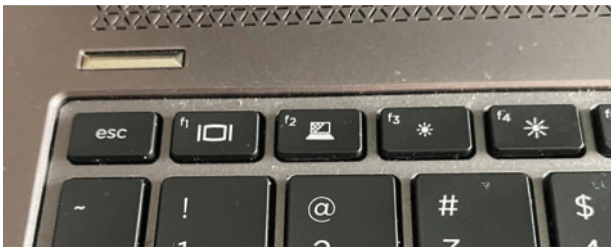
Privacy Screens

To keep curious and prying eyes from viewing a laptop (or desktop) monitor, users can install a privacy screen over the front of their display. It's a thin sheet of semi-transparent plastic that reduces the viewing angle so that a display can be read only by someone directly in front of it. An example made by `TechStar.com` is shown in Figure 4.41.

FIGURE 4.41 Laptop privacy screen

Some laptops will have a built-in privacy screen setting that can be activated by the function (Fn) keys. For example, it would be Fn+F2 on the laptop keyboard shown in Figure 4.42. In our experience, the built-in privacy screens aren't quite as effective at cutting down the viewing angle as add-on privacy screens are, but they are better than nothing.

FIGURE 4.42 Fn+F2 enables the privacy screen.



Setting Up and Configuring Accessories and Ports

Other than needing to plug in and charge for a while, laptops and mobile devices don't require anything to be plugged into them to provide a user with full functionality. With that said, there are a number of accessories that can be used to enhance functionality. These include touch pens, headsets, speakers, webcams, trackpads/drawing pads, docking stations, and port replicators. We'll examine each one of them after we take a quick look at accessory connection methods.

Connection Methods

Smaller mobile devices such as smartphones and tablets generally don't have very many physical ports on them. One or two at the most is all you're going to get. That of course limits expandability unless you are connecting the accessory using wireless methods. Laptops often give you more physical connectors—you're likely to have at least a few USB ports to play with, along with an audio jack and the power port.

Exam Objective 1.2, "Compare and contrast accessories and connectivity options for mobile devices," lists five connection methods you should be familiar with. They are USB and its variants, Lightning, NFC, Bluetooth, and tethering/hotspot.

We've covered all the connection methods listed in the A+ Exam Objective 1.2 for Exam 220-1201 in previous chapters with the exception of tethering/hotspot. Because of that, here we'll just do a quick refresh of the ones you need to know.

USB USB absolutely lives up to its name of being universal. There are multiple USB types to know, including microUSB, miniUSB, and the oval-shaped USB-C. Just about any type of accessory you could possibly want will come with a USB adapter.

Lightning Proprietary to Apple devices, the Lightning connector was the only external connector you would find on iPhones and iPads for several years. Macs and MacBooks might or might not have Lightning ports as well—if they do, they are intended mostly for device charging. Apple started moving away from Lightning in favor of USB-C in 2022, but expect Lightning to still be common for several years.

Near-Field Communication NFC is primarily used for mobile commerce and applications like scanning RFID tags, but it could be used for accessories as well. It's more likely to be used *by* an accessory, though, such as a mobile commerce scanner, than it is to connect one. Of the wireless technologies commonly used with mobile devices (Wi-Fi, Bluetooth, NFC), it has the shortest range (about 10 centimeters, or 4″).

Bluetooth As far as mobile accessories go, Bluetooth is a very popular connection technology. Mobile headsets, speakers, keyboards, and mice are all commonly connected to mobile devices using Bluetooth. Do keep in mind that if you have enabled Bluetooth on a small device such as a smartphone, it will eat up battery life a lot faster.

Tethering/Hotspot A mobile *hotspot* is when a device shares its cellular Internet connection with Wi-Fi capable devices. For example, a Wi-Fi-enabled laptop could use Wi-Fi to connect to a smartphone hotspot and get Internet access through the smartphone's cellular Internet connection. This tends to be relatively slow Internet, given cellular connection speeds, but it's better than nothing.

Tethering is when you have connected a device to a mobile hotspot. The term used to be reserved only for when you connected via USB cable, as opposed to connecting via wireless. Some devices will not function as a mobile hotspot but will allow you to tether a laptop (or other device) to it, so that the mobile device can share the cellular Internet connection.

Mobile Accessories

We could spend an entire chapter talking about mobile device accessories. There are dozens of mobile accessories on the market, including security devices, input/output tools, communication enablers, and mobile commerce endpoints. Here we will focus on the ones you need to know for the A+ exam. Most of them have already been covered in this book, so this will be a quick review of those and an introduction to the stylus.

Mobile accessories you need to be familiar with, per Exam Objective 1.2, are stylus, headset, speakers, and webcam. Docking stations, port replicators, and trackpad/drawing pad/track points are also listed in the same objective.

Stylus A *touch pen*, also known as a *stylus*, is a pen-shaped accessory used to write with or as a pointer. Touch pens come in a variety of shapes and sizes, although most have either a narrow tip (like a pen) or a soft rubber ball-like tip, kind of like a pencil eraser. The idea is that a touch pen will act as an input device on a touchscreen, enabling

freeform writing, drawing, or clicking through answers to a questionnaire, such as at a medical office. Several touch pens are shown in Figure 4.43.

FIGURE 4.43 Several touch pens

Source: Craig Spurrier / Wikimedia Commons / CC BY 2.5

Headset A *headset* is a common mobile device that delivers both output and input. The output is in the form of speakers, either in-ear or over-ear, and input comes from the attached microphone. Headsets come in a huge range of options, both wired and wireless (usually Bluetooth), from cheap models for conference calls to high-end, audiophile-grade gems. Figure 4.44 shows an inexpensive USB headset.

FIGURE 4.44 USB headset

Speakers *Speakers* are common audio output devices. Like headsets, a wide variety of speakers are available. USB and Bluetooth are the two most common options.

Webcam *Webcams* are video input devices. Most mobile devices have a built-in webcam. Stand-alone USB and Bluetooth ones are available as well.

Trackpad/Drawing Pad/Track Point We covered these accessories earlier in the chapter in the "Input Devices" section. Recall that these are input devices. Most of the time, trackpads and track points are built into laptop keyboards. Drawing pads can be added as an accessory for those who need larger freeform drawing spaces. Some have buttons, like the one shown in Figure 4.45, to provide additional functionality such as on-pad menu management or erase and undo features.

FIGURE 4.45 Drawing pad accessory with stylus

Source: Selftaken / Wikimedia Commons / CC BY SA 3.0

Docking Stations and Port Replicators

While users love laptops for their portability, sometimes a user simply needs to work with a full-sized keyboard or monitor. In addition, some users might have other external peripherals they need to plug in, such as external storage or a printer. It can be a pain to plug and unplug these devices constantly.

A *docking station* (shown in Figure 4.46) is a laptop-only peripheral that is basically an extension of the motherboard of a laptop. Because a docking station is designed to stay behind when the laptop is removed, it can contain things like a full-sized drive bay and expansion slots. The docking station in Figure 4.46 has the following ports: four USB (two are on the side), two DisplayPort, one HDMI, one RJ45, and a cable lock. It connects to the PC through a USB-C connection. It's about 5″ square and 2″ high.

FIGURE 4.46 A USB-C docking station

The docking station functions as a *port replicator*. A port replicator reproduces the functions of the ports on the back of a laptop so that peripherals such as monitors, keyboards, printers, and so on that don't travel with the laptop can remain connected to the dock and don't all have to be unplugged physically each time the laptop is taken away.

What's the Difference?

The difference between a docking station and a port replicator today is miniscule, because they basically provide the same functionality. Technically speaking, the only real difference is in how they connect to the laptop. (Some docking stations will contain built-in bays for storage or optical drives, but those are becoming increasingly rare.)

Older laptops will have a connector on the bottom to connect to a specific type of docking port, making docking ports proprietary. These ports work only with docking stations designed by the laptop's manufacturer, and vice versa. An example from the bottom of an HP laptop is shown in Figure 4.47. The HP docking station would have a corresponding connector.

FIGURE 4.47 HP laptop docking port

Today, almost all laptop port-expanding devices connect via USB-C, typically in the same port that would have been otherwise used for the power cord. These are technically port replicators, and they are not proprietary to one manufacturer.

In the real world, the terms get used interchangeably. Some manufacturers will call USB-C devices docking stations. It really doesn't matter which term is used, as long as the end user gets the functionality they need.

Summary

In this chapter, you learned about laptop and mobile device hardware. We discussed differences between laptops, mobile devices, and desktops, including the various components that make up smaller devices and how they differ in appearance and function from those on a desktop. We also talked about principles for disassembling and reassembling smaller devices in order to not lose parts.

We broke the chapter into several sections to discuss different components. We started with the case, and then moved on to displays, motherboards and processors, memory, storage, input devices, internal expansion, batteries and power adapters, physical privacy and security, other internal components, and external peripherals. In each section, we covered specifics related to laptop versions of these components and included exercises to show you how to service them.

The chapter puts particular emphasis on how to replace various components in the system, which is an important skill for many service technicians.

Finally, we ended the chapter by examining accessories and their connection methods. Connection methods included USB, Lightning, serial, NFC, Bluetooth, and tethering/hotspot. Accessories to remember include touch pens/stylus, trackpads/drawing pads/track points, headsets, speakers, webcams, and docking stations and port replicators.

Exam Essentials

Understand how to install and configure laptop components. Components include the battery, keyboard, RAM, hard drives and solid-state drives, and wireless cards.

Know laptop physical privacy and security components. They include biometrics and near-field scanner features.

Understand the components that make up a display in a mobile device. Display components include the screen, which may be a touchscreen or include a digitizer; Wi-Fi antenna; camera or webcam; microphone; and inverter.

Know how to connect mobile device accessories. Connection methods include USB, Lightning, serial, near-field communication (NFC), Bluetooth, and tethering/hotspot.

Be familiar with mobile device accessories. Ones to know include touch pens, trackpads/drawing pads/track points, stylus, headsets, speakers, webcams, docking stations, and port replicators.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. A client has a laptop with which you are unfamiliar. You are asked to perform a memory upgrade. How can you obtain the service manual for this laptop computer?
   - **A.** By pressing F1 while in Windows
   - **B.** By pressing F2 while the system is booting up
   - **C.** By reading the paper copy that comes with the laptop
   - **D.** By searching the manufacturer's website
2. Your manager tells you to replace the inverter on a broken laptop. Where will you find this component?
   - **A.** Underneath the keyboard
   - **B.** In the display
   - **C.** Next to the heat sink and fan
   - **D.** Near the CPU
3. You need to replace an LCD on a defective laptop. Which of the following are components of an LCD? (Choose all that apply.)
   - **A.** Inverter
   - **B.** Screen
   - **C.** Wi-Fi card
   - **D.** Backdrop
4. A user complains that his mouse cursor randomly jumps to different places on the screen when he types. Which device might you want to disable to prevent this from happening?
   - **A.** Trackpad
   - **B.** Mouse
   - **C.** Point stick
   - **D.** Trackball
5. A user needs to get their laptop connected to the Internet, but no Wi-Fi connections are available. They do have their smartphone with them. What is the best option for them to get Internet access for their laptop?
   - **A.** NFC
   - **B.** Tethering
   - **C.** Lightning
   - **D.** Bluetooth

6. You need to replace defective DDR4 memory in a laptop. How many pins will the replacement memory module have?
   - **A.** 200
   - **B.** 204
   - **C.** 260
   - **D.** 288

7. A user complains that his 2-in-1 laptop screen does not properly detect where he places his finger on the screen. What should you do first to resolve the issue?
   - **A.** Replace the touchscreen.
   - **B.** Replace the trackpad.
   - **C.** Recalibrate the touchscreen.
   - **D.** Recalibrate the trackpad.
   - **E.** Degauss the touchscreen.
   - **F.** Degauss the trackpad.

8. You need to wirelessly connect two accessories to a smartphone. Which of the following could be valid options to connect these wireless accessories? (Choose two.)
   - **A.** USB
   - **B.** Lightning
   - **C.** NFC
   - **D.** Bluetooth

9. There has recently been a string of hardware thefts in your office building. What should you recommend that your users implement to help avoid this problem?
   - **A.** Enable GPS tracking.
   - **B.** Use a docking station.
   - **C.** Use a cable lock.
   - **D.** Use the screensaver lock.

10. Which of the following describes the job of an inverter in a laptop?
    - **A.** It changes the screen orientation when the laptop is rotated.
    - **B.** It converts AC power into DC power for the display backlight.
    - **C.** It converts DC power into AC power for the display backlight.
    - **D.** It allows the laptop display to function in tablet mode.

11. A user is getting a new laptop. Which of the following is *not* a valid option to get his files and data from the old hard drive to the new hard drive?
    - **A.** Manually copy files from the old drive to the cloud, and then to the new drive.
    - **B.** Manually copy files from the old drive to an external hard drive, and then to the new drive.
    - **C.** Use a hard drive migration app.
    - **D.** Use the NFC disk transfer utility.

12. A user on your network wants to be able to draw images on a screen and have the laptop capture them. What type of device needs to be installed on their laptop?

A. Inverter

B. Capturer

C. Digitizer

D. Trackpad

13. A user has requested a laptop memory upgrade. The technician attempts to put in new memory, but the memory module is barely too wide for the slot. What is the most likely scenario?

A. The technician is trying to install DDR4 into a motherboard with DDR3 slots.

B. The technician is trying to install DDR3 into a motherboard with DDR4 slots.

C. The technician is trying to install MicroDIMMs into SODIMM slots.

D. The technician is trying to install DIMMs into SODIMM slots.

14. You have a user who needs to keep desktop devices such as keyboard, monitor, and mouse permanently connected so that they can be used by an attached laptop. What type of device do you recommend that they use?

A. Docking station

B. Keyboard, video, mouse (KVM) switch

C. Input/output virtual server

D. USB hub

15. The process by which the processor slows down to conserve power is officially called _____.

A. Underclocking

B. Cooling

C. Disengaging

D. Throttling

16. You need to replace a failed AC adapter for a client's laptop. When replacing the adapter, which of the following purchases is acceptable to obtain the same or better results?

A. An AC adapter with a higher voltage rating than the original

B. An AC adapter with a higher wattage rating than the original

C. A DC adapter with the same voltage rating as the original

D. An AC adapter with a lower voltage and wattage rating than the original

17. What should you do for a Li-ion battery that appears to charge fully but does not last as long as the battery's meter indicates that it will last?
    - **A.** Replace the battery.
    - **B.** Exercise the battery.
    - **C.** Calibrate the battery.
    - **D.** Short the terminals to discharge the battery.

18. You are giving a seminar to new technicians on replacing laptop hard drives. A student asks how do laptop hard drives differ from desktop hard drives. What do you tell them?
    - **A.** Laptop hard drives use completely different standards from those used by desktop hard drives for communication with the host.
    - **B.** Laptop hard drives are solid-state, whereas desktop hard drives have spinning platters.
    - **C.** Laptop hard drives require a separate power connection, whereas desktop hard drives are powered through the drive interface.
    - **D.** The most common form factors for laptop hard drives is smaller than that of a desktop hard drive.

19. One of your network users has an older laptop with no Bluetooth support. He recently received a Bluetooth headset that he needs to use with his laptop. What is the quickest and cheapest way to make the headset compatible with his laptop?
    - **A.** Buy a Wi-Fi–compatible headset.
    - **B.** Buy a USB Bluetooth adapter.
    - **C.** Buy a Mini PCIe Bluetooth adapter.
    - **D.** Replace the laptop.

20. A user wants her laptop to have a full-sized optical drive and external hard drive. Which of the following accessories is most likely to have full-sized bays for these devices?
    - **A.** Laptop base
    - **B.** Port replicator
    - **C.** Docking station
    - **D.** Mini PCIe dock

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answer compares to the authors', refer to Appendix B.

The hard drive on a Dell Inspiron 13 7000 computer failed. You have an extra hard drive of the exact same type. What would you do to replace it?

Chapter

5

Printers and Multifunction Devices

THE FOLLOWING COMPTIA A+ 220-1201 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

- ✓ **3.0 Hardware**
- ✓ **3.7 Given a scenario, deploy and configure multifunction devices/printers and settings.**
  - Properly unbox device and consider set-up location
  - Use appropriate drivers for a given operating system
    - Printer Control Language (PCL) vs. postscript
  - Firmware
  - Device connectivity
    - USB
    - Ethernet
    - Wireless
  - Public/shared devices
    - Printer share
    - Print server
  - Configuration settings
    - Duplex
    - Orientation
    - Tray settings
    - Quality
  - Security
    - User authentication
    - Badging

- Audit logs
- Secured prints

- Network scan services
  - Email
  - SMB
  - Cloud services
- Automatic document feeder (ADF)/flatbed scanner

✓ **3.8 Given a scenario, perform appropriate printer maintenance.**

- Laser
  - Maintenance: Replace toner, apply maintenance kit, calibrate, and clean
- Inkjet
  - Ink cartridge, printhead, roller, and feeder
  - Maintenance: Clean printheads, replace cartridges, calibrate, and clear jams
- Thermal
  - Feed assembly
  - Special thermal paper
  - Maintenance: Replace paper, clean heating element, and remove debris
- Impact
  - Multipart paper
  - Maintenance: Replace ribbon, printhead, and paper

Even as technology progresses and almost all of our lives seem digitized, our society is still reliant on paper. When we conduct business, we use different types of paper documents, such as contracts, letters, and, of course, money. And because most of those documents are created on computers, printers are inherently important. Even with electronic business being the norm in many situations, you will likely still have daily situations that require an old-fashioned hard copy of something.

Printers are electromechanical output devices that are used to put information from the computer onto paper. They have been around since the introduction of the computer. Other than the display monitor, the printer is the most popular output device purchased for a computer because a lot of people want and sometimes need to have paper copies of the documents they create.

In this chapter, we will discuss the details of each major type of printing technology, including impact printers, inkjet printers, laser printers, and thermal printers. Once we cover the different types, we'll talk about installing and configuring printers and finish up with a section on printer maintenance.

Printer troubleshooting is covered in Volume 1 Chapter 13, "Mobile Device, Display, and Printer Troubleshooting."

Understanding Print Technologies and Imaging Processes

Several types of printers are available on the market today. As with all other computer components, there have been significant advancements in printer technology over the years. Most of the time, when faced with the decision of purchasing a printer, you're going to be weighing performance versus cost. Some of the higher-quality technologies, such as color laser printing, are relatively expensive for the home user. Other technologies are less expensive but don't provide the same level of quality.

In the following sections, you will learn about the various types of print technologies that you will see as a technician as well as their basic components and how they function. Specifically, we are going to look at four classifications of printing—impact, inkjet, laser, and thermal.

Impact Printers

The most basic type of printer is in the category known as an *impact printer*. Impact printers, as their name suggests, use some form of impact and an inked *printer ribbon* to make an imprint on the paper. Impact printers also use a paper feed mechanism called a *tractor feed* that requires special paper. Perhaps you've seen it before—it's continuous-feed paper with holes running down both edges.

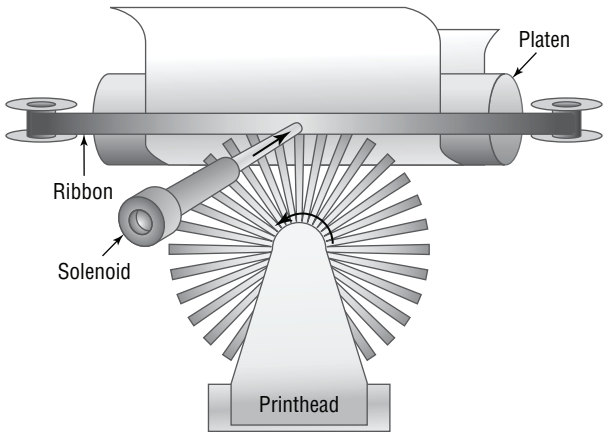
There are two major types of impact printers: daisy-wheel and dot-matrix. Each type has its own service and maintenance issues.

For Exam Objective 3.8, "Given a scenario, perform appropriate printer maintenance," you need to be familiar with multipart paper as it relates to impact printers. We will cover more maintenance concepts in the "Performing Appropriate Printer Maintenance" section later in this chapter.

Daisy-Wheel Printers

The first type of impact printer to know about is the *daisy-wheel printer*. This is one of the oldest printing technologies in use. These impact printers contain a wheel (called the daisy wheel because it looks like a daisy) with raised letters and symbols on each "petal" (see Figure 5.1). When the printer needs to print a character, it sends a signal to the mechanism that contains the wheel. This mechanism is called the *printhead*. The printhead rotates the daisy wheel until the required character is in place. An electromechanical hammer (called a *solenoid)* then strikes the back of the petal containing the character. The character pushes up against an inked ribbon that ultimately strikes the paper, making the impression of the requested character.

FIGURE 5.1 A daisy-wheel printer mechanism



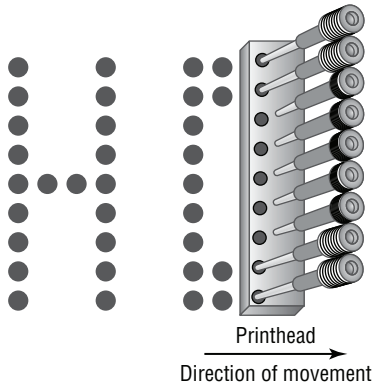
Daisy-wheel printers were among the first types of impact printer developed. Their speed is rated by the number of *characters per second (cps)* they can print. The earliest printers could print only two to four characters per second. Aside from their poor speed, the main disadvantage of this type of printer is that it makes a lot of noise when printing—so much so, in fact, that special enclosures were developed to contain the noise. There is also no concept of using multiple fonts; the font is whatever the character on the wheel looks like.

The daisy-wheel printer has a few advantages, of course. First, because it is an impact printer, you can print on multipart forms (like carbonless receipts), assuming that they can be fed into the printer properly. Sometimes, you will hear this type of paper referred to as *multipart paper* or *impact paper*. Second, it is relatively inexpensive compared to the price of a laser printer of the same vintage. Finally, the print quality is easily readable; the level of quality was given a name: *letter quality (LQ)*. Today, LQ might refer to quality that's better than an old-school typewriter (if you're familiar with them) but not up to inkjet standards.

Dot-Matrix Printers

The other type of impact printer to understand is the *dot-matrix printer*. These printers work in a manner similar to daisy-wheel printers, but instead of a spinning, character-imprinted wheel, the printhead contains a row of pins (short, sturdy stalks of hard wire). These pins are triggered in patterns that form letters and numbers as the printhead moves across the paper (see Figure 5.2).

FIGURE 5.2 Formation of images in a dot-matrix printer



The pins in the printhead are wrapped with coils of wire to create a solenoid and are held in the rest position by a combination of a small magnet and a spring. To trigger a particular pin, the printer controller sends a signal to the printhead, which energizes the wires around the appropriate print wire. This turns the print wire into an electromagnet, which repels the print pin, forcing it against the ink ribbon and making a dot on the paper. The arrangement of the dots in columns and rows creates the letters and numbers that you see on the page. Figure 5.2 illustrates this process.

The main disadvantage of dot-matrix printers is their image quality, which can be quite poor compared to the quality produced with a daisy wheel. Dot-matrix printers use patterns of dots to make letters and images, and the early dot-matrix printers used only nine pins to make those patterns. The output quality of such printers is referred to as *draft quality*—good mainly for providing your initial text to a correspondent or reviser. Each letter looked fuzzy because the dots were spaced as far as they could be and still be perceived as a letter or image. As more pins were crammed into the printhead (17-pin and 24-pin models were eventually developed), the quality increased because the dots were closer together. Dot-matrix technology ultimately improved to the point that a letter printed on a dot-matrix printer was *almost* indistinguishable from daisy-wheel output. This level of quality is known as *near letter quality (NLQ)*.

Most impact printers have an option to adjust how close the printhead rests from the ribbon. So if your printing is too light, you may be able to adjust the printhead closer to the ribbon. If it's too dark or you get smeared printing, you may be able to move the printhead back.

Dot-matrix printers are noisy, but the print wires and printhead are covered by a plastic dust cover, making them quieter than daisy-wheel printers. They also use a more efficient printing technology, so the print speed is faster (typically starting around 72 cps). Some dot-matrix printers (like the Epson DFX series, which can run up to 1550 cps) can print close to a page per second! Finally, because dot-matrix printers are also impact printers, they can use multipart forms. Because of these advantages, dot-matrix printers quickly made daisy-wheel printers obsolete.

Inkjet Printers

One of the most popular types of printers in use today is the *inkjet printer*. As opposed to impact printers, which strike the page, these printers spray ink on the page to form the image. Inkjet printers typically use a reservoir of ink, a pump, and a nozzle to accomplish this. Older inkjet printers were messy, noisy, and inefficient, but the technology is good enough now that you see plenty of photo printers using inkjet technology. You might also hear these types of printers referred to as bubble-jet printers, but that term is copyrighted by Canon. You can think of inkjets as spraying droplets of ink in a very high-definition dot-matrix pattern, although printer manufacturers would likely scoff at the comparison to an older technology.

In the following sections, you will learn the parts of an inkjet printer as well as how inkjet printers work.

Parts of a Typical Inkjet Printer

Inkjet printers are simple devices. They contain very few parts (even fewer than dot-matrix printers) and, as such, are inexpensive to manufacture. It's common today to have a $40 to $50 inkjet printer with print quality that rivals that of basic laser printers.

Inkjet parts you need to know for Exam Objective 3.8 are the ink cartridge, printhead, roller, and feeder. We will cover a few additional components as well. They aren't specifically called out in the exam objectives, but we cover them for two reasons. One, so you understand how the entire printing process works and can troubleshoot problems in real life. Two, you will be tested on troubleshooting printers, and it helps to understand the process from start to finish to know where the problem is occurring.

The printer parts can be divided into the following categories:

- Printhead/ink cartridge
- Head carriage, belt, and stepper motor
- Paper feed mechanism
- Control, interface, and power circuitry

Why So Cheap?

The cost of inkjet printers is pretty low—one would think that the printer companies make very little money on them (which is true). So why do they price them so low?

The answer is that printer companies make the vast majority of their profits from ink cartridge sales. They figure that if they can get you into their printer, you will buy their ink refills—which is true—so you are then a captive consumer. Ink cartridges are very cheap to produce and have a high profit margin. Industry experts estimate that it costs about $2 to make an ink cartridge. On average, black cartridges cost about $20–$50, and color cartridges are about $20–$60.

Printhead/Ink Cartridge

The first part of an inkjet printer is the one that people see the most: the *printhead*. This part of a printer contains many small nozzles (usually 100 to 200) that spray the ink in small droplets onto the page. Many times, the printhead is part of the *ink cartridge*, which contains a reservoir of ink and the printhead in a removable package. Most color inkjet printers include multiple printheads. Either there will be one for the black cartridge and one for the color one, or there will be one for each of the *CMYK* (cyan, magenta, yellow, and black) print inks. The print cartridge must be replaced as the ink supply runs out.

Inside the ink cartridge are several small chambers. At the top of each chamber are a metal plate and a tube leading to the ink supply. At the bottom of each chamber is a small pinhole. These pinholes are used to spray ink on the page to form characters and images as patterns of dots, similar to the way a dot-matrix printer works but with much higher resolution.

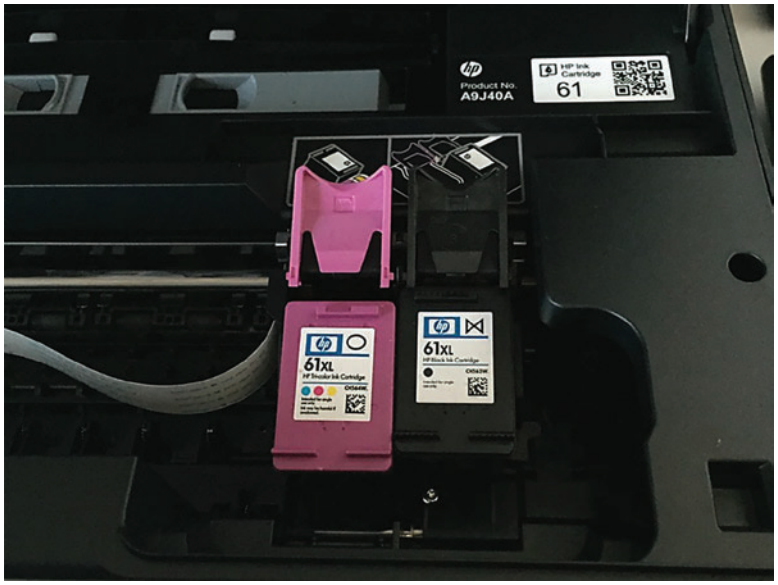
There are two methods of spraying the ink out of the cartridge. Hewlett-Packard (HP) popularized the first method: when a particular chamber needs to spray ink, an electric signal is sent to the heating element, energizing it. The elements heat up quickly, causing the ink to vaporize. Because of the expanding ink vapor, the ink is pushed out of the pinhole and forms a bubble. As the vapor expands, the bubble eventually gets large enough to break off into a droplet. The rest of the ink is pulled back into the chamber by the surface tension of the ink. When another drop needs to be sprayed, the process begins again. The second method, developed by Epson, uses a piezoelectric element (either a small rod or a unit that looks like a miniature drumhead) that flexes when energized. The outward flex pushes the ink from the nozzle; on the return, it sucks more ink from the reservoir.

When the printer is done printing, the printhead moves back to its maintenance station. The *maintenance station* contains a small suction pump and ink-absorbing pad. To keep the ink flowing freely, before each print cycle the maintenance station pulls ink through the ink nozzles using vacuum suction. The pad absorbs this expelled ink. The station serves two functions: to provide a place for the printhead to rest when the printer isn't printing and to keep the printhead in working order.

Head Carriage, Belt, and Stepper Motor

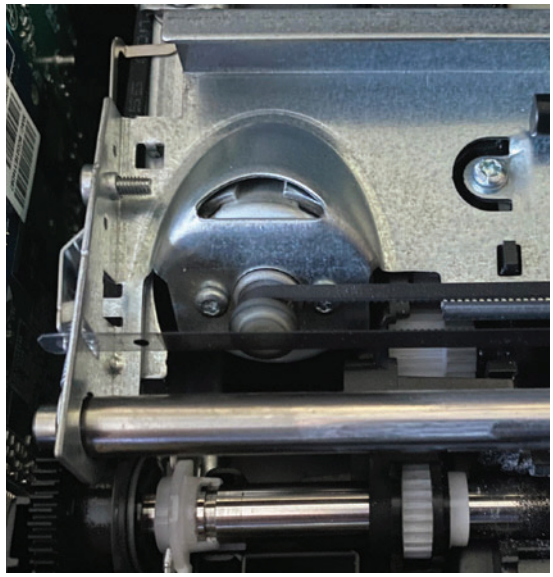
Another major component of the inkjet printer is the head carriage and the associated parts that make it move. The *printhead carriage* is the component of an inkjet printer that moves back and forth during printing. It contains the physical as well as electronic connections for the printhead and (in some cases) the ink reservoir. Figure 5.3 shows an example of a head carriage. Note the clips that keep the ink cartridge in place and the electronic connections for the ink cartridge. These connections cause the nozzles to fire, and if they aren't kept clean, you may have printing problems.

FIGURE 5.3 A printhead carriage (holding two ink cartridges) in an inkjet printer



The stepper motor and belt make the printhead carriage move. A *stepper motor* is a precisely made electric motor that can move in the same very small increments each time it is activated. That way, it can move to the same position(s) time after time. The motor that makes the printhead carriage move is also often called the *carriage motor* or *carriage stepper motor*. Figure 5.4 shows an example of a stepper motor.

FIGURE 5.4 A carriage stepper motor



In addition to the motor, a belt is placed around two small wheels or pulleys and attached to the printhead carriage. This belt, called the *carriage belt*, is driven by the carriage motor and moves the printhead back and forth across the page while it prints. To keep the printhead carriage aligned and stable while it traverses the page, the carriage rests on a small metal *stabilizer bar*. Figure 5.5 shows the entire system—the stepper motor, carriage belt, stabilizer bar, and printhead carriage. We will cover inkjet printer maintenance in the "Performing Appropriate Printer Maintenance" section later in this chapter. For now, though, know that if any part of the print carriage assembly breaks in an inkjet printer, it will likely be faster and less expensive to just go buy a new printer rather than trying to fix it. The only maintenance you should need to do is clean the printheads, replace ink cartridges, and clear paper jams.

Paper Feed Mechanism

In addition to getting the ink onto the paper, the printer must have a way to get the paper into the printer. That's where the *paper feed mechanism* (or *feeder*) comes in. The paper feed mechanism picks up paper from the paper drawer and feeds it into the printer. This component consists of several smaller assemblies. First are the *pickup rollers* (see Figure 5.6), which

are one or more rubber rollers with a slightly grippy texture; they rub against the paper as they rotate and feed the paper into the printer. They work against small cork or rubber patches known as *separation pads* (see Figure 5.7), which help keep the rest of the paper in place so that only one sheet goes into the printer. The pickup rollers are turned on a shaft by the *pickup stepper motor*.

FIGURE 5.5 Carriage stepper motor, carriage belt, stabilizer bar, and printhead carriage in an inkjet printer

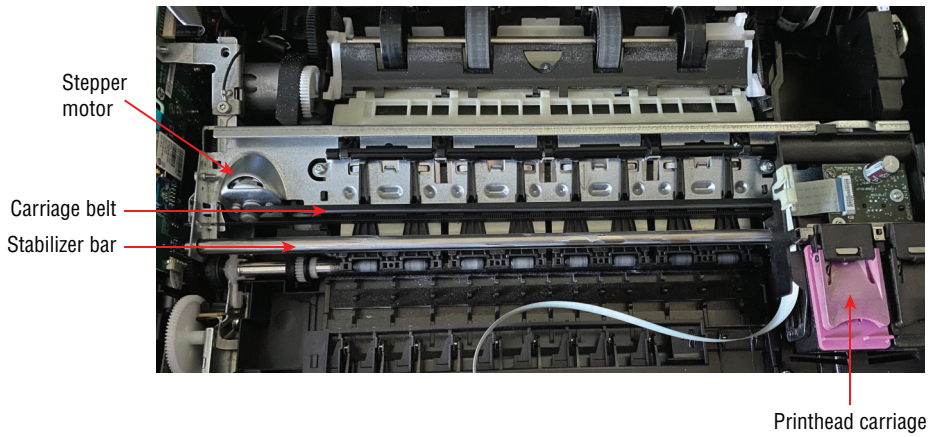
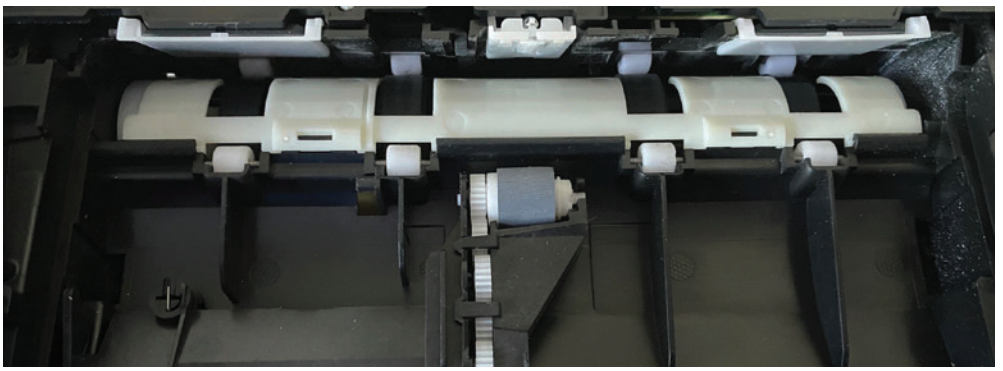


FIGURE 5.6 Inkjet pickup roller (center darker roller)



Clean pickup rollers (and other rubber rollers) with mild soap and water, not with alcohol. Alcohol can dry out the rollers, making them brittle and ineffective.

Sometimes the paper that is fed into an inkjet printer is placed into a *paper tray*, which is simply a small plastic tray in the front of the printer that holds the paper until it is fed into the printer by the paper feed mechanism. On smaller printers, the paper is placed vertically into a *paper feeder* at the back of the printer; it uses gravity, in combination with feed rollers and separation pads, to get the paper into the printer. No real rhyme or reason dictates which manufacturers use these different parts; some models use them, and some don't. Generally, more expensive printers use paper trays because they hold more paper. Figure 5.8 shows an example of a paper tray on an inkjet printer.

FIGURE 5.7 Inkjet separation pads

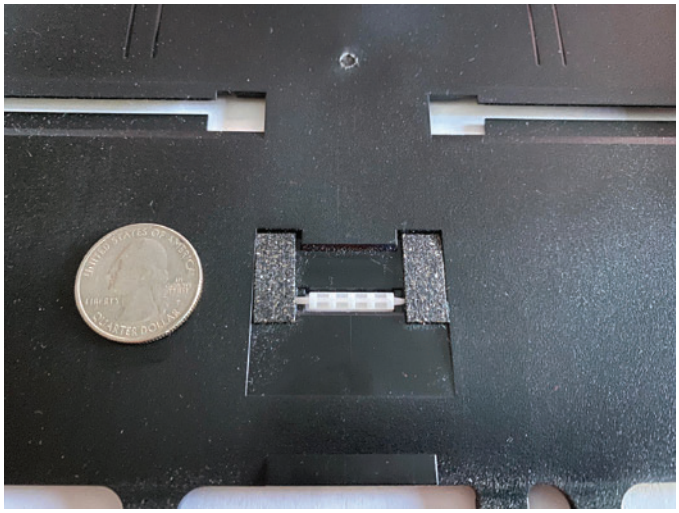
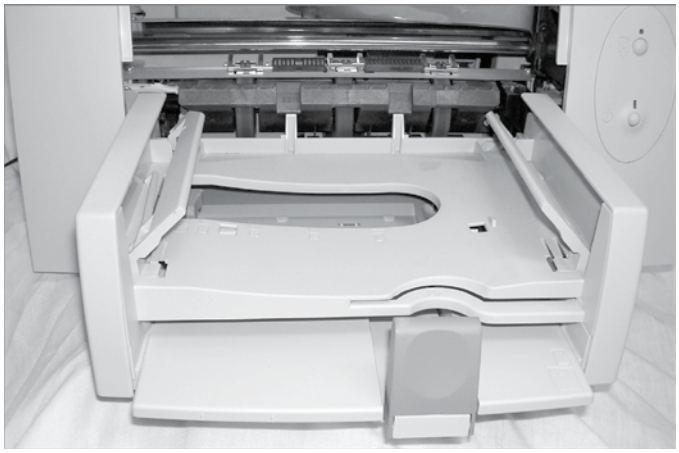


FIGURE 5.8 A paper tray on an inkjet printer



Next are the *paper feed sensors*. These components tell the printer when it is out of paper as well as when a paper jam has occurred during the paper feed process. Figure 5.9 shows an example of a paper feed sensor. Finally, there is the *duplexing assembly*, which allows for printing on both sides of a page. After the first side is printed and has a few seconds to dry, the duplexing assembly will pull the paper back in and flip it over to print the second side. Not all inkjet printers have one, but those that do will usually have them near the back bottom of the printer. Figure 5.10 shows the duplexing assembly rollers. This area is the most likely one to incur a paper jam, so there's usually a removable panel inside the printer for easy access to clear the problem.

FIGURE 5.9 A paper feed sensor on an inkjet printer

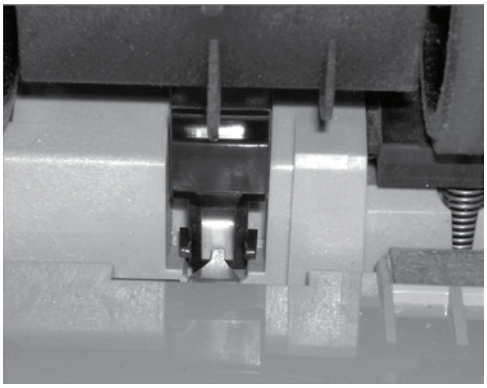
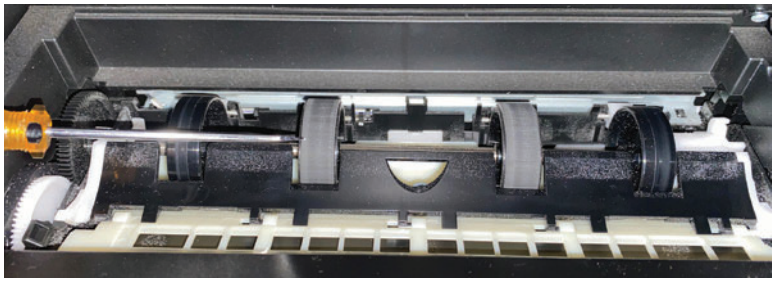


FIGURE 5.10 Duplexing assembly rollers



Being able to identify the parts of an inkjet printer is an important skill for an A+ candidate. In Exercise 5.1, you will identify the parts of an inkjet printer. For this exercise, you'll need an inkjet printer.

Many of the exercises in this chapter require printer hardware. If you don't have the proper hardware on which to practice, you can find many helpful videos online showing how to identify parts and install and perform maintenance on printers. Sites such as `IFixit.com` and YouTube `.com` are good places to start.

EXERCISE 5.1

Identifying the Parts of an Inkjet Printer

1. Unplug the inkjet printer from the power source and the computer.
2. Open the top cover to expose the inner print mechanism.
3. Locate and identify the paper tray.
4. Locate and identify the paper feed sensor.
5. Locate and identify the pickup roller(s).
6. Locate and identify the separation pad(s).
7. Locate and identify the printhead, carriage assembly, and belt.
8. Locate and identify the ink cartridge(s).

Control, Interface, and Power Circuitry

The final component group is the electronic circuitry for printer control, printer interfaces, and printer power. The *printer control circuits* are usually on a small circuit board that contains all the circuitry to run the stepper motors the way the printer needs them to work (back and forth, load paper and then stop, and so on). These circuits are also responsible for monitoring the health of the printer and for reporting that information back to the PC.

The second power component, the *interface circuitry* (commonly called a *port*), makes the physical connection to whatever signal is coming from the computer (USB, Ethernet, wireless, etc.) and also connects the physical interface to the control circuitry. The interface circuitry converts the signals from the interface into the data stream that the printer uses.

The last set of circuits the printer uses is the *power circuits*. Essentially, these conductive pathways convert 110V (in the United States) or 220V (in most of the rest of the world) from a standard wall outlet into the voltages that the inkjet printer uses, usually 12V and 5V, and distribute those voltages to the other printer circuits and devices that need it. This is accomplished through the use of a *transformer*. A transformer, in this case, takes the 110V AC current and changes it to 12V DC (among others). This transformer can be either internal (incorporated into the body of the printer) or external. Either design can be used in today's inkjets, although the integrated design is preferred because it is simpler and doesn't show the bulky transformer.

The Inkjet Printing Process

Before you print to an inkjet printer, you must ensure that the device is calibrated. *Calibration* is the process by which a device is brought within functional specifications. For example, inkjet printers need their printheads aligned so that they print evenly and don't print funny-looking letters and unevenly spaced lines. The process is part of the installation for all inkjet printers. Printers will typically run a calibration routine every time you install new ink cartridges. You will only need to manually initiate a calibration if the printing alignment appears off.

Just as with other types of printing, the inkjet printing process consists of a set of steps that the printer must follow in order to put the data onto the page being printed. The following steps take place whenever you click the Print button in your favorite software (like Microsoft Word or Google Chrome):

1. You click the Print button (or similar) that initiates the printing process.
2. The software from which you are printing sends the data to be printed to the printer driver that you have selected.

The function and use of the printer driver and page-description language are discussed later in this chapter.

3. The printer driver uses a page-description language to convert the data being printed into the format that the printer can understand. The driver also ensures that the printer is ready to print.
4. The printer driver sends the information to the printer via whatever connection method is being used (USB, Ethernet, wireless, and so on).
5. The printer stores the received data in its onboard print buffer memory.

   A *print buffer* is a small amount of memory (typically 512 KB to 16 MB) used to store print jobs as they are received from the printing computer. This buffer allows several jobs to be printed at once and helps printing to be completed quickly.
6. If the printer has not printed in a while, the printer's control circuits activate a cleaning cycle.

   A *cleaning cycle* is a set of steps the inkjet printer goes through to purge the printheads of any dried ink. It uses a special suction cup and sucking action to pull ink through the printhead, dislodging any dried ink or clearing stuck passageways.
7. Once the printer is ready to print, the control circuitry activates the paper feed motor.

   This causes a sheet of paper to be fed into the printer until the paper activates the paper feed sensor, which stops the feed until the printhead is in the right position and the leading edge of the paper is under the printhead. If the paper doesn't reach the paper feed sensor in a specified amount of time after the stepper motor has been activated, the Out Of Paper light is turned on and a message is sent to the computer.
8. Once the paper is positioned properly, the printhead stepper motor uses the printhead belt and carriage to move the printhead across the page, little by little.
9. The motor is moved one small step, and the printhead sprays the dots of ink on the paper in the pattern dictated by the control circuitry.

   Typically, this is either a pattern of black dots or a pattern of CMYK inks that are mixed to make colors.

10. Then the stepper motor moves the printhead another small step; the process repeats all the way across the page.

    This process is so quick, however, that the entire series of starts and stops across the page looks like one smooth motion.

11. At the end of a pass across the page, the paper feed stepper motor advances the page a small amount. Then the printhead repeats step 8.

12. Depending on the model, either the printhead returns to the beginning of the line and prints again in the same direction or it moves backward across the page so that printing occurs in both directions. This process continues until the page is finished.

13. Once the page is finished, the feed stepper motor is actuated and ejects the page from the printer into the output tray. (On printers with a duplexing assembly, the paper is only partially ejected so the duplexing assembly can grab it, pull it back in, and flip it over for printing on the second side.)

    If more pages need to print, the process for printing the next page begins again at step 7.

14. Once printing is complete and the final page is ejected from the printer, the printhead is *parked* (locked into rest position) and the print process is finished.

Laser Printers

Laser printers and inkjet printers are referred to as *page printers* because they receive their print job instructions one page at a time rather than receiving instructions one line at a time. There are two major types of page printers that use the electrophotographic (EP) imaging process. The first uses a laser to scan the image onto a photosensitive drum, and the second uses an array of light-emitting diodes (LEDs) to create the image on the drum. Even though they write the image in different ways, both types still follow the laser printer imaging process.

Xerox, Hewlett-Packard, and Canon were pioneers in developing the laser printer technology we use today. Scientists at Xerox developed the electrophotographic (EP) imaging process in 1971. HP introduced the first successful desktop laser printer in 1984, using Canon hardware that used the EP process. This technology uses a combination of static electric charges, laser light, and a black powdery ink-like substance called *toner*. Printers that use this technology are called EP process laser printers, or just *laser printers*. Every laser printer technology has its foundations in the EP printer imaging process.

Let's discuss the basic components of the EP laser printer and how they operate so that you can understand the way an EP laser printer works.

In past versions of the A+ exam, laser printer components were explicitly listed in the exam objectives. With Exam 220-1201 they have been removed. Much like we did for inkjet printers, we've chosen to cover them here anyway because of two reasons. First, Exam Objective 5.6, "Given a scenario, troubleshoot printer issues," still wants you to troubleshoot

laser printer problems, and you may need to troubleshoot them in real life. It's challenging to do that if you don't know how the laser printer is supposed to work in the first place. (As noted earlier, troubleshooting is covered in Volume 1 Chapter 13.) Second, if a question that requires knowledge of laser printer parts slips through to this version of the exam, you will be ready!

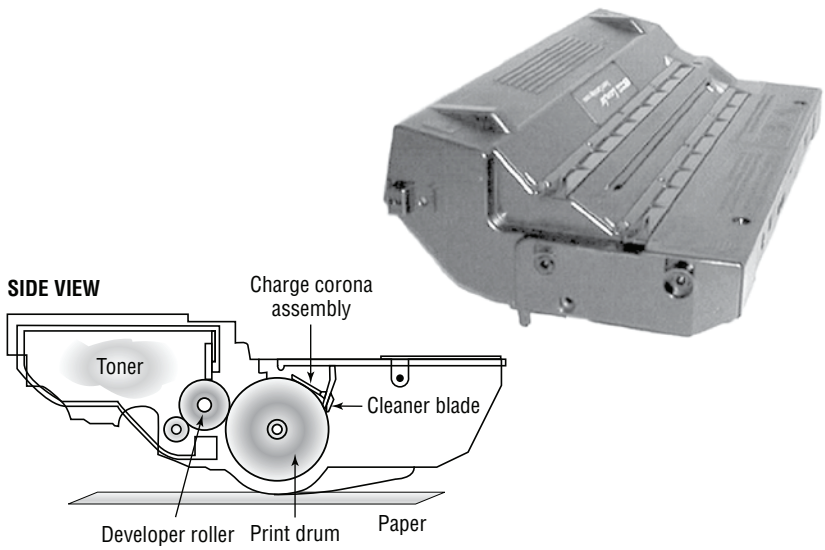
Basic Components

Most printers that use the EP imaging process contain nine standard assemblies: the toner cartridge, laser scanner, high-voltage power supply, DC power supply, paper transport assembly (including paper-pickup rollers and paper-registration rollers), transfer corona, fusing assembly, printer controller circuitry, and ozone filter. Let's discuss each of the components individually, along with a duplexing assembly, before we examine how they all work together to make the printer function.

The Toner Cartridge

The EP toner cartridge (see Figure 5.11), as its name suggests, holds the toner. Toner is a black carbon substance mixed with polyester resins to make it flow better and iron oxide particles to make it sensitive to electrical charges. These two components make the toner capable of being attracted to the photosensitive drum and of melting into the paper. In addition to these components, toner contains a medium called the *developer* (also called the *carrier*), which carries the toner until it is used by the EP process.

FIGURE 5.11 An EP toner cartridge



The toner cartridge also contains the EP print drum. This drum is coated with a photosensitive material that can hold a static charge when not exposed to light but *cannot* hold a charge when it *is* exposed to light—a curious phenomenon and one that EP printers exploit for the purpose of making images. Finally, the drum assembly contains a cleaning blade that continuously scrapes the used toner off the photosensitive drum to keep it clean.

Exposing a photosensitive drum to dust or light can damage it, but touching it will most likely render the drum inoperable! It's best to just not mess around with them.

In most laser printers, *toner cartridge* means an EP toner cartridge that contains toner and a photosensitive drum in one plastic case. In some laser printers, however, the toner and photosensitive drum can be replaced separately instead of as a single unit. If you ask for a toner cartridge for one of these printers, all you will receive is a cylinder full of toner. Consult the printer's manual to find out which kind of toner cartridge your laser printer uses.

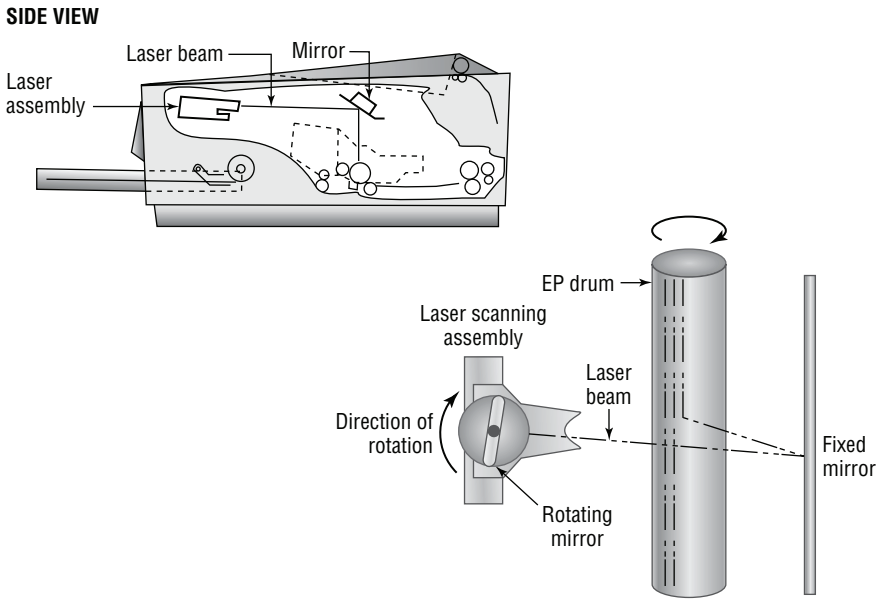
Never ship a printer anywhere with a toner cartridge installed! The jostling that happens during shipping could cause toner to spill out of the cartridge and all over the inside of the printer. This will be a huge mess to clean up! If the printer is a laser printer, remove the toner cartridge first. You can put it in a sealed, airtight bag to ship, if needed. If it's an LED page printer, there is a method to remove the photosensitive drum and toner cartridge. Also be careful to avoid too much jostling or turning the printer on its side or upside down when unboxing it. (And remember that A+ Exam Objective 3.7 calls out properly unboxing a printer!) Check your manual for details.

The Laser Scanning Assembly

As we mentioned earlier, the EP photosensitive drum can hold a charge if it's not exposed to light. It is dark inside an EP printer, except when the laser scanning assembly shines on particular areas of the photosensitive drum. When it does that, the drum discharges, but only in the area that has been exposed. As the drum rotates, the laser scanning assembly scans the laser across the photosensitive drum, exposing the image onto it. Figure 5.12 shows the laser scanning assembly.

Laser light is damaging to human eyes. Therefore, the laser is kept in an enclosure and will operate only when the laser printer's cover is closed.

FIGURE 5.12 The EP laser scanning assembly (side view and simplified top view)



High-Voltage Power Supply

The EP process requires high-voltage electricity. The high-voltage power supply (HVPS) provides the high voltages used during the EP process. This component converts AC current from a standard wall outlet (120V and 60 Hz) into higher voltages that the printer can use. This high voltage is used to energize both the charging corona and the transfer corona.

Anything with the words *high voltage* in it should make you pause before opening a device and getting your hands into it. The HVPS can hurt or kill you if you're working inside a laser printer and don't know what you're doing.

DC Power Supply

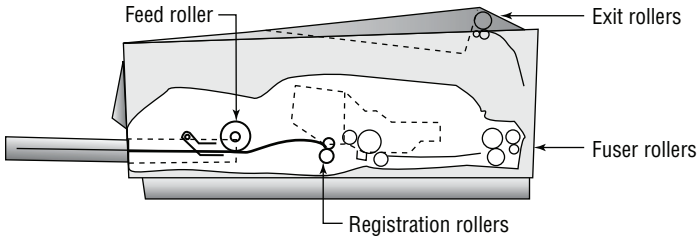
The high voltages used in the EP process can't power the other components in the printer (the logic circuitry and motors). These components require low voltages, between +5VDC and +24VDC. The DC power supply (DCPS) converts house current into three voltages: +5VDC and –5VDC for the logic circuitry and +24VDC for the paper transport motors. This component also runs the fan that cools the internal components of the printer.

Paper Transport Assembly

The paper transport assembly is responsible for moving the paper through the printer. It consists of a motor and several rubberized rollers that each performs a different function.

The first type of roller found in most laser printers is the *feed roller*, or *paper pickup roller* (see Figure 5.13). This D-shaped roller, when activated, rotates against the paper and pushes one sheet into the printer. This roller works in conjunction with a special rubber separation pad to prevent more than one sheet from being fed into the printer at a time.

FIGURE 5.13 Paper transport rollers



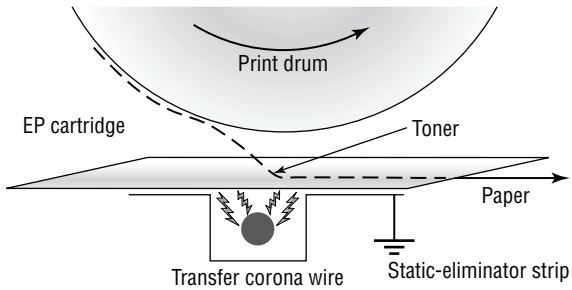
Another type of roller that is used in the printer is the *registration roller* (also shown in Figure 5.13). There are actually two registration rollers, which work together. These rollers synchronize the paper movement with the image-formation process in the EP cartridge. The rollers don't feed the paper past the EP cartridge until the cartridge is ready for it.

Both of these rollers are operated with a special electric motor known as an *electronic stepper motor*. This type of motor can accurately move in very small increments. It powers all the paper transport rollers as well as the fuser rollers.

The Transfer Corona Assembly

When the laser writes (exposes) the images on the photosensitive drum, the toner then sticks to the exposed areas. (We'll cover this later in the "Electrophotographic Imaging Process" section.) How does the toner get from the photosensitive drum onto the paper? The *transfer corona assembly* (see Figure 5.14) is given a high-voltage charge, which is transferred to the paper, which, in turn, pulls the toner from the photosensitive drum.

FIGURE 5.14 The transfer corona assembly



Included in the transfer corona assembly is a *static-charge eliminator strip* that drains away the charge imparted to the paper by the corona. If you didn't drain away the charge, the paper would stick to the EP cartridge and jam the printer.

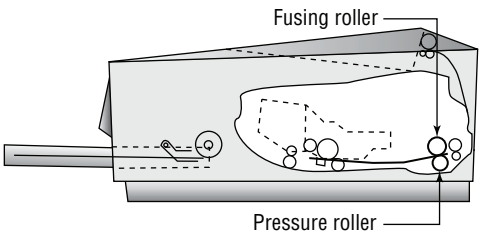
There are two types of transfer corona assemblies: those that contain a transfer *corona wire* and those that contain a transfer *corona roller*. The transfer corona wire is a small-diameter wire that is charged by the HVPS. The wire is located in a special notch in the floor of the laser printer (under the EP print cartridge). The transfer corona roller performs the same function as the transfer corona wire, but it's a roller rather than a wire. Because the transfer corona roller is directly in contact with the paper, it supports higher speeds. For this reason, the transfer corona wire is used infrequently in laser printers today.

Fusing Assembly

The toner in the EP toner cartridge will stick to just about anything, including paper. This is true because the toner has a negative static charge and most objects have a net positive charge. However, these toner particles can be removed by brushing any object across the page. This could be a problem if you want the images and letters to stay on the paper permanently.

To solve this problem, EP laser printers incorporate a device known as a *fuser* (see Figure 5.15), which uses two rollers that apply pressure and heat to fuse the plastic toner particles to the paper. You may have noticed that pages from either a laser printer or a copier (which uses a similar device) come out warm. This is because of the fuser.

FIGURE 5.15 The fuser



The fuser is made up of three main parts: a halogen heating lamp, a Teflon-coated aluminum-fusing roller, and a rubberized pressure roller. The fuser uses the halogen lamp to heat the fusing roller to between 329° F (165° C) and 392° F (200° C). As the paper passes between the two rollers, the pressure roller pushes the paper against the fusing roller, which melts the toner into the paper.

The fuser can cause severe burns! Be careful when working with it.

Printer Controller Circuitry

Another component in the laser printer that we need to discuss is the *printer controller assembly*. This large circuit board converts signals from the computer into signals for the various assemblies in the laser printer using a process known as *rasterizing*. This circuit board is usually mounted under the printer. The board has connectors for each type of interface and cables to each assembly.

When a computer prints to a laser printer, it sends a signal through a cable to the printer controller assembly. The controller assembly formats the information into a page's worth of line-by-line commands for the laser scanner. The controller sends commands to each of the components, telling them to wake up and begin the EP imaging process.

Ozone Filter

Laser printers use various high-voltage biases inside the case. As anyone who has been outside during a lightning storm can tell you, high voltages create ozone. Ozone is a chemically reactive gas that is created by the high-voltage coronas (charging and transfer) inside the printer. Because ozone is chemically reactive and can severely reduce the life of laser printer components, many older laser printers contain a filter to remove ozone gas from inside the printer as it is produced. This filter must be removed and cleaned with compressed air periodically. (Cleaning it whenever the toner cartridge is replaced is usually sufficient.) Most newer laser printers don't have ozone filters. This is because these printers don't use transfer corona wires but instead use transfer corona rollers, which dramatically reduce ozone emissions.

Duplexing Assembly

Any laser printer worth its money today can print on both sides of the paper (as can some nicer models of inkjet printers, mentioned earlier). This is accomplished through the use of a *duplexing assembly*. Usually located inside or on the back of the printer, the assembly is responsible for taking the paper, turning it over, and feeding back into the printer so the second side can be printed.

Electrophotographic Imaging Process

The *electrophotographic (EP) imaging process* is the process by which an EP laser printer forms images on paper. It consists of seven major steps, each designed for a specific goal. Although many different manufacturers word these steps differently or place them in a different order, the basic process is still the same. Here are the steps in the order in which you should know them:

1. Processing
2. Charging
3. Exposing
4. Developing
5. Transferring
6. Fusing
7. Cleaning

Before any of these steps can begin, however, the controller must sense that the printer is ready to start printing (toner cartridge installed, fuser warmed to temperature, and all covers in place). Printing cannot take place until the printer is in its ready state, usually indicated by an illuminated Ready LED light or a display that says something like `00 READY` (on HP printers). The computer sends the print job to the printer, which begins processing the data as the first step to creating output.

In past versions of the A+ exam, the laser printer imaging process was central to the printing exam objectives. Much like laser printer parts, exam 220-1201 objectives don't list the process. Again, we've chosen to cover them here anyway because of the same two reasons we covered the parts. First, you need to know how to troubleshoot laser printer problems, and second, to prepare you if a question on the process slips through.

Step 1: Processing

The *processing step* consists of two parts: receiving the image and creating the image. The computer sends the print job to the printer, which receives it via its print interface (USB, wireless, etc.). Then, the printer needs to create the print job in such a way that it can accurately produce the output.

If you think back to our discussion of impact printing earlier in this chapter, you might recall that impact printers produce images by creating one strip of dots at a time across the page. Laser printers use the same concept of rendering one horizontal strip at a time to create the image. Each strip across the page is called a *scan line* or a *raster line*.

A component of the laser printer called the *Raster Image Processor* (RIP) manages raster creation. Its responsibility is to generate an image of the final page in memory. How the raster gets created depends on the page-description language that your system is using, such as PostScript (PS) or Printer Control Language (PCL). (We will get into the details of PS and PCL in the "Driver Software" section later in the chapter.) Ultimately, this collection of lines is what gets written to the photosensitive drum and onto the paper.

Step 2: Charging

The next step in the EP process is the *charging step* (see Figure 5.16). In this step, a special wire or roller (called a *charging corona*) within the EP toner cartridge (above the photosensitive drum) gets high voltage from the HVPS. It uses this high voltage to apply a strong, uniform negative charge (around –600VDC) to the surface of the photosensitive drum.

Step 3: Exposing

Next is exposing the drum to the image, often referred to as the *exposing step*. In this step, the laser is turned on and scans the drum from side to side, flashing on and off according to the bits of information that the printer controller sends it as it communicates the individual bits of the image. Wherever the laser beam touches, the photosensitive drum's charge is severely reduced from –600VDC to a slight negative charge (around –100VDC). As the

drum rotates, a pattern of exposed areas is formed, representing the image to be printed. Figure 5.17 shows this process.

FIGURE 5.16 The charging step of the EP process

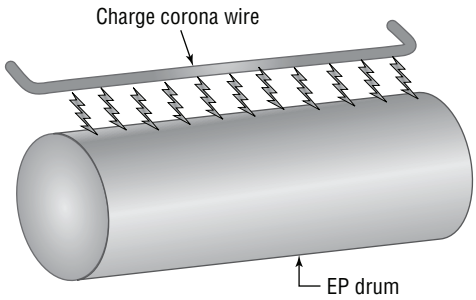
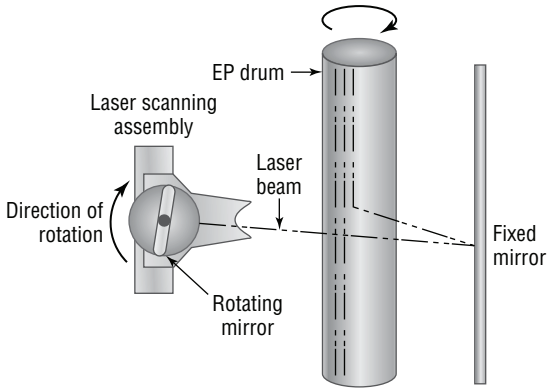


FIGURE 5.17 The exposing step of the EP process

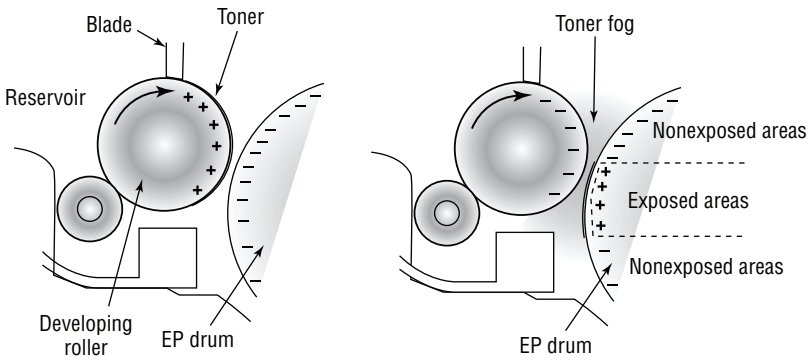


At this point, the controller sends a signal to the pickup roller to feed a piece of paper into the printer, where it stops at the registration rollers.

Step 4: Developing

Now that the surface of the drum holds an electrical representation of the image being printed, its discrete electrical charges need to be converted into something that can be transferred to a piece of paper. The EP process step that accomplishes this is the *developing step* (see Figure 5.18). In this step, toner is transferred to the areas that were exposed in the exposing step.

FIGURE 5.18 The developing step of the EP process



A metallic roller called the *developing roller* inside an EP cartridge acquires a –600VDC charge (called a *bias voltage*) from the HVPS. The toner sticks to this roller because there is a magnet located inside the roller and because of the electrostatic charges between the toner and the developing roller. While the developing roller rotates toward the photosensitive drum, the toner acquires the charge of the roller (–600VDC). When the toner comes between the developing roller and the photosensitive drum, the toner is attracted to the areas that have been exposed by the laser (because these areas have a lesser charge, –100VDC). The toner also is repelled from the unexposed areas (because they are at the same –600VDC charge and like charges repel). This toner transfer creates a fog of toner between the EP drum and the developing roller.

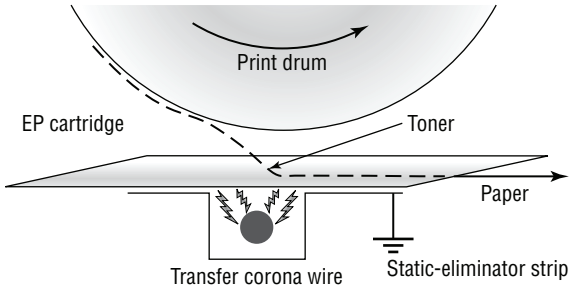
The photosensitive drum now has toner stuck to it where the laser has written. The photosensitive drum continues to rotate until the developed image is ready to be transferred to paper in the next step.

Step 5: Transferring

At this point in the EP process, the developed image is rotating into position. The controller notifies the registration rollers that the paper should be fed through. The registration rollers move the paper underneath the photosensitive drum, and the process of transferring the image can begin; this is the *transferring step*.

The controller sends a signal to the charging corona wire or roller (depending on which one the printer has) and tells it to turn on. The corona wire/roller then acquires a strong *positive* charge (+600VDC) and applies that charge to the paper. Thus charged, the paper pulls the toner from the photosensitive drum at the line of contact between the roller and the paper because the paper and toner have opposite charges. Once the registration rollers move the paper past the corona wire, the static-eliminator strip removes all charge from that line of the paper. Figure 5.19 details this step. If the strip didn't bleed this charge away, the paper would be attracted to the toner cartridge and cause a paper jam.

FIGURE 5.19 The transferring step of the EP process



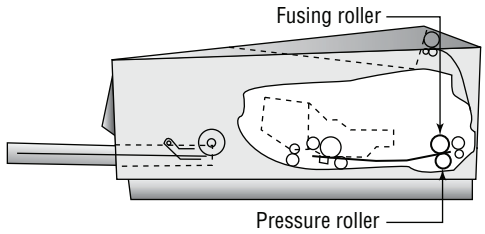
The toner is now held in place by weak electrostatic charges and gravity. It will not stay there, however, unless it is made permanent, which is the reason for the fusing step.

Step 6: Fusing

The penultimate step before the printer produces the finished product is called the *fusing step*. Here the toner image is made permanent. The registration rollers push the paper toward the fuser rollers. Once the fuser grabs the paper, the registration rollers push for only a short time longer. The fuser is now in control of moving the paper.

As the paper passes through the fuser, the 350° F fuser roller melts the polyester resin of the toner, and the rubberized pressure roller presses it permanently into the paper (see Figure 5.20). The paper continues through the fuser and eventually exits the printer.

FIGURE 5.20 The fusing step of the EP process

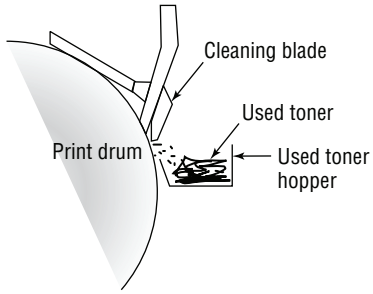


Once the paper completely exits the fuser, it trips a sensor that tells the printer to finish the EP process with the cleaning step.

Step 7: Cleaning

In the last part of the laser imaging process, a rubber blade inside the EP cartridge scrapes any toner left on the drum into a used toner receptacle inside the EP cartridge, and a fluorescent lamp discharges any remaining charge on the photosensitive drum. (Remember that the drum, being photosensitive, loses its charge when exposed to light.) This step is called the *cleaning step* (see Figure 5.21).

FIGURE 5.21 The cleaning step of the EP process



The EP cartridge is constantly cleaning the drum. It may take more than one rotation of the photosensitive drum to make an image on the paper. The cleaning step keeps the drum fresh for each use. If you didn't clean the drum, you would see ghosts of previous pages printed along with your image.

The amount of toner removed in the cleaning process is quite small, and the cartridge will run out of toner before the used toner receptacle fills up. The toner that's in the receptacle is useless because the imaging process has already chemically altered it. In addition, it's considered a hazardous substance. Recycle the print cartridge and don't pay attention to the leftover toner.

At this point, the printer can print another page, and the EP process can begin again.

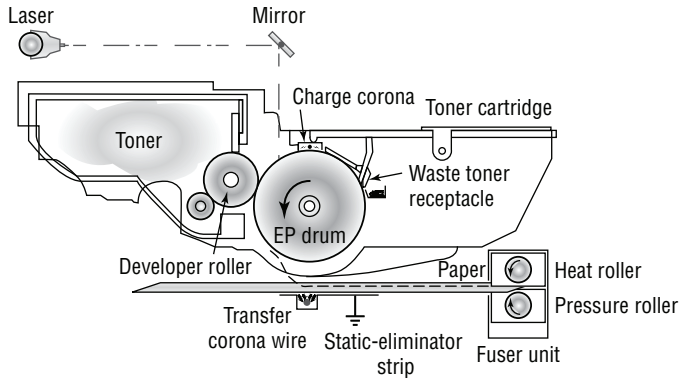
Summary of the EP Imaging Process

Figure 5.22 provides a diagram of all the parts involved in the EP printing process. Here's a summary of the process, which you should commit to memory:

1. The printer receives and processes the image and stores a page in memory.
2. The printer places a uniform –600VDC charge on the photosensitive drum by means of a charging corona.
3. The laser "paints" an image onto the photosensitive drum, discharging the image areas to a much lower voltage (–100VDC).
4. The developing roller in the toner cartridge has charged (–600VDC) toner stuck to it. As it rolls the toner toward the photosensitive drum, the toner is attracted to (and sticks to) the areas of the photosensitive drum that the laser has discharged.
5. The image is then transferred from the drum to the paper at its line of contact by means of the transfer corona wire (or corona roller) with a +600VDC charge. The static-eliminator strip removes the high positive charge from the paper, and the paper, now holding the image, moves on.

6. The paper then enters the fuser, where a fuser roller and the pressure roller make the image permanent. The paper exits the printer.
7. The printer uses a rubber scraper to clean the photosensitive drum. At that point, it is ready to print the next page or it returns to the ready state.

FIGURE 5.22 The EP imaging process



Thermal Printers

The types of printers that you have learned about so far in this chapter account for 90 percent of all paper printers that are used with home or office computers and that you will see as a repair technician. The remaining 10 percent consist of other types of printers that primarily differ by the method they use to put colored material on the paper to represent what is being printed. Examples of these include solid ink, dye sublimation, and thermal printers. Keep in mind that, for the most part, these printers operate like other paper printers in many ways: they all have a paper feed mechanism (sheet-fed or roll); they all require consumables such as ink or toner and paper; they all use the same interfaces, for the most part, as other types of printers; and they are usually about the same size.

Thermal printer concepts you need to know for A+ Exam Objective 3.8 include the feed assembly and special thermal paper. We'll cover the maintenance topics in the "Performing Appropriate Printer Maintenance" section later in this chapter.

Thermal printing technology is primarily used in point-of-sale (POS) terminals. They print on special *thermal paper*, which is a kind of waxy paper that comes on a roll; it's heat-sensitive paper that turns black when heat passes over it. *Thermal printers* work by using a printhead that is the width of the paper. When it needs to print, a heating element heats certain spots on

the printhead. The paper below the heated printhead turns black in those spots. As the paper moves through the printer, the pattern of blackened spots forms an image on the page of what is being printed. Another type of thermal printer uses a heat-sensitive ribbon instead of heat-sensitive paper. A thermal printhead melts wax-based ink from the ribbon onto the paper. These are called *thermal transfer printers* or *thermal wax-transfer printers*.

Thermal direct printers typically have long lives because they have few moving parts. The only unique part that you might not be as familiar with is the paper feed assembly, which often needs to accommodate a roll of paper instead of sheets. The paper is somewhat expensive, doesn't last long (especially if it is left in a very warm place, like a closed car in summer), and produces poorer-quality images than the paper used by most of the other printing technologies.

Installing and Maintaining Printers

Odds are that everyone either owns a printer or has easy access to a printer at a library, work, or some other place. Many retailers and computer manufacturers make it incredibly easy to buy a printer because they often bundle a printer with a computer system as an incentive to get you to buy.

The CompTIA A+ 220-1201 exam will test your knowledge of the procedures to install and maintain printers. In the following sections, we will discuss installing and sharing local and networked printers, implementing network printer security and scan services, and performing printer maintenance.

Many inkjet and laser printers today provide several functions, including copying, scanning, and faxing. These devices go by several names, including multifunctional printer, multifunction device (MFD), multifunction product (MFP), or all-in-one printer.

Installing and Sharing Local Printers

Although every device is different, there are certain accepted methods used for installing almost all of them. The following procedure works for installing many kinds of devices:

1. Choose the proper setup location for the device.
2. Attach the device using a local port (generally USB, but may be something else) and connect the power.
3. Install and update the device driver, update the firmware, and calibrate the device.
4. Configure options and settings.
5. Print a test page.
6. Verify compatibility with the operating system and applications.
7. Educate users about basic functionality.

Before installing any device, read your device's installation instructions. There are exceptions to every rule.

Step 1: Choose the Proper Setup Location for the Device

Before installing the printer, scout out the best location for it. If it's a home-based printer, you may want to choose a convenient but inconspicuous location. In an office setting, having the printer centrally located may save a lot of headaches. Consider the users and how easy or difficult it will be for them to get to the printer. Also consider connectivity. For a wireless printer, how close is it to an access point? (It should be close.) If it's a wired printer, it will need to be near an RJ45 wall jack. Finally, always choose a flat, stable surface. After you determine the location, be sure to properly unbox the device. Most printers will come with a quick setup guide—often a glossy poster-sized printout—that will show you how to properly unbox and connect the device. You may need a box cutter, scissors, or other types of tools to get the box open without destroying it. In general, unboxing steps include:

1. Cutting the tape on the box, taking care as to not damage anything important.
2. Removing any packing materials.
3. Lifting the printer out gently. Some printers are heavy so employ team lifting if needed.
4. Placing it on a stable surface.
5. Removing any protective tape or plastic from the printer itself, including checking inside the paper trays and under the scanner lid (if applicable).
6. Installing paper trays and then ink or toner cartridges according to the manufacturer's instructions.
7. Checking for any additional packaging materials inside the printer that need to be removed.
8. Recycling packing materials and the box.

Always consult the user manual for specific details regarding the printer model.

Exam Objective 3.7, "Given a scenario, deploy and configure multifunction devices/printers and settings," wants you to be able to properly unbox the device and consider setup location.

Step 2: Attach the Device Using a Local Port and Connect the Power

After you have unboxed the printer, with the device powered off, connect it to the host computer. Connecting a printer is generally done with one of three methods: Universal Serial Bus (USB), Ethernet (RJ45), or wireless. Today, the vast majority of local printers are USB, but wireless options supporting Bluetooth or Wi-Fi are common as well.

For Exam Objective 3.7, know the three common connectivity methods: USB, Ethernet, and wireless.

Universal Serial Bus We covered USB in Volume 1 Chapter 3, "Peripherals, Cables, and Connectors," so review there if you need technical details. Given that USB is so popular, it shouldn't surprise you that it's the most common type of wired printer interface as well. Printers often use the USB Type-B connector (it's the square one). The convenience for printers is that it has a higher transfer rate than older serial or parallel connections, and it automatically recognizes new devices. And, of course, USB is physically very easy to connect.

Older printers might connect using a DB-9 serial cable or a DB-25 parallel cable. Most printers with those connections are fortunately retired at this point.

Ethernet Many printers sold today have a wired Ethernet interface that allows them to be hooked directly to an Ethernet cable. These printers have an internal network interface card (NIC) and ROM-based software that allow them to communicate on the network with servers and workstations.

As with any other networking device, the type of network interface used on the printer depends on the type of network to which the printer is being attached. It's likely that the only connection type that you will run into is RJ45 for an Ethernet connection.

Wireless Clearly, people love their Wi-Fi because it enables them to roam around their home or office and still remain connected to one another and to their network. It logically follows that someone came up with the brilliant idea that it would be nice if printers could be that mobile as well—after all, many are on carts with wheels. Most printers today have built-in Wi-Fi interfaces, while others can accept wireless network cards. Wi-Fi–enabled printers support nearly all 802.11 standards (a, b, g, n, ac, ax, be), and the availability of devices will mirror the current popularity of each standard.

The wireless technology that is especially popular among peripheral manufacturers is *Bluetooth*. Bluetooth is a short-range wireless technology; most devices are specified to work within 10 meters (33 feet). Printers such as the HP Sprocket series and OfficeJet 150 mobile printers have Bluetooth capability. We will cover Bluetooth and Wi-Fi in more detail in Volume 1 Chapter 9, "Wireless and SOHO Networking."

When printing with a Bluetooth-enabled device (like a smartphone or tablet) and a Bluetooth-enabled printer, all you need to do is get within range of the device (that is, move closer), select the printer driver from the device, and choose Print. The information is transmitted wirelessly through the air using radio waves and is received by the device.

When Wi-Fi is used to connect printers to a network on a more permanent basis, it is known as *infrastructure mode*. Wi-Fi and Bluetooth can be used to connect a printer temporarily to a single computer (or mobile device), and the connection does not have permanent status. This type of configuration is known as an ad hoc network connection.

Once you have connected the device, connect power to it using whatever supplied power adapter comes with it. Some devices have their own built-in power supply and just need an AC power cord connecting the device to the wall outlet, whereas others rely on an external transformer and power supply. Then, turn on the device.

Step 3: Install and Update the Device Driver, Update the Firmware, and Calibrate the Device

Once you have connected and powered up the device, Windows should automatically recognize it. When it does, a screen will pop up saying that Windows is installing the driver. If a driver is not found, you will be given the option to specify the location of the driver. You can insert the driver media (flash drive, DVD, etc.) that came with the device or point it to the location of the driver files on your hard drive, and the wizard will guide you through the device driver installation.

If Windows fails to recognize the device, you can start the process manually by initiating the Add Printer Wizard to troubleshoot the installation and to install the device drivers. To start the wizard in Windows 11, type **`printer`** into the Windows search bar and then click Printers & Scanners when it appears. Click the Add Device button next to Add A Printer Or Scanner, as shown in Figure 5.23. If a printer is not detected, a link will appear with the words "The printer that I want isn't listed." Click that link, and it will start the wizard shown in Figure 5.24.

FIGURE 5.23 Printers & Scanners

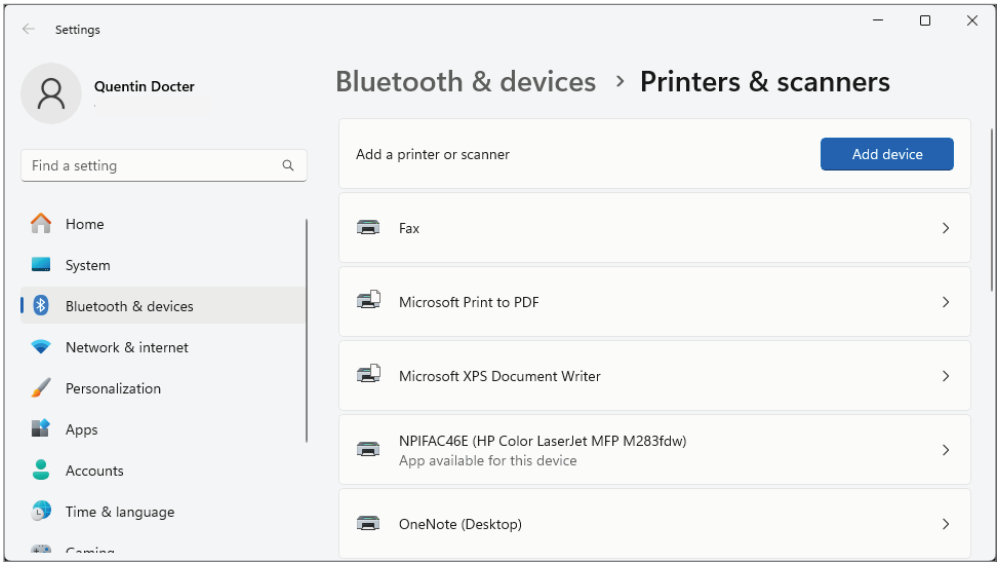
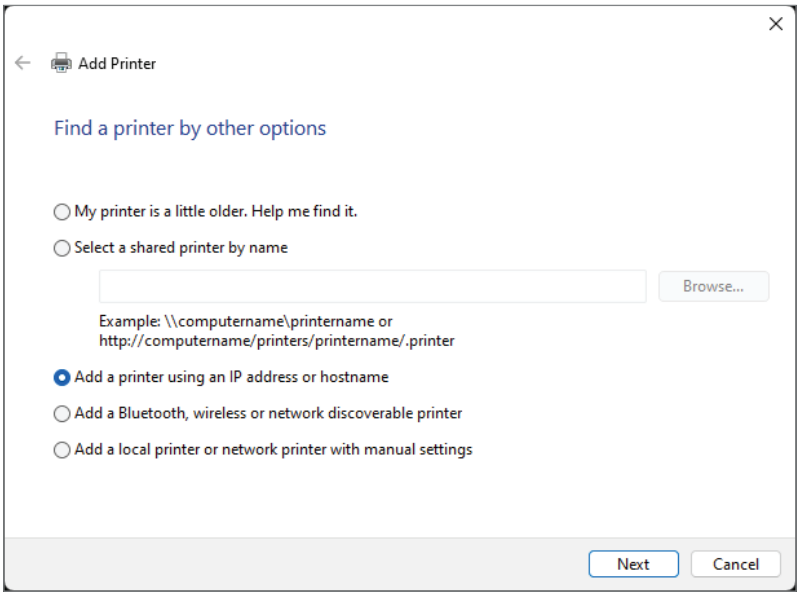


FIGURE 5.24 Add Printer wizard



Driver Software

The *driver* software controls how the printer processes the print job. When you install a printer driver for the printer you are using, it allows the computer to print to that printer correctly. The driver must be written specifically for the operating system the computer is using and for the printer being used. In other words, macOS clients need a different driver than Windows clients need, even to print to the same printer. If you need a refresher on how to install a driver, it's covered in Volume 1 Chapter 2, "Expansion Cards, Storage Devices, and Power Supplies."

If you're working with a Windows-based operating system, Microsoft refers to the software that is installed on the computer and lets you print as the "printer." The physical device where the paper comes out is referred to as the "print device." Here, when we say "printer," we mean the physical device.

When you need to print, you select the printer driver for your printer from a preconfigured list. The driver that you select has been configured for the type, brand, and model of printer, as well as the computer port to which it is connected. You can also select which paper tray the printer should use as well as any other features the printer has (if applicable). Also, each printer driver is configured to use a particular page-description language.

If the wrong printer driver is selected, the computer will send commands in the wrong language. If that occurs, the printer will print several pages full of garbage (even if only one page of information was sent). This "garbage" isn't garbage at all but the printer page-description language commands printed literally as text instead of being interpreted as control commands.

A *page-description language* works just as its name implies: it describes the whole page being printed by sending commands that describe the text as well as the margins and other settings. The controller in the printer interprets these commands and turns them into laser pulses (or pin strikes). Several printer communication languages exist, but the two most common are *PostScript (PS)* and *Printer Control Language (PCL)*.

Hewlett-Packard (HP) developed PCL in 1984, and HP's name for the acronym is Printer Command Language. The CompTIA A+ exam objectives say Printer Control Language. They're not going to ask you to define the acronym on the exam. Be familiar with the basics of what it does, where it's used, and the problems caused by using the wrong page description language.

The first page-description language was PostScript, developed by Adobe in 1984. It made printing graphics fast and simple. Here's how PostScript works. The PostScript printer driver describes the page in terms of "draw" and "position" commands. The page is divided into a very fine grid (as fine as the resolution of the printer). When you want to print a square, a communication like the following takes place:

```
POSITION 1,42%DRAW 10%POSITION 1,64%DRAW10D% . . .
```

These commands tell the printer to draw a line on the page from line 42 to line 64 (vertically). In other words, a page-description language tells the printer to draw a line on the page and gives it the starting and ending points—and that's that. Rather than send the printer the location of each and every dot in the line and an instruction at each and every location to print that location's individual dot, PostScript can get the line drawn with fewer than five instructions. As you can see, PostScript uses commands that are more or less in English. The commands are interpreted by the processor on the printer's controller and converted into the print-control signals.

When HP developed PCL, it was originally intended for use with inkjet printers as a competitor to PostScript. Since then, its role has been expanded to virtually every printer type, and it's a de facto industry standard.

Most newer printers can handle both PS and PCL and will automatically translate for you. Therefore, it's less likely that you'll install the wrong printer driver than it was several years ago.

The main advantage of page-description languages is that they move some of the processing from the computer to the printer. With text-only documents, they offer little benefit. However, with documents that have large amounts of graphics or that use numerous fonts, page-description languages make the processing of those print jobs happen much faster. This makes them an ideal choice for laser printers, although nearly every type of printer uses them.

Exam Objective 3.7 wants you to be familiar with using the appropriate printer drivers, PCL (more common and used for typical office printing) and PostScript (less common and used for professional graphics), and updating device firmware.

Updating Firmware

Once the driver is installed, the device will function. But before anyone uses it, a good next step to take is to check for any firmware upgrades for the printer. Many printers will have a BIOS with firmware, similar to a computer, and manufacturers will occasionally release new versions with new features or security. Firmware updates can be found on the printer manufacturer's website.

Calibrating the Printer

Some print devices must be calibrated before use, particularly inkjets. If the printer requires this step, it will tell you. You'll need to walk through a few steps before the printer will print, but instructions will be provided either on your computer screen or on the printer's display.

When you are working with print media, it is especially important to calibrate all your hardware, including your monitor, scanner, printer, and digital camera, to ensure color matching.

Each manufacturer's process is different, but a typical alignment/calibration works like this:

1. During software installation, the installation wizard asks you if you would like to calibrate now, to which you will respond Yes or OK.
2. The printer prints out a sheet with multiple sets of numbered lines. Each set of lines represents an alignment instance.
3. The software will ask you which set(s) looks the best. Enter the number and click OK or Continue.
4. Some alignment routines end at this point. Others will reprint the alignment page and see if the alignment "took." If not, you can reenter the number of the one that looks the best.
5. Some devices with built-in scanners will ask you to scan the printed image to ensure that calibration looks correct.
6. Click Finish to end the alignment routine.

Real World Scenario

Life Without a Page-Description Language

Page-description languages make printing an efficient process. But what about when they are not used? In situations like this, the computer sends all the instructions the printer needs in a serial stream, like so: Position 1, print nothing; Position 2, strike pins 1 and 3; Position 3, print nothing. This type of description language works well for dot-matrix printers, but it can be inefficient for laser printers. For example, if you wanted to print a page using a standard page-description language and only one character was on the page, there would be a lot of signals wasted on the "print nothing" commands.

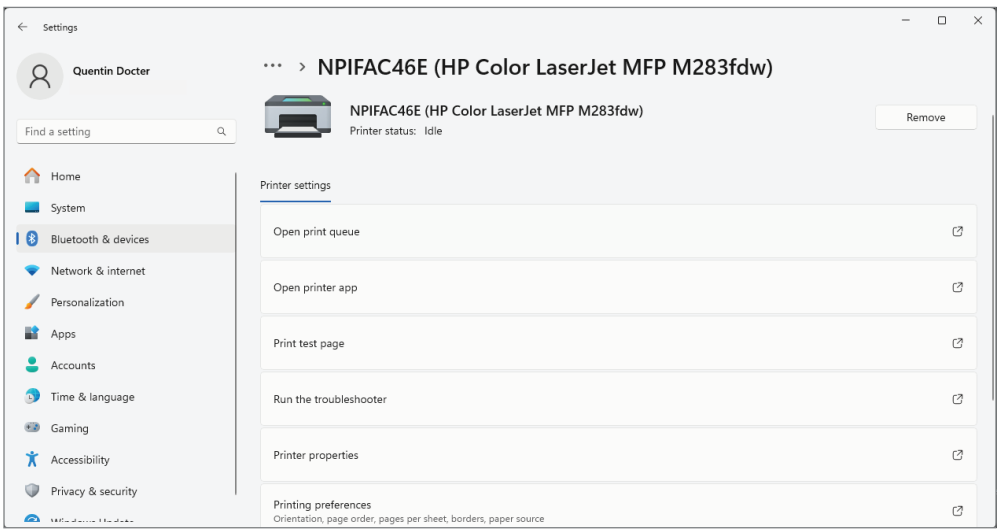
With graphics, the commands to draw a shape on the page are relatively complex. For example, to draw a square, the computer (or printer) has to calculate the size of the square and convert that into a lot of "strike pin *x*" (or "turn on laser") and "print nothing" commands. This is where the other types of page-description languages come into the picture.

Step 4: Configure Options and Settings

Once you have installed the drivers and calibrated the device, you can configure any options that you would like for the printer. All the settings and how to change them can be found online or in your user manual.

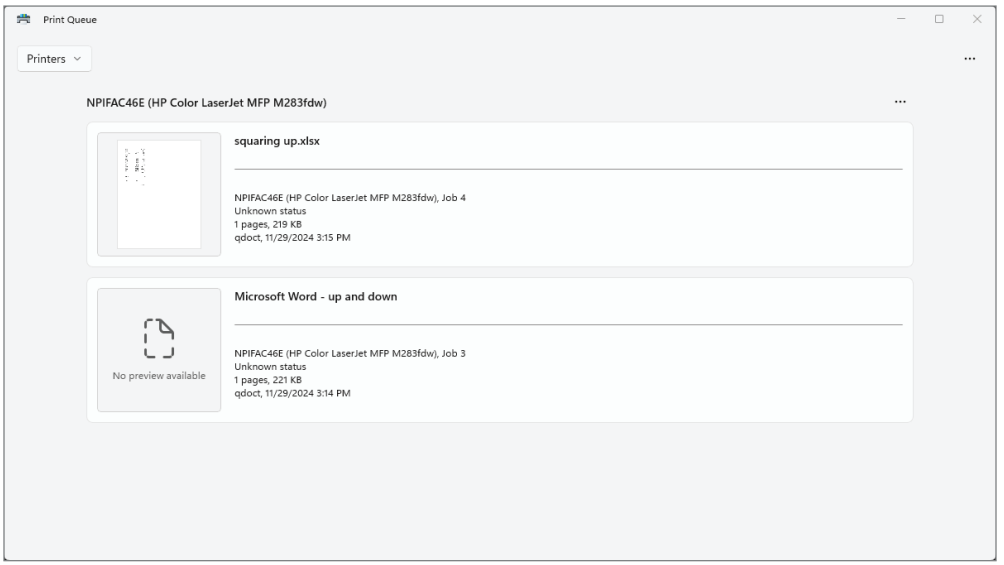
Where you configure specific printer properties depends a lot on the printer itself. As a rule of thumb, you're looking for the Printer Properties or Printing Preferences applet. In Windows 11, if you open Printers & Scanners, you will see the list of printers that are installed. Clicking an installed printer will reveal several options, shown in Figure 5.25, five of which we'll examine here. There's also a button at the top to remove the device if you choose.

FIGURE 5.25 Printer options



Open Print Queue The print queue (Figure 5.26) is the list of jobs waiting to be printed. This option lets you manage print jobs by pausing, restarting, and canceling print jobs that are in the queue. For example, one could be paused to let others go first, restarted if there was a mishap mid-print, or canceled if no longer needed. Jobs are managed by right-clicking them, or by clicking the three dots in the upper-right corner.

FIGURE 5.26 Print Queue



Open Printer App The Open printer app will differ depending on the printer, but in general it will give you settings provided by the manufacturer. In the case of HP printers, it will open HP Smart (Figure 5.27), which among other things lets you configure scan options, order ink or toner, or change printer settings.

Print Test Page Printer test page is a good diagnostic tool to check printer connectivity and basic functionality. If a test page won't print, nothing else will either and you should check your connections.

Printer Properties Printer Properties General tab (Figure 5.28) is incredibly useful, and the dialog box will be very similar for any printer that you install. Here the most important tabs are Sharing, Ports, Advanced, and Security. The Sharing tab (Figure 5.29) is where the printer is shared, meaning it's made available to other users on the network. On the Ports tab (Figure 5.30), you can configure the port the printer operates on as well as set up printer pooling. This would be used if you have multiple physical printers that operate under the same printer name.

FIGURE 5.27 HP Smart printer app

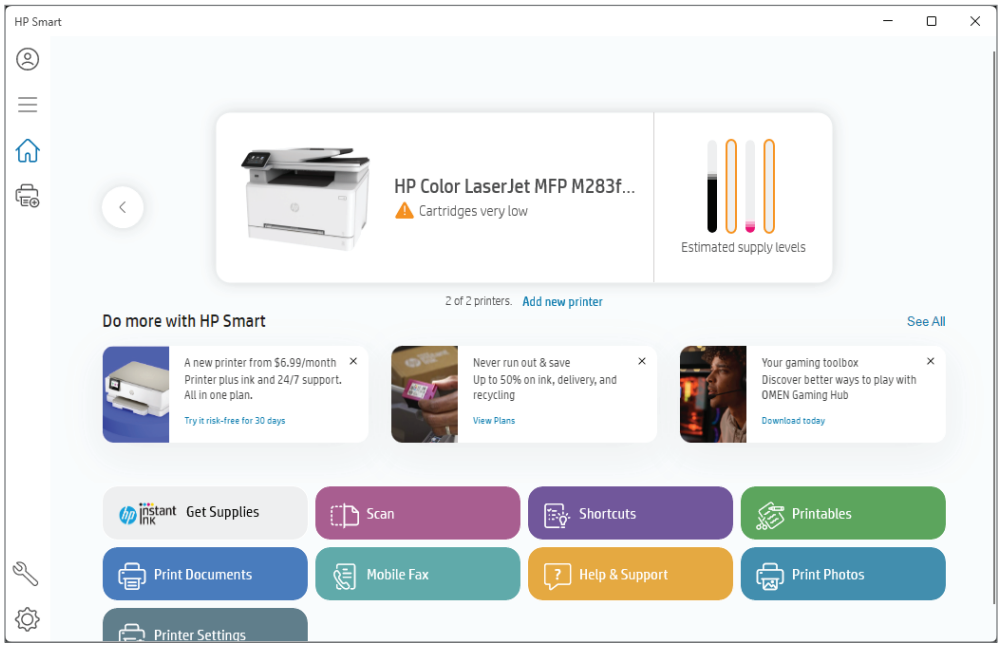


FIGURE 5.28 Printer Properties, General tab

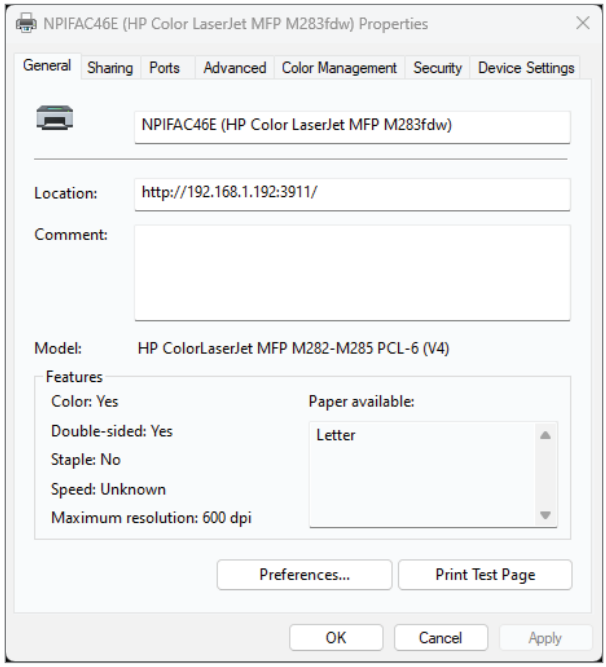


FIGURE 5.29 Sharing tab

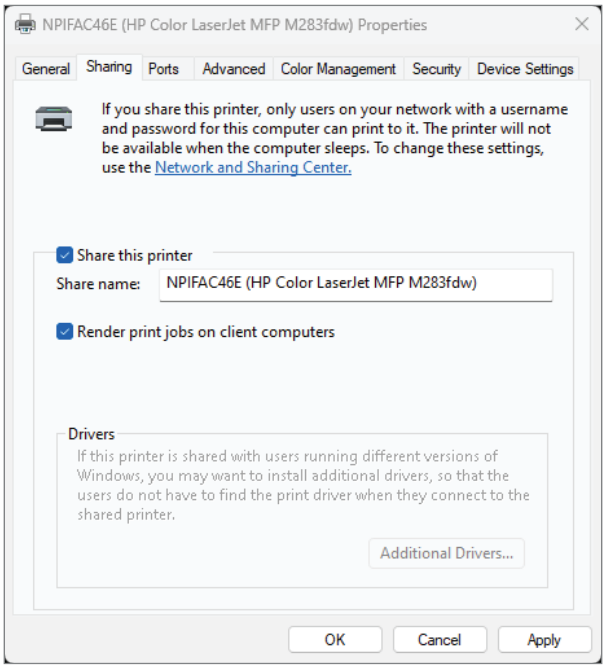
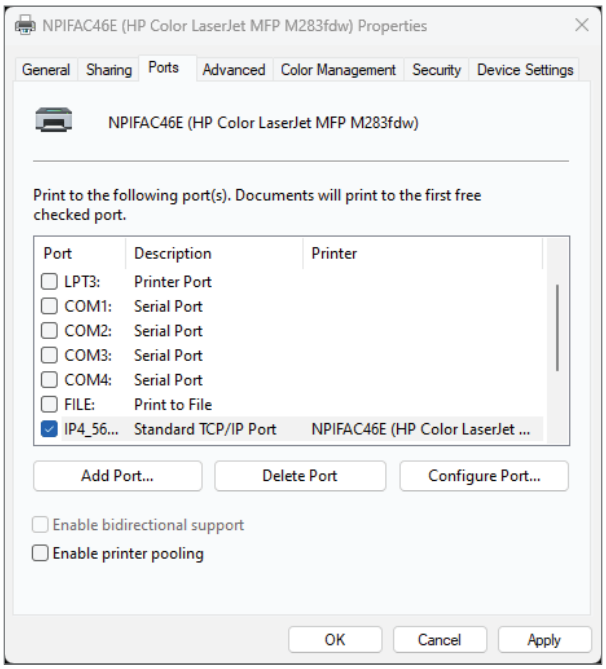


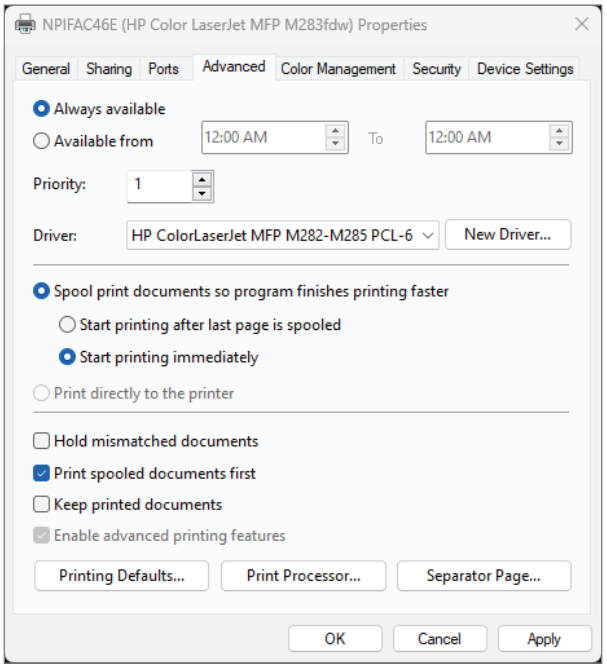
FIGURE 5.30 Ports tab



If you're going to configure a printer pool, remember that all of the output can appear on any of the devices that are part of that pool. Make sure that all the printers in that pool are in the same physical location. Otherwise, you will have people wandering all over the office trying to find their printouts. That might be entertaining for you, but not so much for them.

Advanced (Figure 5.31) contains control options. On this tab, you can configure the printer to be available during only certain hours of the day. This might be useful if you're trying to curtail after-hours printing of non–work-related documents, for example. You can also configure the spool settings. For faster printing, you should always spool the jobs instead of printing directly to the printer. However, if the printer is printing garbage, you can try printing directly to it to see if the spooler is causing the problem. (Spooling a job means it gets entirely processed in memory so by the time it starts printing the job goes faster versus needing to pause in the middle to retrieve more of the job.)

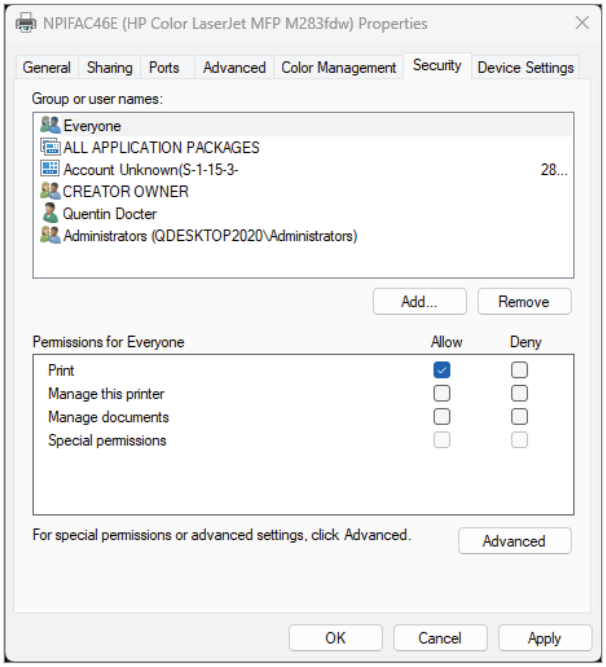
FIGURE 5.31 Advanced tab



Regarding the options at the bottom, you will always want to print spooled documents first because that speeds up the printing process. If you need to maintain an electronic copy of all printed files, select Keep Printed Documents. Keep in mind that doing so will

eat up a lot of hard disk space and could potentially create a security risk. Finally, there is the Security tab (Figure 5.32), where you control who can print to the printer.

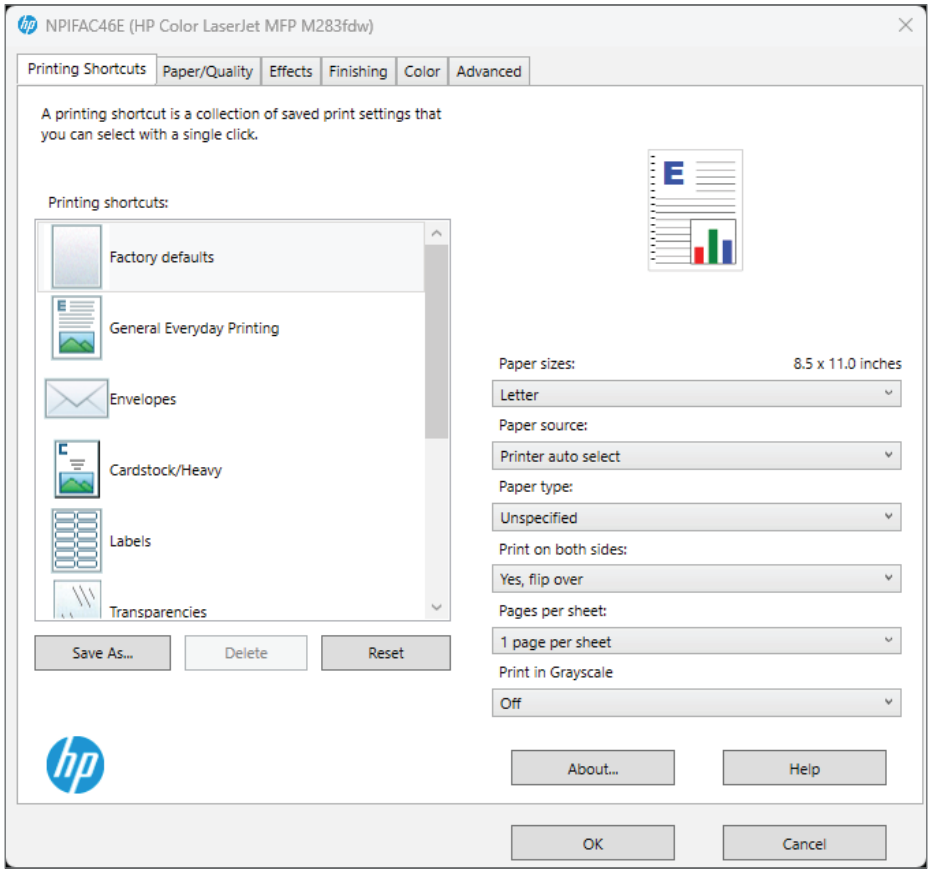
FIGURE 5.32 Security tab



Printing Preferences If you click Printing Preferences in the window shown earlier in Figure 5.25, a window named based on the printer opens. Options in this window (Figure 5.33) will differ based on the installed printer. Here you can configure options such as dual-sided printing and color—of course the available options will depend on what the printer is capable of. Within Printing Preferences you can find configuration settings such as *orientation* (portrait or landscape), *duplex* (double sided), quality, color, and paper tray settings (if applicable) for the printer. Paper tray settings are useful when an MFD supplies multiple sizes of paper, such as letter-sized (8.5″ × 11″) paper and legal-sized (8.5″× 14″) paper in the United States. Or perhaps the MFD has a tray designated for the most common paper size in Europe (and the international standard for office documents), A4, which measures 210mm × 297mm, or 8.3″× 11.7″.

Printer configuration settings, including duplex, orientation, tray settings, and quality, are part of Exam Objective 3.7. Know where to find them!

FIGURE 5.33 Printer-specific options



Step 5: Print a Test Page

Once you have configured your printer, you are finished and can print a test page to test its output. Windows has a built-in function for doing just that—you saw links to do so in Figure 5.25 and Figure 5.28. Click the link or button, and Windows will send a test page to the printer. If the page prints, your printer is working. If not, double-check all your connections. If they appear to be in order, then skip ahead to Volume 1 Chapter 13, for troubleshooting tips.

Step 6: Verify Compatibility with Operating System and Applications

Once your printer is installed and you have printed a test page, everything else should work well, right? That's usually true, but it's good practice to verify compatibility with applications before you consider the device fully installed.

With printers, this process is rather straightforward. Open the application you're wondering about and print something. For example, open Microsoft Word, type in some gibberish (or open a real document, if you want), and print it. If you are running non-Microsoft applications (such as a computer-aided drafting program or accounting software) and have questions about their compatibility with the printer, try printing from those programs as well.

Step 7: Educate Users About Basic Functionality

Most users today know how to print, but not everyone knows how to install the right printer or how to print efficiently. This can be a significant issue in work environments.

Say your workplace has 10 different printers, and you just installed number 11. First, your company should use a naming process to identify the printers in a way that makes sense. Calling a printer HPLJ4 on a network does little to help users understand where that printer is in the building. Naming it after its physical location might make more sense.

After installing the printer, offer installation assistance to those who might want to use the device. Show users how to install the printer in Windows (or if printer installation is automated, let them know that they have a new printer and where it is). Also, let users know the various options available on that printer. Can it print double-sided? If so, you can save a lot of paper. Show users how to configure that. Is it a color printer? Do users really need color for rough drafts of documents or presentations? Show users how to print in black and white on a color printer to save the expensive color ink or toner cartridges.

On the printer we've used as an example in this chapter, most of the options involving print output are accessed by clicking Printing Preferences to open the printer-specific settings (shown earlier in Figure 5.33). Duplex (or Print On Both Sides) and Tray settings (or Paper Source) are on the Printing Shortcuts tab. Print Quality (Best, Normal, Draft) is on the Paper/Quality tab, and Orientation (Portrait or Landscape) is set on the Finishing tab. Collation, which enables you to select whether you want it to print pages in order (1, 2, 3. . . 1, 2, 3. . . and so on) or multiple copies of the same page at once (1, 1, 1. . . 2, 2, 2. . . and so forth), is set on the Advanced tab. Sometimes it just takes a bit of poking around to find out where to set the option the user needs.

For the A+ exam, you don't need to memorize which tabs these configuration settings are on. Know what the options do (particularly duplex, orientation, tray settings, and quality), and if you're presented with a screenshot or a series of screenshots, be able to click on the right option to enable the proper feature for the situation.

In Exercise 5.2, we'll step through the process of installing a USB printer in Windows 11.

EXERCISE 5.2

Installing a USB Printer in Windows 11

For this exercise, you will need the following items:

- A USB printer
- A USB printer cable
- The software driver that came with the printer (flash drive, DVD, or downloaded)
- A computer with a free USB port and the ability to access the driver

1. Turn on the computer.
2. Plug the printer into the wall AC outlet and turn it on.
3. Access the driver file to begin the installation process. Typically this is done by inserting the flash drive provided by the manufacturer, and then double-clicking the `setup.exe` or `install.exe` program on the drive. Or, if you downloaded it from the manufacturer's website, find the file and run it from the location where you saved it.
4. Follow the prompts in the installation program to install the driver.
5. Once the software has been installed, plug one end of the USB cable into the printer and the other end into the free USB port. Some installation programs will prompt you for this step.
6. Windows will automatically detect the new printer, install the driver, and configure it.

   Windows will display a balloon in the lower-right corner of the screen that says, "Your hardware is now installed and is ready to use." If Windows doesn't properly detect the printer, open the Add Printer wizard to begin the installation process again, and manually specify the location of the printer driver (such as the flash drive or download location).
7. Print a test page to see if the printer can communicate and print properly.

Real World Scenario

Which Printer Did That Go To?

One of the authors used to work at a satellite office in Salt Lake City for a company whose headquarters were in Houston. Because of printer problems, a new network printer had been installed, which had a different network name from the previous printer.

At the end of the month, one of the accountants printed her monthly reconciliation report, which typically ran about 400 pages. (A hard copy was required for regulatory reasons.)

Puzzled when it didn't come out of the printer, she printed it again. And again. And again. After the fourth failed attempt, and several hours later, she decided to ask someone in IT what the problem was.

It turns out that she had mapped (installed) the new network printer but had gotten a few letters wrong in the printer name. Instead of being at her office, all her print jobs were sent to a printer in the Houston office. And, of course, there were people in Houston trying to print similar reports and who just kept refilling the printer with paper because they didn't want to cut someone else's report off in the middle.

While this wasn't a catastrophic failure, it was annoying. She had unintentionally wasted three reams of paper, the associated toner, and hours of printer life. It wasn't a malicious act, and she was a literate computer user, but it's illustrative of the need to educate and help users with installing and configuring devices. Had the printer been mapped correctly the first time, the waste could have been avoided.

Installing and Sharing Networked Printers

The previous section was about installing a printer attached to your local computer. There are advantages to that approach, such as being able to manage and control your own printer, not to mention having a printer at your own desk. That doesn't happen often in the business world these days!

There are some big disadvantages as well. First, it means that all users who need to print to your device may need local user accounts on your computer, unless you are on a network domain. If so, you will need to manage security for these accounts and the printer. Second, your computer is the print server. The *print server* is the device that hosts the printer and processes the necessary printer commands. This can slow your system down. Third, because your computer is the print server, if for any reason it's turned off, no one will be able to print to that device.

Exam Objective 3.7 lists two subobjectives under public/shared devices. The first is printer share, which simply means a shared printer on the network. The second is print server. Be sure to understand the difference between the two for the exam.

There is another option, though. Instead of needing a specific computer to be the print server, why not make the print server part of the printer itself, or make it a separate network device that hosts the printers? That is exactly the principle behind network printing. Next, we will cover two types of network printing—local network printing and cloud printing—as well as talk about data privacy concerns with printing to public or shared printers.

Local Network Printing

The key to local network printing is that you are moving the print server from your computer to another location, accessible to other users on the network. Therefore, the print server needs a direct attachment to the network, via either a wired (RJ45) or wireless connection. You will find two major varieties of print servers. The first, called an *integrated print server*, is incorporated into the printer itself, and the second is a separate hardware print server. If you are using a stand-alone print server, the printers attach to the print server, either physically or logically. In most cases, if a printer is capable of connecting directly to a network, it has the capability to be its own print server.

Installing and using a networked printer is very similar to installing and using a local printer. You need to ensure that both devices are plugged in, turned on, and attached to the network (either with an RJ45 Ethernet connection or by using wireless). Probably the biggest difference is that when you install it, if it's not detected automatically, you need to tell your computer that you are adding a networked printer instead of a local printer. For example, in Windows 11, when you open the Add Printer utility, you choose from several network printer options, such as Add A Bluetooth, Wireless Or Other Network Discoverable Printer. From there, you will be asked to install the printer driver, just as if the printer were directly attached to your computer. Once it's installed, you use it just as you would use a local printer, including setting the configuration options that we looked at in earlier sections. Every computer on the local network should be able to see and add the printer in the same way.

The print server needs to have drivers available (installed) for all the types of clients that will connect to its printers. For example, if the network has macOS, Linux, and Windows clients, the server will need to have all three drivers. If not, users may not be able to install the printer properly and may not be able to print.

There are a few other ways that you can add shared networked printers: by using TCP, Bonjour, or AirPrint.

TCP Printing

Printers that are network-aware need IP addresses (every device on a network needs a unique address), so it makes sense that you can add a networked printer by using TCP/IP, also known as *TCP printing*. Exercise 5.3 walks you through the general process of installing a TCP printer, using Windows 11 as an example.

EXERCISE 5.3

Installing a TCP/IP Printer in Windows 11

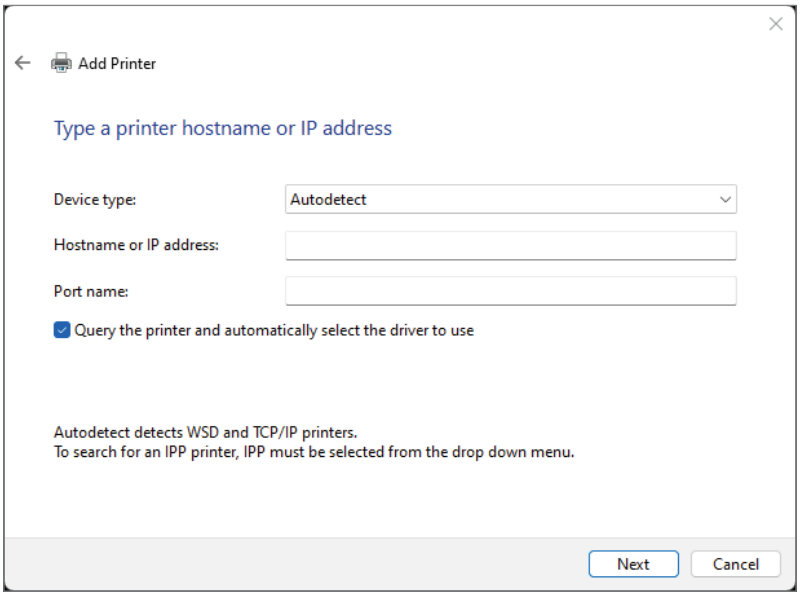
1. Connect the printer to the network and power it on.
2. Configure the printer with an IP address, if it does not already have one.

EXERCISE 5.3 *(continued)*

Most network-aware printers will have their own display panel where you can configure or verify network settings. The IP address needs to be on the same subnet as the computer trying to add the printer.

3. From your Windows 11 computer, start the Add Printer wizard. Open Printers & Scanners, then click Add Device. Even if it finds a printer, ignore that and click the Add Manually link. It will open the Add Printer wizard (shown earlier in Figure 5.24).
4. Select Add A Printer Using A TCP/IP Address Or Hostname, and then click Next.
5. On the next screen, shown in Figure 5.34, you have a few options. One is to let the system automatically detect the device. If so, set the Device Type as Autodetect and click Next. Otherwise, set Device Type to TCP/IP Device and move to the next step. For this exercise, let's do the latter.

FIGURE 5.34 Adding an IP printer



6. Enter the IP address (or hostname) of the printer that you want to add, add a port name (it's just for identification), and then click Next.
7. Select the make and model of your printer from the list.
8. You will be prompted to install the driver; continue installing the printer as you would any other local or networked printer.

Some installations will ask you which TCP printing protocol you want to use, RAW or LPR. RAW (also called the Standard TCP/IP Port Monitor) is the default, and it uses TCP port 9100. It also uses the Simple Network Management Protocol (SNMP) for bidirectional communication between the computer and the printer. LPR is older, and the protocol is included for use with legacy systems. It's limited to source ports 721–731 and the destination port 515.

We will discuss ports and port numbers in Volume 1 Chapter 7, "Introduction to TCP/IP."

After the printer is installed, it will appear in your Printers & Scanners window, just as any other printer would.

There are a few advantages to using TCP printing. First, it sends the print jobs directly to the printer, so your system does not need to act as the print server or spend processing time dealing with formatting the print job. Second, it allows clients with different OSs, such as Linux or macOS, to add printers without worrying about intra-OS conflicts.

Bonjour

Apple introduced *Bonjour* in 2002 (then under the name Rendezvous) as an implementation of zero configuration networking. It's designed to enable automatic discovery of devices and services on local networks using TCP/IP as well as to provide hostname resolution. Currently, it comes installed by default on Apple's macOS and iOS operating systems. Bonjour makes it easy to discover and install printers that have been shared by other Bonjour-enabled clients on the network.

Even though Apple developed Bonjour, it does work on other operating systems. For example, it comes with iTunes and the Safari browser, so if you have either of those installed on a Windows computer, odds are that you have Bonjour as well. Once installed, the Bonjour service starts automatically and scans the network looking for shared devices. Exercise 5.4 shows you how to see if Bonjour is installed in Windows.

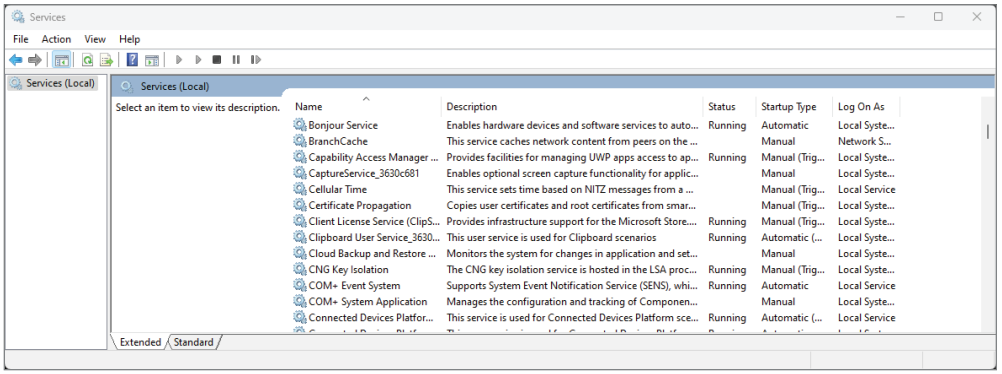
EXERCISE 5.4

Determining if Bonjour Is Installed in Windows

1. Type **services** into the Windows 11 search bar and then click Services when the system app appears. The Services window will open.
2. Sort the list by name and look for the Bonjour service, as shown in Figure 5.35.

EXERCISE 5.4 *(continued)*

FIGURE 5.35 Bonjour service is running and set to start automatically.



Bonjour works only on a single broadcast domain, meaning that it will not find a printer or other device if it's on the other side of a router from your computer. All major printer manufacturers support Bonjour technology.

If you are using macOS, adding a Bonjour printer is easy. You open System Preferences ➢ Print And Scan, click the plus sign under Printers to open the Add Printer window, and look for the printer on the list. If macOS doesn't have the driver available, you will be asked to provide it. Otherwise, you're done.

In order to add or share a Bonjour printer from Windows, if it doesn't have the service, you need to download Bonjour Print Services for Windows. It's found on Apple's support site at `https://support.apple.com/en-us/106380`.

AirPrint

The one big complaint that Apple aficionados had about Bonjour was that it didn't support printing from iPhones or iPads. In 2010, Apple introduced *AirPrint* to meet that need.

The idea behind AirPrint is quite simple. Mobile devices can automatically detect AirPrint-enabled printers on their local network and print to them without requiring the installation of a driver. To be fair, what Apple really did was eliminate the need for a specific printer driver to be installed on the client and replaced it with the AirPrint concept. Then it was up to the printer manufacturers to develop their own drivers that talked to AirPrint. HP was happy to oblige with its Photosmart Plus series, and other manufacturers soon followed. The list of AirPrint-enabled printers is available at `https://support.apple.com/en-us/HT201311`. From the end-user standpoint, though, no driver is required.

There really is no installation process, and printing is easy. Just be sure that your mobile device is on the same local network as an AirPrint printer. When you attempt to print from your device, select the printer to which you want to print, and it should work.

You can also purchase AirPrint servers, which are small print servers that enable you to connect almost any printer to the network and make them AirPrint-compatible.

Network Printer Security

When printing to a public printer, or one that is shared in a common workspace, there may be data privacy concerns. For example, employees in Human Resources (HR) might need to print confidential personnel files, or someone in the Mergers group might have a highly restricted contract to print. Let's take a look at some security options for networked printers.

User Authentication

Requiring *user authentication* means that users to log in (authenticate) to the printer is one step that can improve printer security. Not all printers have this capability, but most newer laser printers and MFDs designed for office use will. Figure 5.36 shows the front of a Xerox AltaLink MFD. In the center of the picture is a touchscreen display where a user can copy, scan to email, look at printer jobs, configure the device, and use a secure print feature called SafeQ. If a user taps the LogIn button in the upper-left corner, they will be presented with a keyboard so that they can enter their username and password. While functional, this is a bit old-school. An easier way for many users is to scan their work badge (this is called *badging*) on the badge reader at the left of the unit. Doing so will automatically log them in and provide access to secure printing features.

FIGURE 5.36 Xerox AltaLink badge scanner and touchscreen

Exam Objective 3.7 wants you to know four printer security options: user authentication, badging, secured prints, and audit logs.

Secured Prints

Printing a document is usually pretty straightforward. On the computer, the user hits some sort of Print button, chooses the printer to send it to, and presses Print again (or OK, or something similar). Then they get up and walk to the printer to retrieve the hard copy. Most office denizens have been well trained in this process.

Some print jobs might contain sensitive information, though, so the user wants to ensure that they are physically at the printer before it starts printing. Or, perhaps the user has printed to a device in a different building and wants the printer to wait to start printing until they can get there. In cases like these, a *secured prints* feature can be used to hold the print job until the user is ready for it.

Looking back at Figure 5.36, you can see a feature on the bottom center of the touchscreen called SAFEQ Print. YSoft SAFEQ is an industry-standard enterprise print management suite adopted by many organizations, and it works seamlessly with many printers. The administrator sets it up on the printer (and we mean logical printer here, not necessarily the physical device), and when the user prints to that printer, the job waits for them until they authenticate on the physical printer and tell it to start. Figure 5.37 shows the SAFEQ user authentication screen. Once logged in, the user will be presented with the job(s) to print, as shown in Figure 5.38.

FIGURE 5.37 SAFEQ authentication screen

Audit Logs

Being able to see who used (or perhaps abused) a printer after the fact can come in handy. Some printers have the ability to save a list of documents that have been printed as an *audit log*. The Xerox printer we've used as an example in this section does just that, and the log is shown in Figure 5.39.

FIGURE 5.38 Secured print job

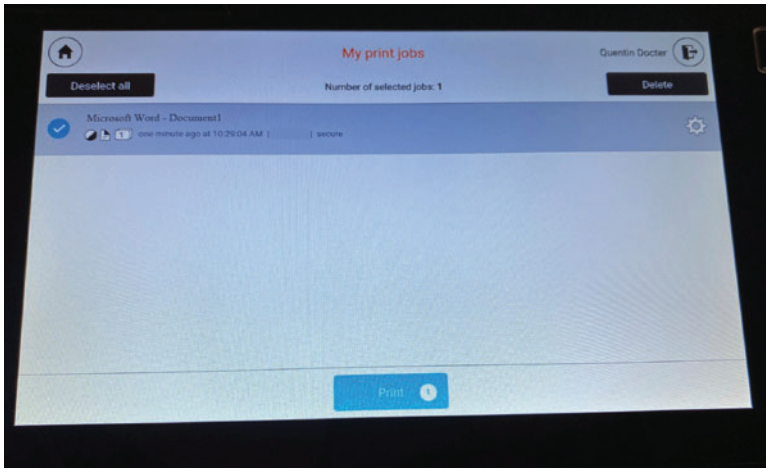
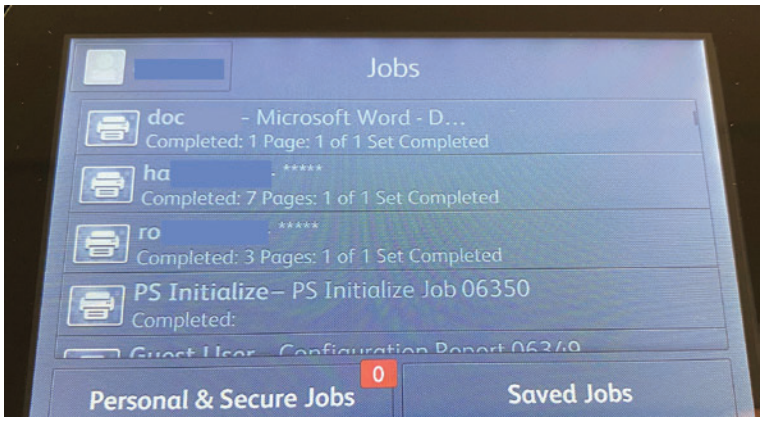


FIGURE 5.39 Xerox print log



Other printers will integrate logging software into the operating system's standard logging utilities. For example, some HP printers will install an audit log into Windows Event Viewer. (We will cover Event Viewer in Volume 2 Chapter 2, "Windows Configuration.") Third-party audit software is also available for use.

Another security issue is storing print jobs on print servers. Print jobs are cached on the print server (it's called *hard drive caching*), and they can be saved indefinitely. We looked at the check box to keep printed documents in Figure 5.31. If you are configuring a printer for a public area, do not select this option. If you're printing to a public printer (such as at

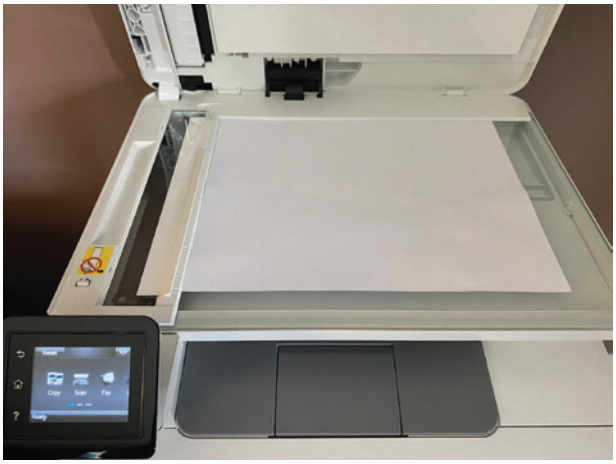
a hotel), and you can see the printer properties, check to make sure that this option is not selected. That will help ensure that your files will not be cached on the hotel computer's hard drive.

Network Scan Services

Scanning is really the opposite of printing. Instead of printing to get electronic information onto paper, scanning takes printed information and stores it electronically. In the office environment, the two most popular types of scanners are flatbed scanners and automatic document feeder (ADF) scanners.

To use a *flatbed scanner*, simply open the lid and lay the document to be scanned on the scanner glass, aligning it to the proper corner as shown in Figure 5.40. There will be an icon on the scanning bed that shows where the document should go. Close the lid and press Scan on the touchscreen, and scanning will begin.

FIGURE 5.40 A flatbed scanner



If you have more than one document to scan, using a flatbed scanner can be a real pain. Fortunately, some scanners have an attachment called an *automatic document feeder (ADF)* that lets you scan multiple pieces of paper in one job. Figure 5.41 shows an example with paper loaded in it. It's common to have ADFs that allow for up to 50 pages to be scanned at once.

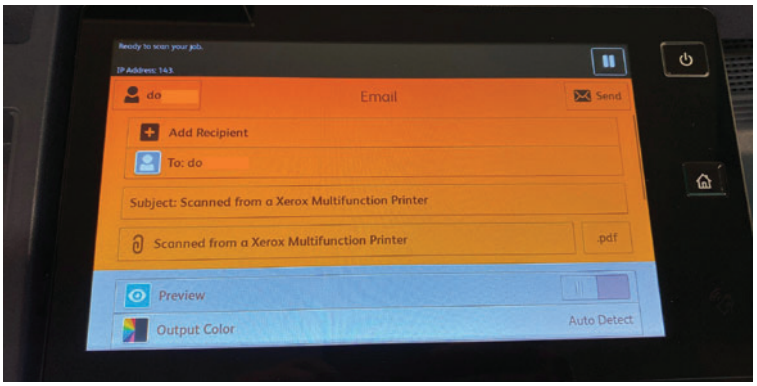
Automatic document feeder (ADF), flatbed scanner, and network scan services (email, SMB, cloud services) are part of exam objective 3.7. Understand how they are used.

FIGURE 5.41 Automatic document feeder (ADF) on an MFD

Whenever you scan a document, you need to figure out where to send it. MFDs don't usually have the memory to save images of scans—besides, the point of scanning a document is usually to email it to someone or save it on a hard drive for later retrieval. Let's take a look at three different ways to send or save scanned materials. For all three of these options, it's assumed that the scanner is connected to the network.

Scan to Email Using a *scan to email* option, the scanner will simply email the file, usually in PDF format, to an email recipient. This is probably the most common method that people use when scanning files. Figure 5.42 shows an example of setting up scan to email.

FIGURE 5.42 Scan to email



Scan to Folder An alternative option to emailing a scanned file is to save it in a network folder. This is a particularly viable solution if the scanned file is too large to be emailed.

The protocol the printer uses to transport the file from itself to the network folder is called *Server Message Block (SMB)*. In addition, the administrator must set up the MFD to support SMB scanning, and the recipient folder needs to be properly shared and secured too. Performing the scan from the MFD is done via a screen similar to the one shown in Figure 5.42, except instead of selecting an email recipient, you would navigate to the folder where you want to save the file.

Scan to Cloud The final option is to use *cloud services* to save the scan. Various file saving and sharing services support scanning to the cloud, such as Google Drive, OneDrive, Box, and Dropbox, among others. In addition, some printer manufacturers such as HP and Epson offer scan to cloud services. The exact process to follow depends on which cloud service you use to save the file. Exercise 5.5 has you "scan" a document to Google Drive—and it doesn't require you to have an actual scanner! Of course, you do need an Android device to make it work.

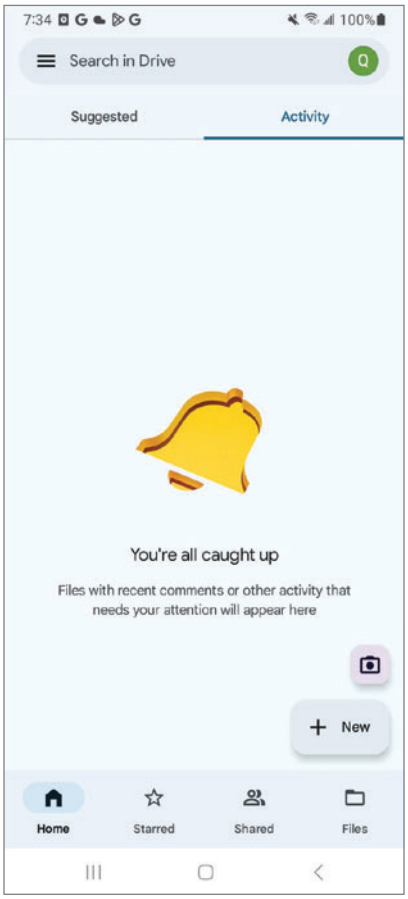
EXERCISE 5.5

Scanning a Document to Google Drive

This exercise requires the use of an Android phone or tablet.

1. On the Android device, open the Google Drive app.
2. In the bottom-right corner, tap New (like in Figure 5.43). It should have a plus sign next to it.

FIGURE 5.43 New button in Google Drive



3. Tap Scan. It should have a camera icon next to it.

You might need to give Drive permission to use the camera.

4. Take a photo of the object you want to scan.

5. Tap Done to save the scanned document to Google Drive.

To make the process even shorter, you can install the Drive scan widget on your Home screen.

Performing Appropriate Printer Maintenance

Considering the amount of work they do, printers last a pretty long time. Some printers handle over 100,000 pages per month, yet they're usually pretty reliable devices. You can help your printers live long and fulfilling lives by performing the right maintenance, and smoothly running printers always make your officemates happy. After all, going to get your print job from the printer and discovering that the printer is in the shop is a very frustrating experience!

Regardless of the type of printer you use, giving it a regular checkup is a good idea. You're probably familiar with some of the activities that fall under maintenance, such as replacing paper, ink, or toner cartridges. We'll look at those as well as some additional, more involved maintenance procedures.

Performing appropriate printer maintenance is a major part of Exam Objective 3.8. Be sure to understand how to do the following:

- **Laser:** Replace toner, apply maintenance kit, calibrate, and clean.
- **Inkjet:** Clean printheads, replace cartridges, calibrate, and clear jams.
- **Thermal:** Replace paper, clean heating element, and remove debris.
- **Impact:** Replace ribbon, printhead, and paper.

Replacing Consumables

To maintain a printer properly, you need to replace consumables such as toner or ink cartridges, assemblies, filters, and rollers on occasion. Trying to cut costs by buying cheaper supplies rarely pays off.

Whenever purchasing supplies for your printer, always get supplies from the manufacturer or from an authorized reseller. This way, you'll be sure that the parts are of high quality. Using unauthorized parts can damage your printer and possibly void your warranty.

Printer Paper

Most people don't give much thought to the kind of paper they use in their printers. It's a factor that can have a tremendous effect on the quality of the hard-copy printout, however, and the topic is more complex than people think. For example, the wrong paper can cause frequent paper jams and possibly even damage components.

Several aspects of paper can be measured; each gives an indication as to the paper's quality. The first factor is *composition*. Paper is made from a variety of substances. Paper used to be made from cotton and was called rag stock. It can also be made from wood pulp, which is cheaper. Most paper today is made from the latter or a combination of the two.

Another aspect of paper is the property known as *basis weight* (or simply *weight*). The weight of a particular type of paper is the actual weight, in pounds, of a ream (500 sheets) of the standard size of that paper made of that material. For regular bond paper, that size is 17 × 22.

The final paper property we'll discuss is the *caliper* (or thickness) of an individual sheet of paper. If the paper is too thick, it may jam in feed mechanisms that have several curves in the paper path. On the other hand, a paper that's too thin may not feed at all.

These are just three of the categories we use to judge the quality of paper. Because there are so many different types and brands of printers as well as paper, it would be impossible to give the specifications for the "perfect" paper. However, the documentation for any printer will give specifications for the paper that should be used in that printer.

Many impact printers need to use special paper that has tractor feed perforations on the side, or they will not work properly. When replacing tractor feed paper, it's very easy to get it misaligned, and it will feed crookedly and ultimately jam the printer. Similarly, thermal printers also require special paper that needs to be loaded properly. In many cases, if you load it upside down, the unit will not produce images. By comparison, adding paper to a laser or inkjet printer is usually very easy.

For best results, store paper in an area where it will not get wet or be exposed to excessive humidity.

Ink and Toner

The area in which using recommended supplies is the biggest concern is ink and toner cartridges. Using the wrong ink or toner supplies is the easiest way to ruin a perfectly good printer.

Dot-matrix printers use a cloth or polyester ribbon soaked in ink and coiled up inside a plastic case. This assembly is called a *printer ribbon* (or *ribbon cartridge*). Once the ribbon has run out of ink, it must be discarded and replaced. Ribbon cartridges are developed closely with their respective printers. For this reason, ribbons should be purchased from the same manufacturer as the printer. The wrong ribbon could jam in the printer as well as cause quality problems.

It is possible to re-ink a ribbon. Some vendors sell a bottle of ink solution that can be poured into the plastic casing, where the cloth ribbon will soak up the solution. This can be a messy process, and you should do this only if the manufacturer recommends it.

Inkjet cartridges have a liquid ink reservoir. The ink in these cartridges is sealed inside. Once the ink runs out, the cartridge must be removed and discarded. A new, full cartridge is installed in its place. Because the ink cartridge contains the printing mechanism as well as ink, it's like getting a new printer every time you replace the ink cartridge.

In some inkjet printers, the ink cartridge and the printhead are in separate assemblies. This way, the ink can be replaced when it runs out, and the printhead can be used several times. This approach works fine if the printer is designed to work this way. However, some people think that they can do this on their integrated cartridge/printhead system, using special ink cartridge refill kits. These kits consist of a syringe filled with ink and a long needle. The needle is used to puncture the top of an empty ink cartridge, and the syringe is then used to refill the reservoir.

Do not use ink cartridge refill kits! There are several problems with these kits (the ones you see advertised with a syringe and a needle). First, the kits don't use the same kind of ink that was originally in the ink cartridges. The new ink may be thinner, causing it to run out or not print properly. Also, the printhead is often supposed to be replaced around this same time. Refilling the cartridge doesn't replace the printhead, so you'll have print-quality problems. Finally, the hole the syringe leaves cannot be plugged and may allow ink to leak out. These problems can happen with do-it-yourself kits as well as with cartridges refilled by office supply stores or private printer supply sellers. Here's the bottom line: *Buy new ink cartridges from the printer manufacturer*. Yes, they are a bit more expensive, but in the long run you will save money because you won't have any of the problems described here.

The final type of consumable is toner. Each model of laser printer uses a specific toner cartridge. You should check the printer's manual to see which toner cartridge your printer needs. Many businesses will recycle your toner or ink cartridges for you, refill them, and sell them back to you at a discount. Don't buy them. While some businesses that perform this "service" are more legitimate than others, using recycled parts is more dangerous to your hardware than using new parts. The reason for this is that refilled cartridges are more likely to break or leak than new parts, and this leakage could cause extensive damage to the inside of your printer. And again, using secondhand parts can void your warranty, so you're left with a broken printer that you have to pay for. Avoid problems like this by buying new parts.

Real World Scenario

Think Before You Refill

Just as with ink cartridges, you should always buy the exact model of toner cartridge recommended by the manufacturer. The toner cartridges have been designed specifically for a particular model. Additionally, *never* refill toner cartridges, for most of the same reasons we don't recommend refilling ink cartridges. The printout quality will be poor, and the fact that you're just refilling the toner means that you might *not* be replacing the photosensitive drum (which is usually inside the cartridge), and the drum might *need* to be replaced.

Simply replacing refilled toner cartridges with proper, name-brand toner cartridges has solved most laser printer quality problems that we have run across. We keep recommending the right ones, but clients keep coming back with the refilled ones. The result is that we take our clients' money to solve their print-quality problems when all it involves is a toner cartridge, our (usually repeat) advice to buy the proper cartridge next time, and the obligatory minimum charge for a half hour of labor (even though the job of replacing the cartridge takes all of 5 minutes).

Always properly recycle your used ink and toner cartridges. Just don't buy recycled cartridges!

Performing Scheduled Maintenance

When shopping for a printer, one of the characteristics you should look for is the printer's capacity, which is often quoted in monthly volume. This is particularly important if the printer will be serving in a high-load capacity. Every printer needs periodic maintenance, but printers that can handle a lot of traffic typically need it less frequently. Check the printer specifications to see how often scheduled maintenance is suggested. Never, ever fall behind on performing scheduled maintenance on a printer.

Many laser printers have LCD displays that provide useful information, such as error messages or notices that you need to replace a toner cartridge. The LCD display will also tell you when the printer needs scheduled maintenance. How does it know? Printers keep track of the number of pages they print, and when the page limit is reached, they display a message, usually something simple like *Perform user maintenance*. The printer will still print, but you should perform the maintenance.

Being the astute technician that you are, you clean the printer with the recommended cleaning kit or install the maintenance kit that you purchased from the manufacturer. Now, how do you get the maintenance message to go away? Reset the page count using a menu option. For example, on many HP laser printers, you press the Menu button until you get to the Configuration menu. Once there, you press the Item key until the display shows *Service Message = ON*. Then press the plus key (+) to change the message to *Service Message = OFF*. Bring the printer back online, and you're ready to go.

When performing maintenance on an impact printer, always carefully inspect the printhead for damage. Replace damaged printheads with authorized parts from the manufacturer.

Performing routine maintenance will keep the printer clean, make it last longer, and help prevent annoying paper jams.

Using Cleaning Solutions

With all of the ink or toner they use, printers get dirty. If printers get too dirty or if the printheads get dirty, you'll notice print problems. No one wants this to happen.

Most printers have a self-cleaning utility that is activated through a menu option or by pressing a combination of buttons on the printer itself. It's recommended that you run the cleaning sequence every time you replace the toner or ink cartridges. If you experience print-quality problems, such as lines in the output, run the cleaning routine.

Sometimes, the self-cleaning routines aren't enough to clear up the problem. If you are having print-quality issues, you might want to consider purchasing a cleaning or maintenance kit, which frequently comes with a cleaning solution.

Cleaning kits are often designed for one specific type of printer and should be used only on that type of printer. For example, don't apply an inkjet cleaning solution to a laser printer.

Each cleaning kit comes with its own instructions for use. Exercise 5.6 walks you through the steps of using an inkjet cleaning solution. Note that the steps for your printer might differ slightly; please consult your manual for specific instructions. After using a cleaning kit on a laser or inkjet printer, it's best to perform a calibration per the printer's instructions.

EXERCISE 5.6

Using an Inkjet Cleaning Solution

1. Power on the printer, and open the top cover to expose the area containing the print cartridges.
2. Initiate a self-cleaning cycle. When the printhead moves from its resting place, pull the AC power plug. This lets you freely move the printheads without damaging them.
3. Locate the sponge pads on which to apply the cleaning solution.

   They'll be in the area where the printheads normally park.
4. Use a cotton swab or paper towel to gently soak up any excess ink in the pads.
5. Using the supplied syringe, apply the cleaning solution to the sponge pads until they are saturated.
6. Plug the printer back into the wall outlet, and turn it on. The printheads will park themselves.
7. Turn the printer back off. Let the solution sit for at least 3 hours.
8. Power the printer back on, and run three printer cleaning cycles.
9. Print a nozzle check pattern (or a test page) after each cleaning cycle to monitor the cleaning progress.

That should take care of it. If not, refer to your printer's manual for more instructions.

Thermal printers require special attention because they contain a heating element. Always unplug the device and ensure that it's cooled off before trying to clean it. Thermal printer cleaning cards, cleaning pens, and kits are widely available in the marketplace. If you need to remove any debris (from any printer), use compressed air or a specialized computer vacuum.

When using compressed air, take care to not just blow the dust or debris all over the place or onto other components. Have a plan in place to contain it all before squeezing the trigger!

Ensuring a Suitable Environment

Printers won't complain if the weather outside is too hot or too cold, but they are susceptible to environmental issues. Here are some things to watch out for in your printer's environment:

Heat Laser printers and thermal printers can generate a lot of heat. Because of this, ensure that your printer is in a well-ventilated area. Resist the temptation to put the printer in the little cubbyhole in your desk. Overheating will reduce the shelf life of your printer.

Humidity High humidity can cause printer paper to stick together. Sticky paper leads to paper jams. Humidity over 80 or 90 percent can cause issues.

Light The laser printer's toner cartridge contains a photosensitive drum. Exposing that drum to light could ruin the drum. While the drum is encased in plastic, it's best to avoid exposing the printer or toner cartridges to extreme light sources. Under no circumstance should you open the toner cartridge, unless you're ready to get rid of it as well as clean up a big mess.

Ozone Laser printers that use corona wires produce ozone as a by-product of the printing process. In offices, ozone can cause respiratory problems in small concentrations, and it can be seriously dangerous to people in large amounts. Ozone is also a very effective oxidizer and can cause damage to printer components.

Fortunately, laser printers don't produce large amounts of ozone, and most laser printers have an ozone filter. Ozone is another reason to ensure that your printer area has good ventilation. Also, replace the ozone filter periodically; check your printer's manual for recommendations on when to do this.

Ammonia A printer doesn't produce ammonia, but it is contained in many cleaning products. Ammonia can greatly reduce the printer's ability to neutralize ozone and can cause permanent damage to toner cartridges. It's best to avoid using ammonia-based cleaners near laser printers.

Summary

In this chapter, we discussed how different types of printers and multifunction devices work as well as the most common methods of connecting them to computers. You learned how computers use page-description languages to format data before they send it to printers and drivers to talk to them. You also learned about the various types of consumable supplies and how they relate to each type of printer.

The most basic category of printer currently in use is the impact printer. Impact printers form images by striking something against a ribbon, which in turn makes a mark on the paper. You learned how these printers work and the service concepts associated with them.

One of the most popular types of printer today is the inkjet printer, so named because of the mechanism used to put ink on the paper.

The most complex type of printer is the laser printer. You learned about the steps in the electrophotographic (EP) imaging process, the process that explains how laser printers print. We also explained the various components that make up this printer and how they work together.

The last printer type we looked at was thermal, which uses special heat sensitive thermal paper that's heat sensitive to create output. Be sure to store the special thermal paper in a cool place or it could get ruined.

You then learned about the interfaces used to connect printers to PCs and how to install and share a printer. Common interfaces include USB, Ethernet, and wireless. Proper steps include connecting the device, installing the driver, configuring options, validating application and operating system compatibility, and educating users on how to use the device. Installing the device is the first step, but you're not done until you ensure that it works properly and that users know how to access it.

Installing network printers usually involves a few more steps than are needed to install local printers, and the device is connected to the network instead of to a host. Networked MFDs are often used for scan services, such as scanning to email, SMB, and the cloud. Security becomes critical here as well, so you should be familiar with user authentication, badging, secured prints, and audit logs.

Finally, we looked at how to perform printer maintenance, including the importance of using recommended supplies.

Exam Essentials

Know the differences between types of printer technologies (for example, laser, inkjet, thermal, impact). Laser printers use a laser and toner to create the page. Inkjet printers spray ink onto the page. Thermal printers use heat to form the characters on the page. Impact printers use a mechanical device to strike a ribbon, thus forming an image on the page.

Know the three most common ways to connect a printer. The methods are USB, Ethernet, and wireless.

Be familiar with printer configuration settings. Know how duplex, orientation, tray settings, and print quality are configured.

For networked printers, understand security and scan services options. Security can include user authentication and badging, audit logs, and secured prints. Network scan services include scan to email, scan to a folder (using the SMB protocol), and scan to cloud.

Understand what print servers and printer shares are. A shared printer is one that's made available to users across the network. The print server is the computer or system that hosts the shared printer.

Know how to install and configure printers. The basic procedure is as follows:

1. Find a suitable location for the printer and properly unbox the device.
2. Attach the device using a local or network port and connect the power.
3. Install and update the device driver and calibrate the device.
4. Configure options and default settings.
5. Print a test page.
6. Verify compatibility with the operating system and applications.
7. Educate users about basic functionality.

Know the seven steps in the laser imaging process. The seven steps are processing, charging, exposing, developing, transferring, fusing, and cleaning.

Know the key parts in a laser printer and appropriate maintenance procedures. Key parts are the imaging drum, fuser assembly, transfer belt, transfer roller, pickup rollers, separation pads, and duplexing assembly. Maintenance includes replacing toner, applying a maintenance kit, calibrating, and cleaning.

Know the key parts in an inkjet printer and appropriate maintenance procedures. Inkjet parts include the ink cartridge, printhead, roller, feeder, duplexing assembly, and carriage belt. Maintenance items include cleaning printheads, replacing cartridges, calibrating, and clearing paper jams.

Know the key components in a thermal printer and appropriate maintenance procedures. The feed assembly and heating element are important thermal printer parts. The paper is also important here because it's special heat-sensitive thermal paper. Maintenance includes replacing paper, cleaning the heating element, and removing debris.

Know the key parts in an impact printer and appropriate maintenance procedures. Impact printer parts to know include the printhead, ribbon, tractor feed (and special tractor feed paper), and impact or multipart paper. Maintenance includes replacing the ribbon, printhead, and paper.

Understand the importance of using recommended supplies. Using consumables (paper, ink, toner) that are recommended for your printer is important. Using bad supplies could ruin your printer and void your warranty.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. A user has a printer attached to their computer. They configured the printer so other users on the network can print to it. What does that make their computer?
   - **A.** Print server
   - **B.** Printer share
   - **C.** Network scan service
   - **D.** ADF

2. Which types of printers are referred to as page printers because they receive their print job instructions one page at a time? (Choose two.)
   - **A.** Impact
   - **B.** Inkjet
   - **C.** Laser
   - **D.** Thermal

3. Which of the following is not an advantage of a Universal Serial Bus (USB) printer interface?
   - **A.** It has a higher transfer rate than a serial connection.
   - **B.** It has a higher transfer rate than a parallel connection.
   - **C.** It automatically recognizes new devices.
   - **D.** It allows the printer to communicate with networks, servers, and workstations.

4. You have a finance manager who needs to print contracts in triplicate. Which type of printers are best for printing on multipart forms?
   - **A.** Inkjet printers
   - **B.** Laser printers
   - **C.** Thermal printers
   - **D.** Dot-matrix printers

5. Which step in the EP imaging process uses a laser to discharge selected areas of the photosensitive drum, thus forming an image on the drum?
   - **A.** Exposing
   - **B.** Transferring
   - **C.** Developing
   - **D.** Cleaning

6. Which of the following is a type of device that lets users scan in 10-page double-sided documents all at once?
   - **A.** SMB
   - **B.** MFD
   - **C.** ADF
   - **D.** Flatbed

7. Recently, employees have printed off unauthorized materials which may contain prohibited hate speech. Which of the following can be used to see which users printed specific jobs?
   - **A.** Audit logs
   - **B.** Badging
   - **C.** User authentication
   - **D.** Secured prints

8. Which device in an inkjet printer contains the printhead?
   - **A.** Toner cartridge
   - **B.** Ink cartridge
   - **C.** Daisy wheel
   - **D.** Paper tray

9. What is the correct order of the steps in the EP imaging process?
   - **A.** Developing, exposing, transferring, fusing, charging, cleaning, processing
   - **B.** Charging, processing, exposing, developing, transferring, fusing, cleaning
   - **C.** Processing, transferring, exposing, developing, charging, cleaning, fusing
   - **D.** Processing, charging, exposing, developing, transferring, fusing, cleaning

10. The network administrator has set up security on office MFDs that requires users to authenticate before printing. Which of the following are valid methods for users to authenticate to the MFD? (Choose two.)
    - **A.** Use an authentication log.
    - **B.** Enable secured print.
    - **C.** Type their username and password on the MFD's touchscreen.
    - **D.** Use SMB.
    - **E.** Use their company ID for badging.

11. What is typically included in the EP laser printer toner cartridge? (Choose three.)
    - **A.** Toner
    - **B.** Print drum
    - **C.** Laser
    - **D.** Cleaning blade

12. What happens during the developing stage of laser printing?
    - **A.** An electrostatic charge is applied to the drum to attract toner particles.
    - **B.** Heat is applied to the paper to melt the toner.
    - **C.** The laser creates an image of the page on the drum.
    - **D.** An electrostatic charge is applied to the paper to attract toner particles.

13. Which of the following are possible interfaces for printers? (Choose three.)

A. Parallel

B. SATA

C. USB

D. Network

14. You have just installed a new printer, but it prints only garbled text. Which of the following is most likely the problem?

A. Wrong IP address

B. Worn printhead

C. Incorrect printer drivers

D. Unsupported printer

15. Which printer contains a wheel that looks like a flower with raised letters and symbols on each "petal"?

A. Inkjet printer

B. Daisy-wheel printer

C. Dot-matrix printer

D. Laser printer

16. What part of a laser printer supplies the voltages for charging and transferring corona assemblies?

A. High-voltage power supply (HVPS)

B. DC power supply (DCPS)

C. Controller circuitry

D. Transfer corona

17. Which printer part gets the toner from the photosensitive drum onto the paper?

A. Laser-scanning assembly

B. Fusing assembly

C. Corona assembly

D. Drum

18. Which step in the laser printer printing process occurs immediately after the exposing phase?

A. Charging

B. Fusing

C. Transferring

D. Developing

19. Which laser printer component permanently presses the toner into the paper?

- **A.** Transfer corona
- **B.** Fuser assembly
- **C.** Printer controller circuitry
- **D.** Paper transport assembly

20. Which of the following most accurately describes how to obtain a firmware upgrade for your laser printer?

- **A.** Download the firmware upgrade for free from the manufacturer's website.
- **B.** Pay to download the firmware upgrade from the manufacturer's website.
- **C.** Have a certified laser printer technician come to your site and install a new firmware chip.
- **D.** Contact the manufacturer of the printer, and they will send you the firmware upgrade on an SSD.